A Course in PDE

Mark Williams

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Introduction

We begin by giving an informal overview of some of the material we'll cover; in the process we'll introduce some notation.

We can write a quite general PDE for an unknown $u : \mathbb{R}^n_x \to \mathbb{R}$ in the form[1]

$$F(D^k u(x), D^{k-1}u(x), \dots, Du(x), u, x) = 0,$$

where F takes values in $\mathbb{R}$. Here $D^k u$ represents the vector of all partials $\partial^\alpha u$ with $|\alpha| = k$, where $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ is a multi-index, and[2]

$$\partial^\alpha u = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} u.$$

To obtain a *system* of q scalar equations in p scalar unknowns, we can use the same formula but now let F take values in $\mathbb{R}^q$ and $u : \mathbb{R}^n \to \mathbb{R}^p$. When F is linear in $(D^k u, D^{k-1}, \dots, u)$, we say the PDE is linear. One can allow u or x to be complex-valued, or to take values in more general manifolds. In this course we'll study mostly linear PDEs on subsets of $\mathbb{R}^n_x$.

The general scalar linear partial differential operator of order $m \in \mathbb{N}_0$ acting on functions of $x \in \mathbb{R}^n$ takes the form

$$P(x, \partial) = \sum_{|\alpha| \le m} c_\alpha(x) \partial^\alpha \text{ where } c_\alpha : \mathbb{R}^n \to \mathbb{R}. \tag{0.1}$$

The *symbol* $p(x, \xi)$ and *principal symbol* $p_m(x, \xi)$ are given respectively by

$$p(x, \xi) = \sum_{|\alpha| \le m} c_\alpha(x) \xi^\alpha \text{ and } p_m(x, \xi) = \sum_{|\alpha| = m} c_\alpha(x) \xi^\alpha. \tag{0.2}$$

A partial classification of PDEs can be given based on properties of the symbol or principal symbol. Consider for example the classical equations:

(a) $\Delta u = 0$ (Laplace's equation), where $\Delta = \partial_1^2 + \cdots + \partial_n^2$, $u = u(x)$, $x \in \mathbb{R}^n$;
(b) $u_t - \Delta u = 0$ (heat equation), $u = u(x, t)$, $(x, t) \in \mathbb{R}^n \times \mathbb{R}$;
(c) $\Box u := u_{tt} - \Delta u = 0$ (wave equation), $u = u(x, t)$, $(x, t) \in \mathbb{R}^n \times \mathbb{R}$.[3]

[1] We write $\mathbb{R}^n_x$ to indicate that we denote the variable point in $\mathbb{R}^n$ by x.
[2] Here $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and we write ∂_j for ∂_{x_j}.
[3] The abuse of notation "$u = u(x)$" is often used to indicate that the function u has independent variable x.

The gravitational potential $c/|x|$ satisfies Laplace's equation in $\mathbb{R}^3 \setminus \{0\}$ when $n = 3$; the temperature $u(x,t)$ at position x and time t in a conducting solid satisfies the heat equation when $n = 3$; and the displacement $u(x,t)$ of a vibrating drumhead at position x and time t satisfies the wave equation when $n = 2$.[4]

Without attempting to give careful definitions right now, we note that Laplace's equation is classified as *elliptic*, since the principal symbol $p_2(\xi)$ is *nonzero* for all $\xi \neq 0$. More generally, an operator $P(x,\partial)$ with principal symbol $p_m(x,\xi)$ is elliptic in $\mathcal{O} \subset \mathbb{R}^n$ if $p_m(x,\xi) \neq 0$ for all $x \in \mathcal{O}$, $\xi \in \mathbb{R}^n \setminus \{0\}$.

The wave equation is classified as *hyperbolic* because the equation $p_2(\xi,\tau) = \tau^2 - |\xi|^2 = 0$ has two distinct roots $\tau = \pm|\xi|$ for each $\xi \neq 0$. The heat equation is classified as *parabolic* (for $t \geq 0$), since the principal symbol has the form $\tau - h(\xi)$, where $h(\xi)$ is homogeneous of degree two in $\xi \neq 0$ and satisfies $h(\xi) \geq C|\xi|^2$ for some $C > 0$. [5]

Another important classical equation is

(d) $u_t - i\Delta u = 0$ (Schrödinger's equation), where now $u(x,t)$ must be complex-valued. The function $|u(x,t)|^2$ gives the probability density for finding a free particle near position x at time t when $n = 3$.

The equations (b), (c), and (d) have a distinguished variable denoted t, which we often think of as representing time. Thus, we call these equations "evolution" equations.

A large part of linear PDE is devoted to understanding properties of solutions like existence, uniqueness, and regularity (or smoothness). As we should expect from the study of ODEs, uniqueness generally holds only when some extra boundary or initial conditions are imposed.

We will see that, unlike most PDEs, solutions to the classical equations (a)-(d) can be written down rather explicitly, and the questions of existence, uniqueness, and regularity can be addressed rather directly.[6] For more general elliptic, hyperbolic, or parabolic problems "a priori estimates" and functional analysis arguments are often used to study these questions.[7]

A fundamental problem is to understand how properties of solutions of $P(x,\partial)u = 0$ are related to algebraic properties of the symbol or principal symbol of P. Let's focus on *regularity* for now. It turns out that when $P(x,\partial)$ is elliptic, solutions of $P(x,\partial)u = 0$ in an open set $\mathcal{O}$ necessarily lie in $C^\infty(\mathcal{O})$! When $n = 2$ and $P(x,\partial) = \Delta$, you can prove this easily using complex analysis:

Exercise A: Let $\mathcal{O} \subset \mathbb{R}^2$ be open and assume $u \in C^2(\mathcal{O},\mathbb{R})$ satisfies $\Delta u = 0$ in $\mathcal{O}$. Show that $u \in C^\infty(\mathcal{O},\mathbb{R})$.

[4] More precisely, the solutions of these equations *approximate* the corresponding physical quantities.

[5] As in the elliptic case, one can define more general classes of parabolic and hyperbolic operators $P(x,\partial)$ using properties of $p_m(x,\xi)$ that generalize those of $\tau - |\xi|^2$ and $\tau^2 - |\xi|^2$.

[6] Sometimes we must settle for a less then fully explicit formula that involves an unevaluated integral, for example.

[7] Examples of a priori estimates are given sections 2.2.1 and 3.2.

Later we'll prove an analogue of this result by very different methods for general elliptic operators on $\mathcal{O} \subset \mathbb{R}^n$.

This kind of regularity result fails for hyperbolic operators. Consider the wave equation for $n = 1$:

$$\Box u := u_{tt} - u_{xx} = (\partial_t + \partial_x)(\partial_t - \partial_x)u = 0. \tag{0.3}$$

Observe that if $f : \mathbb{R} \to \mathbb{R}$ lies in $C^2(\mathbb{R}) \setminus C^3(\mathbb{R})$, then $u(x,t) = f(x-t)$ is a solution which lies in $C^2(\mathbb{R}^2) \setminus C^3(\mathbb{R}^2)$. Moreover, if $f(x)$ fails to be C^3 at $x = 0$, then $u(x,t)$ fails to be C^3 in x at $x = t$. In other words, a "C^3-singularity" propagates to the right at speed one as t increases. Similarly, if $f(x)$ has support in $(-1,1)$, then $u(x,t)$ has $x-$support in $(t-1, t+1)$; its $x-$support propagates to the right at speed one as t increases.

Exercise B. Show that the general C^2 solution of (0.3) has the form

$$u(x,t) = f(x-t) + g(x+t), \text{ where } f, g \in C^2(\mathbb{R}).$$

Hint: Change variables: $u = t + x$, $v = t - x$ and use calculus.

The fact that features of the solution to the wave equation like support or singularities propagate at a finite speed justifies our describing these solutions as "waves". This fact allows us to use waves to transmit information. Nothing of the sort is exhibited by solutions of elliptic equations; recall the solution $u(x) = c/|x|$ of $\Delta u = 0$.

Observe that the *characteristic set* of $\Box$ (0.3), defined as $\mathcal{C} := \{(\xi, \tau) : p_2(\xi, \tau) = 0\}$, is

$$\mathcal{C} = \{(\xi, \tau) : \tau = \xi\} \cup \{(\xi, \tau) : \tau = -\xi\}.$$

It is no coincidence that the slopes of these lines in (ξ, τ) space are equal to the propagation speeds observed in the solutions $f(x-t)$ or $g(x+t)$. Later we will explain this in a more general setting. We will also see that solutions of the heat equation exhibit infinite propagation speed; in that case $\mathcal{C} = \{(\xi, \tau) : -|\xi|^2 = 0\} = \{(\tau, 0) : \tau \in \mathbb{R}\}$.

It is often desirable to make sense of derivatives of functions that are not differentiable in the classical sense. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous except for a jump discontinuity at $x = 0$. Then $u(x,t) = f(x-t)$ describes a jump discontinuity propagating to the right at speed one as t increases. Moreover, we can make sense of u as a solution to $\Box u = 0$ on $\mathbb{R}^2$ in a *weak (or distributional) sense* as follows. First note that when $f \in C^2(\mathbb{R})$ and $\phi \in C_c^2(\mathbb{R}^2)$ is *any* C^2 function with compact support, setting $u(x,t) := f(x-t)$ and integrating by parts twice in both x and t gives

$$0 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\Box u)(x,t)\ \phi(x,t) dx dt = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x-t)(\Box \phi)(x,t) dx dt. \tag{0.4}$$

When f is merely continuous, the right side still makes sense for any $\phi \in C_c^2(\mathbb{R}^2)$, even though the integrations by parts are no longer justified. We now take the vanishing of the right side

for *all* $\phi \in C_c^2(\mathbb{R}^2)$, a quite strong condition, to *define* what it means for $u(x,t) = f(x-t)$ to satisfy $\Box u = 0$ in the weak sense.

Exercise C. Show that when $f \in C(\mathbb{R}) \setminus C^1(\mathbb{R})$, the function $u(x,t) = f(x-t)$ satisfies $\Box u = 0$ in the weak sense. Hint: use an approximation argument.

Note that even though u in Exercise C is only a weak solution, it has a clear "physical" interpretation as a wave capable of transmitting information.

Notations 0.1.

1. Let $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.
2. We write $\mathbb{R}^n_x$ to indicate that we denote the variable point in $\mathbb{R}^n$ by x.
3. When $x \in \mathbb{R}^n$ and $u = u(x)$, we denote by $D^k u$ the vector of all partial derivatives $\partial^\alpha f$ where $|\alpha| = k$. We also write $\partial_{x_j} u = \partial_j u$.
4. Unless otherwise stated $\Omega \subset \mathbb{R}^n$ and $U \subset \mathbb{R}^n$ will always denote *open* sets and $I \subset \mathbb{R}$ will denote some open interval.
5. We write $\Omega \subset\subset \mathbb{R}^n$ to indicate that $\overline{\Omega}$ is compact.
6. For $\Omega \subset \mathbb{R}^n$ we write

$$\begin{aligned} C^k(\Omega, \mathbb{R}^m) &= \{f : \Omega \to \mathbb{R}^m : \partial^\alpha f \text{ is continous for all } |\alpha| \le k\}; \\ C_c^k(\Omega, \mathbb{R}^m) &= \{f \in C^k(\Omega, \mathbb{R}^m) \text{ with supp} f \textit{ compact}\}. \end{aligned} \tag{0.5}$$

When $m = 1$ we'll often write $C^k(\Omega)$ or $C_c^k(\Omega)$.

7. When $u = u(x,t)$ we write ∇u for $\nabla_x u$.
8. If $p \in \mathbb{R}^n$ and $r > 0$, we set $B(p,r) = \{x \in \mathbb{R}^n : |x-p| < r\}$.
9. Let $x \in \mathbb{R}^n$. We use dS_r or $dS_r(y)$ to represent the standard surface measure on $\{y \in \mathbb{R}^n : |x-y| = r\}$; see (8.5).
10. We say $f \in L^p_{loc}(U)$ if $f \in L^p(K)$ for all compact sets $K \subset U$. Unless otherwise indicated, we always take Lebesgue measure m on U.
11. The symbol $d'x$ occurs sometimes in integrals. If the integral is on a subset of $\mathbb{R}^n$, then $d'x := (2\pi)^{-n/2} dx$, where dx is Lebesgue measure. Using $d'x$ helps to simplify formulas involving the Fourier transform.
12. If $T : V \to W$ is any linear transformation, we let $N(T)$ and $R(T)$ denote the null space and range of T.
13. In estimates C (resp. C_ϵ) denotes a constant (resp. constant depending on ϵ) that may change from line to line or term to term.

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1 Classical examples of PDEs

1.1 Transport equations

Perhaps the simplest type of PDE is the transport equation $Xu = f$, where X is a vector field. In this section we use the *method of characteristics* to solve initial value problems for transport equations.

Let $\Omega \subset \mathbb{R}^n$. We can think of a vector field on Ω either as a map $X : \Omega \to \mathbb{R}^n$, where $X = (a_1, \dots, a_n)$, or as a partial differential operator $X = a_1\partial_1 + \cdots + a_n\partial_n$.

An *integral curve* (or "characteristic curve") of X is a curve $\gamma : I \to \Omega$ such that

$$\dot{\gamma}(s) = X(\gamma(s)). \tag{1.1}$$

The equation (1.1) is an $n \times n$ system of in general nonlinear ODEs for the unknown γ. When X is C^1, the basic existence and uniqueness theorem for ODEs, Theorem 8.1, tells us that for any $c \in \mathbb{R}^n$, there exists a unique solution of (1.1) $\gamma \in C^1(I, \mathbb{R}^n)$ satisfying $\gamma(0) = c$ on some open interval $I \ni 0$.

Let's introduce a distinguished "time" variable t and consider vector fields of the form

$$X = \partial_t + a_1\partial_1 + \cdots + a_n\partial_n, \text{ where } a_i \in C^1(\mathbb{R}^n_x \times [0,\infty)_t).$$

The method of characteristics uses integral curves of X to study solutions of

$$\begin{aligned} &(a) Xu = f \in C^1(\mathbb{R}^n_x \times [0,\infty)_t) \\ &(b) u|_{t=0} = g \in C^1(\mathbb{R}^n). \end{aligned} \tag{1.2}$$

for a given "forcing term" f and boundary datum g. To make clear the main ideas let's first study the case where X is a constant vector field, that is,

$$X = \partial_t + a \cdot \nabla, \quad \text{where } a \in \mathbb{R}^n.$$

In this case the equations (1.1) with initial condition $\gamma(0) = (x,t)$ are easily integrated to give

$$\gamma(s) = (x(s), t(s)) = (x + as, t + s).$$

Observe that if u is a C^1 solution of (1.2) with $f = 0$, then u is *constant* along γ. Indeed,

$$\begin{aligned} \frac{d}{ds}u(\gamma(s)) = \frac{d}{ds}(u(x(s),t(s)) = (\nabla u)(\gamma(s)) \cdot \dot{x}(s) + u_t(\gamma(s))\dot{t}(s) = \\ (\nabla u)(\gamma(s)) \cdot a + u_t(\gamma(s)) = (Xu)(\gamma(s)) = 0. \end{aligned} \tag{1.3}$$

This implies for such a u that $u(\gamma(0)) = u(\gamma(-t))$, that is,

$$u(\gamma(0)) = u(x,t) = u(x-at,0) = g(x-at). \tag{1.4}$$

We have shown that *if* a C^1 solution exists to (1.2) with $f = 0$, then $u(x,t)$ must be $g(x-at)$. A short computation shows that the C^1 function $u(x,t) = g(x-at)$ *is* a solution.

To solve (1.2) for general $f \in C^1$, suppose u is a C^1 solution. Setting $v(s) = u(\gamma(s))$, we now obtain

$$\dot{v}(s) = (Xu)(\gamma(s)) = f(\gamma(s)). \tag{1.5}$$

So we can construct u by "integrating f along the characteristic curve". We have

$$\begin{aligned} &v(0) = u(x,t) \text{ and } v(-t) = u(x-at,0) = g(x-at), \text{ so} \\ &v(0) - v(-t) = \int_{-t}^{0} \dot{v}(s)ds \Rightarrow u(x,t) - g(x-at) = \int_{-t}^{0} f(x+as,t+s)ds = \\ &\int_0^t f(x+(r-t)a,r)dr. \end{aligned} \tag{1.6}$$

Thus, if u is a C^1 solution of (1.2), it must be given by

$$u(x,t) = g(x-at) + \int_0^t f(x+(s-t)a,s)ds := u_1 + u_2. \tag{1.7}$$

A short computation shows that this u *is* a solution of (1.2); in fact,

$$\begin{aligned} &(a) Xu_1 = 0,\ u_1|_{t=0} = g \\ &(b) Xu_2 = f,\ u_2|_{t=0} = 0. \end{aligned} \tag{1.8}$$

Duhamel's principle. Set $w(x,t;s) := f(x+(s-t)a,s)$ and note that for a given s this function of (x,t) satisfies

$$\begin{aligned} &Xw = (\partial_t + a\cdot\nabla)w = -\nabla f\cdot a + a\cdot\nabla f = 0 \\ &w|_{t=s} = f(x,s), \end{aligned} \tag{1.9}$$

while $u_2(x,t) = \int_0^t w(x,t;s)ds$ satisfies (1.8)(b). Thus, the solution of the inhomogeneous problem (1.8)(b) can be expressed as the integral superposition $\int_0^t w(x,t;s)ds$ of solutions $w(x,t;s)$ to the family of homogeneous problems (1.9).

More generally, suppose X is any C^1 vector field $\partial_t + a(x)\cdot\nabla$ and $w(x,t;s) \in C^1$ satisfies

$$Xw = 0, w|_{t=s} = f(x,s) \text{ for } s \in [0,\infty). \tag{1.10}$$

Then $u(x,t) := \int_0^t w(x,t;s)ds$ satisfies

$$\begin{aligned} &Xu = (\partial_t + a(x)\cdot\nabla)u = w(x,t;t) + \int_0^t w_t ds + \int_0^t a(x)\cdot\nabla w ds = f(x,t) + 0 = f(x,t). \\ &u|_{t=0} = 0. \end{aligned} \tag{1.11}$$

This principle applies more generally to yield solutions to inhomogeneous evolution equations from solutions to families of homogeneous ones. For example, in section 1.4 we apply this to solve the heat equation with a nonzero forcing term.

Exercise 1.1a. Let $X = \partial_t + a(x) \cdot \nabla$, where $a \in C^1(\mathbb{R}^n, \mathbb{R}^n)$. Use the method of characteristics to construct a short time C^1 solution to

$$\begin{aligned} Xu &= f \in C_c^1(\mathbb{R}^n \times [0, \infty) \\ u|_{t=0} &= g \in C_c^1(\mathbb{R}^n). \end{aligned} \tag{1.12}$$

Hint: See Theorem 8.1.

1.2 Laplace's equation

1.2.1 Fundamental solution

The fundamental solution of Δ is a particular solution of $\Delta u = 0$ in $|x| > 0$ which can be used to construct solutions to various problems involving Δ; see Theorem 1.2 for example.[1]

For $n \geq 2$ let $\Delta = \partial_1^2 + \cdots + \partial_n^2$. We start by constructing a solution of $\Delta u = 0$ in $x \neq 0$. Considering the radial symmetry of the symbol $|\xi|^2$, it is natural to look for radial solutions, that is, solutions of the form $u(x) = v(|x|)$ for some function $v : (0, \infty) \to \mathbb{R}$. Setting $r = |x|$, we have $r_{x_i} = \frac{x_i}{r}$, so if $u(x) = v(|x|)$ we obtain

$$\begin{aligned} u_{x_i} &= v' \frac{x_i}{r} \\ u_{x_i x_i} &= v'' \left(\frac{x_i}{r}\right)^2 + v' \left(\frac{r - \frac{x_i^2}{r}}{r^2}\right) = v'' \left(\frac{x_i}{r}\right)^2 + v' \left(\frac{1}{r} - \frac{x_i^2}{r^3}\right) \Rightarrow \\ \Delta u &= v'' + v' \frac{n}{r} - v' \frac{r^2}{r^3} = v'' + v' \left(\frac{n-1}{r}\right). \end{aligned} \tag{1.13}$$

To solve $\Delta u = 0$ in $|x| > 0$ we solve the ODE $v'' + v' \left(\frac{n-1}{r}\right) = 0$ in $r > 0$. We reduce the order by setting $w = v'$ to obtain:[2]

$$\begin{aligned} w' + w \left(\frac{n-1}{r}\right) = 0 \Leftrightarrow \frac{w'}{w} = \frac{1-n}{r} \Leftrightarrow \ln|w| = (1-n)\ln r + c_1 \Leftrightarrow \\ |w| = c_2 r^{1-n}, c_2 > 0 \Leftrightarrow w = c_3 r^{1-n}, c_3 \in \mathbb{R} \setminus 0. \end{aligned} \tag{1.14}$$

Thus, $v = \begin{cases} a \ln r + b, & n = 2 \\ \frac{a}{r^{n-2}} + b, & n > 2 \end{cases}$ for some $a, b \in \mathbb{R}$.

[1] Later we'll redefine the fundamental solution as a Schwartz distribution on $\mathbb{R}^n$.

[2] The equivalences hold provided w is nonvanishing.

The "fundamental solution" for Δ is obtained by making a special choice of the constants a, b to be explained shortly. Let

$$\alpha(n) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)},$$

the volume of $B(0,1) \subset \mathbb{R}^n$. The area of $S^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$ is then $A(S^{n-1}) = n\alpha(n)$.

Definition 1.1. *The fundamental solution of* Δ *is* $\Phi(x) := \begin{cases} -\frac{1}{2\pi} \ln |x|, & n = 2 \\ \frac{1}{n(n-2)\alpha(n)|x|^{n-2}}, & n > 2 \end{cases}$.

Since $\Delta\Phi = 0$ in $\mathbb{R}^n \setminus \{0\}$, we say Φ is *harmonic* in $|x| > 0$. Observe that for a given y, the function $x \to \Phi(x-y)f(y)$ is harmonic in $\{x \in \mathbb{R}^n : x \neq y\}$.

Theorem 1.2. *For* $n \geq 2$ *let* $f \in C_c^2(\mathbb{R}^n)$ *and set* $u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy$. *Then* $u \in C^2(\mathbb{R}^n)$ *and* $-\Delta u = f$.

Remark 1.3. It is clear just from the statement of the theorem that the singularity of Φ at $x = 0$ must play a big role here. If Φ were C^2 on $\mathbb{R}^n$ we could apply Proposition 8.2 to obtain

$$\Delta u(x) = \int \Delta_x \Phi(x-y)f(y)dy = \int 0 \cdot f(y)dy = 0.$$

Proof of Theorem 1.2. We'll prove the case $n > 2$; the case $n = 2$ is similar.

1. We can make a change of variable to write

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy = \int_{\mathbb{R}^n} \Phi(y)f(x-y)dy, \tag{1.15}$$

provided both integrals converge absolutely (why?). Since f is compactly supported, the only issue is integrability near the singularities of $\Phi(y)$ and $\Phi(x-y)$. Example 8.8 shows that $\frac{1}{|y|^\alpha}$ is integrable near 0 when $\alpha < n$, so $\Phi(y) = c_n \frac{1}{|y|^{n-2}}$ is integrable near 0. Hence, $y \to \Phi(x-y)$ is integrable near x.

2. Show $u \in C^2(\mathbb{R}^n)$**.** Fix $x_0 \in \mathbb{R}^n$ and henceforth assume x is near x_0. By Proposition 8.2 we have

$$\begin{aligned} u_{x_i}(x) &= \int \Phi(y)\partial_{x_i} f(x-y)dy, \text{ since for some } K, M > 0, \\ |\Phi(y)\partial_{x_i} f(x-y)| &\leq \begin{cases} K|\Phi(y)|, & |y| \leq M \\ 0, & |y| > M \end{cases} \in L^1(\mathbb{R}^n). \end{aligned} \tag{1.16}$$

Taking a sequence $x_n \to x_0$, we obtain continuity of u_{x_i} at x_0 using the same estimate and the dominated convergence theorem. A similar estimate and argument yields

$$u_{x_i x_i}(x) = \int \Phi(y) f_{x_i x_i}(x-y)dy$$

along with continuity of $u_{x_ix_i}$ at x_0.

3. Compute Δu. By step **2** we have for any $\epsilon > 0$:

$$\Delta u(x) = \int \Phi(y)\Delta_x f(x-y)dy = \int_{B(0,\epsilon)}(...) + \int_{\mathbb{R}^n\setminus B(0,\epsilon)}(...). \tag{1.17}$$

Now since $f \in C_c^2(\mathbb{R}^n)$:

$$\begin{aligned}\left|\int_{B(0,\epsilon)}\Phi(y)\Delta_x f(x-y)dy\right| \le |\Delta f|_{L^\infty}\int_{B(0,\epsilon)}|\Phi(y)|dy = C_n\int_{B(0,\epsilon)}\frac{1}{|y|^{n-2}}dy = \\ C_n\int_0^\epsilon\int_{S^{n-1}}\frac{1}{r^{n-2}}r^{n-1}dS_1dr = C_n n\alpha(n)\int_0^\epsilon rdr \to 0 \text{ as } \epsilon\to 0.\end{aligned} \tag{1.18}$$

Thus, to finish it is enough to show

$$\int_{\mathbb{R}^n\setminus B(0,\epsilon)}\Phi(y)\Delta_x f(x-y)dy \to -f(x) \text{ as } \epsilon\to 0. \tag{1.19}$$

4. Show (1.19). Let $U_\epsilon := B(0,R)\setminus B(0,\epsilon)$. For x near x_0 and R large enough, $f(x-y)$ vanishes in $|y| > R-1$, so the integral in (1.19) equals

$$\begin{aligned}\int_{U_\epsilon}\Phi(y)\Delta_x f(x-y)dy = \int_{U_\epsilon}\Phi(y)\Delta_y f(x-y)dy = \\ -\int_{U_\epsilon}\nabla_y\Phi(y)\nabla_y f(x-y)dy + \int_{\partial B(0,\epsilon)}\Phi(y)\frac{\partial f}{\partial\nu_y}(x-y)dS_\epsilon := K_\epsilon + L_\epsilon.\end{aligned} \tag{1.20}$$

Here we used Corollary 8.6 to get the second equality. Now

$$|L_\epsilon| \le |\nabla f|_{L^\infty}\int_{\partial B(0,\epsilon)}|\Phi(y)|dS_\epsilon = C_n\frac{1}{\epsilon^{n-2}}A(\partial B(0,\epsilon)) = O(\epsilon)\to 0 \text{ as } \epsilon\to 0. \tag{1.21}$$

5. Show $K_\epsilon \to -f(x)$. Applying Corollary 8.6 again, we obtain

$$\begin{aligned}K_\epsilon = -\int_{U_\epsilon}\nabla_y\Phi(y)\nabla_y f(x-y)dy = \int_{U_\epsilon}\Delta\Phi(y)f(x-y)dy - \int_{\partial B(0,\epsilon)}\frac{\partial\Phi}{\partial\nu}(y)f(x-y)dS_\epsilon = \\ -\int_{\partial B(0,\epsilon)}\frac{\partial\Phi}{\partial\nu}(y)f(x-y)dS_\epsilon.\end{aligned} \tag{1.22}$$

For $|y| = \epsilon$ we compute

$$\frac{\partial\Phi}{\partial\nu}(y) = \nabla\phi\cdot\nu = \left(\frac{-1}{n\alpha(n)}\frac{y}{|y|^n}\right)\cdot\frac{-y}{|y|} = \frac{1}{n\alpha(n)|y|^{n-1}} = \frac{1}{n\alpha(n)\epsilon^{n-1}}. \tag{1.23}$$

Substituting in (1.22) gives[3]

$$K_\epsilon = -\frac{1}{n\alpha(n)\epsilon^{n-1}}\int_{\partial B(0,\epsilon)} f(x-y)dS_\epsilon = -\frac{1}{n\alpha(n)\epsilon^{n-1}}\int_{\partial B(0,1)} f(x-\epsilon\omega)\epsilon^{n-1}dS_1(\omega) =$$
$$-\frac{1}{n\alpha(n)}\int_{\partial B(0,1)} f(x-\epsilon\omega)dS_1(\omega) \to -f(x) \text{ as } \epsilon \to 0. \tag{1.24}$$

□

Exercise 1.2.1a. Explain carefully what is wrong with the following argument disproving Theorem 1.2.

Let $f \in C_c^2(\mathbb{R}^n)$ and set $u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy$. Observe that for any given x, the function $h(x,y) := \Delta_x\Phi(x-y)$ vanishes everywhere except on the set of Lebesgue measure zero given by $\{y \in \mathbb{R}^n : y = x\}$. Differentiating under the integral sign we obtain

$$\Delta u(x) = \int_{\mathbb{R}^n} \Delta_x\Phi(x-y)f(y)dy = \int_{\mathbb{R}^n} h(x-y)f(y)dy = 0,$$

since the integrand $h(x,y)f(y)$ is 0 almost everywhere as a function of y.

Exercise 1.2.1b. (a) Let $u \in C(\mathbb{R}^n)$ and for $x \in \mathbb{R}^n$, $r > 0$ define the averages of u on $B(x,r)$ and $\partial B(x,r)$ by

$$\fint_{B(x,r)} u(y)dy := \frac{1}{\text{vol}(B(x,r))}\int_{B(x,r)} u(y)dy$$
$$\fint_{\partial B(x,r)} u(y)dS_r := \frac{1}{\text{area}(\partial B(x,r))}\int_{\partial B(x,r)} u(y)dS_r. \tag{1.25}$$

Show that $\lim_{r\to 0}\fint_{B(x,r)} u(y)dy = \lim_{r\to 0}\fint_{\partial B(x,r)} u(y)dS_r = u(x)$.

1.2.2 Mean value property

In complex analysis one shows that harmonic functions on $\mathbb{R}^2$ satisfy special properties like the mean value property, Liouville's theorem, the maximum principle, and being C^∞ (an example of elliptic regularity). Proving the same results for harmonic functions on $\mathbb{R}^n$ requires completely different methods. First we prove the mean value property and then derive the other properties from that.

[3] For the final limit we use that $|f(x-\epsilon\omega)-f(x)| \to 0$ uniformly on $\partial B(0,1)$ as $\epsilon \to 0$.(how?) Alternatively, the dominated convergence theorem can be used here.

Theorem 1.4 (Mean value property)**.** *Suppose $u \in C^2(U)$ satisfies $\Delta u = 0$ in U. Then if $\overline{B}(x,r) \subset U$ we have*

$$\begin{aligned}(a)u(x) &= \fint_{\partial B(x,r)} u(y)dS_r(y)\\ (b)u(x) &= \fint_{B(x,r)} u(y)dy.\end{aligned} \tag{1.26}$$

Proof. **1.** For any $x \in U$, $r > 0$ such that $\overline{B}(x,r) \subset U$, define $\phi(r) := \fint_{\partial B(x,r)} u(y)dS_r(y)$. We'll show $\phi'(r) = 0$. This implies (1.26)(a) since

$$\phi(r) = \lim_{r\to 0} \phi(r) = u(x)$$

by Exercise 1.2.1b.

2. Make the change of variable $y = x + rz$, $z \in \partial B(0,1)$, and use $dS_r = r^{n-1}dS_1$ to rewrite $\phi(r)$ as

$$\phi(r) = \frac{\int_{\partial B(0,1)} u(x+rz)r^{n-1}dS_1}{r^{n-1}n\alpha(n)} = \frac{\int_{\partial B(0,1)} u(x+rz)dS_1}{n\alpha(n)}. \tag{1.27}$$

Thus, using that $z = \frac{y-x}{r} = \nu(y)$, the outer normal to $\partial B(x,r)$, we obtain

$$\begin{aligned}\phi'(r) = \frac{\int_{\partial B(0,1)} \nabla u(x+rz)\cdot z dS_1(z)}{n\alpha(n)} = \frac{\int_{\partial B(x,r)} \nabla u(y)\cdot \nu(y) dS_r(y)}{r^{n-1}n\alpha(n)} =\\ \frac{\int_{\partial B(x,r)} \frac{\partial u}{\partial \nu}(y)dS_r(y)}{r^{n-1}n\alpha(n)} = \frac{\int_{B(x,r)} \Delta u\, dy}{r^{n-1}n\alpha(n)} = 0.\end{aligned} \tag{1.28}$$

Here we used Corollary 8.6 for the second to last equality.

3. To prove (1.26)(b) we use polar coordinates to write

$$\begin{aligned}\int_{B(x,r)} u(y)dy = \int_0^r \int_{S^{n-1}} u(x+sz)s^{n-1}dS_1(z)ds = \int_0^r \int_{\partial B(x,s)} u(y)dS_s(y)ds =\\ \int_0^r u(x)s^{n-1}n\alpha(n)ds = u(x)r^n\alpha(n) = u(x)\mathrm{vol}(B(x,r)).\end{aligned} \tag{1.29}$$

Here we used (1.26)(a) for the third equality. □

Exercise 1.2.2a. Suppose $u \in C(U)$. Step **3** shows that: if (1.26)(a) holds whenever $\overline{B}(x,r) \subset U$, then (1.26)(b) holds whenever $\overline{B}(x,r) \subset U$. Prove the converse of this assertion.

Next we show that the mean value property implies harmonicity.

Theorem 1.5. *If $u \in C^2(U)$ and $u(x) = \fint_{\partial B(x,r)} u(y)dS_r(y)$ whenever $\overline{B}(x,r) \subset U$, then $\Delta u = 0$ in U.*

Proof. The mean value property implies that $\phi(r) := \fint_{\partial B(x,r)} u(y) dS_r(y)$ is constant for small $r > 0$.

$$0 = \phi'(r) = \frac{\int_{B(x,r)} \Delta u \, dy}{r^{n-1} n\alpha(n)}, \tag{1.30}$$

where the second equality follows from the fourth equality in (1.28). Thus, for all x, r such that $B(x,r) \subset U$ we have

$$\int_{B(x,r)} \Delta u \, dy = 0. \tag{1.31}$$

It is easy to check this implies $\Delta u = 0$ in U.(check)

□

Using the mean value property, we now deduce the maximum principle for harmonic functions on $\mathbb{R}^n$.

Theorem 1.6 (Maximum principle)**.** *Let $U \subset \mathbb{R}^n$ be open and bounded, and suppose $u \in C^2(U) \cap C(\overline{U})$ satisfies $\Delta u = 0$ in U. Then*

(a) if U is connected and if there exists $x_0 \in U$ such that $u(x_0) = \max_{\overline{U}} u$, then u is constant on U;

(b) $\max_{\overline{U}} u = \max_{\partial U} u$.

Proof. **(a)** Suppose $x_0 \in U$ and $u(x_0) = M := \max_{\overline{U}} u$. Let $S = \{x \in U : u(x) = M\}$. Then S is nonempty and obviously closed. We'll show S is also open to conclude using connectedness of U that $S = U$. Let $x_1 \in S$. For $r > 0$ such that $B(x_1, r) \subset U$ the mean value property implies

$$M = u(x_1) = \fint_{B(x_1,r)} u(y) dy. \tag{1.32}$$

Since $u \leq M$ on $B(x_1, r)$, this implies $u = M$ on $B(x_1, r)$ (why?). Hence S is open.

(b) Suppose $\max_{\overline{U}} u$ occurs at an interior point $x_0 \in U$. Let U_c be the connected component of U that contains x_0. Part (a) implies $u = u(x_0)$ on U_c, hence $u = u(x_0)$ on $\overline{U}_c$, and thus $u = u(x_0)$ on $\partial \overline{U}_c \subset \partial U$.

□

Exercise 1.2.2b. Uniqueness for the Dirichlet problem. Let $U \subset \mathbb{R}^n$ be bounded and open. Consider the Dirichlet problem

$$\begin{aligned} \Delta u &= f \in C(U) \\ u|_{\partial U} &= g \in C(\partial U). \end{aligned} \tag{1.33}$$

(a) Show that there exists at most one solution $u \in C^2(U) \cap C(\overline{U})$.

(b) Show that uniqueness can fail if U is unbounded. Hint: Take $U = \mathbb{R}^2 \setminus \{0\}$.

Remark 1.7. Exercise 1 in section 1.7 shows that a bounded harmonic function on $\mathbb{R}^n \setminus \{0\}$ must be constant. This together with (b) indicates that we should think of ∞ as part of the boundary in a problem like (1.33). Doing that allows us to explain the failure of uniqueness in (b) as being due to an incomplete boundary condition. Adding the "boundary condition at infinity" that u is bounded on $\mathbb{R}^n \setminus \{0\}$ implies uniqueness in (1.33).

The proof of the next result uses a special Friedrichs mollifier (Definition 8.9). Define $\eta \in C_c^\infty(\mathbb{R}^n)$ by

$$\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2-1}\right), & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}, \tag{1.34}$$

where C is chosen so that $\int_{\mathbb{R}^n} \eta(x)dx = 1$. We'll write $\eta(x) = \tilde{\eta}(|x|)$, for the obvious function $\tilde{\eta}(r)$.

Theorem 1.8 (Regularity of harmonic functions)**.** *Suppose $u \in C^2(U)$ and $\Delta u = 0$ in U. Then $u \in C^\infty(U)$.*

Theorem 1.8 is an immediate corollary of the following surprising theorem.

Theorem 1.9 (MVP $\Rightarrow C^\infty$)**.** *Suppose $u \in C(U)$ satisfies $u(x) = \fint_{\partial B(x,r)} u dS_r$ whenever $\overline{B}(x,r) \subset U$. Then $u \in C^\infty(U)$.*

Proof. **1.** If we set $U^\epsilon := \{x \in U : \text{dist}(x, \partial U) > \epsilon\}$ and for $x \in U^\epsilon$ define

$$u^\epsilon(x) := \int_U \eta_\epsilon(x-y)u(y)dy,$$

then $u^\epsilon \in C^\infty(U^\epsilon)$.[4] Thus, it suffices to show $u = u^\epsilon$ in U^ϵ for small $\epsilon > 0$.

2. For $x \in U^\epsilon$ we compute, using at one point the change of variable $y = x + z$,

$$\begin{aligned} u^\epsilon(x) &= \int_{B(x,\epsilon)} \eta_\epsilon(x-y)u(y)dy = \frac{1}{\epsilon^n}\int_{B(x,\epsilon)} \tilde{\eta}\left(\frac{|x-y|}{\epsilon}\right)u(y)dy = \\ &\frac{1}{\epsilon^n}\int_{B(0,\epsilon)} \tilde{\eta}\left(\frac{|z|}{\epsilon}\right)u(x+z)dz = \frac{1}{\epsilon^n}\int_0^\epsilon \int_{\partial B(0,r)} \tilde{\eta}\left(\frac{r}{\epsilon}\right)u(x+z)dS_r(z)dr = \\ &\frac{1}{\epsilon^n}\int_0^\epsilon \tilde{\eta}\left(\frac{r}{\epsilon}\right)[u(x)r^{n-1}n\alpha(n)]dr = u(x)\int \eta_\epsilon(x)dx = u(x). \end{aligned} \tag{1.35}$$

Here we used the mean value property for the second to last equality. □

Question. How does this proof fail if we use a more general *non*-radial approximate identity η_ϵ?

The following proposition is familiar from complex analysis.

[4] See section 8.5 for more on approximate identities.

Proposition 1.10. *Suppose $u \in C^2(\mathbb{R}^2)$ is bounded and harmonic. Then u is constant.*

Proof. For $(x, y) \in \mathbb{R}^2$ write $z = x + iy$. Let v be a harmonic conjugate of u, so that $u + iv$ is an entire function of z. Define the entire function $F(z) = e^{u+iv}$ and note that $|F(z)| = e^u$ is bounded. Liouville's theorem implies F is constant, and hence u is constant.

□

We now extend this result to harmonic functions on $\mathbb{R}^n$. The key is the following Cauchy-type estimate.

Lemma 1.11. *Suppose $u \in C^2(U)$ and $\Delta u = 0$ in U. Then if $\overline{B}(x_0, r) \subset U$ we have*

$$|u_{x_i}(x_0)| \leq \frac{n}{r} |u|_{L^\infty(\partial B(x_0,r))}. \tag{1.36}$$

Proof. Since $u \in C^\infty(U)$ we see that $\Delta u_{x_i} = \partial_i \Delta u = 0$, so u_{x_i} is harmonic in U. Using the mean value property of u_{x_i} we estimate:

$$\begin{aligned} |u_{x_i}(x_0)| = \left| \fint_{B(x_0,r)} u_{x_i} dx \right| = \frac{\left| \int_{\partial B(x_0,r)} u \nu_i dS_r \right|}{r^n \alpha(n)} \leq \\ \frac{|u|_{L^\infty(\partial B(x_0,r))} r^{n-1} n\alpha(n)}{r^n \alpha(n)} = \frac{n}{r} |u|_{L^\infty(\partial B(x_0,r))}. \end{aligned} \tag{1.37}$$

Here we used Theorem 8.4 for the second equality. □

Theorem 1.12 (Liouville's theorem)**.** *Suppose $u \in C^2(\mathbb{R}^n)$ is harmonic on $\mathbb{R}^n$ and there exists $M > 0$ such that $|u(x)| \leq M$ for all $x \in \mathbb{R}^n$. Then u is constant.*

Proof. Fix $x_0 \in \mathbb{R}^n$ and for any $r > 0$ apply Lemma 1.11 on $B(x_0, r)$ to obtain

$$|u_{x_i}(x_0| \leq \frac{n}{r} M. \tag{1.38}$$

Letting $r \to \infty$ we obtain $u_{x_i}(x_0) = 0$, so $\nabla u = 0$ on $\mathbb{R}^n$, and thus u is constant on $\mathbb{R}^n$ (why?).

□

Exercise 1.2.2c. Suppose $u \in C^2(\mathbb{R}^n)$, $\Delta u = 0$ on $\mathbb{R}^n$, and there exists $C > 0$, $m \in \mathbb{N}$ such that

$$|u(x)| \leq C|x|^m \text{ for all } x \in \mathbb{R}^n.$$

Show that u is a polynomial of degree $\leq m$.

1.2.3 The Dirichlet problem

Let U be open and bounded. The classical Dirichlet problem is the boundary value problem

$$\begin{aligned} \Delta u &= f \text{ in } U \\ u &= g \text{ on } \partial U, \end{aligned} \tag{1.39}$$

where f and g are given. When $f = 0$, the solution u represents the electrostatic potential inside a region U devoid of charges when fixed voltages g are applied on a surrounding conducting surface. The solution u also represents the steady-state or equilibrium temperature distribution inside a heat-conducting solid when a given temperature g is maintained on the outer boundary of the solid. The problem was formulated around 1828 by George Green who studied it using potential theory. The many different approaches to this problem directly stimulated developments in analysis including Fourier series, integral equations (Fredholm integral equations, compact operators, Fredholm operators, spectral theory), the calculus of variations (Dirichlet's principle), numerical methods (finite difference, finite element, Monte Carlo), conformal mapping, distribution theory (fundamental solutions), stochastic calculus (Brownian motion, Feynman-Kac formula), harmonic analysis (singular integrals, Hardy spaces), and the use of maximum principles, weak solutions (Lax-Milgram), and energy estimates (Sobolev spaces, Hölder spaces) in elliptic PDE.

Assuming for now that u is an *arbitrary* given function in $C^2(\overline{U})$, we proceed to derive a formula that expresses u at $x \in U$ in terms of $f := \Delta u$ and $g := u|_{\partial U}$.

Proposition 1.13. *Let $u \in C^2(\overline{U})$ and assume ∂U is C^3.* [5] *Defining $f := \Delta u$ and $g = u|_{\partial U}$, we have*

$$u(x) = -\int_{\partial U} g \partial_{\nu_y} G(x,y) dS(y) - \int_U G(x,y) f(y) dy, \tag{1.40}$$

where G, defined in (1.48), is called the "Green's function" for U.

Proof. **1.** For any $x \in U$, $\epsilon > 0$ such that $\overline{B}(x,\epsilon) \subset U$, define $V_\epsilon = U \setminus \overline{B(x,\epsilon)}$. An application of Gauss-Green (Cor. 8.7) yields

$$\begin{aligned} &\int_{V_\epsilon} \left[u(y) \Delta_y \Phi(y-x) - \Phi(y-x) \Delta u(y) \right] dy = \\ &\int_{\partial V_\epsilon} \left[u(y) \frac{\partial \Phi}{\partial \nu_y}(y-x) - \Phi(y-x) \frac{\partial u}{\partial \nu}(y) \right] dS(y) = \int_{\partial U} (...) + \int_{\partial B(x,\epsilon)} (...). \end{aligned} \tag{1.41}$$

We claim $\int_{\partial B(x,\epsilon)} (...) \to u(x)$ as $\epsilon \to 0$. Indeed, we have

$$\left| \int_{\partial B(x,\epsilon)} \Phi(y-x) \frac{\partial u}{\partial \nu}(y) dS \right| \leq C_n(u) \frac{\epsilon^{n-1}}{\epsilon^{n-2}} \to 0 \text{ as } \epsilon \to 0, \tag{1.42}$$

[5] We make this assumption to guarantee the existence of a function ϕ^x as in (1.45); see Remark 1.14.

and the computation (1.22)-(1.24) gives

$$\int_{\partial B(x,\epsilon)} u(y)\frac{\partial \Phi}{\partial \nu_y}(y-x)dS \to u(x) \text{ as } \epsilon \to 0. \tag{1.43}$$

Using that $\Delta_y \Phi(y-x) = 0$ and taking the limit as $\epsilon \to 0$ in (1.41), we obtain

$$u(x) = \int_{\partial U} \left[\Phi(y-x)\frac{\partial u}{\partial \nu}(y) - u(y)\frac{\partial \Phi}{\partial \nu_y}(y-x)\right] dS(y) - \int_U \Phi(y-x)\Delta u(y)dy. \tag{1.44}$$

2. The bad term in (1.44) is $\int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial \nu}(y)$, since it can't obviously be expressed in terms of f and g. To cancel this term out, we introduce a function $\phi^x(y)$ which satisfies for $x \in U$ fixed:

$$\begin{aligned} &\Delta_y \phi^x(y) = 0 \text{ in } U \\ &\phi^x(y) = \Phi(y-x) \text{ for } y \in \partial U \\ &\phi^x \in C^2(\overline{U}). \end{aligned} \tag{1.45}$$

Remark 1.14. The existence and regularity results of chapter 3 imply that $\phi^x \in C^\infty(\overline{U})$ exists when ∂U is C^∞. In Remark 1.15(a) we explain why ϕ^x always exists under the assumptions of Proposition 1.13.

Applying Gauss-Green (Cor. 8.7) again yields

$$\begin{aligned} \int_U u(y)\Delta_y \phi^x(y)dy - \int_U \phi^x(y)\Delta u(y)dy = \int_{\partial U} \left[u(y)\frac{\partial \phi^x}{\partial \nu}(y) - \phi^x(y)\frac{\partial u}{\partial \nu}(y)\right] dS(y) = \\ \int_{\partial U} \left[u(y)\frac{\partial \phi^x}{\partial \nu}(y) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\right] dS. \end{aligned} \tag{1.46}$$

The first term on the left in (1.46) is zero, so adding (1.44) and (1.46) we obtain

$$u(x) = -\int_{\partial U} u(y)\partial_{\nu_y}[\Phi(y-x) - \phi^x(y)]dS - \int_U [\Phi(y-x) - \phi^x(y)]\Delta u(y)dy. \tag{1.47}$$

Setting

$$G(x,y) := \Phi(y-x) - \phi^x(y) \tag{1.48}$$

in (1.47) gives (1.40). □

Observe that for $x \in U$, the map $y \to G(x,y)$ is C^∞ except at $y = x$ and satisfies

$$\begin{aligned} &\Delta_y G(x,y) = 0 \text{ for } y \in U \setminus \{x\} \\ &G(x,y) = 0 \text{ for } y \in \partial U. \end{aligned} \tag{1.49}$$

Remark 1.15. (a) Let $0 < \gamma < 1$. In the Dirichlet problem (1.39) when U is open and bounded, ∂U has Hölder regularity $C^{2,\gamma}$, $g \in C^{2,\gamma}(\partial U)$, and $f \in C^{0,\gamma}(\overline{U})$, classical Schauder theory yields a unique solution $u \in C^{2,\gamma}(\overline{U})$ such that[6]

$$|u|_{C^{2,\gamma}(\overline{U})} \leq C\left(|f|_{C^{0,\gamma}(\overline{U})} + |g|_{C^{2,\gamma}(\partial U)}\right). \tag{1.50}$$

This implies the existence of ϕ^x satisfying (1.45) and hence the existence of $G(x,y)$ when ∂U is $C^{2,\gamma}$. Since $C^3 \subset C^{2,\gamma}$, we conclude that G must exist under the assumptions of Proposition 1.13.

(b) When U is open and bounded, ∂U is C^3, $f \in C(\overline{U})$ and $g \in C^2(\partial U)$, Proposition 1.13 implies that *if* there exists a solution $u \in C^2(\overline{U})$ to problem (1.39), then it must be given by the formula (1.40), where G is the Green's function of U. These assumptions on the problem (1.39) are not the same as those in part (a) of this Remark, so even when G can be found explicitly, a separate analysis is needed to check that the formula (1.40) gives a solution.

(c) Under the assumptions of part (a) on the problem (1.39), we know a unique solution $u \in C^{2,\gamma}(\overline{U}) \subset C^2(\overline{U})$ exists and we know G exists, so the proof of Proposition 1.13 shows that u *must be* given by the formula (1.40).

Poisson kernel for $\mathbb{R}^n_+$.

For $n > 2$ consider the Dirichlet problem on $\mathbb{R}^n_+ = \{x = (x', x_n) \in \mathbb{R}^n : x_n > 0\}$:

$$\begin{aligned} \Delta u &= 0 \text{ in } \mathbb{R}^n_+ \\ u &= g \text{ on } x_n = 0. \end{aligned} \tag{1.51}$$

In this case the Green's function can be found by inspection to be

$$G(x,y) = \Phi(y-x) - \Phi(y-\tilde{x}) \text{ where } \tilde{x} = (x', -x_n). \tag{1.52}$$

Indeed, recalling that $\Phi(y-x) = \frac{c_n}{|y-x|^{n-2}}$ and using that $|y-x| = |y-\tilde{x}|$ for $y \in \partial\mathbb{R}^n_+$, we see that for $x \in \mathbb{R}^n_+$ the function $\phi^x(y) := \Phi(y-\tilde{x}) = \frac{c_n}{|y-\tilde{x}|^{n-2}}$ satisfies[7]

$$\begin{aligned} \Delta_y \phi^x(y) &= 0 \text{ in } \mathbb{R}^n_+ \\ \phi^x(y) &= \Phi(y-x) \text{ on } \partial\mathbb{R}^n_+. \end{aligned} \tag{1.53}$$

When $n = 3$, since $\Phi(y-x)$ is the electrostatic potential at $y \neq x$ due to some charge q placed at $x \in \mathbb{R}^n_+$, we can view the choice of $G(x,y)$ in (1.52) as dictated by the need to place an equal but opposite charge at $\tilde{x}$ in order to cancel the potential on $\partial\mathbb{R}^n_+$ due to the charge at x.[8]

[6] See section 8.15 for the definition of Hölder spaces. Schauder theory is presented in [GT01].

[7] Recall $c_n = \frac{1}{n(n-2)\alpha(n)}$.

[8] George Green provided a similar argument using charges and potentials in attempting to establish that Green's functions exist for any bounded U. Imagine that ∂U is perfectly conducting, encloses a vacuum, and is "grounded" to have potential zero. A negative charge placed at some $x \in U$ induces a distribution of positive charge on ∂U to keep the potential on ∂U zero. Then for $y \neq x$, $G(x,y)$ is the potential at $y \in U$ produced by the charge at x and those on ∂U.

Although $\mathbb{R}^n_+$ is unbounded, given the result of Proposition 1.13 we might still hope that a solution to (1.51) is given by the formula

$$u(x) = -\int_{\partial\mathbb{R}^n_+} g(y)\partial_{\nu_y} G(x,y)dS(y). \tag{1.54}$$

For $x \in \mathbb{R}^n_+$ and $y \in \partial\mathbb{R}^n_+$ we compute

$$\begin{aligned} -\frac{\partial G}{\partial \nu_y}(x,y) = \frac{\partial G}{\partial y_n}(x,y) = \frac{\partial \Phi}{\partial y_n}(y-x) - \frac{\partial \Phi}{\partial y_n}(y-\tilde{x}) = \\ -\frac{1}{n\alpha(n)}\left[\frac{y_n - x_n}{|y-x|^n} - \frac{y_n + x_n}{|y-\tilde{x}|^n}\right] = \frac{2x_n}{n\alpha(n)|y-x|^n} := K(x,y). \end{aligned} \tag{1.55}$$

We call $K(x,y)$ the *Poisson kernel* for the upper half plane. Writing points $y \in \partial\mathbb{R}^n_+$ as $y = (y', 0)$, we see that (1.54) has the form

$$u(x) = \int_{\partial\mathbb{R}^n_+} K(x,y)g(y)dS(y) = \frac{2x_n}{n\alpha(n)}\int_{\mathbb{R}^{n-1}} \frac{1}{|x-y|^n} g(y)dy', \tag{1.56}$$

since now $dS(y) = dy'$.

Theorem 1.16. *Let $g \in C(\partial\mathbb{R}^n_+) \cap L^\infty(\partial\mathbb{R}^n_+)$ and let u be given by (1.56). Then*
(i) $u \in C^\infty(\mathbb{R}^n_+) \cap L^\infty(\mathbb{R}^n_+)$;
(ii) $\Delta u = 0$ in $\mathbb{R}^n_+$;
(iii) for $x_0 \in \partial\mathbb{R}^n_+$, $\lim_{x\to x_0, x\in\mathbb{R}^n_+} u(x) = g(x_0)$. Thus, u extends continuously to $\overline{\mathbb{R}}^n_+$.

Proof. **1. Show $\Delta_x K = 0$ for $x \in \mathbb{R}^n_+$.** For $y \in \partial\mathbb{R}^n_+$ the map

$$x \to G(x,y) = \Phi(y-x) - \Phi(y-\tilde{x})$$

is clearly harmonic on $\mathbb{R}^n_+$. Moreover,

$$\Delta_x K(x,y) = \Delta_x \partial_{y_n} G(x,y) = \partial_{y_n}\Delta_x G(x,y) = 0. \tag{1.57}$$

2. (ii). Fix $x_0 \in \mathbb{R}^n_+$. For x near x_0 the integral (1.56) is absolutely convergent, and so are the integrals obtained by repeated differentiations with respect to x under the integral sign; recall Example 8.8. Thus, by Proposition 8.2 $u \in C^\infty(\mathbb{R}^n_+)$. Applying Δ_x under the integral sign and using (1.57), we obtain (ii).

3. (i). We show by a direct computation in section 8.6 that

$$\int_{\mathbb{R}^{n-1}} K(x,y)dy' = 1 \text{ for } x \in \mathbb{R}^n_+,\ y = (y',0). \tag{1.58}$$

Thus, for $x \in \mathbb{R}^n_+$ we can estimate (1.56) to obtain

$$|u(x)| \le |g|_{L^\infty}\int K(x,y)dy' = |g|_{L^\infty} \Rightarrow |u|_{L^\infty(\mathbb{R}^n_+)} \le |g|_{L^\infty(\partial\mathbb{R}^n_+)}. \tag{1.59}$$

This finishes the proof of (i).

4. (iii). Fix $x_0 \in \partial\mathbb{R}^n_+$ and $\epsilon > 0$. For $x \in \mathbb{R}^n_+$ near x_0 we use (1.58) to write

$$|u(x) - g(x_0)| = \left| \int_{\mathbb{R}^{n-1}} K(x,y)(g(y) - g(x_0))dy' \right|. \tag{1.60}$$

Choose $\delta > 0$ such that $y \in \partial\mathbb{R}^n_+$, $|y - x_0| < \delta \Rightarrow |g(y) - g(x_0| < \epsilon$. Then

$$\begin{aligned} &\left| \int_{\mathbb{R}^{n-1}} K(x,y)(g(y) - g(x_0))dy' \right| \leq \\ &\epsilon \int_{\mathbb{R}^{n-1} \cap B(x_0,\delta)} K(x,y)dy' + 2|g|_{L^\infty} \int_{\mathbb{R}^{n-1} \setminus B(x_0,\delta)} K(x,y)dy'. \end{aligned} \tag{1.61}$$

In the second integral $|y - x_0| \geq \delta$. Thus, if $x \in \mathbb{R}^n_+$ and $|x - x_0| < \frac{\delta}{2}$, we have

$$|x - x_0| < \frac{\delta}{2} \leq \frac{|y - x_0|}{2} \Rightarrow |y - x| \geq ||y - x_0| - |x_0 - x|| > \frac{|y - x_0|}{2}. \tag{1.62}$$

Thus,

$$\frac{2x_n}{n\alpha(n)} \int_{\mathbb{R}^{n-1} \setminus B(x_0,\delta)} \frac{1}{|x-y|^n} dy' \leq \frac{2^{n+1}x_n}{n\alpha(n)} \int_{\mathbb{R}^{n-1} \setminus B(x_0,\delta)} \frac{1}{|y - x_0|^n} dy' = D_n x_n \tag{1.63}$$

for some D_n depending on x_0. This implies that for

$$x \in \mathbb{R}^n_+, \ |x - x_0| < \frac{\delta}{2}, \ x_n < \frac{\epsilon}{2|g|_{L^\infty} D_n},$$

the right side of (1.61) is $< \epsilon + \epsilon$. This proves (iii).

□

Exercise 1.2.3a. Let U be the open first quadrant of $\mathbb{R}^2$, $U = \{(x,y) : x, y > 0\}$. Use a charge placement argument to construct a Green's function for U; recall the discussion after (1.53). For $n > 2$ do the same for $U = \{x \in \mathbb{R}^n : x_i > 0, i = 1, \ldots, n\}$.

Poisson kernel for the unit ball

Consider the Dirichlet problem for the unit ball $B(0,1) \subset \mathbb{R}^n$:

$$\begin{aligned} &\Delta u = 0 \text{ on } B(0,1) \\ &u = g \in C(\partial B(0,1)). \end{aligned} \tag{1.64}$$

We look for a solution of the form

$$u(x) = -\int_{\partial B(0,1)} g(y) \partial_{\nu_y} G(x,y) dS(y), \tag{1.65}$$

where $G(x,y) = \Phi(x-y) - \phi^x(y)$ and ϕ^x is constructed to satisfy

$$\begin{aligned} &\Delta_y \phi^x = 0 \text{ in } B(0,1) \\ &\phi^x(y) = \Phi(x-y) \text{ for } |y| = 1. \end{aligned} \tag{1.66}$$

One checks that if $x \in B(0,1)$, $x \neq 0$, then the "reflected" point $\tilde{x} := \frac{x}{|x|^2}$ lies in $|x| > 1$ and satisfies

$$||x|(y-\tilde{x})| = |y-x| \text{ when } |y| = 1. \tag{1.67}$$

This suggests setting $\phi^x(y) := \Phi(|x|(y-\tilde{x}))$ to obtain

$$G(x,y) = \Phi(x-y) - \Phi(|x|(y-\tilde{x})), \text{ where } \tilde{x} = \frac{x}{|x|^2} \text{ for } x \in B(0,1), x \neq 0. \tag{1.68}$$

Let us ignore for a moment the fact that $\phi^x(y) = \Phi(|x|(y-\tilde{x}))$ is undefined when $x = 0$.[9] One can proceed anyway to compute the Poisson kernel

$$K_B(x,y) = -\partial_{\nu_y} G(x,y) = \frac{1-|x|^2}{n\alpha(n)} \frac{1}{|x-y|^n} \text{ for } |y| = 1. \tag{1.69}$$

Observe that as a function of x for a given y such that $|y| = 1$, this is well-defined, C^∞, and harmonic in $B(0,1)$.[10] An argument very similar to the one we gave for the half-space shows that

$$u(x) = \frac{1-|x|^2}{n\alpha(n)} \int_{\partial B(0,1)} \frac{g(y)}{|x-y|^n} dS(y) \tag{1.70}$$

is a solution of (1.64). In particular, the boundary values are taken in the same sense as before.

Exercise 1.2.3b. Check that u as in (1.70) satisfies

$$\begin{aligned} &u \in C^\infty(B(0,1)) \\ &\Delta u = 0 \text{ in } B(0,1) \\ &\lim_{x \in B(0,1), x \to x_0} u(x) = g(x_0) \text{ for each } x_0 \in \partial B(0,1). \end{aligned} \tag{1.71}$$

You will need to use $\int_{\partial B(0,1)} K_B(x,y) dS(y) = 1$. Establish this *without* doing an explicit computation.

[9] But note that $\lim_{x \to 0} \phi^x(y) = c_n$, where $\Phi(x) = c_n/|x|^{n-2}$. If we define $\phi^0(y) = c_n$, then (1.66) is satisfied when $x = 0$.

[10] Harmonicity of $x \to K_B(x,y)$ in $B(0,1)$ can be checked using (1.69) by direct computation.

1.2.4 Energy methods: Dirichlet's principle

Here we illustrate an approach to the Dirichlet problem based on "energy methods". Let $U \subset \mathbb{R}^n$ be bounded and open with C^1 boundary, and consider again the problem

$$\begin{aligned} -\Delta u &= f \text{ in } U \\ u &= g \text{ on } \partial U, \end{aligned} \tag{1.72}$$

where $u \in C^2(\overline{U})$, $f \in C(\overline{U})$, $g \in C(\partial U)$.

Proposition 1.17 (Uniqueness)**.** *If $v \in C^2(\overline{U})$ also satisfies (1.72), then $u = v$.*

Proof. Set $w = u - v$, so $\Delta w = 0$ in U and $w = 0$ on ∂U. Thus,

$$0 = -\int_U w \Delta w dx = \int_U |\nabla w|^2 dx \Rightarrow |\nabla w| = 0 \text{ on } U. \tag{1.73}$$

Thus w is constant on each component of U. Those constants must be zero since $w = 0$ on ∂U. □

Dirichlet's principle is the idea, central to the calculus of variations, that certain partial differential equations can be solved by minimizing an "energy functional". We now illustrate the principle as it applies to (1.72).

For any $w \in C^2(\overline{U})$ define the *energy functional*

$$I(w) = \int_U \left(\frac{1}{2} |\nabla w|^2 - wf \right) dx. \tag{1.74}$$

For u as in (1.72) this represents the total electrostatic potential energy of the electric field $E = -\nabla u$ in a region U with charge density f. Define an admissible set A:

$$A = \{w \in C^2(\overline{U}) : w = g \text{ on } \partial U\}. \tag{1.75}$$

Theorem 1.18. *Let $u \in A$. Then*

$$I(u) = \min_{w \in A} I(w) \Leftrightarrow -\Delta u = f \text{ in } U,\ u|_{\partial U} = g.$$

Proof. [11] **1.** $\Leftarrow$ Suppose $-\Delta u = f$, $u \in A$ and let $w \in A$. Then[12]

$$\begin{aligned} 0 &= \int_U (-\Delta u - f)(u - w) = \int_U [\nabla u \cdot \nabla (u - w) - f(u - w)] dx \Rightarrow \\ \int_U (|\nabla u|^2 - uf) dx &= \int_U (\nabla u \cdot \nabla w - wf) dx \leq \frac{1}{2} \int_U |\nabla u|^2 dx + \int_U \left(\frac{|\nabla w|^2}{2} - wf \right) dx \Rightarrow \\ \int_U \left(\frac{|\nabla u|^2}{2} - uf \right) dx &\leq \int_U \left(\frac{|\nabla w|^2}{2} - wf \right) dx \Leftrightarrow I(u) \leq I(w). \end{aligned} \tag{1.76}$$

[11] For the direction $\Rightarrow$ we assume for now the existence of a minimizer of I in A. The existence of minimizers is discussed in Chapter 4.

[12] For the inequality in the second line we used $ab \leq \frac{a^2+b^2}{2}$.

2. $\Rightarrow$ Suppose $u \in A$ minimizes $I(w)$ over A. Let $v \in C_c^\infty(U)$ (so $v = 0$ near ∂U), and for $\tau \in \mathbb{R}$ define $i(\tau) = I(u + \tau v)$. Observe that $i : \mathbb{R} \to \mathbb{R}$, $u + \tau v \in A$, and $i(\tau)$ has a global min when $\tau = 0$. Applying Calc I to i we conclude $i'(0) = 0$. Let's compute $i'(\tau)$. We have

$$\begin{aligned} i(\tau) = \int_U \left[\frac{1}{2}|\nabla u + \tau \nabla v|^2 - (u + \tau v)f\right] dx = \\ \int_U \frac{1}{2}|\nabla u|^2 + \tau \nabla u \cdot \nabla v + \frac{1}{2}|\nabla v|^2\tau^2 - (u + \tau v)f dx \Rightarrow \\ i'(\tau) = \int_U (\nabla u \cdot \nabla v + \tau|\nabla v|^2 - vf)dx \Rightarrow 0 = i'(0) = \int_U (\nabla u \cdot \nabla v - vf)dx. \end{aligned} \tag{1.77}$$

Applying Gauss-Green we obtain

$$\int_U (-\Delta u - f)v dx = 0 \text{ for all } v \in C_c^\infty(U) \Leftrightarrow -\Delta u = f \text{ in } U.$$

□

The equation $-\Delta u = f$ is called the *Euler-Lagrange equation* associated to the functional $I(w)$.

1.2.5 Single and double layer potentials.

Here we give a short informal sketch of the method of layer potentials for solving the Dirichlet and Neumann problems. This method played an important role in the development of functional and harmonic analysis as well as PDE, since it stimulated the study of integral equations, compact operators, Fredholm operators, Hilbert spaces like $L^2(\partial U)$, and singular integrals.[13]

Let U be a bounded connected open set with C^2 boundary. Consider the Dirichlet problem

$$\begin{aligned} \Delta u &= 0 \text{ in } U \\ u &= g \in C(\partial U) \end{aligned} \tag{1.78}$$

and the Neumann problem

$$\begin{aligned} \Delta u &= 0 \text{ in } U \\ \partial_{\nu_-} u &= g \in C(\partial U) \end{aligned} \tag{1.79}$$

where ∂_{ν_-} is the inward normal derivative on ∂U.

Recall the formula (1.40)

$$u(x) = -\int_{\partial U} \partial_{\nu_y} G(x, y)g(y)dS(y), \text{ where } G(x, y) = \Phi(x, y) - \phi^x(y). \tag{1.80}$$

[13] An excellent reference for the mathematics of layer potentials is chapter 3 of [Fol95]; for the history see [Die81].

for the solution to (1.78). The usefulness of (1.40) is limited by the fact that for most domains the Green's function is not known explicitly, since ϕ^x is not known explicitly. It is natural to ask "how close" we come to a solution of (1.78) if we just ignore ϕ^x, and consider the (explicit) function u defined for $x \in U$ by

$$u(x) = -\int_{\partial U} \partial_{\nu_y} \Phi(x,y) g(y) dS(y). \tag{1.81}$$

It is clear that u is harmonic in U, and it can be shown that for $x_0 \in \partial U$

$$u|_{\partial U}(x_0) := \lim_{x \in U, x \to x_0} u(x) = \left(\frac{1}{2} I + T\right) g, \tag{1.82}$$

where $T : L^2(\partial U) \to L^2(\partial U)$ is a *compact* operator.[14] Thus, in order to solve (1.78) we should try first to solve

$$\left(\frac{1}{2} I + T\right) \phi = g \tag{1.83}$$

and then take

$$u = -\int_{\partial U} \partial_{\nu_y} \Phi(x,y) \phi(y) dS(y). \tag{1.84}$$

The operator

$$\frac{1}{2} I + T : L^2(\partial U) \to L^2(\partial U)$$

is *Fredholm* with index zero, and (1.83) can be studied using Fredholm theory.[15] The compact operator T is an integral operator of the form

$$(T\phi)(x) = \int_{\partial U} K(x,y) \phi(y) dS(y), \ x \in \partial U \tag{1.85}$$

for an appropriate kernel $K(x,y)$.

When $U' := \mathbb{R}^n \setminus \overline{U}$ has just one (necessarily unbounded) component, the equation (1.83) can be solved for any $g \in C(S)$. When U' has more than one component, the map $\frac{1}{2} I + T$ is not surjective, but has *finite* codimension. The form of the solution to (1.78) in this case is described in [Fol95], Theorem 3.40.

Similarly, it is fruitful to look for a solution to the Neumann problem (1.79) of the form

$$u(x) = -\int_{\partial U} \Phi(x,y) \phi(y) dS(y). \tag{1.86}$$

[14] A compact operator on a Hilbert space H is a bounded linear operator that maps bounded sets to relatively compact sets. A compact operator can be viewed as "small" in a useful sense.

[15] See [Wil26], Chapter 9) for definitions of these terms.

Remark 1.19 (Terminology)**.** The integral (1.86) is called a *single layer potential with moment* ϕ. When $n = 3$ it gives the potential at $x \in U$ due to a charge distribution on ∂U with density ϕ. The integral in (1.84) is called a *double layer potential with moment* ϕ. Indeed, that integral is the limit as $t \to 0$ of difference quotients

$$-\int_{\partial U} \frac{\Phi(x, y + t\nu(y)) - \Phi(x, y)}{t} \phi(y) dS(y), \tag{1.87}$$

where one term gives the potential at x due to a charge distribution on ∂U with density $-\phi(y)/t$, and the other term gives the potential at x due to a charge distribution on the translated surface $\{y + t\nu(y) : y \in \partial U\}$ with density $\phi(y)/t$.

1.3 Distributions I

When we were studying the fundamental solution $\Phi(x) = \frac{c_n}{|x|^{n-2}}$ of the Laplacian, you might have been puzzled by a couple of questions. We showed $\Delta\Phi = 0$ for $|x| > 0$, but what can we say about $\Delta\Phi$ in neighborhood of 0, say in $B(0, 1)$? The function Φ itself is locally integrable near $x = 0$, and so is Φ_{x_i}, but $\Phi_{x_i x_i}$ is not (check). What equation does $\Delta\Phi$ satisfy in $B(0, 1)$? The theory of distributions provides simple answers to these questions.

1.3.1 Basic definitions and examples

Definition 1.20. *(a)Let $U \subset \mathbb{R}^n$ be open. The space $C_c^\infty(U)$ is often denoted $\mathcal{D}(U)$ and referred to as the space of* test functions *on U.*

(b)A distribution *u is a linear functional $u : \mathcal{D}(U) \to \mathbb{C}$ which is continuous in a sense to be defined below; see Definition 1.23. We denote the space of distributions on U by $\mathcal{D}'(U)$.*

Recall the following basic fact about continuity of linear operators on normed spaces.

Proposition 1.21. *Suppose V and W are normed spaces and $T : V \to W$ is linear. Then T is continuous $\Leftrightarrow$ there exists $C > 0$ such that $|Tv| \le C|v|$ for all $v \in V$.*[16]

Proof. ($\Leftarrow$)The estimate clearly implies continuity at 0, which implies continuity at any $v \in V$ since T is linear.

($\Rightarrow$) Since T is continuous at 0, given $\epsilon > 0$ there exists $\delta > 0$ such that $|v| \le \delta$ implies $|Tv| < \epsilon$. Suppose $v \neq 0$. Then we have

$$|T(\delta v/|v|)| < \epsilon \Rightarrow |Tv| \le (\epsilon/\delta)|v|.$$

□

[16]Here as usual we refrain from writing $|\cdot|_V$ and $|\cdot|_W$.

For T as in Proposition 1.21 we define the *operator norm* of T to be

$$|T| = \sup_{|v|\leq 1} |Tv|. \tag{1.88}$$

When $W = \mathbb{R}$ or $\mathbb{C}$, we call T a linear functional. The set of all continuous linear functionals on V is called the *dual space* of V and denoted V'.

The standard topology on the space of test functions $\mathcal{D}(U)$ is neither normable nor metrizable; see section 8.9. But we can use a family of norms defined on subspaces of $\mathcal{D}(U)$ to characterize continuous linear functionals on $\mathcal{D}(U)$.

Definition 1.22. *(a) If $K \subset U$ is compact, let $\mathcal{D}_K \subset \mathcal{D}(U)$ be the subspace of test functions with support in K.*

(b) For $m \in \mathbb{N}_0$ let $\mathcal{D}_K^m = \{\phi \in C_c^m(U) : \operatorname{supp} \phi \subset K\}$ with the norm

$$|\phi|_{K,m} := \sup_{x\in K, |\alpha|\leq m} |\partial^\alpha \phi(x)|. \tag{1.89}$$

Clearly $\mathcal{D}_K \subset \mathcal{D}_K^m$ for all m, so we can give $\mathcal{D}_K$ any of the norms $|\cdot|_{K,m}$, $m \in \mathbb{N}_0$. Now we make precise the sense in which a distribution $u \in \mathcal{D}'(U)$ is "continuous" on $\mathcal{D}(U)$.

Definition 1.23. *A linear functional $u : \mathcal{D}(U) \to \mathbb{C}$ is a distribution on U if for any compact $K \subset U$, there exists an $m = m(K) \in \mathbb{N}_0$ such that u is continuous on $\mathcal{D}_K$ when it carries the $|\cdot|_{K,m}$ norm.*

Observe that if u is continuous on $\mathcal{D}_K$ when it carries the $|\cdot|_{K,m}$ norm, the same holds for any $m' \geq m$.(why?) The next proposition, which just reformulates Definition 1.23, follows immediately from Proposition 1.21.

Proposition 1.24. *A linear functional $u : \mathcal{D}(U) \to \mathbb{C}$ is a distribution on U if for any compact $K \subset U$, there exist $m(K) \in \mathbb{N}_0$ and $C(K) > 0$ such that*

$$|u(\phi)| \leq C(K)|\phi|_{K,m(K)} = C(K) \sup_{x\in K, |\alpha|\leq m(K)} |\partial^\alpha \phi(x)| \text{ for all } \phi \in \mathcal{D}_K. \tag{1.90}$$

Remark 1.25. (1) The space $\mathcal{D}_K$ can be given the metric.

$$d(\phi, \psi) := \sum_{m=1}^{\infty} 2^{-m} \frac{|\phi - \psi|_{K,m}}{1 + |\phi - \psi|_{K,m}}. \tag{1.91}$$

and it turns out $\mathcal{D}_K$ is complete with this metric; see Exercise 1.3.1(a). A neighborhood base at $0 \in \mathcal{D}_K$ for the metric topology is given by either of the sets

$$\begin{aligned} &\{O_\epsilon : \epsilon > 0\}, \text{ where } O_\epsilon := \{\phi \in \mathcal{D}_K : d(\phi, 0) < \epsilon\}, \text{ or} \\ &\{\Omega_{\epsilon,m} : \epsilon > 0, m \in \mathbb{N}_0\}, \text{ where } \Omega_{\epsilon,m} := \{\phi \in \mathcal{D}_K : |\phi|_{K,m} < \epsilon\}. \end{aligned} \tag{1.92}$$

The space $\mathcal{D}_K$ with this metric is an example of a *Fréchet space*, that is, a locally convex topological vector space with its topology defined by a complete translation-invariant metric; see [Wil26], section 3.2. Another example of a Fréchet topology is the standard topology on $C(U, \mathbb{C})$, where $U \subset \mathbb{C}$ is open, defined in complex analysis texts in discussions of normal families.

(2) It is not obvious that there exists a topology τ on $\mathcal{D}(U)$ such that a linear functional u on $\mathcal{D}(U)$ is a distribution in the sense of Definition 1.23 if and only if $u : (\mathcal{D}(U), \tau) \to \mathbb{C}$ is continuous. Such a topology is described in section 8.9; it can be defined as the "inductive limit" of the metric topologies on the spaces $\mathcal{D}_K$ for $K \subset U$ compact.

Exercise 1.3.1a. Show that a sequence $\phi_n \to \phi$ in $\mathcal{D}_K$ with the above metric if and only if for any $m \in \mathbb{N}_0$ we have $|\phi_n - \phi|_{K,m} \to 0$, that is, if and only if

$$\text{for any } m \text{ and } |\alpha| \leq m,\ \partial^\alpha \phi_n \to \partial^\alpha \phi \text{ uniformly on } K.$$

Similiarly, show that (ϕ_n) is Cauchy in $\mathcal{D}_K$ if and only if it is Cauchy in every norm $|\cdot|_{K,m}$. Deduce the completeness of $\mathcal{D}_K$ with this metric.

The next "definition" is actually a consequence of the definition of the inductive limit topology τ on $\mathcal{D}(U)$; see [Wil26], Theorem 12.9.

Definition 1.26. *A sequence (ϕ_n) in $\mathcal{D}(U)$ is Cauchy (resp. convergent to $\phi \in \mathcal{D}(U)$) if there exists a compact set $K \subset U$ such that*
(i) $\phi_n \in \mathcal{D}_K$ for all n;
(ii) (ϕ_n) is Cauchy (resp. convergent to ϕ) in $\mathcal{D}_K$.

It follows immediately from this definition and Exercise 1.3.1a that $(\mathcal{D}(U), \tau)$ is complete.

Notations 1.27. *When $u \in \mathcal{D}'(U)$ and $\phi \in \mathcal{D}(U)$, we'll often write the distributional pairing as $\langle u, \phi \rangle := u(\phi)$.*

Examples 1.28. 1. If $f \in L^1_{loc}(U)$, then f defines a distribution "by integration", and we write $f \in \mathcal{D}'(U)$. To prove this let $K \subset U$ be compact, and observe that for $\phi \in \mathcal{D}_K$ we have

$$|\langle f, \phi \rangle| = |\int_U f\phi dx| = |\int_K f\phi dx| \leq \sup_{x \in K} |\phi| \left(\int_K |f| dx \right) = C_K(f) \sup_{x \in K} |\phi|.$$

2. Let μ be any Borel measure on U that is finite on compact sets. Then μ defines a distribution by integration; indeed, for any compact $K \subset U$, we have for $\phi \in \mathcal{D}_K$:

$$|\langle \mu, \phi \rangle| = |\int_U \phi d\mu| = |\int_K \phi d\mu| \leq \mu(K) \sup_{x \in K} |\phi|.$$

3. For $a \in U$ define $\delta_a(\phi) := \phi(a)$. Then $\delta_a \in \mathcal{D}'(U)$ since for compact $K \subset U$, we have for $\phi \in \mathcal{D}_K$:

$$|\delta_a(\phi)| = |\phi(a)| \leq \sup_{x \in K} |\phi|.$$

Observe that if we define a measure μ_a on the power set of U by $\mu(E) = \begin{cases} 1, & \text{if } a \in E \\ 0, & \text{if } a \neq E \end{cases}$, then $\int_U \phi d\mu_a = \phi(a) = \delta_a(\phi)$; hence, $\delta_a = \mu_a$ as distributions. We refer to δ_a (correctly) as the "Dirac measure at a", or (incorrectly) as the Dirac "delta function" at a. We usually write $\delta_0 = \delta$.

Exercise 1.3.1b. Show that there is *no* $f \in L^1_{loc}(\Omega)$ such that $\delta = f$ as distributions.

1.3.2 Operations on distributions

We now define *distribution derivatives*, which generalize the "weak derivatives" mentioned in the Introduction.

Proposition 1.29. *Let $u \in \mathcal{D}'(U)$, $f \in C^\infty(U)$, and $\phi \in \mathcal{D}(U)$.*
(a) Define the linear functional $\partial^\alpha u$ by $\langle \partial^\alpha u, \phi \rangle := (-1)^{|\alpha|} \langle u, \partial^\alpha \phi \rangle$. Then $\partial^\alpha u \in \mathcal{D}'(U)$.
(b) Define the linear functional fu by $\langle fu, \phi \rangle := \langle u, f\phi \rangle$. Then $fu \in \mathcal{D}'(U)$.

Proof. **(a).** Fix a compact subset $K \subset U$. Since $u \in \mathcal{D}'(U)$ there exists N and C such that for $\phi \in \mathcal{D}_K$ we have

$$|\langle \partial^\alpha u, \phi \rangle| = |\langle u, \partial^\alpha \phi \rangle| \leq C \sup_{|\beta| \leq N, x \in K} |\partial^\beta (\partial^\alpha \phi)| \leq C \sup_{|\gamma| \leq N + |\alpha|, x \in K} |\partial^\gamma \phi|.$$

(b). Left to you. □

Remark 1.30.
Given $u \in \mathcal{D}'(U)$ and any linear partial differential operator of order $m \in \mathbb{N}$,

$$P(x, \partial) := \sum_{|\alpha| \leq m} c_\alpha(x) \partial^\alpha \text{ where } c_\alpha \in C^\infty(U),$$

by Proposition 1.29 it makes sense to apply $P(x, \partial)$ to u and we have $P(x, \partial)u \in \mathcal{D}'(U)$. One can now study partial differential equations with distributional "forcing terms" $f \in \mathcal{D}'(U)$, $P(x, \partial)u = f$, and look for solutions $u \in \mathcal{D}'(U)$. When f is regular, finding a distributional solution is sometimes a useful step toward finding a regular solution.

An important early application of distribution theory was the proof of the Malgrange-Ehrenpreis theorem, which states that if $P(D)$ is any constant coefficient linear differential operator, then $P(D)E = \delta$ has a solution $E \in \mathcal{D}'(\mathbb{R}^n)$ [Yos65]. The solution E is called a *fundamental solution*.

Examples 1.31. 1. Consider $H \in L^1_{loc}(\mathbb{R})$ given by $H(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$, the Heaviside function. We claim $H' = \delta$. Indeed if $\phi \in \mathcal{D}(\mathbb{R})$ we have

$$\langle H', \phi \rangle = -\langle H, \phi' \rangle = -\int_{\mathbb{R}} H\phi' dx = -\int_0^\infty \phi'(x)dx = \phi(0) = \langle \delta, \phi \rangle.$$

Similarly, for $a \in \mathbb{R}$ if $H_a = \begin{cases} 1, & x > a \\ 0, & x < a \end{cases}$, then $H_a' = \delta_a$.

2. Let $x_+ = \begin{cases} x, & x > 0 \\ 0, & x < 0 \end{cases}$. Then $(x_+)' = H$ (check).

Exercise 1.3.2a. Define $H \in \mathcal{D}'(\mathbb{R}^2)$ by $H(x,y) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$. Compute the distribution derivatives $\partial_x H$ and $\partial_y H$.

3. The almost everywhere defined function $u(x) = \ln(|x|)$ is locally integrable on $\mathbb{R}$, and hence defines an element $u \in \mathcal{D}'(\mathbb{R})$. Let us find $u' \in \mathcal{D}'(\mathbb{R})$. For $x \neq 0$ the pointwise classical derivative of u is $1/x$. But this can't be the distribution derivative, since $1/x$ is not locally integrable near $x = 0$. To find u' we compute for $\phi \in \mathcal{D}(\mathbb{R})$

$$\begin{aligned} \langle u', \phi \rangle = -\langle u, \phi' \rangle = -\lim_{\epsilon \to 0} \left(\int_\epsilon^\infty \ln x \, \phi'(x)dx + \int_{-\infty}^{-\epsilon} \ln(-x)\phi'(x)dx \right) = \\ -\lim_{\epsilon \to 0} \left(-\ln(\epsilon)\phi(\epsilon) - \int_\epsilon^\infty \frac{\phi(x)}{x}dx + \ln(\epsilon)\phi(-\epsilon) - \int_{-\infty}^{-\epsilon} \frac{\phi(x)}{x}dx \right) = \\ \lim_{\epsilon \to 0} \ln(\epsilon)[\phi(\epsilon) - \phi(-\epsilon)] + \lim_{\epsilon \to 0} \int_{|x|>\epsilon} \frac{\phi(x)}{x}dx = 0 + \langle p.v.\frac{1}{x}, \phi \rangle. \end{aligned} \tag{1.93}$$

Thus, $\frac{d}{dx}\ln(|x|) = p.v.\frac{1}{x}$, the element of $\mathcal{D}'(\mathbb{R})$ defined by $\lim_{\epsilon \to 0} \int_{|x|>\epsilon} \frac{\phi(x)}{x}dx$. We call $p.v.\frac{1}{x}$ the *principal value* of $\frac{1}{x}$.

Exercise 1.3.2b. The function $\Phi(x) = \frac{c_n}{|x|^{n-2}}$ lies in $L^1_{loc}(\mathbb{R}^n)$ and hence defines a distribution $\Phi \in \mathcal{D}'(\mathbb{R}^n)$.

(a) Show that $-\Delta\Phi = \delta$ in $\mathcal{D}'(\mathbb{R}^n)$.

(b) More generally, show that if $a \in \mathbb{R}^n$, then $-\Delta\Phi(\cdot - a) = \delta_a$ in $\mathcal{D}'(\mathbb{R}^n)$. (Hint: The main work was done in the proof of Theorem 1.2.)

Definition 1.32 (Localization and support)**.** *(a) Let $\omega \subset U$ be open and suppose $u_1, u_2 \in \mathcal{D}'(U)$. We say $u_1 = u_2$ in ω if $\langle u_1, \phi \rangle = \langle u_2, \phi \rangle$ for all $\phi \in \mathcal{D}(\omega)$. Here we regard $\mathcal{D}(\omega)$ as a subspace of $\mathcal{D}(U)$ in the obvious way.*

(b) Let $u \in \mathcal{D}'(U)$. Let $\mathcal{O} \subset U$ be the largest open set in which $u = 0$.[17] *Define the* support *of u to be supp $u := \mathcal{O}^c \subset U$.*

Examples 1.33. (1) Suppose $a \in U$. The support of δ_a is $\{a\}$. Similarly for any multiindex α the support of $\partial^\alpha \delta_a$ is $\{a\}$. One can show that if $u \in \mathcal{D}'(U)$ and supp $u = \{a\}$, then u has the form $u = \sum_{|\alpha| \le N} c_\alpha \partial^\alpha \delta_a$ for some N and $c_\alpha \in \mathbb{C}$; for a proof see [Rud91], Theorem 6.25.

(2) If $f \in C(U)$, the support of f as a distribution equals the support of f as a function, namely, $\overline{\{x \in U : f(x) \neq 0\}}$.(check)

(3) Suppose $f \in L^1_{loc}(U)$. The definition of supp f as a function used in (2) makes no sense now, but the support of f as a distribution is well-defined. What is it?

Definition 1.34 (Topology on $\mathcal{D}'(U)$)**.** *The standard topology on $\mathcal{D}'(U)$ is the weak*-topology; it is defined by taking the following local subbase $s\mathcal{B}$ at $0 \in \mathcal{D}'(U)$:*

$$s\mathcal{B} := \{\mathcal{O}(\phi, \epsilon) : \phi \in \mathcal{D}(U), \epsilon > 0\}, \text{ where } \mathcal{O}(\phi, \epsilon) := \{u \in \mathcal{D}'(U) : |\langle u, \phi \rangle| < \epsilon\}.$$

A local base $\mathcal{B}$ at 0 is given by the set of all finite intersections of sets in $s\mathcal{B}$, and a local base at $v \in \mathcal{D}'(U)$ is $v + \mathcal{B}$.

Remark 1.35. (1) For each $\phi \in \mathcal{D}(U)$ the function $p_\phi : \mathcal{D}' \to \mathbb{R}$ given by $p_\phi(u) := |\langle u, \phi \rangle|$ is an example of a *seminorm*. A *seminorm* $p : V \to \mathbb{R}$ on a vector space V is a nonnegative function such that $p(\alpha v) = |\alpha| p(v)$ and $p(v + w) \le p(v) + p(w)$, but $p(v) = 0$ does not imply $v = 0$.

(2) Observe that $u_n \to u$ in $\mathcal{D}'(U)$ with this topology if and only if $\langle u_n, \phi \rangle \to \langle u, \phi \rangle$ for all $\phi \in \mathcal{D}(U)$.

Exercise 1.3.2c. (i) Show that the topology of $\mathcal{D}'(U)$ is Hausdorff. Use that fact that the family of seminorms $\mathcal{F} = \{p_\phi : \phi \in \mathcal{D}(U)\}$ is *separating*; that is, given $u \in \mathcal{D}'(U)$, $u \neq 0$, there exists an element $p_\phi \in \mathcal{F}$ such that $p_\phi(u) \neq 0$. The Hausdorff property implies that limits of sequences are unique.

(ii) Suppose $\rho \in C_c^\infty(\mathbb{R}^n)$ satisfies $\rho \ge 0$ and $\int \rho(x) dx = 1$. For $k \in \mathbb{N}$ let $\rho_k(x) := k^n \rho(kx)$. Show $\rho_k \to \delta$ in $\mathcal{D}'(\mathbb{R}^n)$. The sequence (ρ_k) is called an *approximate identity*.

(iii) Let $P(x, \partial_x) = \sum_{|\alpha| \le m} c_\alpha(x) \partial^\alpha$, where $c_\alpha \in C^\infty(U)$ for each α. Show that $P(x, \partial_x) : \mathcal{D}'(U) \to \mathcal{D}'(U)$ is continuous.

1.3.3 The Fourier transform

The Fourier transform is useful for many reasons. One is that it sometimes allows us to solve differential equations by doing some simple algebra. Motivation comes from the theory of Fourier series, where $2\pi-$periodic functions are expressed as infinite sums of oscillatory functions like $e^{in\theta}$, $n \in \mathbb{N}$; see the Fourier inversion formula, Theorem 1.40.

We begin by defining the Fourier transform for functions in $L^1(\mathbb{R}^n)$ and eventually "use duality" to extend the definition to the class of tempered distributions $\mathcal{S}'(\mathbb{R}^n)$.

[17] That is, $\mathcal{O}$ is the union of all the open $\omega \subset U$ such that $u = 0$ in ω.

Definition 1.36 (Fourier transform)**.** *Let* $f \in L^1(\mathbb{R}^n)$*. For* $\xi \in \mathbb{R}^n$ *define* the Fourier tranform of f*, denoted* $\mathcal{F}f$ *or* $\hat{f}$*, by*

$$\mathcal{F}f(\xi) = \hat{f}(\xi) := \int_{\mathbb{R}^n} e^{-ix\cdot\xi} f(x) d'x, \text{ where } d'x := (2\pi)^{-n/2} dx. \tag{1.94}$$

Observe that $|\hat{f}(\xi)| \leq \int |e^{-ix\cdot\xi}||f(x)|d'x = \int |f(x)|d'x < \infty$ and the dominated convergence theorem implies $\hat{f}$ is continuous.

Notations 1.37. 1. Let $D_{x_j} := \frac{1}{i}\partial/\partial x_j = \frac{1}{i}\partial_{x_j}$ and for a multiindex α set

$$D^\alpha = (D_{x_1})^{\alpha_1}(D_{x_2})^{\alpha_2}\cdots(D_{x_n})^{\alpha_n}.$$

So we have $D^\alpha = \left(\frac{1}{i}\right)^{|\alpha|}\partial^\alpha = (-1)^{|\alpha|} i^{|\alpha|}\partial^\alpha$.

2. We set $(-D)^\alpha := (-1)^{|\alpha|}D^\alpha$. Sometimes we write $(-D_\xi)^\alpha := (-1)^{|\alpha|}D_\xi^\alpha$ to indicate differentiation with respect to ξ.

Define $\mathcal{S}(\mathbb{R}^n)$ to be the space of *Schwartz functions* $f \in C^\infty(\mathbb{R}^n)$ such that for all $N \in \mathbb{N}_0$ we have $q_N(f) < \infty$, where

$$q_N(f) = \sup_{x\in\mathbb{R}^n, |\alpha|\leq N} (1+|x|)^N |\partial^\alpha f(x)|. \tag{1.95}$$

Parallel to (1.91) the norms q_N can be assembled to define the metric

$$d(\phi,\psi) := \sum_{N=0}^{\infty} 2^{-N} \frac{q_N(\phi-\psi)}{1+q_N(\phi-\psi)}, \phi,\psi \in \mathcal{S}(\mathbb{R}^n). \tag{1.96}$$

The space $\mathcal{S}(\mathbb{R}^n)$ is a complete metric space with this metric, in fact, a Fréchet space.

Next we show that the Fourier transform changes D^α into ξ^α.

Proposition 1.38. *Let* $f \in \mathcal{S}(\mathbb{R}^n) =: \mathcal{S}$*. We have*

(i) $\widehat{D^\alpha f} = \xi^\alpha \hat{f}$*;*

(ii) $\widehat{x^\alpha f} = (-D_\xi)^\alpha \hat{f}$*.*

Proof. **(i)** Using (1.94) and integrating by parts we have

$$\widehat{D^\alpha f}(\xi) = \int e^{-ix\xi} D^\alpha f(x) d'x = \xi^\alpha \int e^{-ix\xi} f(x) d'x = \xi^\alpha \hat{f}(\xi). \tag{1.97}$$

There are no boundary terms since f decays rapidly, that is, faster than $(1+|x|)^{-N}$ for all N.

(ii) Using (1.94) and differentiating under the integral sign we have

$$\partial_\xi^\alpha \hat{f}(\xi) = \int e^{-ix\xi}(-ix)^\alpha f(x) d'x. \tag{1.98}$$

Multiply by $i^{|\alpha|}$ to finish. □

Proposition 1.39. *The Fourier transform is a continuous linear map $\mathcal{F} : \mathcal{S} \to \mathcal{S}$.*

Proof. Linearity is clear. We first prove that $f \in \mathcal{S} \Rightarrow \hat{f} \in \mathcal{S}$. For any multiindices α, β we have

$$|\xi^\alpha \partial_\xi^\beta \hat{f}(\xi)| = |\widehat{D_x^\alpha(x^\beta f)}| = \left| \int e^{-ix\xi} D_x^\alpha (x^\beta f)\, d'x \right| \leq C_{\alpha\beta}(f)$$

for some constant $C_{\alpha\beta}(f)$. To prove continuity note that

$$\begin{aligned} |\xi^\alpha \partial_\xi^\beta \hat{f}(\xi)| = \left| \int e^{-ix\xi} D_x^\alpha (x^\beta f)\, d'x \right| \leq \int (1+|x|)^{n+1} |D_x^\alpha (x^\beta f)| (1+|x|)^{-(n+1)} d'x \leq \\ C_n \sup_{x \in \mathbb{R}^n, |\gamma| \leq |\alpha|} (1+|x|)^{n+1+|\beta|} |\partial^\gamma f| \leq C_n q_N(f), \end{aligned} \tag{1.99}$$

where $N = \max(n+1+|\beta)|, |\alpha|)$. Now apply part (b) of Exercise 1.3.3a.

□

The next theorem will allow us to show that the Fourier transform is a homeomorphism of $\mathcal{S}$ onto itself.

Theorem 1.40 (Fourier inversion formula)**.** *For $g \in \mathcal{S}(\mathbb{R}^n)$ we have $g(x) = \int e^{ix\cdot\xi} \hat{g}(\xi) d'\xi$.*

We sometimes describe the integral giving g as the "inverse Fourier transform" of $\hat{g}$. The proof of Theorem 1.40 uses the next proposition.

Proposition 1.41. *Let $\phi \in \mathcal{S}(\mathbb{R}^n)$, $a \in \mathbb{R}^n$, and $\lambda > 0$. We have:*
(i) if $h(x) = \phi(x+a)$, then $\hat{h}(\xi) = e^{i\xi\cdot a}\hat{\phi}(\xi)$;
(ii) if $g(x) = e^{ix\cdot a}\phi(x)$, then $\hat{g}(\xi) = \hat{\phi}(\xi - a)$;
(iii) if $p(x) = \phi(x/\lambda)$, then $\hat{p}(\xi) = \lambda^n \hat{\phi}(\lambda\xi)$.

Proof. These are immediate from the definition of $\mathcal{F}$. For example,

$$\hat{p}(\xi) = \int e^{-ix\xi} \phi(x/\lambda) d'x = \int e^{-i\lambda y\cdot\xi} \phi(y) \lambda^n d'y = \lambda^n \hat{\phi}(\lambda\xi).$$

□

Proof of Theorem 1.40. **1.** The main step is to prove the special case

$$g(0) = \int \hat{g}(\xi) d'\xi \text{ for } g \in \mathcal{S}. \tag{1.100}$$

Given this we finish by taking $k(x) := g(x+a)$ and using

$$k(0) = \int \hat{k}(\xi) d'\xi \Leftrightarrow g(a) = \int e^{i\xi a} \hat{g}(\xi) d'\xi.$$

2. Let $f, g \in \mathcal{S}$. By considering $\int\int f(x)g(\xi)e^{-ix\xi}d'xd'\xi$ and applying Fubini's theorem, we obtain

$$\int \hat{f}(\xi)g(\xi)d'\xi = \int f(x)\hat{g}(x)d'x, \tag{1.101}$$

an important identity in its own right.

3. Let $\phi, g \in \mathcal{S}$ and $\lambda > 0$, and set $f(x) = \phi(x/\lambda)$. Then by (1.101)

$$\int \hat{f}(\xi)g(\xi)d'\xi = \int \lambda^n \hat{\phi}(\lambda\xi)g(\xi)d'\xi = \int \phi(x/\lambda)\hat{g}(x)d'x. \tag{1.102}$$

Setting $\eta = \lambda\xi$ the latter equation becomes

$$\begin{gathered} \int \hat{\phi}(\eta)g(\eta/\lambda)d'\eta = \int \phi(x/\lambda)\hat{g}(x)d'x, \text{ which gives} \\ g(0)\int \hat{\phi}(\eta)d'\eta = \phi(0)\int \hat{g}(x)d'x \end{gathered} \tag{1.103}$$

upon letting $\lambda \to \infty$ and applying the dominated convergence theorem. Now if we can choose ϕ such that $\int \hat{\phi}(\eta)d'\eta = 1$ and $\phi(0) = 1$, we see that (1.103) implies (1.100).

4. I claim that $\phi(x) = e^{-|x|^2/2} \in \mathcal{S}(\mathbb{R}^n)$ works. Clearly $\phi(0) = 1$. Next we show $\hat{\phi}(\xi) = e^{-|\xi|^2/2}$ and $\int \hat{\phi}(\xi)d'\xi = 1$. For this it suffices to prove both assertions in the case $n = 1$ (why?).[18]

For $x \in \mathbb{R}^1$ observe that $\phi(x) = e^{-x^2/2}$ satisfies the initial value problem:

$$\phi'(x) = -x\phi(x), \ \phi(0) = 1.$$

Proposition 1.38 implies $\frac{d}{d\xi}\hat{\phi} = -\xi\hat{\phi}$, the same ODE. Moreover, $\hat{\phi}(0) = \int_{\mathbb{R}} e^{-ix\cdot 0}e^{-x^2/2}d'x = 1$.[19] By uniqueness for initial value problems we conclude $\phi = \hat{\phi}$. The previous integral also shows $\int_{\mathbb{R}} e^{-\xi^2/2}d'\xi = 1$, so we are done.

□

Corollary 1.42. *The Fourier transform $\mathcal{F}$ is a homeomorphism of $\mathcal{S}(\mathbb{R}^n)$ onto itself.*

Proof. The continuity of $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ was proved earlier. Observe that if $\phi \in \mathcal{S}$ Theorem 1.40 implies

$$\hat{\hat{\phi}}(x) = \int e^{-ix\xi}\hat{\phi}(\xi)d'\xi = \phi(-x) \Rightarrow \hat{\hat{\hat{\phi}}} = \phi. \tag{1.104}$$

This implies $\mathcal{F}$ is bijective on $\mathcal{S}$. Since $\mathcal{F}^{-1} = \mathcal{F} \circ \mathcal{F} \circ \mathcal{F}$, we see that $\mathcal{F}^{-1}$ is continuous. □

[18] Answer: To show $\hat{\phi}(\xi) = e^{-|\xi|^2/2}$, write $e^{-|x|^2/2}$ and $e^{-ix\cdot\xi}$ as products of n factors and use Fubini's theorem.

[19] This integral, call it I, is done in multivariable calculus by computing I^2 using polar coordinates.

Definition 1.43 (Convolutions I)**.** *For $f, g \in L^1(\mathbb{R}^n)$ define the convolution $f * g(x) = \int f(x-y)g(y)d'y$.*

Proposition 1.44. *The function $f * g$ lies in $L^1(\mathbb{R}^n)$. Moreover,*

$$\widehat{f * g} = \hat{f}\hat{g}.$$

Proof. **1.** To conclude $f * g \in L^1(\mathbb{R}^n)$ by Fubini's theorem, we estimate

$$\int\int |f(x-y)|g(y)|dydx = \int\int |f(x-y)|g(y)|dxdy = |f|_{L^1}|g|_{L^1}. \tag{1.105}$$

2. We compute

$$\begin{aligned}\widehat{f * g} = \int e^{-ix\xi}\int f(x-y)g(y)d'yd'x = \int g(y)\int e^{-ix\xi}f(x-y)d'xd'y = \\ \int g(y)\int e^{-i(y+z)\xi}f(z)d'zd'y = \hat{f}\hat{g}.\end{aligned} \tag{1.106}$$

□

Exercise 1.3.3a (a) Suppose $u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is a linear functional. Use the definition of the metric (1.96) to show that u is continuous on $\mathcal{S}(\mathbb{R}^n)$ if and only if there exists N, C_N such that

$$|u(\phi)| \leq C_N q_N(\phi) \text{ for all } \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.107}$$

See [Wil26], Proposition 4.3 for a more general version of this.

(b) Similarly, show that a linear map $h : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous if and only if given any fixed q_M, there exists N, C_N such that

$$q_M(\phi) \leq C_N q_N(\phi) \text{ for all } \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.108}$$

1.4 The heat equation

The heat equation governs diffusion processes, such as temperature changes in a heat conductor or the mixing of two fluids by Brownian motion. Our first task is to derive the "heat kernel" $H(t, x)$, which plays a role for the heat equation similar to that of $\Phi(x)$ for Laplace's equation.

1.4.1 The heat kernel and the initial value problem

Consider the parabolic problem

$$\begin{aligned} &u_t - \Delta u = 0 \text{ on } (0,\infty)_t \times \mathbb{R}^n_x \\ &u|_{t=0} = g \in \mathcal{S}(\mathbb{R}^n). \end{aligned} \tag{1.109}$$

Assume for the moment that there exists a solution $u \in C^1([0,\infty), \mathcal{S})$. Taking the Fourier transform of (1.109) in x and writing $\hat{u} = \hat{u}(t,\xi)$, we obtain the family of ODEs in t parametrized by ξ:

$$\begin{aligned} &\hat{u}_t + |\xi|^2 \hat{u} = 0 \\ &\hat{u}(0,\xi) = \hat{g}(\xi). \end{aligned} \tag{1.110}$$

For each $\xi \in \mathbb{R}^n$ (1.110) has the unique solution $\hat{u}(t,\xi) = \hat{g}(\xi) e^{-t|\xi|^2}$.

Remark 1.45. This formula for $\hat{u}$ shows that $\hat{u} \in C([0,\infty), \mathcal{S}(\mathbb{R}^n_\xi))$, which implies $(\hat{u})^\vee \in C([0,\infty), \mathcal{S}(\mathbb{R}^n_x))$, where $\vee$ denote the inverse Fourier transform in x. In fact, the formula shows that for any $m \in \mathbb{N}$, $\hat{u} \in C^m([0,\infty), \mathcal{S})$, so $(\hat{u})^\vee \in C^m([0,\infty), \mathcal{S})$. Taking $u = (\hat{u})^\vee$ we see that our starting assumption is actually satisfied.

Using Proposition 1.44 we can recover $u(t,x)$ in $t > 0$ by taking

$$u(t,x) = \left(\hat{g}(\xi) e^{-t|\xi|^2}\right)^\vee = \left(g *_x (e^{-t|\xi|^2})^\vee\right)(x) \tag{1.111}$$

Step **4** of the proof of Theorem 1.40 showed $(e^{-|x|^2/2})^\wedge = e^{-|\xi|^2/2}$, so a change of variable implies

$$(e^{-t|\xi|^2})^\vee = (2t)^{-n/2} e^{-|x|^2/4t} := h(t,x) \text{ for } t > 0. \tag{1.112}$$

Thus we obtain[20]

$$\begin{aligned} &u(t,x) = (g *_x h(t,\cdot)(x) = (4\pi t)^{-n/2} \int g(y) e^{-|x-y|^2/4t} dy, \ t > 0 \\ &u(0,x) = g(x). \end{aligned} \tag{1.113}$$

Definition 1.46. *The function $H(t,x) := (4\pi t)^{-n/2} e^{-|x|^2/4t} = (2\pi)^{-n/2} h(t,x)$, $t > 0$ is generally referred to as the "heat kernel".*

Next we study the problem (1.109) under a much weaker assumption on the initial datum g.

Theorem 1.47. *Consider the problem (1.109) with $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, and let $u = g *_x h(t,\cdot)$ as in (1.113). Then*

(a) $u \in C^\infty((0,\infty)_t \times \mathbb{R}^n) \cap L^\infty((0,\infty)_t \times \mathbb{R}^n)$;

(b) $u_t - \Delta u = 0$ in $t > 0$;

(c) $\lim_{(t,x)\to(0,x_0)} u(t,x) = g(x_0)$ for $x_0 \in \mathbb{R}^n$, and the convergence is uniform for x_0 in any compact $K \subset \mathbb{R}^n$;

(d) if $g \in \mathcal{S}$, then u is the unique solution in $C^1([0,\infty), \mathcal{S})$.[21]

[20] Recall that $d'y = (2\pi)^{-n/2} dy$ appears in the definition of convolution.

[21] Recall Remark 1.45.

Proof. **a.** Applying Proposition 8.2, to see that $u \in C^\infty$ in $t > 0$ we can differentiate repeatedly under the integral sign in (1.113). Observe that for $t > 0$

$$(4\pi t)^{-n/2} \int e^{-|x-y|^2/4t} dy = (4\pi t)^{-n/2} \int e^{-|y|^2/4t} dy = \int h(t,y) d'y = \hat{h}(t,\xi)|_{\xi=0} = 1. \quad (1.114)$$

Thus, the formula (1.113) implies $|u|_{L^\infty((0,\infty)_t \times \mathbb{R}^n)} \leq |g|_{L^\infty(\mathbb{R}^n)}$.

b. Applying $\partial_t - \Delta$ under the integral sign in (1.113) proves (b), since $(\partial_t - \Delta)H = 0$ in $t > 0$.

c. Part (c) follows directly from Exercise 8.5a and Remark 8.12, since the heat kernel is an approximate identity.

d. If $v \in C^1([0,\infty), \mathcal{S})$ is another solution, then $w = u - v$ satisfies

$$w_t - \Delta w = 0, \; w(0,x) = 0 \quad (1.115)$$

Setting $E(t) = \int |w|^2(t,x)dx = \int w\overline{w}dx$, we compute[22]

$$\begin{aligned} E'(t) = \int (w_t \overline{w} + w\overline{w}_t)dx = 2\mathrm{Re} \int w_t \overline{w} dx = \\ 2\mathrm{Re} \int \Delta w \; \overline{w} dx = 2\mathrm{Re}(-\int |\nabla w|^2 dx) \leq 0. \end{aligned} \quad (1.116)$$

Thus, $E(t)$ is decreasing, $E(t) \geq 0$, and $E(0) = 0$, so $E(t)$ must be zero for all $t \geq 0$. Hence $w = 0$.

□

Remark 1.48. (i) Note the *regularizing property* of $u_t - \Delta u = 0$. The initial data was merely continuous, but u is C^∞ in $t > 0$.

(ii) The solution u as in (1.113) exhibits *infinite propagation speed*. Suppose $g \geq 0$ is not identically zero and has compact support. Then for any $t > 0$, no matter how small, $u(t,x) > 0$ for all x.

(iii) Check that the uniqueness argument for part (d) still works if we assume that u and v are solutions of (1.110) in

$$C^1([0,\infty), L^2(\mathbb{R}^n)) \cap C([0,\infty), H^2(\mathbb{R}^n)),$$

where for $m \in \mathbb{N}_0$:

$$H^m(\mathbb{R}^n) = \{u \in L^2 : \partial^\alpha u \in L^2, |\alpha| \leq m\} \text{ with } |u|_{H^m} = \sum_{|\alpha| \leq m} |\partial^\alpha u|_{L^2}\}.$$

[22] There is no boundary term after integration by parts since $w(t,\cdot) \in \mathcal{S}$.

Exercise 1.4.1a: fundamental solution. Let $H(t,x)$ be the heat kernel on $(0,\infty)\times\mathbb{R}^n$. Define its extension to $\mathbb{R}\times\mathbb{R}^n$ by

$$H_e(t,x)=\begin{cases}H(t,x),\ t>0\\ 0,\ t\le 0\end{cases}.$$

(a) Show that $H_e\in L^1_{loc}(\mathbb{R}\times\mathbb{R}^n)$, and thus $H_e\in\mathcal{D}'(\mathbb{R}\times\mathbb{R}^n)$.
(b) Show that H_e is a fundamental solution for the heat operator; that is,

$$(\partial_t-\Delta)H_e=\delta\in\mathcal{D}'(\mathbb{R}^n\times\mathbb{R}).$$

Hint: For $\eta>0$ define $H_\eta(x,t)=\begin{cases}H_e,\ t>\eta\\ 0,\ t\le\eta\end{cases}$, and show:
(c) $H_\eta\to H_e$ in $\mathcal{D}'(\mathbb{R}\times\mathbb{R}^n)$ as $\eta\to 0$;
(d) $(\partial_t-\Delta)H_\eta\to\delta$ in $\mathcal{D}'(\mathbb{R}\times\mathbb{R}^n)$ as $\eta\to 0$.[23]
(e) Where does your argument use the fact that the topology of $\mathcal{D}'(\mathbb{R}\times\mathbb{R}^n)$ is Hausdorff?

Consider next the inhomogeneous problem

$$\begin{aligned}&(a)u_t-\Delta u=f\text{ in }(0,\infty)\times\mathbb{R}^n\\ &(b)u(0,x)=0\text{ for }x\in\mathbb{R}^n.\end{aligned}\tag{1.117}$$

Duhamel's principle suggests we should look for a solution of the form

$$\begin{aligned}u(t,x)=\int_0^t w(t,x;s)ds=\int_0^t\int_{\mathbb{R}^n}H(t-s,x-y)f(s,y)dyds=\\ \int_0^t\int_{\mathbb{R}^n}H(s,y)f(t-s,x-y)dyds\end{aligned}\tag{1.118}$$

since $w(t,x;s):=\int_{\mathbb{R}^n}H(t-s,x-y)f(s,y)dy$ satisfies

$$\begin{aligned}&w_t-\Delta w=0\text{ in }t>s\\ &w(s,x;s)=f(s,x).\end{aligned}\tag{1.119}$$

Theorem 1.49. *Suppose*

$$f\in C_c^{1,2}([0,\infty)\times\mathbb{R}^n):=\{f\in C_c([0,\infty)\times\mathbb{R}^n):f_t,D_xf,D_x^2f\in C([0,\infty)\times\mathbb{R}^n)\},$$

and let u be as in (1.118). Then $u\in C^{1,2}((0,\infty)\times\mathbb{R}^n)$ and satisfies

$$\begin{aligned}&(a)u_t-\Delta u=f\ \textit{in}\ (0,\infty)\times\mathbb{R}^n\\ &(b)\lim_{t\searrow 0}u(t,x)=0\ \textit{uniformly for}\ x\in\mathbb{R}^n.\end{aligned}\tag{1.120}$$

In particular, u has a continuous extension to $[0,\infty)\times\mathbb{R}^n$.

[23] Recall Exercise 1.3.2c.

Proof. **1.** For (t,x) near a fixed $(t_0,x_0) \in \{t > 0\}$, the last integrand in (1.118) has $(s,y)-$support in a fixed compact set. The function $H(t,x)$ is locally integrable in $t \geq 0$, so the integral defining $u(x,t)$ converges absolutely. Continuity of u near (t_0,x_0) follows by the dominated convergence theorem.

Applying Proposition 8.2 and the dominated convergence theorem, we conclude that $D_x u$ and $D_x^2 u$ are continuous near (t_0,x_0) since $f \in C_c^{1,2}([0,\infty)\times\mathbb{R}^n)$. To analyze u_t we compute

$$u_t = \int_0^t \int_{\mathbb{R}^n} H(s,y) f_t(t-s,x-y)dyds + \int_{\mathbb{R}^n} H(t,y) f(0,x-y)dy, \tag{1.121}$$

and this implies u_t is continuous near (t_0,x_0) by the dominated convergence theorem.

Since $H \geq 0$ and $\int H(t,x)dx = 1$ in $t > 0$, the formula (1.118) implies

$$|u(t,x)| \leq t|f|_{L^\infty} \Rightarrow (1.120)(b). \tag{1.122}$$

2. Denote by $K(t,x)$ the second integral in (1.121). It remains to compute

$$\begin{aligned}(u_t - \Delta u)(t,x) = \int_0^t \int_{\mathbb{R}^n} H(t,y)[\partial_t - \Delta_x) f(t-s,x-y)]dyds + K(t,x) = \\ \int_0^t \int_{\mathbb{R}^n} H(t,y)[-\partial_s - \Delta_y) f(t-s,x-y)]dyds + K(t,x) = \\ \int_\epsilon^t \int_{\mathbb{R}^n} (\cdots)dyds + \int_0^\epsilon \int_{\mathbb{R}^n} (\cdots)dyds + K(t,x) := I(t,x) + J(t,x) + K(t,x).\end{aligned} \tag{1.123}$$

Clearly,

$$|J(t,x)| \leq \epsilon|(-\partial_s - \Delta_y) f(t-s,x-y)|_{L^\infty} \to 0 \text{ as } \epsilon \to 0.$$

Integrating by parts in $I(x,t)$ and using $(\partial_s - \Delta_y)H(s,y) = 0$ in $s > 0$ yields

$$I(t,x) = \int_{\mathbb{R}^n} H(\epsilon,y) f(t-\epsilon,x-y)dy - K(t,x). \tag{1.124}$$

Letting $\epsilon \to 0$ in (1.123) and canceling the K terms, we obtain (1.120)(a).(why?)[24]

□

1.4.2 Uniqueness, regularity, maximum principle

Consider a classical solution u of

$$u_t - \Delta u = 0, \quad u|_{t=0} = 0 \text{ in } t > 0. \tag{1.125}$$

In this section we show that if u does not grow too fast, it must be zero.

[24] Answer: Use that $H(\epsilon,y)$ is an approximate identity.

Theorem 1.50 (Uniqueness). *Suppose* $u \in C([0,\infty)\times\mathbb{R}^n)\cap C^{1,2}((0,\infty)\times\mathbb{R}^n)$ *satisfies* (1.125). *Assume that for all* $\epsilon>0$ *there exists* C_ϵ *such that*

$$|u(t,x)|\le C_\epsilon e^{\epsilon|x|^2},\quad |\nabla u(t,x)|\le C_\epsilon e^{\epsilon|x|^2}\ \text{on}\ (0,\infty)\times\mathbb{R}^n. \tag{1.126}$$

Then $u=0$.

Proof. **1. An identity.** Suppose $f(t,x)$, $g(t,x)$ are $C^{1,2}$. Then we check easily that

$$g(\partial_t f-\Delta f)+f(\partial_t g+\Delta g)=\sum_{j=1}^n \partial_j(f\partial_j g-g\partial_j f)+\partial_t(fg)=\nabla_{x,t}\cdot F, \tag{1.127}$$

where $F=(fg, f\partial_1 g-g\partial_1 f,\dots, f\partial_n g-g\partial_n f)$.

2. Pick (t_0,x_0) with $t_0>0$. We will show $u(t_0,x_0)=0$. For any $r>0$ and $0<a<b<t_0$ let Ω be the open cylinder[25]

$$\Omega=\{(t,x):|x-x_0|<r, a<t<b\}.$$

Then the outer unit normals $\nu=(\nu_t,\nu_x)$ on the top and bottom flat sides are $\pm(1,0)$, while that on the lateral side is $(0,\frac{x-x_0}{r})$.

Now take $u(t,x)=f(t,x)$ and $g(t,x)=H(t_0-t,x-x_0)$ in (1.127).[26] Observe

$$(\partial_t-\Delta_x)f=0\ \text{in}\ t>0\ \text{and}\ (\partial_t+\Delta_x)g=0\ \text{for}\ t<t_0. \tag{1.128}$$

Using (1.127), (1.128) and applying Gauss-Green (or the divergence theorem), we obtain

$$\begin{aligned}0=\int_\Omega \nabla_{t,x}\cdot F\,dxdt=\int_{\partial\Omega}F\cdot\nu\,dS=\\ \int_{|x-x_0|\le r}u(b,x)H(t_0-b,x-x_0)dx-\int_{|x-x_0|\le r}u(a,x)H(t_0-a,x-x_0)dx+\\ \sum_{j=1}^n\int_a^b\int_{|x-x_0|=r}[u(t,x)\partial_j H(t_0-t,x-x_0)-H(t_0-t,x-x_0)\partial_j u(t,x)]\,\frac{(x-x_0)_j}{r}dS(x)dt\end{aligned} \tag{1.129}$$

3. The assumption (1.126) implies that for any $\epsilon>0$ there exists C_ϵ such that

$$|u(t,x)\partial_j H(t_0-t,x-x_0)|+|H(t_0-t,x-x_0)\partial_j u(t,x)|\le C_\epsilon e^{\epsilon|x|^2}e^{-\frac{|x-x_0|^2}{4(t_0-t)}} \tag{1.130}$$

for $|x-x_0|=r$, $t\in[a,b]$. Choosing $\epsilon<\frac{1}{4(t_0-a)}\le\frac{1}{4(t_0-t)}$ for $t\in[a,b]$ and letting $r\to\infty$ in (1.129), we see that the integrals on the lateral side of $\partial\Omega$ approach zero.(why?)[27] This gives

$$0=[u(b,\cdot)*H(t_0-b,\cdot)](x_0)-[u(a,\cdot)*H(t_0-a,\cdot)](x_0). \tag{1.131}$$

Letting $b\nearrow t_0$ and $a\searrow 0$ we obtain $0=u(x_0,t_0)$.

□

[25] Draw it.

[26] Here $H(t,x)=\frac{1}{(4\pi t)^{n/2}}e^{-|x|^2/4t}$.

[27] Answer: This would be clear if $e^{\epsilon|x|^2}$ were replaced by $e^{\epsilon|x-x_0|^2}$ in (1.130). Use $|x|\le|x_0|+|x-x_0|$.

Remark 1.51. The theorem can be strengthened. It remains true as stated when the phrase "for all $\epsilon > 0$ there exists C_ϵ such that" in the assumption (1.126) is replaced by "there exists $\epsilon > 0$ and C_ϵ such that". (why?)[28] We can also replace $(0,\infty)\times\mathbb{R}^n$ by $(0,T)\times\mathbb{R}^n$ for any $T > 0$ in these results.

Theorem 1.52 (Regularity)**.** *Let $u \in C^{1,2}((0,\infty)\times\mathbb{R}^n)$ satisfy $u_t - \Delta u = 0$ on $(0,\infty)\times\mathbb{R}^n$. Then u is C^∞ on that set.*

Proof. **1.** First we assume $u \in C^{1,3}((0,\infty)\times\mathbb{R}^n)$ and satisfies $u_t - \Delta u = 0$. Fix (t_0, x_0) with $t_0 > 0$ and choose $\zeta(t,x) \in C_c^\infty((0,\infty)\times\mathbb{R}^n)$ such that $\zeta = 1$ on an open ball $U \ni (t_0,x_0)$. Defining $v = \zeta u$ we compute

$$\begin{aligned} &v_t - \Delta v = \zeta_t u - 2\nabla\zeta\cdot\nabla u - u\Delta\zeta := \tilde{f}(t,x) \in C_c^{1,2}([0,\infty)\times\mathbb{R}^n) \\ &v|_{t=0} = 0, \end{aligned} \tag{1.132}$$

where $\tilde{f} = 0$ on U. By Theorem 1.49 a bounded solution to (1.132) with the regularity assumed in the uniqueness theorem, Theorem 1.50, is given by

$$\tilde{v}(t,x) = \int_0^t \int_{\mathbb{R}^n} H(t-s, x-y)\tilde{f}(s,y)dyds.$$

The uniqueness theorem then implies $v = \tilde{v}$. But $\tilde{v}(t,x)$ is C^∞ for (t,x) near (t_0,x_0), since $\tilde{f}(s,y) = 0$ near (t_0,x_0).(check). Thus u is C^∞ near (t_0,x_0).

2. If $u \in C^{1,2}((0,\infty)\times\mathbb{R}^n)$, use an approximate identity $\eta_\epsilon(x)$ to define $u_\epsilon(t,x) = (\eta_\epsilon *_x u)(t,x)$, and observe that $\partial_t u_\epsilon - \Delta u_\epsilon = 0$. Now choose the cutoff ζ supported near (t_0,x_0) with $\zeta = 1$ near (t_0,x_0) as we did in step **1**, set $v = \zeta u$ and $v^\epsilon = \zeta u_\epsilon$, and note that

$$\begin{aligned} &v^\epsilon_t - \Delta v^\epsilon = \tilde{f}^\epsilon := \zeta_t u_\epsilon - 2\nabla\zeta\cdot\nabla u_\epsilon - u_\epsilon\Delta\zeta \\ &v^\epsilon|_{t=0} = 0. \end{aligned} \tag{1.133}$$

Repeating the argument based on Theorem 1.50 given in step **1** yields

$$v^\epsilon(t,x) = \tilde{v}^\epsilon(t,x) := \int_0^t \int_{\mathbb{R}^n} H(t-s,x-y)\tilde{f}^\epsilon(s,y)dyds \text{ for each } \epsilon. \tag{1.134}$$

Let $\epsilon \to 0$ to conclude

$$v(t,x) = \tilde{v}(t,x) := \int_0^t \int_{\mathbb{R}^n} H(t-s,x-y)\tilde{f}(y,s)dyds, \text{ where } \tilde{f} := \zeta_t u - 2\nabla\zeta\cdot\nabla u - u\Delta\zeta.$$

Now finish as before, using that $v = u$ near (t_0,x_0) and that $\tilde{v}$ is C^∞ for (t,x) near (t_0,x_0) since $\tilde{f} = 0$ on U.

□

[28] Hint: for a given ϵ such that (1.126) holds, consider the problem on $[0,T]\times\mathbb{R}^n$ for a small enough T. Next consider $[T,2T]\times\mathbb{R}^n$.

Remark 1.53. The same proof applies if $(0,\infty)\times\mathbb{R}^n$ is replaced by $(0,T)\times\Omega$ for some $T>0$ in the statement of the theorem.

Theorem 1.54 (Maximum principle)**.** *Let $\Omega\subset\mathbb{R}^n$ be open and bounded and let $0<T<\infty$. Suppose*

$$\begin{aligned} &u\in C([0,T]\times\overline{\Omega})\cap C^{1,2}((0,T)\times\Omega)\\ &u_t-\Delta u=0 \text{ on } (0,T)\times\Omega. \end{aligned} \tag{1.135}$$

Then $u(t,x)$ attains its max on

$$\Gamma_T:=(\{0\}\times\Omega)\cup([0,T]\times\partial\Omega).$$

Imagine a circular two-dimensional metal plate Ω, whose temperature at time 0 is given by $u(0,x)$ and whose boundary is kept at temperature $u(t,x)$ for $t\in[0,T]$. We would expect that the temperature inside the plate is never higher than the max of the highest temperature at $t=0$ and the highest temperature on the sides during $[0,T]$.

Proof of Theorem 1.135. **1.** For $\epsilon>0$ set $v(t,x):=u(t,x)+\epsilon|x|^2$. Then

$$(\partial_t-\Delta)v=(\partial_t-\Delta)u+\epsilon(\partial_t-\Delta)|x|^2=-\epsilon\Delta|x|^2=-\epsilon 2n<0 \text{ in } (0,T)\times\Omega. \tag{1.136}$$

We claim that if $0<T'<T$, the max of v on $C_{T'}:=[0,T']\times\overline{\Omega}$ occurs on $\Gamma_{T'}$.

Indeed, it v has a max at an interior point P of $C_{T'}$, then Calc I implies $\partial_t v$, ∇v are zero at P, and $\partial_j^2 v\le 0$ at P. Thus, $\Delta v\le 0$ at P, so $(\partial_t-\Delta)v\ge 0$ at P, contradiction.

If v has its max on $C_{T'}$ at a point $P=(T',x_0)$ of $\{T'\}\times\Omega$, then Calc I implies $\partial_j^2 v\le 0$ at P, so $\Delta v\le 0$ at P. We must have $\partial_t v\ge 0$ at P, since $v(t,x_0)\le v(T',x_0)$ for $t\le T'$. Thus, $(\partial_t-\Delta)v\ge 0$ at P, contradiction. This establishes the claim.

2. Step **1** implies

$$\max_{[0,T']\times\overline{\Omega}} u\le\max_{[0,T']\times\overline{\Omega}} v=\max_{\Gamma_{T'}} v\le\max_{\Gamma_{T'}} u+\epsilon\max_{\Gamma_{T'}}|x|^2. \tag{1.137}$$

Letting $\epsilon\to 0$ we obtain $\max_{[0,T']\times\overline{\Omega}} u\le\max_{\Gamma_{T'}} u$. Finally, letting $T'\to T$ and using continuity of u, we obtain

$$\max_{[0,T]\times\overline{\Omega}} u\le\max_{\Gamma_T} u.$$

□

1.4.3 Backward ill-posedness

Observe that $w(t,x)$ solves the "backward" heat equation:

$$\begin{aligned} &(\partial_t-\Delta)w=0 \text{ in } t<0\\ &w(0,x)=g(x) \end{aligned} \tag{1.138}$$

if and only if $u(t,x) := w(-t,x)$ satisfies the problem in $t > 0$:

$$\begin{aligned} &(\partial_t + \Delta)u = 0 \text{ in } t > 0 \\ &\tilde{u}(0,x) = g(x). \end{aligned} \tag{1.139}$$

Thus, we also refer to (1.139) as the backward heat equation.

You should check that when we try to solve (1.139) using the Fourier transform in x, we are faced with the puzzling problem of trying to compute $\hat{u}(t,\xi)$ as the inverse Fourier transform of $\hat{g}(\xi)e^{t|\xi|^2}$, where $t > 0$! Here are a few handwaving arguments that the backward heat equation should be ill-posed.[29]

(a) Even when $g \in \mathcal{S}$ and thus $\hat{g} \in \mathcal{S}$, the function $\hat{g}(\xi)e^{t|\xi|^2}$ may still grow exponentially in ξ when $t > 0$; thus we expect singularities in a solution $u(t,x)$ to (1.139) when $t > 0$.

(b) We have seen that the heat equation takes nonsmooth initial data and smooths it out as t increases. So we should not expect to obtain a solution to the heat equation (1.138) in $t < 0$ unless the data at $t = 0$ is already C^∞.

(c) For the heat equation time is irreversible, and the ill-posedness of the backward problem is related to the second law of thermodynamics.[30]

Here is a rigorous ill-posedness result that can be proved easily using our earlier results.

Proposition 1.55. *Fix any $N \in \mathbb{N}$ and suppose $f \in C_c^N(\mathbb{R}^n) \setminus C^{N+1}(\mathbb{R}^n)$ in (1.139). Then for every $T > 0$ it is impossible to find a solution to (1.139) on $[0,T] \times \mathbb{R}^n$ with the properties*

$$\begin{aligned} &u \in C([0,T] \times \mathbb{R}^n) \cap C^{1,2}((0,T) \times \mathbb{R}^n) \\ &u, \nabla u \text{ are bounded on } (0,T] \times \mathbb{R}^n. \end{aligned} \tag{1.140}$$

Proof. Suppose such a solution u exists and define $v(t,x) = u(T-t,x)$. Then

$$\begin{aligned} &v_t - \Delta v = 0 \text{ on } (0,\infty) \\ &v(0,x) = u(T,x). \end{aligned} \tag{1.141}$$

By Theorem 1.47 the problem (1.141) has a solution w given by

$$w(t,x) = (4\pi t)^{-n/2} \int_{\mathbb{R}^n} u(T, x-y) e^{-|y|^2/4t} dy = (4\pi t)^{-n/2} \int_{\mathbb{R}^n} u(T,y) e^{-|x-y|^2/4t} dy. \tag{1.142}$$

These formulas show that w, like v, also satisfies (1.140), so our uniqueness result, Theorem 1.50 (with Remark 1.51) applies to show that $v = w$. But (1.142) also shows that $w(T,x)$ is C^∞, and hence $v(T,x) = u(x,0) = f$ is C^∞. Contradiction. □

[29] J. Hadamard declared a problem to be ill-posed when any of the following hold: solutions do not always exist for reasonably chosen data, solutions exist but are not unique, or small perturbations in the data can cause large changes in the solution.

[30] Extra apologies for (c).

1.5 Distributions II

Let $\mathcal{S}$ be the Fréchet space of Schwartz functions defined in section 1.3.3.

Definition 1.56. *A* tempered distribution *is a continuous linear functional* $u : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$, *and we denote by* $\mathcal{S}'(\mathbb{R}^n)$ *the space of tempered distributions. If* $u : \mathcal{S} \to \mathbb{C}$ *is linear, then*

$$u \in \mathcal{S}' \Leftrightarrow \exists N, C_N \text{ such that } |\langle u, \phi\rangle| \leq C_N q_N(\phi) \ \ \forall \phi \in \mathcal{S}, \tag{1.143}$$

where $q_N(\phi) = \sup_{x\in\mathbb{R}^n, |\alpha|\leq N}(1+|x|)^N |\partial^\alpha \phi|$. .

Examples 1.57. 1. If f is a polynomial or if $f \in L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$, then $f \in \mathcal{S}'(\mathbb{R}^n)$, while $e^x \notin \mathcal{S}'(\mathbb{R})$. Use Hölder's inequality to see this for $f \in L^p(\mathbb{R}^n)$.

2. Any Lebesgue measurable function $f : \mathbb{R}^n \to \mathbb{C}$ for which there exist N, C_N such that $|f(x)| \leq C_N(1+|x|)^N$ a.e. defines an element of $\mathcal{S}'$.

3. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ have supp $u \subset K$ for some compact set K. Then u determines a unique element u_s of $\mathcal{S}'$ as follows. Choose $\rho \in \mathcal{D}$ such that $\rho = 1$ on a neighborhood of K. For any $\phi \in \mathcal{S}$ define with obvious notation

$$\langle u_s, \phi\rangle_{\mathcal{S}'} := \langle u, \rho\phi\rangle_{\mathcal{D}'}.$$

One checks that u_s is well-defined and $u_s \in \mathcal{S}'$.(check) From now on we'll drop the s on u_s.

Remark 1.58 (Topology on $\mathcal{S}'$)**.**

1. We give $\mathcal{S}'$ the weak*-topology, which is similar to the topology on $\mathcal{D}'(U)$ defined in Definition 1.34. Subbasic neighborhoods of zero have the form

$$\Omega(\phi, \epsilon) = \{u \in \mathcal{S}' : |\langle u, \phi\rangle| < \epsilon\}, \text{ where } \phi \in \mathcal{S}, \epsilon > 0.$$

This is the seminorm topology determined by the seminorms $p_\phi(u) := |\langle u, \phi\rangle|$, where $u \in \mathcal{S}'$, $\phi \in \mathcal{S}$. Observe that $u_n \to u$ in $\mathcal{S}' \Leftrightarrow \langle u_n, \phi\rangle \to \langle u, \phi\rangle$ for all $\phi \in \mathcal{S}$. The family of seminorms $\{p_\phi : \phi \in \mathcal{S}\}$ is separating, so the weak* topology on $\mathcal{S}'$ is Hausdorff.

2. Linear partial differential operators with coefficients that are in $C^\infty(\mathbb{R}^n)$ with *polynomial growth* define continuous maps from $\mathcal{S}'$ to $\mathcal{S}'$ by the obvious analogue of Exercise 1.3.2c, part (iii).(check)

To justify the use of the term "distribution" for the elements of $\mathcal{S}'(\mathbb{R}^n)$, we now show that $\mathcal{S}'$ can be identified with a subspace of $\mathcal{D}'(\mathbb{R}^n)$. By Proposition 8.24 we see that the inclusion $i : \mathcal{D} \to \mathcal{S}$ is continuous, since each inclusion $i : \mathcal{D}_K \to \mathcal{S}$ is continuous (check). We can then define the "transpose" (or adjoint) map $i^t : \mathcal{S}' \to \mathcal{D}'$ by

$$\langle i^t u, \phi\rangle = \langle u, i\phi\rangle, \text{ or equivalently } i^t u = u \circ i = u|_{\mathcal{D}}. \tag{1.144}$$

The map i^t is continuous when both $\mathcal{S}'$ and $\mathcal{D}'$ carry their weak* topologies (check).

Proposition 1.59. *The map* $i^t : \mathcal{S}' \to \mathcal{D}'$ *is injective, and hence we can identify* $\mathcal{S}'$ *with a subspace of* $\mathcal{D}'$.

Proof. Observe that $\mathcal{D}$ is dense in $\mathcal{S}$. Indeed, if $f \in \mathcal{S}$ and if we choose $\phi \in \mathcal{D}$ such that $\phi = 1$ on $|x| \leq 1$, we have

$$f_n(x) := f(x)\phi(x/n) \to f(x) \text{ in } \mathcal{S}.$$

Thus if $u \in \mathcal{S}'$ and $i^t u = u|_{\mathcal{D}} = 0$, we must have $u = 0$.

□

Remark 1.60. Proposition 1.59 allows us to identify $\mathcal{S}'$ with

$$i^t(\mathcal{S}') = \{v \in \mathcal{D}' : v \text{ has a continuous extension from } \mathcal{D} \text{ to } \mathcal{S}\}.$$

Note for example that $e^x \in \mathcal{D}'$ has no such extension.

1.5.1 Extension of the Fourier transform

In this section we always work on $\mathbb{R}^n$. We have defined $\mathcal{F}$ on L^1 by an absolutely convergent integral. We show first how to extend from $L^1 \cap L^2$ to L^2, and then extend $\mathcal{F}$ to $\mathcal{S}'$.

Recall that the L^2 pairing is $(f, g)_{L^2} = \int_{\mathbb{R}^n} f(x)\overline{g(x)}dx$.

Proposition 1.61. *(a) Let $f, g \in \mathcal{S}$. We have*

$$(f, g)_{L^2} = (\hat{f}, \hat{g})_{L^2}. \tag{1.145}$$

(b) Consequently, $\mathcal{F}$ extends by continuity to an isometric isomorphism $\mathcal{F} : L^2 \to L^2$.

Proof. **(a)** We have by the Fourier inversion formula and Fubini:

$$\begin{gathered}(2\pi)^{-n/2}(f, g)_{L^2} = \int f\overline{g}d'x = \int \overline{g(x)} \left(\int e^{ix\xi} \hat{f}(\xi)d'\xi \right) d'x = \\ \int \overline{\hat{g}(\xi)}\hat{f}(\xi)d'\xi = (2\pi)^{-n/2}(\hat{f}, \hat{g})L^2.\end{gathered} \tag{1.146}$$

(b) In (1.145) take $f = g \in \mathcal{S}$ to obtain

$$|f|_{L^2}^2 = |\hat{f}|_{L^2}^2. \tag{1.147}$$

Since $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ is bijective and $\mathcal{S}$ is dense in L^2, this implies that $\mathcal{F}$ extends by continuity to an isometric isomorphism of L^2 onto itself. Explicitly, given $f \in L^2$ to define $\hat{f} \in L^2$, choose $f_n \in \mathcal{S}$ such that $f_n \to f$ in L^2. Then (1.147) implies $(\hat{f}_n)$ is Cauchy in L^2 and hence converges to some $h \in L^2$. Now *define* $\hat{f} := h$. I leave you to check that this map is well-defined, surjective, and that $|\hat{f}|_{L^2} = |f|_{L^2}$.[31]

□

[31] Note that when $f \in L^2$ it makes no sense to define $\hat{f}(\xi) := \int e^{-ix\xi} f(x)d'x$. The fact that the Fourier transform is an isometry on L^2 is important in quantum mechanics, where for any time t, $|\psi(t, x)|^2$ has an interpretation as a probability density for position when ψ is a solution of Schrödinger's equation, and $|\hat{\psi}(t, \xi)|^2$ is interpreted as a probability density for momentum. See [Wil26], Chapter 14, for example.

We now use the dual pairing to extend $\mathcal{F}$ to all of $\mathcal{S}'$.

Proposition 1.62. *For $u \in \mathcal{S}'$ define the linear map $\mathcal{F}u = \hat{u} : \mathcal{S} \to \mathbb{C}$ by*

$$\langle \hat{u}, \phi \rangle := \langle u, \hat{\phi} \rangle \text{ for } \phi \in \mathcal{S}. \tag{1.148}$$

Then $\mathcal{F}u \in \mathcal{S}'$ and the map $\mathcal{F} : \mathcal{S}' \to \mathcal{S}'$ is a homeomorphism.

Proof. **1.** Observe that $\mathcal{F}u = u \circ \mathcal{F} = \mathcal{F}^t u$, where $\mathcal{F}^t : \mathcal{S}' \to \mathcal{S}'$ is the "transpose" (or adjoint) map of $\mathcal{F} : \mathcal{S} \to \mathcal{S}$. As a composition of two continuous maps $\mathcal{F}^t u$ is continuous on $\mathcal{S}$.[32] That is, $\mathcal{F}u \in \mathcal{S}'$.

2. Continuity of $\mathcal{F} : \mathcal{S}' \to \mathcal{S}'$ follows easily from the definition of the weak*-topology on $\mathcal{S}'$.

3. To see that $\mathcal{F}$ is a homeomorphism of $\mathcal{S}'$ observe that for $u \in \mathcal{S}'$, $\phi \in \mathcal{S}$ we have:

$$\langle \hat{\hat{u}}, \phi \rangle = \langle u, \hat{\hat{\phi}} \rangle = \langle u, \phi \rangle, \text{ so } \hat{\hat{u}} = u.$$

This implies $\mathcal{F} : \mathcal{S}' \to \mathcal{S}'$ is bijective and that $\mathcal{F}^{-1}$ is continuous. (why?).

□

Remark 1.63. The identity (1.101) both motivates the definition of $\mathcal{F}u$ for $u \in \mathcal{F}'$ and shows that this definition is consistent with Definition 1.36 when $u \in \mathcal{S}$.

Exercise 1.5.1a. Check that the definition of $\mathcal{F}u$ for $u \in \mathcal{S}'$ is *consistent* with the Definition 1.36 when $u \in L^1 \subset \mathcal{S}'$ and with the definition of Proposition 1.61 for $u \in L^2 \subset \mathcal{S}'$.

Examples 1.64. 1. We claim $\hat{\delta} = (2\pi)^{-n/2}$. Indeed, for $\phi \in \mathcal{S}$ we have

$$\langle \hat{\delta}, \phi \rangle = \langle \delta, \hat{\phi} \rangle = \hat{\phi}(0) = \int \phi(x) d'x = \langle (2\pi)^{-n/2}, \phi \rangle. \tag{1.149}$$

2. We claim $\hat{1} = (2\pi)^{n/2}\delta$. Indeed, for $\phi \in \mathcal{S}$ we have

$$\langle \hat{1}, \phi \rangle = \langle 1, \hat{\phi} \rangle = \int \hat{\phi}(\xi) d\xi = (2\pi)^{n/2} \int \hat{\phi}(\xi) d'\xi = (2\pi)^{n/2} \phi(0). \tag{1.150}$$

3. Suppose $E \in \mathcal{S}'$ is a fundmental solution of the constant coefficient partial differential operator $P(D)$; that is, E satisfies $P(D)E = \delta$. If $f \in \mathcal{S}$, a solution of $P(D)u = f$ is given by $E * f$ (check).

Proposition 1.65. *Let $u \in \mathcal{S}'$. We have*
(i) $\widehat{D^\alpha u} = \xi^\alpha \hat{u}$;
(ii) $\widehat{x^\alpha u} = (-D_\xi)^\alpha \hat{u}$.

[32] Caution: In the equation $\mathcal{F}u = u \circ \mathcal{F}$ the symbol $\mathcal{F}$ has two different meanings. The $\mathcal{F}$ on the left is really the transpose map $\mathcal{F}^t : \mathcal{S}' \to \mathcal{S}'$. We hope this slight abuse is not confusing.

Proof. For $u \in \mathcal{S}'$ and $\phi \in \mathcal{S}$ we compute

$$\begin{aligned} \langle \widehat{D^\alpha u}, \phi \rangle = \langle D^\alpha u, \hat{\phi} \rangle = \langle u, (-D)^\alpha \hat{\phi} \rangle = \\ \langle u, \widehat{\xi^\alpha \phi} \rangle = \langle \hat{u}, \xi^\alpha \phi \rangle = \langle \xi^\alpha \hat{u}, \phi \rangle. \end{aligned} \tag{1.151}$$

The proof of (ii) is similar. □

1.5.2 Convolutions

Convolutions can be used to approximate irregular functions or distributions by smooth functions. We illustrate this below in Proposition 1.69. Again in this section we work on $\mathbb{R}^n$.

To motivate the next definition note that if $f, g \in \mathcal{S}$

$$f * g(x) = \int f(y) g(x-y) d'y = \langle f, g(x - \cdot) \rangle (2\pi)^{-n/2},$$

where $\langle \cdot, \cdot \rangle$ is the $\mathcal{S}' \times \mathcal{S}$ pairing.

Definition 1.66. *Let $u \in \mathcal{S}'$, $\phi \in \mathcal{S}$. We define*

$$(u * \phi)(x) := \langle u, \phi(x - \cdot) \rangle (2\pi)^{-n/2}. \tag{1.152}$$

Proposition 1.67. *Let $u \in \mathcal{S}'$, $\phi \in \mathcal{S}$. Then*
(a) $u * \phi \in C^\infty$ and has polynomial growth;
(b) $\partial^\alpha (u * \phi) = \partial^\alpha u * \phi = u * \partial^\alpha \phi$;
(c) $\widehat{u * \phi} = \hat{\phi} \hat{u}$;
(d) $\hat{u} * \hat{\phi} = \widehat{\phi u}$.

Proof. **(a) and (b).** First we show $u * \phi$ is continuous. Indeed, if $x_n \to x$ then

$$\phi(x_n - \cdot) \to \phi(x - \cdot) \text{ in } \mathcal{S}.$$

This implies $\langle u, \phi(x_n - \cdot) \rangle \to \langle u, \phi(x - \cdot) \rangle$.

To prove differentiability of $u * \phi$ consider the difference quotients

$$\begin{aligned} Q_\epsilon(x) := \frac{(u * \phi)(x + \epsilon e_1) - (u * \phi)(x)}{\epsilon} = (u * f_\epsilon)(x) = \langle u, f_\epsilon(x - \cdot) \rangle (2\pi)^{-n/2}, \\ \text{where } f_\epsilon(x) = \frac{\phi(x + \epsilon e_1) - \phi(x)}{\epsilon}. \end{aligned} \tag{1.153}$$

Now $f_\epsilon(x - \cdot) \to \partial_{x_1} \phi(x - \cdot)$ in $\mathcal{S}$ as $\epsilon \to 0$.(check) Thus,

$$\lim_{\epsilon \to 0} Q_\epsilon(x) = \langle u, \partial_{x_1} \phi(x - \cdot) \rangle (2\pi)^{-n/2} = (u * \partial_{x_1} \phi)(x) = (\partial_{x_1} u * \phi)(x),$$

a continuous function of x. This shows $u * \phi$ is C^1. Proceed inductively to show $u * \phi$ is C^∞ and satisfies (b).

To see that $u * \phi$ has polynomial growth we let $p_N(\phi) := \sup_{x\in\mathbb{R}^n, |\alpha|\le N} |(1+|x|^2)^N||\partial^\alpha \phi|$ and recall that since $u \in \mathcal{S}'$ we have

$$\exists N, C_N \text{ such that } |\langle u, \phi\rangle| \le C_N p_N(\phi) \text{ for all } \phi \in \mathcal{S}. \tag{1.154}$$

Thus, for $\phi \in \mathcal{S}$

$$\begin{gathered}|(u*\phi)(x)| = |\langle u, \phi(x-\cdot)\rangle|(2\pi)^{-n/2} \le C_N \sup_{y\in\mathbb{R}^n, |\alpha|\le N} (1+|y|^2)^N |\partial_y^\alpha \phi(x-y)| = \\ C_N \sup_{w\in\mathbb{R}^n, |\alpha|\le N} (1+|x-w|^2)^N |\partial_w^\alpha \phi(w)| := B(x) \quad \text{(here we set } w = x-y).\end{gathered} \tag{1.155}$$

Now $1+|x-w|^2 \le 2(1+|x|^2)(1+|w|^2)$, so $B(x) \le C_N'(1+|x|^2)^N p_N(\phi)$.

(c) and (d). We will show

$$\widehat{\langle u * \phi, f\rangle} = \langle \hat\phi \hat u, f\rangle \text{ for all } f \in \mathcal{S}. \tag{1.156}$$

It is enough to show this for $f \in \widehat{\mathcal{D}}$, since $\widehat{\mathcal{D}}$ is dense in $\mathcal{S}$.[33] So let $f = \hat\psi$, where $\psi \in \mathcal{D}$ has supp $\psi \subset K$, a fixed compact set. We have[34]

$$\begin{gathered}\widehat{\langle u*\phi, f\rangle} = \widehat{\langle u*\phi, \hat\psi\rangle} = \langle u*\phi, \hat{\hat\psi}\rangle = \int (u*\phi)(x)\psi(-x)dx = \\ \int_{-K} \langle u, \phi(x-\cdot)\psi(-x)d'x = \langle u, \int_{-K} \phi(x-\cdot)\psi(-x)d'x\rangle = \\ \langle u, \widehat{\widehat{\phi * \psi}}\rangle = \langle \hat u, \widehat{\phi*\psi}\rangle = \langle \hat u, \hat\phi\hat\psi\rangle = \langle \hat\phi \hat u, f\rangle.\end{gathered} \tag{1.157}$$

The equality in the second line can be justified using Riemann sums. One can choose Riemann sums $S_M(y)$ such that

$$S_M(y) \to \int_{-K} \phi(x-y)\psi(-x)d'x \text{ in } \mathcal{S}(\mathbb{R}^n_y) \text{ as } M \to \infty.$$

For this use the fact that for any N and $|\alpha| \le N$, the function $(1+|y|)^N \partial_y^\alpha[\phi(x-y)\psi(-x)]$ is uniformly continuous and bounded on $-K_x \times \mathbb{R}^n_y$.

The proof of (d) follows easily from part (c). □

Remark 1.68. When $u \in \mathcal{D}'(\mathbb{R}^n)$ and $v \in \mathcal{D}(\mathbb{R}^n)$, we can similarly defined $u * \phi$ by (1.66). Arguments like above show that $u * \phi \in C^\infty$ and

$$\partial^\alpha(u*\phi) = \partial^\alpha u * \phi = u * \partial^\alpha \phi.$$

[33] Indeed, let $f \in \mathcal{S}$. Then $f = \hat g$ for some $g \in \mathcal{S}$. Choose $\phi_n \in \mathcal{D}$ such that $\phi_n \to g$ in $\mathcal{S}$. Then $\hat\phi_n \to \hat g = f$ in $\mathcal{S}$.

[34] Here we use $\hat{\hat g}(x) = g(-x)$.

By Exercise 1.6.2b we have

$$\operatorname{supp} u * v \subset \overline{\operatorname{supp} u + \operatorname{supp} v}. \tag{1.158}$$

We end this section with a proposition that illustrates how convolutions can be used to approximate irregular objects (like general elements of $\mathcal{S}'$) by smooth objects (in this case C^∞ functions with polynomial growth). Part (a) is used in our proof of the Spectral Theorem.

Proposition 1.69. *Let $f \in \mathcal{S}(\mathbb{R}^n)$ be such that $\int f dx = (2\pi)^{n/2}$, $f \geq 0$ and set $f_k(x) = k^n f(kx)$ for $k = 1, 2, \ldots$. Let $u \in \mathcal{S}'(\mathbb{R}^n)$. Then*
(a) $u * f_k \to u$ in $\mathcal{S}'$ as $k \to \infty$ and
(b) $f_k \to (2\pi)^{n/2}\delta$ in $\mathcal{S}'$ as $k \to \infty$.[35]

Proof. We prove (a); the proof of (b) is easier. For $\phi \in \mathcal{S}$ we must show $\langle u * f_k, \phi\rangle \to \langle u, \phi\rangle$ as $k \to \infty$. We have

$$\langle u * f_k, \phi\rangle = \int \langle u, f_k(x - \cdot)\rangle \phi(x) d'x = \langle u, \int f_k(x - \cdot)\phi(x) d'x\rangle := \langle u, \psi_k(\cdot)\rangle, \text{ where}$$
$$\psi_k(\cdot) = \int f_k(x - \cdot)\phi(x) d'x = \int k^n f(k(x - \cdot))\phi(x) d'x = \int f(s)\phi(\frac{s}{k} + \cdot) d's \to \phi(\cdot) \text{ in } \mathcal{S}. \tag{1.159}$$

Thus, $B_k \to \langle u, \phi\rangle$.

□

1.5.3 Tensor products and the jump formula

The jump formula allows us to differentiate $H(t)u$, where H is the Heaviside function and u is a distribution with some regularity in t. The formula involves tensor products of distributions, which we now define.

Proposition 1.70 (Tensor product of distributions)**.** *Let $X \subset \mathbb{R}^n$ be open, $Y \subset \mathbb{R}^m$ be open, and let $u \in \mathcal{D}'(X)$, $v \in \mathcal{D}'(Y)$. We can define a distribution $u \otimes v \in \mathcal{D}'(X \times Y)$ by*

$$\langle u \otimes v, \phi(x, y)\rangle := \langle u(x), \langle v(y), \phi(x, y)\rangle\rangle, \text{ for } \phi \in \mathcal{D}(X \times Y). \tag{1.160}$$

The tensor product has the following properties:
(i) $\langle u \otimes v, \phi(x, y)\rangle = \langle v \otimes u, \phi(x, y)\rangle = \langle v(y), \langle u(x), \phi(x, y)\rangle\rangle$
(ii) $\operatorname{supp} u \otimes v = \operatorname{supp} u \times \operatorname{supp} v$
(iii) $\partial_x^\alpha \partial_y^\beta (u \otimes v) = \partial_x^\alpha u \otimes \partial_y^\beta v$.

We will sketch the main ideas of the proof. Full details may be found in [FJ98], section 4.3.

[35] Recall that for each k, $u * f_k$ is C^∞ with polynomial growth.

Sketch of proof. **1.** First observe that $\langle v(y), \phi(x,y)\rangle$ in (1.160) is a test function, so the right side of (1.160) defines a linear map $u \otimes v$ on $\mathcal{D}(X \times Y)$. We have

$$\partial_x^\alpha \langle v(y, \phi(x,y)\rangle = \langle u(x), \partial_x^\alpha \phi(x,y)\rangle. \tag{1.161}$$

With this the necessary seminorm estimates for $u \otimes v$ follow readily from those for u and v.

2. The subspace of $\mathcal{T} \subset \mathcal{D}(X \times Y)$ generated by functions of the form

$$(\phi \otimes \psi)(x,y) := \phi(x)\psi(y),$$

where $\phi \in \mathcal{D}(X)$, $\psi \in \mathcal{D}(Y)$ is dense in $\mathcal{D}(X \times Y)$. Indeed, if $\chi \in \mathcal{D}(X \times Y)$, we can find a sequence $\chi_j \in \mathcal{T}$ converging to χ in $\mathcal{D}(X \times Y)$ by smoothly extending χ by periodicity outside some open cube $I \subset \mathbb{R}^n \times \mathbb{R}^m$ containing supp χ, constructing the Fourier series, $\sum_{\alpha=(\alpha_1,\alpha_2)} c_\alpha e^{i\alpha_1 \cdot x + i\alpha_2 \cdot y}$, of the periodic function thus obtained, and then defining χ_j as

$$\rho_1(x)\rho_2(y) \sum_{|\alpha| \le j} c_\alpha e^{i\alpha_1 \cdot x + i\alpha_2 \cdot y}$$

for appropriate cutoffs $\rho_1 \in \mathcal{D}(X)$, $\rho_2 \in \mathcal{D}(Y)$.

3. The distributions $u \otimes v$ and $v \otimes u$ in (i) obviously agree on the dense subset $\mathcal{T}$, so they must be equal.

It is clear from (i) that $\langle u \otimes v, \phi\rangle = 0$ if the support of ϕ is disjoint from supp $u \times$ supp v, so

$$\text{supp } u \otimes v \subset \text{supp } u \times \text{supp } v.$$

To prove the opposite inclusion, observe that if $x \in$ supp u, $y \in$ supp v, we can find $\phi \in \mathcal{D}(X)$, $\psi \in \mathcal{D}(Y)$ such that $\langle u, \phi\rangle \neq 0$ and $\langle v, \psi\rangle \neq 0$. Thus

$$\langle u \otimes v, \phi \times \psi\rangle \neq 0,$$

so $(x,y) \in$ supp $u \otimes v$. This gives (ii).

Finally, (iii) is a direct consequence of (i).

□

Example 1.71. Writing $x \in \mathbb{R}^n$ as $x = (x', x'')$, we have $\delta(x) = \delta(x') \otimes \delta(x'')$.

The jump formula is a "product rule" that works for certain products $H(t)u$, where

$$H(t) = \begin{cases} 1, & t \ge 0 \\ 0, & t < 0 \end{cases}$$

is the Heaviside function. Recall that $\partial_t H(t) = \delta(t)$. Suppose now that $u \in C(\mathbb{R}_t, \mathcal{S}'(\mathbb{R}^n)$ and define $H(t)u(t,x) \in \mathcal{D}'(\mathbb{R}_t \times \mathbb{R}^n)$ by

$$\langle H(t)u(t,x), \phi(t,x)\rangle := \int_0^\infty \langle u(t,\cdot), \phi(t,\cdot)\rangle dt \text{ for } \phi \in \mathcal{D}(\mathbb{R}_t \times \mathbb{R}^n).$$

Proposition 1.72 (Jump formula)**.** *Suppose* $u(t,x) \in C^1(\mathbb{R}_t, \mathcal{S}'(\mathbb{R}^n))$*. Then*

$$\partial_t(H(t)u) = \delta(t) \otimes u(0,x) + H(t)\partial_t u. \tag{1.162}$$

Proof. **1.** Choose $\rho(t) \in \mathcal{D}(\mathbb{R})$ such that $\rho \geq 0$, $\int \rho(t)dt = 1$ and define an approximate identity by $\rho_k(t) = k\rho(kt)$ for $k \in \mathbb{N}$. We have shown $\rho_k \to \delta$ in $\mathcal{D}'(\mathbb{R})$ as $k \to \infty$. Observe that if $v \in C(\mathbb{R}_t, \mathcal{S}'(\mathbb{R}^n))$ we have[36]

$$(H * \rho_k)v \to H(t)v \in \mathcal{D}' \text{ as } k \to \infty. \tag{1.163}$$

Here and below convolution is in the t variable.

2. We have

$$\partial_t((H * \rho_k)u) = (\delta * \rho_k)u + (H * \rho_k)\partial_t u = \rho_k u + (H * \rho_k)\partial_t u. \tag{1.164}$$

Let $k \to \infty$, use (1.163), and use $\rho_k u \to \delta(t) \otimes u(0,x)$ in $\mathcal{D}'(\mathbb{R}^{1+n})$ to finish.[37]

□

Remark 1.73. *Note that Proposition 1.72 works just as well if* $u \in C^1([0,\infty)_t, \mathcal{S}'(\mathbb{R}^n))$*.*

An application of the jump formula to constructing fundamental solutions in given in problem 3 of section 1.7.

1.6 The wave equation

We begin with some results on uniqueness, domains of dependence, and finite propagation speed that can be proved by energy methods.

1.6.1 Energy methods

Theorem 1.74. *Let* $\Box = \partial_t^2 - \Delta$*, suppose* $U \subset \mathbb{R}^n$ *is bounded and open with* C^∞ *boundary* ∂U*, and for* $T > 0$ *set* $U_T = (0,T) \times U$*. Consider the problem*

$$\begin{aligned} &\Box u = f \text{ on } U_T \\ &u = g \text{ on } [0,T] \times \partial U \\ &u|_{t=0} = h_1,\ u_t|_{t=0} = h_2. \end{aligned} \tag{1.165}$$

Then there exists at most one solution $u \in C^2(\overline{U_T})$*.*

[36] To see that (1.163) holds, consider action on a test function $\phi(t,x)$.
[37] Consider action on a test function to see $\rho_k u \to \delta(t) \otimes u(0,y)$.

Proof. Suppose $w \in C^2(\overline{U_T})$ and

$$\begin{aligned} &\Box w = 0 \text{ on } U_T \\ &w = 0 \text{ on } [0,T] \times \partial U \\ &w|_{t=0} = 0,\ w_t|_{t=0} = 0. \end{aligned} \tag{1.166}$$

For $0 \le t \le T$ define

$$e(t) = \frac{1}{2} \int_U \left(w_t^2(t,x) + |\nabla w(t,x)|^2 \right) dx.$$

Then

$$\dot{e}(t) = \int_U (w_t w_{tt} + \nabla w \cdot \nabla w_t) dx = \int_U w_t (w_{tt} - \Delta w) dx + \int_{\partial U} \partial_\nu w \; w_t dS(x) = 0, \tag{1.167}$$

where we used Gauss-Green and $w(t,\cdot) = 0$ on ∂U to obtain the last equality. Thus, $e(t) = e(0) = 0 \Rightarrow w_t = 0$, $\nabla w = 0 \Rightarrow w$ is constant on U_T, and hence $w = 0$.

□

We now refine the above argument to show that if u is a solution of $\Delta u = 0$, the value of u at any (t_0, x_0) is completely determined by the values of u and u_t on any time slice of the backward cone with vertex at (t_0, x_0). This in turn implies that the $x-$support of $u(t,x)$ propagates at finite speed.

Theorem 1.75 (Domain of dependence)**.** *Fix $x_0 \in \mathbb{R}^n$ and $t_0 > 0$. Suppose $u \in C^2([0,\infty) \times \mathbb{R}^n)$ and $\Box u = 0$. Define the backward cone with vertex at (t_0, x_0):*

$$C(t_0, x_0) := \{(t,x) : 0 \le t \le t_0, |x - x_0| \le t_0 - t\}. \tag{1.168}$$

If u and u_t are zero on $C(t_0,x_0) \cap \{t = 0\}$, then $u = 0$ in $C(t_0,x_0)$; hence, $u(t_0,x_0) = 0$.

Proof. For $0 \le t \le t_0$ let $B(x_0, t_0 - t) = \{x \in \mathbb{R}^n : |x - x_0| \le t_0 - t\}$ and define

$$e(t) = \frac{1}{2} \int_{B(x_0,t_0-t)} (u_t^2 + |\nabla u|^2) dx.$$

We claim

$$\dot{e}(t) = \int_{B(x_0,t_0-t)} (u_t u_{tt} + \nabla u \cdot \nabla u_t) dx - \frac{1}{2} \int_{\partial B(x_0,t_0-t)} (u_t^2 + |\nabla u|^2) dS(x). \tag{1.169}$$

This is evident from the following computation for any $f \in C^2$:

$$\begin{aligned} &\int_{B(0,t)} f(t,x) dx = \int_{S^{n-1}} \int_0^t f(t, r\omega) r^{n-1} dr d\omega \Rightarrow \\ &\frac{d}{dt}(RHS) = \int_{S^{n-1}} \int_0^t f_t(t, r\omega) r^{n-1} dr d\omega + \int_{S^{n-1}} f(t, t\omega) t^{n-1} d\omega. \end{aligned} \tag{1.170}$$

Applying Gauss-Green to the first integral in (1.169) gives

$$\dot{e}(t) = \int_{B(x_0,t_0-t)} u_t(u_{tt} - \Delta u)dx + \int_{\partial B(x_0,t_0-t)} \partial_\nu u \; u_t dS(x) - \frac{1}{2}\int_{\partial B(x_0,t_0-t)} (u_t^2 + |\nabla u|^2)dS(x). \tag{1.171}$$

The first integral in (1.171) is zero and

$$|\partial_\nu u \; u_t| \leq \frac{1}{2}|u_t|^2 + \frac{1}{2}|\nabla u|^2,$$

so $\dot{e}(t) \leq 0$. Thus $0 \leq e(t) \leq e(0) = 0 \Rightarrow e(t) = 0 \Rightarrow u = 0$ in $C(t_0, x_0)$. □

Remark 1.76. In view of Theorem 1.75 we call $C(t_0, x_0)$ the *domain of dependence* of (t_0, x_0). If $S \subset (0,\infty)\times\mathbb{R}^n$ we defined the domain of dependence of S to be $\cup_{(t_0,x_0)\in S} C(t_0, x_0)$.[38] Similarly, if $K \subset \{0\}\times\mathbb{R}^n$ and if $C_f(0, x_0)$ is the forward cone $\{(t, x) : 0 \leq t < \infty, |x-x_0| \leq t\}$ with vertex at $(0, x_0)$, then we define *domain of influence* of K to be $\overline{K_i}$, where K_i is the union

$$K_i = \cup_{(0,x_0)\in K} C_f(0, x_0). \tag{1.172}$$

For any $x_0 \in K$, the ray from $(0, x_0)$ to a point $(t, x) \in \partial C_f(0, x_0)$ has slope one, so Theorem 1.75 implies that the outer edge of the $x-$support of u propagates at speed one. If the wave equation is modified to be $(\partial_t^2 - c^2\Delta)u = 0$, then support propagates at speed c. (check)[39]

The wave equation is invariant under $\partial_t \to -\partial_t$, so parallel statements hold for problems in $t < 0$ with initial data at $t = 0$. If $\Box u = 0$ in $\mathbb{R} \times \mathbb{R}^n$ and $u \in C^2$ (say), the value of u at any point (t_0, x_0) is determined by the values of u on any time slice in $t > t_0$ of the forward cone $C_f(t_0, x_0) = \{(t, x) : t \geq t_0, |x - x_0| \leq |t - t_0|\}$.

1.6.2 Fourier transform methods

We will the Fourier transform to construct solutions of the wave equation and to derive more refined information about domains of dependence and propagation of support. Consider the problem

$$\begin{aligned} &\Box u = 0 \text{ in } \mathbb{R} \times \mathbb{R}^n \\ &u|_{t=0} = f, \; u_t|_{t=0} = g, \text{ where } f, g \in \mathcal{S}(\mathbb{R}^n). \end{aligned} \tag{1.173}$$

Assuming for the moment that there exists a solution $u \in C^2(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n))$, we can take the Fourier transform in x to obtain

$$\begin{aligned} &(\partial_t^2 + |\xi|^2)\hat{u} = 0 \\ &\hat{u}|_{t=0} = \hat{f}, \;\; \hat{u}_t|_{t=0} = \hat{g}. \end{aligned} \tag{1.174}$$

[38] If we had considered the problem $\Box u = 0$ in $\mathbb{R} \times \mathbb{R}^n$, we would use infinite backward cones to define the domain of dependence.

[39] Draw a few pictures of domains of dependence and influence.

The ODE in t has linearly independent solutions $\cos(|\xi|t)$, $\sin(|\xi|t)$, so $\hat{u}$ must have the form

$$\hat{u}(t,\xi) = c_1(\xi)\cos(|\xi|t) + c_2(\xi)\sin(|\xi|t).$$

The coefficients are easily determined using the initial conditions, and we conclude

$$\hat{u}(t,\xi) = \hat{f}(\xi)\cos(|\xi|t) + \hat{g}(\xi)\frac{\sin(|\xi|t)}{|\xi|}. \tag{1.175}$$

From the formula we see that $\hat{u} \in C^\infty(\mathbb{R}, \mathcal{S}')$, so $u \in C^\infty(\mathbb{R}, \mathcal{S}')$.

Using Proposition 1.67(c) we can write

$$u(t,x) = f *_x \Psi_t + g *_x \Phi_t, \text{ where } \hat{\Psi}_t := \cos(|\xi|t),\ \hat{\Phi}_t = \frac{\sin(|\xi|t)}{|\xi|}. \tag{1.176}$$

Since $\hat{\Psi}_t = \partial_t \hat{\Phi}_t$, we can focus on finding $\Phi_t = \mathcal{F}_\xi^{-1}\left(\frac{\sin(|\xi|t)}{|\xi|}\right)$.

For $\epsilon > 0$ define

$$\Phi_t^\epsilon(\xi) = e^{-\epsilon|\xi|}\hat{\Phi}_t(\xi) = e^{-\epsilon|\xi|}\frac{\sin(|\xi|t)}{|\xi|} \in C(\mathbb{R}_t, L^1(\mathbb{R}^n_\xi)) \cap \mathcal{S}'(\mathbb{R}\times\mathbb{R}^n). \tag{1.177}$$

Proposition 1.77. *(a) For $\epsilon > 0$ we have*

$$\begin{gathered}\Phi_t^\epsilon(x) = c_n\left[\frac{1}{[(\epsilon+it)^2+|x|^2]^{\frac{n-1}{2}}} - \frac{1}{[(\epsilon-it)^2+|x|^2]^{\frac{n-1}{2}}}\right],\\ \text{where } c_n = \frac{2^{n/2}}{\sqrt{\pi}}\Gamma\left(\frac{n+1}{2}\right)\frac{i}{2(n-1)}.\end{gathered} \tag{1.178}$$

(b) The functions $\Phi^\epsilon \to \Phi$ in $C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n)) \cap \mathcal{S}'(\mathbb{R}\times\mathbb{R}^n)$ as $\epsilon \to 0$.

(c) When $n \geq 2$ is odd, supp $\Phi_t(x) = \{(t,x) : |x| = |t|\}$. *This is called the* strict Huygens' principle.

(d) When $n \geq 2$ is even, supp $\Phi_t(x) = \{(t,x) : |x| \leq |t|\}$ *and*[40]

$$\Phi_t(x) = \frac{(-1)^{\frac{n}{2}-1}\,2^{n/2}}{(n-1)\sqrt{\pi}}\,\Gamma\left(\frac{n+1}{2}\right) sgn(t)(t^2-|x|^2)^{-\frac{n-1}{2}} \text{ in } |x| < |t|. \tag{1.179}$$

Proof. **(a)** We have for $\epsilon > 0$:

$$\begin{aligned}\hat{\Phi}_t^\epsilon(\xi) &= \frac{e^{-(\epsilon-it)|\xi|} - e^{-(\epsilon+it)|\xi|}}{2i|\xi|} = \frac{1}{2i}\int_{\epsilon-it}^{\epsilon+it} e^{-s|\xi|}ds, \text{ so}\\ \Phi_t^\epsilon(x) &= \frac{1}{2i}\int_{\epsilon-it}^{\epsilon+it}\int_{\mathbb{R}^n} e^{ix\xi}e^{-s|\xi|}d'\xi ds.\end{aligned} \tag{1.180}$$

[40]Here $sgn(t) = \begin{cases}1, & t \geq 0\\ -1, & t < 0\end{cases}$.

Using Proposition 8.26 to compute the $d'\xi$ integral, we obtain[41]

$$\Phi_t^\epsilon(x) = \frac{2^{n/2}}{2i\sqrt{\pi}}\,\Gamma\left(\frac{n+1}{2}\right)\int_{\epsilon-it}^{\epsilon+it}\frac{s}{(s^2+|x|^2)^{\frac{n+1}{2}}}ds = \\ c_n\left[\frac{1}{[(\epsilon+it)^2+|x|^2]^{\frac{n-1}{2}}} - \frac{1}{[(\epsilon-it)^2+|x|^2]^{\frac{n-1}{2}}}\right]. \tag{1.181}$$

(b) From the formula (1.177) we can read off that $\hat\Phi_t^\epsilon(\xi) \to \hat\Phi^\epsilon(\xi)$ in $C^\infty(\mathbb{R}, \mathcal{S}'(\mathbb{R}^n)) \cap \mathcal{S}'(\mathbb{R}\times\mathbb{R}^n)$. Part (b) follows since $\mathcal{F}: \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous.

(c) We show

$$\text{supp}\ \Phi_t(x) \subset \{(t,x) : |x| = |t|\}, \tag{1.182}$$

and leave the proof of equality for Exercise 1.6.2a. For the moment assume $t > 0$ and compute

$$\begin{aligned}(\epsilon+it)^2+|x|^2 &= \epsilon^2 + 2i\epsilon t + (|x|^2 - t^2)\\ (\epsilon-it)^2+|x|^2 &= \epsilon^2 - 2i\epsilon t + (|x|^2 - t^2).\end{aligned} \tag{1.183}$$

Observe that if $|x| < t$, then as $\epsilon \to 0$,

$$\begin{aligned}[(\epsilon+it)^2+|x|^2]^{1/2} &\to i\sqrt{t^2-|x|^2}\\ [(\epsilon-it)^2+|x|^2]^{1/2} &\to -i\sqrt{t^2-|x|^2},\end{aligned} \tag{1.184}$$

while if $|x| > t > 0$ both limits are $\sqrt{|x|^2-t^2}$.

Consider the pairing $\langle \Phi_t^\epsilon(x), \phi(t,x)\rangle$, where ϕ is a test function. If ϕ is supported near a point in $|x| > t > 0$, the bracketed term in (1.181) approaches zero pointwise on the support of ϕ for any n. If n is odd, the bracketed term appoaches zero since $n-1$ is even, and hence the sign on i in (1.184) does not matter. Thus the dominated convergence theorem yields (1.182).

Now let $t < 0$. If $|x| < |t|$, the signs on the right in (1.184) are reversed, while if $|x| > |t|$ both limits in (1.184) are still $\sqrt{|x|^2-t^2}$, so (1.182) holds in this case too.

(d) The argument for part (c) shows supp $\Phi_t(x) = \{(t,x) : |x| \le |t|\}$ when n is even. So it remains just to prove (1.179) in $|x| < |t|$.

Assume $t > 0$ for the moment and $|x| < t$. Then (1.184) implies that the limit as $\epsilon \to 0$ of the bracketed term in (1.181) is

$$\frac{2}{\left(i\sqrt{t^2-|x|^2}\right)^{n-1}}.$$

This implies (1.179) when $t > 0$. When $t < 0$ and $|x| < |t|$, the signs on the right in (1.184) are reversed, leading to the appearance of $sgn(t)$ in (1.179).

□

[41] Recall from Proposition 8.26 that we use here the branch of $\sqrt{z}$ cut along the negative real axis.

Problem 3 of section 1.7 shows how to use the jump formula and Φ_t to obtain a forward fundamental solution of $\Box$. Another approach to constructing fundamental solutions of $\Box$ that uses pullbacks of simple homogeneous distributions is given in chapter 7. This approach yields more explicit formulas for $\Phi_t(x)$.

Exercise 1.6.2a. Complete the proof that supp $\Phi_t(x) = \{(t,x) : |x| = |t|\}$ when $n \geq 2$ is odd.

Remark 1.78. The support result in Proposition 1.77 when $n = 3$ can be viewed as predicting the sharp spherical fronts we observe when light propagates from a point source in space. The result when $n = 2$ predicts the smoothed out circular fronts we observe when a pebble is dropped in a pond.

Exercise 1.6.2b. Suppose $f \in \mathcal{S}'(\mathbb{R}^n)$, $g \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\text{supp } f * g \subset \overline{\text{supp } f + \text{supp } g}. \tag{1.185}$$

Hint: This is obvious when f is continuous with polynomial growth. In fact, if $x_0 \notin \overline{\text{supp } f + \text{supp } g}$, then for x near x_0 there is no y such that both $x - y \in$ supp g and $y \in$ supp f. So $\int f(y)g(x-y)dy = 0$ for x near x_0, and thus $x_0 \notin$ supp $f * g$.

Remark 1.79. Proposition 1.77, the formula (1.176) for u, and Exercise 1.6.2b together give a different proof of the earlier statements about domains of dependence and influence when n is even, and yield a refinement of those statements when n is odd. If n is odd, $K \subset \{0\} \times \mathbb{R}^n$, and $C_f(0,x_0)$ is the forward cone $\{(t,x) : 0 \leq t < \infty, |x - x_0| \leq t\}$ with vertex at $(0,x_0)$, we define the *strict domain of influence* of K to be $\overline{K_{i,s}}$, where $K_{i,s}$ is the union of the *boundaries* of the forward cones with vertices at points of K,

$$K_{i,s} = \cup_{(0,x_0)\in K} \partial C_f(0,x_0). \tag{1.186}$$

The above three results imply that if $u(0,x)$ is supported in K, then for $t > 0$ the solution u of $\Box u = 0$ is supported in $\overline{K_{i,s}}$.

Strict domains of dependence are defined similarly when n is odd. If a solution of $\Box u = 0$ is modified only near an *interior* point of the backward cone with vertex at (t_0,x_0), then u is unaffected near (t_0,x_0).[42]

1.7 Exercises

1. Suppose $u \in C^2(\mathbb{R}^n \setminus 0) \cap C(\mathbb{R}^n)$ is harmonic in $\mathbb{R}^n \setminus 0$ and bounded. Show that u is constant on $\mathbb{R}^n$. (This is a challenging and interesting problem. If you can solve it, please

[42] Draw pictures.

show me your solution.)

2. **Schrödinger's equation.** Consider the Schrödinger initial value problem

$$\begin{aligned} &u_t - i\Delta u = 0 \text{ on } \mathbb{R}_t \times \mathbb{R}^n \\ &u|_{t=0} = g \in \mathcal{S}(\mathbb{R}). \end{aligned} \tag{1.187}$$

(a) Starting with the heat equation, explain how a complex change of variable $t \to it$ can be used to quickly derive the solution formula for (1.187)

$$u(t,x) = (4\pi it)^{-n/2} \int e^{-|x-y|^2/4it} g(y) dy. \tag{1.188}$$

(b) Prove the "dispersive estimate"

$$|u(t,x)| \leq Ct^{-n/2} |g|_{L^1}. \tag{1.189}$$

(c) Define $E(t) = \int |u(t,x)|^2 dx$. Use an energy method like that used in step **d** of the proof of Theorem 1.47 to show that E is a constant function of t.
(d) Rederive the result of part (c) using the Fourier transform.

3. **Fundamental solutions.** (a) Let $\Phi_t(x) = \mathcal{F}^{-1}(\frac{\sin(t|\xi|)}{|\xi|})$ for $t \geq 0$. Use the jump formula to show that

$$H(t)\Phi_t(x)(2\pi)^{-n/2} \in \mathcal{D}'(\mathbb{R}^n_x \times \mathbb{R}_t)$$

is a fundamental solution for the wave operator $\partial_t^2 - \Delta$. This is called the "forward fundamental solution", since its support is contained in the forward light cone $\{(t,x) : t \geq 0, |x| \leq |t|\}$. Similarly,

$$-H(-t)\Phi_t(x)(2\pi)^{-n/2}$$

is the "backward fundamental solution".
(b) Let $K_t(x) = \mathcal{F}^{-1}(e^{-t|\xi|^2})$.[43] Use the jump formula to show that

$$H(t)K_t(x)(2\pi)^{-n/2} \in \mathcal{D}'(\mathbb{R}^n_x \times \mathbb{R}_t)$$

is a fundamental solution for the heat operator $\partial_t - \Delta$.

4. Let $P(D) = 1 + D_1^2 + \cdots + D_n^2 = 1 - \Delta$. Find a fundamental solution $E \in \mathcal{S}'$ of $P(D)$.

5. **Poisson kernel for $\mathbb{R}^{1+n}_+$.** Write points in $\mathbb{R}^{1+n}_+$ as (t,x) with $t > 0$. Rederive the Poisson kernel for the upper half space in $\mathbb{R}^{n+1}$ by applying the Fourier transform to the problem

$$\begin{aligned} &(\partial_t^2 + \Delta_x)u = 0 \text{ in } t > 0 \\ &\hat{u}(0,x) = f(x). \end{aligned} \tag{1.190}$$

Recall Remark 8.28.

[43] So $K_t(x)(2\pi)^{-n/2} = (4\pi t)^{-n/2} e^{-|x|^2/4t}$ is the heat kernel.

2 Hyperbolic systems

In studying the classical PDEs we have taken advantage of the explicit solution formulas that can be derived. For most PDEs such formulas are not available, and we must resort to other methods to get information about solutions. In this chapter we illustrate how results from functional analysis can be combined with a priori estimates to study existence and uniqueness of solutions for a large class of hyperbolic systems of PDEs, the symmetric hyperbolic systems, which are of great importance in mathematical physics.

2.1 Sobolev spaces on $\mathbb{R}^n$.

We begin by defining the function spaces we will use. For $s \in \mathbb{N}_0$ we can define the Sobolev space $H^s(\mathbb{R}^n)$ in a few equivalent ways:

$$\begin{aligned} H^s(\mathbb{R}^n) &= \{u \in L^2(\mathbb{R}^n) : D^\alpha u \in L^2(\mathbb{R}^n), |\alpha| \leq s\} = \\ &\{u \in \mathcal{S}'(\mathbb{R}^n) : D^\alpha u \in L^2(\mathbb{R}^n), |\alpha| \leq s\} = \{u \in \mathcal{S}'(\mathbb{R}^n) : \xi^\alpha \hat{u} \in L^2(\mathbb{R}^n), |\alpha| \leq s\} = \\ &\{u \in \mathcal{S}'(\mathbb{R}^n) : \langle\xi\rangle^s \hat{u} \in L^2(\mathbb{R}^n)\}, \end{aligned} \tag{2.1}$$

where $\langle\xi\rangle := (1 + |\xi|^2)^{1/2}$. Clearly, the last definition can be generalized to $s \in \mathbb{R}$.

Definition 2.1. *For $s \in \mathbb{R}$ define $H^s(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n) : \langle\xi\rangle^s \hat{u} \in L^2(\mathbb{R}^n)\}$ with the norm*

$$|u|_s = |\langle\xi\rangle^s \hat{u}|_{L^2(\mathbb{R}^n)}. \tag{2.2}$$

It is not hard to check that for any $s \in \mathbb{R}$, $H^s(\mathbb{R}^n)$ is a Hilbert space with the inner product

$$(u, v)_s := (\langle\xi\rangle^s \hat{u}, \langle\xi\rangle^s \hat{v})_{L^2(\mathbb{R}^n)} = \int \langle\xi\rangle^{2s} \hat{u}\overline{\hat{v}} d\xi. \tag{2.3}$$

Definition 2.2. *For $s \in \mathbb{R}$ define the linear operator $\Lambda^s : \mathcal{S}' \to \mathcal{S}'$ by*[1]

$$\widehat{\Lambda^s u} := \langle\xi\rangle^s \hat{u} \Leftrightarrow \Lambda^s u = \mathcal{F}^{-1}(\langle\xi\rangle^s \hat{u}). \tag{2.4}$$

If $u \in \mathcal{S}$ we can write

$$(\Lambda^s u)(x) = \int e^{ix\xi} \langle\xi\rangle^s \hat{u}(\xi) d'\xi. \tag{2.5}$$

[1] In this chapter we'll often write $\mathcal{S}$, $\mathcal{S}'$, H^s, ... for $\mathcal{S}(\mathbb{R}^n)$, $\mathcal{S}'(\mathbb{R}^n)$, $H^s(\mathbb{R}^n)$, ...

This shows that Λ^s is a *pseudodifferential operator* with symbol $\langle\xi\rangle^s$. More generally, an operator of the form

$$a(x,D)u := \int e^{ix\xi} a(x,\xi)\hat{u}(\xi)d\xi \tag{2.6}$$

is said to be a pseudodifferential operator with symbol $a(x,\xi)$. The operator Λ^s is sometimes called a *Fourier multiplier*, since its symbol does not depend on x.

Exercise 2a. For $m \in \mathbb{N}$ let $P(x,D) = \sum_{|\alpha|\le m} c_\alpha(x)D^\alpha$, where $c_\alpha \in C^\infty(\mathbb{R}^n)$. Show that $P(x,D)$ can be written in the form (2.6).

Proposition 2.3. . *The spaces H^s satisfy:*
(a) $L^2 = H^0$;
(b) if $s > t$ the inclusion map $i : H^s \to H^t$ is continuous;
(c) for any s,t the map $\Lambda^t : H^s \to H^{s-t}$ is an isometry with inverse Λ^{-t};
(d) for any $s \in \mathbb{R}$ and multiindex α, the map $D^\alpha : H^s \to H^{s-|\alpha|}$ is continuous.

Proof. **(a),(b)** These parts are obvious.
(c) Let $u \in H^s$. Then

$$\begin{aligned} |\Lambda^t u|_{s-t} &= |\langle\xi\rangle^{s-t}\langle\xi\rangle^t\hat{u}|_{L^2} = |\langle\xi\rangle^s\hat{u}|_{L^2} = |u|_s \\ \widehat{\Lambda^{-s}\Lambda^s u} &= \langle\xi\rangle^{-s}\langle\xi\rangle^s\hat{u} = \hat{u}. \end{aligned} \tag{2.7}$$

(d) For $u \in H^s$ we have

$$|D^\alpha u|_{s-|\alpha|} = |\langle\xi\rangle^{s-|\alpha|}\xi^\alpha\hat{u}|_{L^2} \le |\langle\xi\rangle^s\hat{u}|_{L^2} = |u|_s. \tag{2.8}$$

□

Theorem 2.4 (Sobolev embedding)**.** *Let $s \in \mathbb{R}$, $k \in \mathbb{N}_0$ and suppose $s > \frac{n}{2} + k$. Then there is a continuous injective map*

$$i : H^s(\mathbb{R}^n) \to C_b^k(\mathbb{R}^n), \tag{2.9}$$

where $C_b^k(\mathbb{R}^n) = \{u \in C^k(\mathbb{R}^n) : \partial^\alpha u \text{ is bounded for } |\alpha| \le k\}$.

The elements of H^s are equivalence classes with respect to the equivalence relation $f \sim g \Leftrightarrow f = g$ almost everywhere. The theorem asserts that each class in H^s has a unique representative in C_b^k.

Proof. **1.** Let $u \in H^s$ and $|\alpha| \le k$. We claim that $\widehat{D^\alpha u} \in L^1(\mathbb{R}^n)$. Indeed,

$$\begin{aligned} \int |\widehat{D^\alpha u}|\, d\xi = \int |\xi^\alpha||\hat{u}|d\xi \le \int \frac{|\xi|^{|\alpha|}}{\langle\xi\rangle^s}\langle\xi\rangle^s|\hat{u}|d\xi \le \\ \int \frac{1}{\langle\xi\rangle^{s-k}}\,\langle\xi\rangle^s|\hat{u}|d\xi \le \left|\frac{1}{\langle\xi\rangle^{s-k}}\right|_{L^2} |u|_s = C_s|u|_s. \end{aligned} \tag{2.10}$$

2. By the consistency result of Exercise 1.5.1a, we see that the *distribution derivative* $D^\alpha u$ satisfies

$$D^\alpha u(x) = \int e^{ix\xi} \widehat{D^\alpha u}(\xi) d'\xi, \tag{2.11}$$

and hence $D^\alpha u$ has a continuous bounded representative, call it g_α, for each $|\alpha| \le k$. We conclude that u has a representative in C_b^k and that

$$D^\alpha u = D_{cl}^\alpha u \text{ for } |\alpha| \le k, \tag{2.12}$$

where D_{cl}^α denotes the classical derivative. The uniqueness of $u \in C_b^k$ follows from the estimate (2.10) since

$$|D^\alpha u|_{L^\infty} \le |\widehat{D^\alpha u}|_{L^1} \le C_s |u|_s \text{ for } |\alpha| \le k.$$

□

Exercise 2.1b. In step **2** of the proof of Theorem 2.4 we wrote: "We conclude that u has a representative in C_b^k and that

$$D^\alpha u = D_{cl}^\alpha u \text{ for } |\alpha| \le k, \tag{2.13}$$

where D_{cl}^α denotes the classical derivative." Explain this. Hint: It is clear that $D^\alpha u = D_{cl}^\alpha u$ when u is C^k (why?). Consider $(D^\alpha u) * \rho_n$, where ρ_n is an approximate identity.

Exercise 2.1c. Let u be as in Theorem 2.4. Show that, in fact, for each $|\alpha| \le k$,

$$D^\alpha u(x) \to 0 \text{ uniformly as } |x| \to \infty.$$

The Riesz representation theorem, Theorem 8.41, tells us that the map $u \to (\cdot, u)_s$ gives an isometric isomorphism of H^s with its dual $(H^s)'$. We'll finish this section by giving another useful characterization of $(H^s)'$.

Lemma 2.5. *For all $s \in \mathbb{R}$ the space $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$.*

Proof. We know from measure theory that $\mathcal{S}$ is dense in L^2. If $u \in H^s$, then $\Lambda^s u \in L^2$, so there exist $\phi_n \in \mathcal{S}$ such that $\phi_n \to \Lambda^s u$ in L^2. Since $\Lambda^{-s} : L^2 \to H^s$ is continuous, this implies

$$\Lambda^{-s}\phi_n \to u \text{ in } H^s.$$

That gives the result since $\Lambda^{-s}\phi_n \in \mathcal{S}$ (why?).

□

Below we let $\langle \cdot, \cdot \rangle$ denote the $L^2(\mathbb{R}^n)$ pairing

$$\langle u, v \rangle := \int u\bar{v} dx = \int \hat{u}\bar{\hat{v}} d\xi \text{ for } u, v \in L^2. \tag{2.14}$$

Proposition 2.6. *The L^2 pairing $\langle u, v\rangle$ for $(u, v) \in \mathcal{S} \times \mathcal{S}$ has a unique extension by continuity to a sesquilinear pairing on $H^s \times H^{-s}$ such that*

$$|\langle u, v\rangle| \leq |u|_s |v|_{-s}. \tag{2.15}$$

Proof. For $u \in H^s$, $v \in H^{-s}$ *define* the extension by

$$\langle u, v\rangle := \int \hat{u}\overline{\hat{v}} d\xi = \int \langle\xi\rangle^s \hat{u}\, \langle\xi\rangle^{-s}\overline{\hat{v}} d\xi. \tag{2.16}$$

Then Cauchy-Schwarz implies (2.15), and Lemma 2.5 implies that the extension is unique. □

Proposition 2.7. *The map $M : H^{-s} \to (H^s)'$ given by $Mv = \langle \cdot, v\rangle$ is an isometric isomorphism of H^{-s} onto $(H^s)'$.*

1. First we check that M is an isometry onto $R(M) \subset (H^s)'$. For $v \in H^{-s}$, $u \in H^s$ we have[2]

$$|(Mv)(u)| = |\langle u, v\rangle| \leq |u|_s |v|_{-s} \Rightarrow |Mv| \leq |v|_{-s}. \tag{2.17}$$

Now set $u := \Lambda^{-2s} v \in H^s$. We have

$$\begin{aligned} &(Mv)(u) = \langle \Lambda^{-2s} v, v\rangle = \int \langle\xi\rangle^{-s}\hat{v}\, \langle\xi\rangle^{-s}\overline{\hat{v}} d\xi = |v|^2_{-s}, \text{ and thus} \\ &|(Mv)(u)| = |v|^2_{-s} = |v|_{-s}|u|_s \Rightarrow |Mv| = |v|_{-s}. \end{aligned} \tag{2.18}$$

2. To see that M is surjective, let $\ell \in (H^s)'$ and use the Riesz representation theorem to find a unique $f \in H^s$ such that for $u \in H^s$:

$$\ell(u) = (u, f)_s = \int \langle\xi\rangle^s \hat{u} \langle\xi\rangle^s \overline{\hat{f}} d\xi = \int (\langle\xi\rangle^s \hat{u})(\langle\xi\rangle^{-s}\langle\xi\rangle^{2s}\overline{\hat{f}}) d\xi. \tag{2.19}$$

The right side of (2.19) is $\langle u, v\rangle$, where $\hat{v} = \widehat{\Lambda^{2s} f}$. Thus, $Mv = \ell$, where $v = \Lambda^{2s} f \in H^{-s}$.

2.2 First-order symmetric hyperbolic systems

Let $T > 0$ and for $(t, x) \in [0, T] \times \mathbb{R}^n$ define the first-order symmetric hyperbolic operator

$$P(t, x, D_t, D_x) = D_t - A(t, x, D_x), \text{ where } A(t, x, D_x) = \sum_{j=1}^n A_j(t, x) D_j + B(t, x). \tag{2.20}$$

Assumption 2.8. *The coefficients A_j and B are C^∞ $m \times m$ matrices satisfying $A_j = A_j^*$. We also assume that there exists a compact set $K \subset \mathbb{R}^n$ such that the matrices $A_j(t, x)$ and $B(t, x)$ are constant for $t \in [0, T]$, $x \in \mathbb{R}^n \setminus K$.*

[2] Here $|Mv| = \sup_{|u|\leq 1} |Mv(u)|$.

Notations 2.9. 1. We often write D_j for $D_{x_j} = \frac{1}{i}\partial_{x_j}$.

2. If $w(x) = (w_1, \dots, w_m)$ we take $|w|_s^2 := \sum_{j=1}^m |w_j|_s^2$. When $|w|_s$ is finite we write $w \in H^s(\mathbb{R}^n)$.

3. In this section $u(t,x)$ will always be a $\mathbb{C}^m-$valued function. For $s \in \mathbb{R}$ the space $L^2([0,T], H^s)$ is the space of functions $u(t,x)$ such that the function $t \to |u(t,x)|_{H^s(\mathbb{R}^n_x)}$ is measurable and $|u|^2_{L^2([0,T],H^s)} := \int_0^T |u(t,x)|^2_{H^s} dt < \infty$.

4. For $u(t,x) \in C([0,T], H^s)$ we define $|u|_{C([0,T],H^s)} = \sup_{t\in[0,T]} |u(t,x)|_{H^s}$.[3]

5. We often write P for $P(t,x,D_t,D_x)$ and D for D_x.

Exercise 2.2a. Show that for any $s \in \mathbb{R}$ the dual space of $L^2([0,T], H^s)$ is $L^2([0,T], H^{-s})$. Hint: Use Theorem 8.41.

Our goal in this chapter is to prove the following theorem.

Theorem 2.10. *Suppose P satisfies Assumption 2.8 and for any $s \in \mathbb{R}$ consider the Cauchy problem*

$$\begin{aligned} Pu &= f \text{ in } (0,T) \times \mathbb{R}^n \\ u|_{t=0} &= g, \end{aligned} \tag{2.21}$$

where $f \in L^2([0,T], H^s)$ and $g \in H^s$ are given. There exists a unique solution $u \in C([0,T], H^s)$. Moreover, for some constant $C = C(s,T)$, the solution u satisfies the estimate

$$|u(t,\cdot)|_s^2 \le C\left[|g|_s^2 + \int_0^T |f(\tau,\cdot)|_s^2 d\tau\right] \text{ for } t \in [0,T]. \tag{2.22}$$

Remark 2.11. Problems of the type (2.21) arise often in mathematical physics. For example, Maxwell's equations have this form. Also, the linearized versions of the following nonlinear systems of equations can be put in this form: the Euler equations for compressible gases, the magnetohydrodynamic (MHD) equations for plasmas, the ZND equations of combustion, the Einstein field equations of general relativity, and the equations of nonlinear elasticity.

2.2.1 A priori estimate and uniqueness

The estimate in the following proposition is called an *a priori* estimate.

Proposition 2.12 (A priori estimate)**.** *Suppose P satisfies Assumption 2.8. Then for any $s \in \mathbb{R}$, $T > 0$ there exists a constant $C = C(s,T)$ such that for* all *$u \in C([0,T], H^{s+1}) \cap C^1([0,T], H^s)$ we have*

$$(I)_s \quad |u(t,\cdot)|_s^2 \le C\left[|u(0,\cdot)|_s^2 + \int_0^T |Pu(\tau,\cdot)|_s^2 d\tau\right] \text{ for } t \in [0,T]. \tag{2.23}$$

[3] Observe that $C([0,T], H^s) \subset L^2([0,T], H^s)$.

We give the proof after some preparation. An a priori estimate in PDE is a bound on the size or regularity of a potential solution, derived *before* the existence of the solution is actually proven. A remarkable fact, illustrated in this chapter and the next, is that such estimates combined with some functional analysis can often be used to prove existence and uniqueness of solutions.

The fact that the right side of (2.23) is finite is a direct consequence of the next lemma.

Lemma 2.13. *For any $s \in \mathbb{R}$ we have $A(t,x,D) : C([0,T], H^{s+1}) \to C([0,T], H^s)$.*

This lemma is obvious when $s \in \mathbb{N}_0$, but requires some effort for general $s \in \mathbb{R}$.

Proof. **1.** Since $D_j : C([0,T], H^{s+1}) \to C([0,T], H^s)$, we reduce to showing

$$A_j(t,x) : C([0,T], H^s) \to C([0,T], H^s) \text{ for each } j = 1, \dots, n. \tag{2.24}$$

By assumption $A_j(t,x) = c_j$ for some constant c for all $(t,x) \in [0,T] \times K^c$. Writing $A_j = (A_j - c_j) + c_j$, we reduce to showing (2.24) when supp $A_j \subset [0,T] \times K.$. We can further suppose that A_j is scalar and that is acts on $\mathbb{C}-$valued functions (why?).

2. Suppose $a \in C_c^\infty(\mathbb{R}^n) \to \mathbb{C}$. We first show that

$$a(x) : H^s \to H^s. \tag{2.25}$$

For $u \in H^s$ we have $a(x)u(x) = \int e^{ix\eta} \hat{a}(\eta) u(x) d'\eta$. To estimate this we use the following estimate proved in Lemma 8.64:

$$|e^{ix\eta} u(x)|_s \leq C \langle \eta \rangle^{|s|} |u|_s. \tag{2.26}$$

We have

$$|au|_s \leq \int \left| e^{ix\eta} \hat{a}(\eta) u(x) \right|_s d'\eta \leq C \left(\int \langle \eta \rangle^{|s|} |\hat{a}(\eta)| d'\eta \right) |u|_s = C_s |u|_s. \tag{2.27}$$

To finish the proof of (2.24), we can repeat the argument of this step after inserting continuous $t-$dependence in a and u.

□

We can now prove the uniqueness statement in Theorem 2.10.

Proposition 2.14 (Uniqueness)**.** *Assume $u_1, u_2 \in C([0,T], H^s)$ satisfy the problem (2.21). Then $u_1 = u_2$.*

Proof. We set $w = u_1 - u_2$ and show that $w = 0$. We claim

$$w \in C([0,T], H^s) \cap C^1([0,T], H^{s-1}).$$

Indeed, since $Pw = 0$ we have by Lemma 2.13:

$$D_t w = A(t,x,D) w \in C([0,T], H^{s-1}).$$

We can now apply the estimate $(I)_{s-1}$ of Proposition 2.12 to see that $|w(t,\cdot)|_{s-1} \leq 0$ for $t \in [0,T]$. □

In preparation for the proof of Proposition 2.12 we give the following definition.

Definition 2.15. *For $t \in [0,T]$ let $M_j(t,x)$, $j = 1,\dots,n$ and $N(t,x)$ be C^∞ $m \times m$ matrices, not necessarily symmetric. Define*

$$M(t,x,D) = \sum_{j=1}^{n} M_j(t,x)D_j + N(t,x).$$

The formal adjoint (adjoint for short) of $M(t,x,D)$ is the operator

$$M(t,x,D)^* := \sum_{j=1}^{n} M_j^* D_j + \sum_{j=1}^{n} D_j M_j^* + N^*. \tag{2.28}$$

Remark 2.16. (a) The operator $M(t,x,D)^*$ has the property that if $\phi, \psi \in \mathcal{D}(\mathbb{R}^n_x)$ and $\langle \cdot, \cdot \rangle$ denotes the $L^2(\mathbb{R}^n_x)$ pairing, then for any $t \in [0,T]$:

$$\langle M(t,x,D)\phi, \psi \rangle = \langle \phi, M(t,x,D)^*\psi \rangle. \tag{2.29}$$

Indeed, for any j we can integrate by parts to obtain:

$$\begin{aligned} \langle M_j(t,x)D_j\phi, \psi \rangle = \langle D_j\phi, M_j^*(t,x)\psi \rangle = \langle \phi, D_j(M_j^*\psi) \rangle = \\ \langle \phi, M_j^*(D_j\psi) + (D_jM_j^*)\psi \rangle. \end{aligned} \tag{2.30}$$

(b) The equality (2.29) remains true when $\phi, \psi \in H^1(\mathbb{R}^n)$.(why?)[4]
(c) When $M_j = M_j^*$ for all j, observe that

$$M(t,x,D) - M(t,x,D)^* = \sum_{j=1}^{n} D_j M_j^* + N^* \tag{2.31}$$

is an operator of order zero. This point is essential for the proof of Proposition 2.2.1.

Proof of Proposition 2.12. **1.** For any $u \in C([0,T], H^{s+1}) \cap C^1([0,T], H^s)$ we will prove the estimate (I_s) (2.23) by first proving

$$(II)_s \quad \frac{d}{dt}|u(t,\cdot)|_s^2 \le C\left(|u(t,\cdot)|_s^2 + |Pu(t,\cdot)|_s^2\right) \text{ for } t \in [0,T], \tag{2.32}$$

and then applying Gronwall's inequality, Proposition 8.68.
2. We begin by proving $(II)_0$. Define

$$f(t,x) := Pu = (D_t - A)u, \text{ so } if = u_t - iAu.$$

[4] Answer: Use the density of $\mathcal{D}(\mathbb{R}^n)$ in $H^1(\mathbb{R}^n)$ and the continuity of the map $M(t,x,D) : H^1(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ for each t.

With $\langle\cdot,\cdot\rangle$ the $L^2(\mathbb{R}^n_x)$ inner product, $u \in C([0,T],H^1)\cap C^1([0,T],H^0)$ we compute

$$\begin{aligned}\frac{d}{dt}\langle u(t,\cdot),u(t,\cdot)\rangle = \langle u_t,u\rangle + \langle u,u_t\rangle = \\ \langle iAu+if,u\rangle + \langle u,iAu+if\rangle = i\left(\langle f,u\rangle - \langle u,f\rangle\right) + i\langle (A-A^*)u,u\rangle\end{aligned} \tag{2.33}$$

Setting $E(t,x) := A - A^*$ we have[5]

$$\begin{aligned}&|\langle f,u\rangle| \le |f(t,\cdot)|_0^2 + |u(t,\cdot)|_0^2\\ &|\langle Eu,u\rangle| \le |Eu(t,\cdot)|_0|u(t,\cdot)|_0 \le C|u(t,\cdot)|_0^2.\end{aligned} \tag{2.34}$$

This proves $(II)_0$.

3. To prove $(II)_s$ for $u \in C([0,T],H^{s+1})\cap C^1([0,T],H^s)$ we apply $(II)_0$ to[6]

$$\Lambda^s u \in C([0,T],H^1)\cap C^1([0,T],H^0).$$

This gives

$$\begin{aligned}\frac{d}{dt}|\Lambda^s u(t,\cdot)|_0^2 \le C\left(|\Lambda^s u(t,\cdot)|_0^2 + |P\Lambda^s u(t,\cdot)|_0^2\right) \le \\ C\left(|u(t,\cdot)|_s^2 + |Pu(t,\cdot)|_s^2 + |[P,\Lambda^s]u(t,\cdot)|_0^2\right),\end{aligned} \tag{2.35}$$

where $[P,\Lambda^s]$ is the commutator

$$[P,\Lambda^s] = P\Lambda^s - \Lambda^s P = -(A\Lambda^s - \Lambda^s A) = -[A,\Lambda^s].$$

The commutator estimate of Proposition 8.65 shows that

$$|[A,\Lambda^s]u(t,\cdot)|_0^2 \le C|u(t,\cdot)|_s^2, \tag{2.36}$$

and with (2.35) this implies $(II)_s$.

4. Having $(II)_s$ we apply Gronwall's inequality, Proposition 8.68, with

$$y(t) := |u(t,\cdot)|_s^2 \text{ and } g(t) = |Pu(t,\cdot)|_s^2.$$

This gives

$$|u(t,\cdot)|_s^2 \le e^{Ct}\left[|u(0,\cdot)|_s^2 + \int_0^t |Pu(\tau,\cdot)|_s^2 e^{-C\tau}d\tau\right] \text{ for } t\in[0,T], \tag{2.37}$$

which implies $(I)_s$. □

In the next section we will use the fact that $u(0,\cdot)$ can be replaced by $u(T,\cdot)$ in the statement of Proposition 2.12. This is another illustration of the time-reversibility of hyperbolic problems.

[5] We use Remark 2.16(c) here.

[6] Here Λ^s is the Fourier multiplier with symbol $\langle\xi\rangle^s$.

Lemma 2.17. *Under the assumptions of Proposition 2.12, we have*

$$(I)_s \quad |u(t,\cdot)|_s^2 \leq C\left[|u(T,\cdot)|_s^2 + \int_0^T |Pu(\tau,\cdot)|_s^2 d\tau\right] \text{ for } t \in [0,T]. \tag{2.38}$$

Proof. Define $v(t,x) := u(T-t,x)$, so $v(0,x) = u(T,x)$, and set

$$\tilde{P}(t,x,D_t,D) = D_t + A(T-t,x,D) = D_t - \tilde{A}(t,x,D), \tag{2.39}$$

where $\tilde{A}(t,x,D) := -A(T-t,x,D)$. We have

$$\tilde{P}v = -(D_t u)(T-t,x) + A(T-t,x,D)u(T-t,x) = -(Pu)(T-t,x). \tag{2.40}$$

To finish, apply $(I)_s$ to the problem satisfied by v to get

$$|v(t,\cdot)|_s^2 \leq C\left[|v(0,\cdot)|_s^2 + \int_0^T |\tilde{P}v(\tau,\cdot)|_s^2 d\tau\right] \text{ for } t \in [0,T]. \tag{2.41}$$

With (2.40) this implies (2.38).

□

2.2.2 Existence

In this section we complete the proof of Theorem 2.10 by proving the following proposition.

Proposition 2.18. *Under the assumptions of Theorem 2.10 consider the problem (2.21):*

$$\begin{aligned} &Pu = f \text{ on } (0,T)\times\mathbb{R}^n, \\ &u|_{t=0} = g. \end{aligned} \tag{2.42}$$

There exists a solution $u \in C([0,T],H^s)$ which satisfies the estimate

$$|u(t,\cdot)|_s^2 \leq C\left[|u(0,\cdot)|_s^2 + \int_0^T |f(\tau,\cdot)|_s^2 d\tau\right] \text{ for } t \in [0,T]. \tag{2.43}$$

Proof. **1.** Define $E := \{\phi \in C^\infty([0,T],H^\infty(\mathbb{R}^n)) : \phi(T) = 0\}$, where $H^\infty = \cap_{s\in\mathbb{R}} H^s$.[7] Let $P^* = D_t - A^*$ be the formal adjoint of P, and observe that $P^*E \subset L^2([0,T],H^{-s})$ for all $s \in \mathbb{R}$. Using the data f, g in (2.42), we define a linear functional $\ell : P^*E \to \mathbb{C}$ as follows. For $\phi \in E$, set

$$\ell(P^*\phi) := \int_0^T \langle f(t,\cdot),\phi(t,\cdot)\rangle dt + \frac{1}{i}\langle g,\phi(0,\cdot)\rangle := \langle f,\phi\rangle_{t,x} + \frac{1}{i}\langle g,\phi(0,\cdot)\rangle. \tag{2.44}$$

[7] The space $H^\infty(\mathbb{R}^n)$ is a Fréchet space with metric $d(f,g) := \sum_{N=0}^\infty 2^{-N}\frac{|f-g|_N}{1+|f-g|_N}$.

Remark 2.19. Here $\langle \cdot, \cdot \rangle_{t,x}$ is the extension of the L^2 pairing from $C([0,T], \mathcal{S}) \times C([0,T], \mathcal{S})$ to $L^2([0,T], H^s) \times L^2([0,T], H^{-s})$. Parallel to Proposition 2.6 we have

$$|\langle v, w \rangle_{t,x}| \leq |v|_{L^2 H^s} |w|_{L^2 H^{-s}} \tag{2.45}$$

and parallel to Proposition 2.7 we have

$$L^2 H^{-s} = (L^2 H^s)';$$

recall Exercise 2.2a.

It is not immediately clear that ℓ is well-defined. Applying $(I)_{-s}$ we obtain for $t \in [0,T]$:

$$|\phi(t,\cdot)|^2_{-s} \leq C \left(|\phi(T,\cdot)|^2_{-s} + \int_0^T |P^* \phi(\tau,\cdot)|^2_{-s} d\tau \right) \leq C \int_0^T |P^* \phi(\tau,\cdot)|^2_{-s} d\tau, \tag{2.46}$$

so ℓ is well-defined. Moreover, we have for some $C = C(f,g,T)$:

$$\begin{aligned} |\ell(P^*\phi)| &\leq \int_0^T |f(t,\cdot|_s |\phi(t,\cdot)|_{-s} dt + |g|_s |\phi(0,\cdot)|_{-s} \leq \\ & C \sup_{[0,T]} |\phi(t,\cdot)|_{-s} \leq C \left(\int_0^T |P^* \phi(\tau,\cdot)|^2_{-s} d\tau \right)^{1/2} = C |P^* \phi|_{L^2 H^{-s}}. \end{aligned} \tag{2.47}$$

This shows that ℓ is a continuous antilinear functional on P^*E with respect to the $L^2 H^{-s}$ norm. By the Hahn-Banach theorem, Theorem 8.29, ℓ extends to a continuous antilinear functional on $L^2 H^{-s}$. By the Riesz representation theorem, Theorem 8.41, there exists a unique $u \in L^2([0,T], H^s)$ such that

$$\ell(P^*\phi) = \langle u, P^* \phi \rangle_{t,x} \text{ for all } \phi \in E. \tag{2.48}$$

It remains to show that this u has the desired properties. The first step is to show $Pu = f$ in the sense of distributions.

2. For $v \in \mathcal{D}'((0,T) \times \mathbb{R}^n)$ and $\phi \in \mathcal{D}((0,T) \times \mathbb{R}^n)$ define the pairing

$$(v, \phi)_{\mathcal{D}'} := v(\phi). \tag{2.49}$$

If $v \in L^2([0,T], H^s) \subset \mathcal{D}'$ we have

$$\langle v, \phi \rangle_{t,x} = (v, \overline{\phi})_{\mathcal{D}'} \text{ for } \phi \in \mathcal{D}. \tag{2.50}$$

Using Proposition 1.29, if we define operator P^t by the relation

$$(Pv, \phi)_{\mathcal{D}'} = (v, P^t \phi)_{\mathcal{D}'} \text{ for all } \phi \in \mathcal{D},$$

observe that $P^* = \overline{P^t}$.

3. We claim that u satisfies $Pu = f$ in $\mathcal{D}'((0,T)\times\mathbb{R}^n)$. Indeed, (2.44) and (2.48) imply that for all $\phi \in \mathcal{D}((0,T)\times\mathbb{R}^n) \subset E$ we have

$$\langle u, P^*\phi\rangle_{t,x} = \langle f, \phi\rangle_{t,x} \tag{2.51}$$

By (2.50) this is equivalent to

$$(u, \overline{P^*\phi})_{\mathcal{D}'} = (f, \overline{\phi})_{\mathcal{D}'} \Leftrightarrow (u, P^t\overline{\phi})_{\mathcal{D}'} = (f, \overline{\phi})_{\mathcal{D}'} \Leftrightarrow (Pu, \overline{\phi})_{\mathcal{D}'} = (f, \overline{\phi})_{\mathcal{D}'}. \tag{2.52}$$

Hence $Pu = f$ as distributions.

4. We now have $u \in L^2H^s$ and $Pu = f$ as distributions. Next we show $u(0,\cdot) = g$.

Since $D_t u = Au + f \in L^2H^{s-1}$, we have $u \in H^1([0,T], H^{s-1}\mathbb{R}^n) \subset C([0,T], H^{s-1})$ (check).

Lemma 2.20. *For any $v \in H^1H^{s-1}$ and $\phi \in E$ we have*

$$\begin{aligned}\langle v, P^*\phi\rangle_{t,x} = \langle Pv, \phi\rangle_{t,x} + i\langle v(T,\cdot), \phi(T,\cdot)\rangle - i\langle v(0,\cdot), \phi(0,\cdot)\rangle = \\ \langle Pv, \phi\rangle_{t,x} - i\langle v(0,\cdot), \phi(0,\cdot)\rangle.\end{aligned} \tag{2.53}$$

Proof. The equality (2.53) follows by integration by parts when $v \in C^\infty([0,T], \mathcal{S})$. Observe that $Pv \in L^2H^{s-2}$ and $v(T,\cdot)$, $v(0,\cdot)$ lie in H^{s-1}. We obtain (2.53) by choosing $v_j \in C^\infty\mathcal{S}$ such that $v_j \to v$ in H^1H^{s-1} and taking the limit as $j \to \infty$ using the estimate (2.45). □

Applying (2.53) to u we obtain

$$\langle u, P^*\phi\rangle_{t,x} = \langle Pu, \phi\rangle_{t,x} - i\langle u(0,\cdot), \phi(0,\cdot)\rangle. \tag{2.54}$$

But $\langle u, P^*\phi\rangle_{t,x} = \langle f, \phi\rangle_{t,x} - i\langle g, \phi(0,\cdot)\rangle$ by (2.48) and (2.44). Thus,

$$\langle u(0,\cdot), \phi(0,\cdot)\rangle = \langle g, \phi(0,\cdot)\rangle \text{ for all } \phi \in E.$$

Since $u(0,\cdot) \in H^{s-1}$ and H^∞ is dense in $H^{-(s-1)}$, this implies $u(0,\cdot) = g$.

5. We now know that $u \in L^2([0,T], H^s) \cap C([0,T], H^{s-1})$ and is a solution to (2.42). It remains to show $u \in C([0,T], H^s)$ and satisfies the estimate (2.22).

Choose $f_j \in C^\infty H^\infty$ such that $f_j \to f$ in L^2H^s and $g_j \in H^\infty$ such that $g_j \to g$ in H^s. By the results of steps **1-4** and Proposition 2.14, we obtain solutions $u_j \in C([0,T], H^\infty)$ to

$$\begin{aligned}Pu_j &= f_j \\ u_j(0,\cdot) &= g.\end{aligned} \tag{2.55}$$

Since $D_t u_j = Au_j + f_j \in CH^\infty$, we see that $u_j \in C^1H^\infty$.[8] Applying $(I)_s$ to $u_j - u_k$ yields

$$|(u_j - u_k)|_s^2 \le C\left(|g_j - g_k|_s^2 + \int_0^T |f_j(\tau,\cdot) - f_k(\tau,\cdot)|_s^2 d\tau\right). \tag{2.56}$$

[8] We can iterate this argument to obtain $u_j \in C^\infty H^\infty$.

Thus, (u_j) is Cauchy in $C([0,T],H^s)$ and hence $u_j \to \tilde{u}$ converges in CH^s to some $\tilde{u} \in CH^s$. That implies $f_j = Pu_j \to P\tilde{u}$ in $\mathcal{D}'$ (why?), so $P\tilde{u} = f$ as distributions. Similarly, $g_j = u_j(0,\cdot) \to \tilde{u}(0,\cdot)$ in H^s, so $\tilde{u}(0,\cdot) = g$. That is, $\tilde{u} \in CH^s$ satisfies the problem (2.42). Proposition 2.14 with $s-1$ in place of s implies $\tilde{u} = u$. Thus, $u \in C([0,T],H^s)$.

Since u_j satisfies the estimate (2.22) with f_j, g_j on the right and $u \in CH^s$, we can let $j \to \infty$ in that estimate to see that u satisfies (2.22).

□

Example 2.21 (Maxwell's equations in vacuum)**.** For $x \in \mathbb{R}^3$ let $E(t,x)$ and $H(t,x)$ take values in $\mathbb{R}^3$ and represent electric and magnetic fields respectively. Maxwell's equations in vacuum can be written as

$$\begin{aligned} \partial_t E &= \operatorname{curl} H \\ \partial_t H &= -\operatorname{curl} E. \end{aligned} \tag{2.57}$$

If we let $u := \begin{pmatrix} E \\ H \end{pmatrix}$ and define

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \; B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \; C = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{2.58}$$

then (2.57) is the same as the 6×6 symmetric hyperbolic system

$$\partial_t u = \begin{pmatrix} 0 & A \\ A^t & 0 \end{pmatrix} \partial_1 u + \begin{pmatrix} 0 & B \\ B^t & 0 \end{pmatrix} \partial_2 u + \begin{pmatrix} 0 & C \\ C^t & 0 \end{pmatrix} \partial_3 u. \tag{2.59}$$

3 Elliptic problems

In this chapter we'll study problems of the form

$$\begin{aligned} L(x,D)u &= f \text{ in } U, \\ u &= 0 \text{ on } \partial U, \end{aligned} \tag{3.1}$$

where $U \subset \mathbb{R}^n$ is bounded and open with C^1 boundary, and $L(x,D)$ is a second order *elliptic* operator; see Definition 3.17.

3.1 Sobolev spaces on open subsets of $\mathbb{R}^n$.

First we need to develop some of the theory of Sobolev spaces on open subsets $U \subset \mathbb{R}^n$.

Definition 3.1. *Let $U \subset \mathbb{R}^n$ be open. For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$ we define*
(a) $W^{k,p}(U) = \{f \in L^p(U) : \partial^\alpha f \in L^p(U) \text{ for } |\alpha| \le k\}$ with the norm

$$|u|_{W^{k,p}(U)} := \begin{cases} \left(\sum_{|\alpha|\le k} \int_U |\partial^\alpha u|^p dx\right)^{1/p}, & 1 \le p < \infty \\ \sum_{|\alpha|\le k} |\partial^\alpha u|_{L^\infty(U)}, & p = \infty \end{cases}.$$

(b) $W_0^{k,p}(U)$ is the closure of $\mathcal{D}(U)$ in $W^{k,p}(U)$.
(c) $H^k(U) := W^{k,2}(U)$ and $H_0^k(U) := W_0^{k,2}(U)$.

Observe that $u_j \to u$ in $W^{k,p}(U) \Leftrightarrow \partial^\alpha u_j \to \partial^\alpha u$ in $L^p(U)$ for all $|\alpha| \le k$.

3.1.1 Interior approximation by smooth functions

Let $\eta_\epsilon(x)$ be the Friedrichs mollifier of Definition 8.9.[1]

Proposition 3.2. *Let $U \subset \mathbb{R}^n$ be an open set. Assume $f \in L^p(U)$ where $1 \le p < \infty$. For $\epsilon > 0$ define*

$$f_\epsilon(x) = (\eta_\epsilon * f)(x) = \int_U \eta_\epsilon(x-y) f(y) dy \text{ for } x \in \mathbb{R}^n. \tag{3.2}$$

Then (a) $f_\epsilon \in C^\infty(\mathbb{R}^n)$;
(b) $f_\epsilon \to f$ in $L^p_{loc}(U)$; that is, for any $V \subset\subset U$ we have $f_\epsilon \to f$ in $L^p(V)$.

[1] Recall that for $\epsilon > 0$, $\eta_\epsilon(x) = \epsilon^{-n}\eta(\frac{x}{\epsilon})$ where $\eta \in \mathcal{D}(\mathbb{R}^n)$ with $\operatorname{supp}\eta \subset \{|x| \le 1\}$, $\eta \ge 0$, and $\int \eta(x)dx = 1$.

Proof. **(a)** Differentiate under the integral sign and apply Proposition 8.2.[2]

(b) Let V and W be open sets such that $V \subset\subset W \subset\subset U$ and fix $\delta > 0$. We have $f \in L^p(W)$ so we can choose $g \in \mathcal{D}(W)$ such that $|g - f|_{L^p(W)} < \delta$. We have

$$|f_\epsilon - f|_{L^p(V)} \leq |f_\epsilon - g_\epsilon|_{L^p(V)} + |g_\epsilon - g|_{L^p(V)} + |g - f|_{L^p(V)} = A + B + C < A + B + \delta. \tag{3.3}$$

By Proposition 8.11 (or its proof), $B < \delta$ for ϵ small since $g_\epsilon \to g$ uniformly on V. To show $A < \delta$ its enough to show

$$f \in L^p(U) \Rightarrow |f_\epsilon|_{L^p(V)} \leq |f|_{L^p(W)} \text{ for } \epsilon \text{ small.} \tag{3.4}$$

When $p = 1$ this is clear.[3] When $1 < p < \infty$, we estimate for $x \in V$

$$|f_\epsilon(x)| = \left| \int_{B(x,\epsilon)} \eta_\epsilon(x-y) f(y) dy \right| \leq \int_{B(x,\epsilon)} \eta_\epsilon^{1-\frac{1}{p}}(x-y) \eta_\epsilon^{\frac{1}{p}}(x-y) |f(y)| dy \leq$$
$$\left(\int_{B(x,\epsilon)} \eta_\epsilon(x-y) dy \right)^{1-\frac{1}{p}} \left(\int_{B(x,\epsilon)} \eta_\epsilon(x-y) |f(y)|^p dy \right)^{\frac{1}{p}} = \left(\int_{B(x,\epsilon)} \eta_\epsilon(x-y) |f(y)|^p dy \right)^{\frac{1}{p}}. \tag{3.5}$$

Here we used Hölder's inequality with $p = p$, $q = (1 - \frac{1}{p})^{-1}$ for the second inequality. Finally,

$$\begin{aligned} \int_V |f_\epsilon|^p dx &\leq \int_V \left(\int_{B(x,\epsilon)} \eta_\epsilon(x-y) |f(y)|^p dy \right) dx \leq \\ \int_W |f(y)|^p \left(\int_{B(y,\epsilon)} \eta_\epsilon(x-y) dx \right) dy &= \int_W |f(y)|^p dy. \end{aligned} \tag{3.6}$$

For the second inequality we used

$$\{(x,y) : x \in V, y \in B(x,\epsilon)\} \subset \{(x,y) : y \in W, x \in B(y,\epsilon)\}.$$

□

Proposition 3.3 (Interior approximation)**.** *Suppose $U \subset \mathbb{R}^n$ is open, $1 \leq p < \infty$, and $u \in W^{k,p}(U)$. Then $u_\epsilon \to u$ in $W^{k,p}_{loc}(U)$ as $\epsilon \to 0$; that is, for any $V \subset\subset U$ we have $u_\epsilon \to u$ in $W^{k,p}(V)$.*

[2] Let $x_0 \in U$, $r > 0$. For a given $\epsilon > 0$ and for $x \in U \cap B(x_0, r)$, the integrand in (3.2) is zero outside a bounded subset $S = S(x_0, r, \epsilon)$ of U.

[3] For $x \in V$ and ϵ small, $|f_\epsilon(x)| \leq \int_W \eta_\epsilon(x-y)|f(y)|dy$. Thus,

$$|f_\epsilon|_{L^1(V)} \leq \int_W |\eta_\epsilon(x-y)|_{L^1(V)} |f(y)| dy = \int_W |f(y)| dy.$$

Proof. For $|\alpha| \leq k$ we have $\partial^\alpha u \in L^p(U)$. We'll show

$$\partial^\alpha u_\epsilon \to \partial^\alpha u \text{ in } L^p(V) \text{ for } |\alpha| \leq k. \tag{3.7}$$

By Proposition 3.2(b) this is an immediate consequence of

$$\partial^\alpha u_\epsilon = \eta_\epsilon * \partial^\alpha u \text{ for } x \in V, \epsilon > 0 \text{ small }. \tag{3.8}$$

To obtain (3.8) we compute:

$$\begin{aligned} \partial^\alpha u_\epsilon(x) = \int_U \partial_x^\alpha(\eta_\epsilon(x-y))u(y)dy = (-1)^{|\alpha|}\int_U \partial_y^\alpha(\eta_\epsilon(x-y))u(y)dy = \\ \int_U \eta_\epsilon(x-y)\partial_y^\alpha u dy. \end{aligned} \tag{3.9}$$

We used the definition of the distribution derivative $\partial^\alpha u$ for the last equality.

□

Remark 3.4. If $u \in L^p(U)$ we can easily extend it to an element of $L^p(\mathbb{R}^n)$ just by defining the extension u_c to be 0 in $\mathbb{R}^n \setminus U$ (the "zero extension"). But if $u \in W^{1,p}(U)$ the zero extension is usually *not* in $W^{1,p}(\mathbb{R}^n)$ because of the jump at ∂U.

Exercise 3.1.1a. Let $V = \{(x,y) \in \mathbb{R}^2 : |(x,y)| < 1, 0 < y < 1\}$ and define $u = 1$ on V. Check that $u \in W^{1,p}(U)$ but that $u_c \notin W^{1,p}(\mathbb{R}^n)$. Hint: compute $\partial_y u_c$ near $(0,0)$.

Exercise 3.1.1b Let $U \subset \mathbb{R}^n$ be open but not bounded. Suppose $1 \leq p < \infty$ and $u \in W^{k,p}(U)$. Show that there exist functions $u_n \in W^{k,p}(U)$ with bounded supports such that $u_n \to u$ in $W^{k,p}(U)$.

3.1.2 Smooth approximation up to the boundary

When working with Sobolev spaces on a domain U, one often wants to extend an element of $W^{k,p}(U)$ to an element of $W^{k,p}(\mathbb{R}^n)$, or to restrict an element of U to ∂U. This is often possible under appropriate assumptions. An essential step in the proofs is to show the existence of functions $u \in C^\infty(\overline{U})$ such that $u_n \to u$ in $W^{k,p}(U)$, not just in $W^{k,p}_{loc}(U)$.[4]

In preparation for the approximation theorem we prove the following lemma.

Lemma 3.5 (Translation Lemma)**.** *Suppose* $1 \leq p < \infty$ *and* $f \in L^p(\mathbb{R}^n)$. *Set* $f_x(y) := f(x+y)$. *Then*

$$f_x \to f \text{ in } L^p(\mathbb{R}^n) \text{ as } x \to 0.$$

[4]Here $C^\infty(\overline{U})$ is the set of $u \in C^\infty(U)$ such that for every α, $\partial^\alpha u$ extends continuously to $\overline{U}$.

Proof. Let $f \in L^p(\mathbb{R}^n)$ and fix $\delta > 0$. Choose $g \in \mathcal{D}(\mathbb{R}^n)$ such that $|g - f|_{L^p} < \delta$. Then

$$|f_x - f|_{L^p} \leq |f_x - g_x|_{L^p} + |g_x - g|_{L^p} + |g - f|_{L^p} \leq 2\delta + |g_x - g|_{L^p}. \tag{3.10}$$

For $|x|$ small $|g_x - g|_{L^p} < \delta$ since g_x and g are smooth with support in a fixed compact set K, and $g_x \to g$ uniformly on K.(why?).

□

The next example, taken from [Ada78], shows that it is not always possible to approximate $u \in W^{1,p}(U)$ by functions in $C^\infty(\overline{U})$.

Example 3.6. Let $U = \{(x, y) \in \mathbb{R}^2 : 0 < |x| < 1, 0 < y < 1\}$ and set

$$u(x, y) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}.$$

Then $u \in W^{1,p}(U)$. However, for $\epsilon > 0$ small enough there is no $\phi \in C^1(\overline{U})$ such that $|u - \phi|_{W^{1,p}(U)} < \epsilon$.

Exercise 3.1.2a. Prove the assertion in Example 3.6.

The problem with Example 3.6 is that U lies on both sides of part of its boundary. To rule out this situation we make the following definition.

Definition 3.7 (Segment property)**.** *Let $U \subset \mathbb{R}^n$ be open. We say that U has the* segment property *if for every $x \in \partial U$ there is an open set $V_x \ni x$ and a nonzero vector $y_x \in \mathbb{R}^n$ such that*

$$\text{if } x \in V_x \cap \partial U, \text{ then } x + \delta y_x \in \mathbb{R}^n \setminus \overline{U} \text{ for } 0 < \delta < 1. \tag{3.11}$$

Exercise 3.1.2b. (a) Check that Example 3.6 does not have the segment property, and neither does $U = \{x \in \mathbb{R}^n : 0 < |x| < 1$.

(b) Show that the interior of any triangle in $\mathbb{R}^2$ (an "open triangle") has the segment property.

(c) Let U be the union of two open triangles in $\mathbb{R}^2$ whose closures intersect only at a vertex. Does U have the segment property?

Theorem 3.8. *Let $U \subset \mathbb{R}^n$ be a bounded open set with the segment property. Let $1 \leq p < \infty$ and suppose $f \in W^{k,p}(U)$. There exist functions $f_k \in \mathcal{D}(\mathbb{R}^n)$ such that $f_k \to f$ in $W^{k,p}(U)$.*[5]

[5]Here we let f_k also denote the restriction of $f_k \in \mathcal{D}(\mathbb{R}^n)$ to U.

Proof. **1.** By compactness of $\overline{U}$ and ∂U we can find an open cover $\{V_j, j = 1, \dots, N\}$ of $\overline{U}$ such that

(a) $V_0 \subset\subset U$,

(b) for $j \geq 1$ $V_j \cap \partial U \neq \emptyset$,

(c) for each $j \geq 1$ there exists $y_j \in \mathbb{R}^n \setminus \{0\}$ such that if $x \in V_j \cap \partial U$, then $x + \delta y_j \in \mathbb{R}^n \setminus \overline{U}$ for $0 < \delta < 1$.

Choose a partition of unity $\{\rho_j \in \mathcal{D}(\mathbb{R}^n), j = 0, \dots, N\}$ on $\overline{U}$ subordinate to this open cover; see Definition 8.69. Note that if we write $f = \sum_{j=0}^{N} \rho_j f$ on U, it suffices to prove the result for each $\rho_j f \in W^{k,p}(U)$.

2. Since supp $\rho_0 f \subset\subset U$, the result in this case follows directly from Proposition 3.3.

3. Fix $j \geq 1$ and rewrite $\rho_j f$ as $f \in W^{k,p}(U)$. For each $|\alpha| \leq k$ there exists $f_\alpha \in L^p(U)$ such that $\partial^\alpha f = f_\alpha$ as distributions on U. Next extend all the f_α by 0 outside U, and denote the extended functions by f_α. Each f_α lies in $L^p(\mathbb{R}^n)$ and may have a jump on the compact set $S = \overline{V}_j \cap \partial U$. Observe that if we let $(\cdot, \cdot)_{\mathcal{D}'}$ denote the distributional pairing on $\mathbb{R}^n$, then for $|\alpha| \leq k$ we have

$$(\partial^\alpha f, \phi)_{\mathcal{D}'} = (f_\alpha, \phi)_{\mathcal{D}'} \text{ for } \phi \in \mathcal{D}(\mathbb{R}^n \setminus S), \tag{3.12}$$

since each such ϕ vanishes on a neighborhood of S.[6] Thus, $\partial^\alpha f = f_\alpha$ on $\mathbb{R}^n \setminus S$ for $|\alpha| \leq k$, and hence, $f \in W^{k,p}(\mathbb{R}^n \setminus S)$.

4. For $0 < \delta < 1$ define the translates

$$f_{\alpha,\delta}(x) = f_\alpha(x - \delta y_j) \text{ and } S_\delta := S + \delta y_j \subset \mathbb{R}^n \setminus \overline{U}.$$

Each $f_{\alpha,\delta}$ lies in $L^p(\mathbb{R}^n)$ and may have a jump on the compact set S_δ. Now (3.12) implies

$$(\partial^\alpha f_\delta, \phi)_{\mathcal{D}'} = (f_{\alpha,\delta}, \phi)_{\mathcal{D}'} \text{ for } \phi \in \mathcal{D}(\mathbb{R}^n \setminus S_\delta), \tag{3.13}$$

Thus, $\partial^\alpha f_\delta = f_{\alpha,\delta}$ on $\mathbb{R}^n \setminus S_\delta$ for $|\alpha| \leq k$, and hence, $f_\delta \in W^{k,p}(\mathbb{R}^n \setminus S_\delta)$. We have $\overline{U} \subset \mathbb{R}^n \setminus S_\delta$, so

$$f_\delta \in W^{k,p}(U) \text{ and } \partial^\alpha f_\delta = f_{\alpha,\delta} \text{ on } U. \tag{3.14}$$

The translation lemma implies that for $|\alpha| \leq k$ as $\delta \to 0$

$$|f_{\alpha,\delta} - f_\alpha|_{L^p(\mathbb{R}^n)} \to 0 \Rightarrow |f_{\alpha,\delta} - f_\alpha|_{L^p(U)} \to 0 \Rightarrow |\partial^\alpha f_\delta - \partial^\alpha f|_{L^p(U)} \to 0. \tag{3.15}$$

Thus, $f_\delta \to f$ in $W^{k,p}(U)$. Each f_δ has a possible jump on S_δ, which is away from $\overline{U}$.

5. To finish it is enough to show that each f_δ can be approximated in $W^{k,p}(U)$ by elements of $\mathcal{D}(\mathbb{R}^n)$. For $\delta > 0$ fixed, choose $\chi \in \mathcal{D}(\mathbb{R}^n)$ such that $\chi = 1$ on U and $\chi = 0$ near S_δ. By Proposition 3.3 $\chi f_\delta \in W^{k,p}(\mathbb{R}^n \setminus S_\delta)$ is the limit in $W^{k,p}(U)$ of a sequence in $\mathcal{D}(\mathbb{R}^n)$. Since $\chi f_\delta = f_\delta$ on U, so is f_δ.

□

[6] The equality (3.12) fails for general $\phi \in \mathcal{D}(\mathbb{R}^n)$ (why?).

Remark 3.9. The result of Exercise 3.1.1b implies that Theorem 3.8 still holds if we remove the assumption that U is bounded.

Exercise 3.1.2c. Prove the statement in Remark 3.9.

3.1.3 Trace theorem

The next result shows that one can sometimes make sense of the "restriction" of an element of $W^{1,p}(U)$ to ∂U, even though the pointwise restriction generally makes no sense.

Theorem 3.10. *Let $U \subset \mathbb{R}^n$ be bounded and open with C^1 boundary, and let $1 \leq p < \infty$. Then there exists a bounded linear operator $T : W^{1,p} \to L^p(\partial U)$ and $C > 0$ such that*
(i) $Tu = u|_{\partial U}$ if $u \in C^1(\overline{U}) \subset W^{1,p}(U)$.
(ii) $|Tu|_{L^p(\partial U)} \leq C|u|_{W^{1,p}(U)}$ for all $u \in W^{1,p}(U)$.

Proof. **1.** Assume at first that $u \in C^1(\overline{U}, \mathbb{R})$ and that ∂U is flat near some point $x_0 \in \partial U$. more precisely, suppose there exists $r > 0$ such that

$$B^+ := B(x_0, r) \cap \{x \in \mathbb{R}^n : x_n \geq 0\} \subset U \text{ and } B(x_0, r) \cap \partial U \subset \{x_n = 0\}.$$

Set $\Gamma = \partial U \cap B(x_0, r/2)$ and choose $\zeta \in \mathcal{D}(B(x_0, r)$ such that $\zeta \geq 0$ and $\zeta = 1$ on $B(x_0, r/2)$.[7]

2. Writing $x = (x', x_n)$, for $p > 1$ we compute[8]

$$\begin{aligned} &\int_\Gamma |u|^p dx' \leq \int_{x_n=0} \zeta |u|^p dx' = -\int_{B^+} (\zeta |u|^p)_{x_n} dx = \\ &-\int_{B^+} \left(|u|^p \zeta_{x_n} + p|u|^{p-1} \mathrm{sgn}(u) u_{x_n} \zeta \right) dx \leq C \int_{B^+} \left(|u|^p + |\nabla u|^p \right) dx \leq C|u|^p_{W^{k,p}(U)}. \end{aligned} \tag{3.16}$$

When $p = 1$ we get the same estimate by applying the above argument with $|u|^p$ replaced by $(\epsilon^2 + |u|^2)^{p/2}$ and letting $\epsilon \to 0$.

3. Keeping $u \in C^1(\overline{U})$, we now drop the flatness assumption, and reduce to the flat case by a local change of variables. Let $x_0 \in \partial U$. Since ∂U is C^1 we can suppose that near x_0, U is given by $x_n - \rho(x') > 0$, where ρ is C^1. Define the local change of variables $\Phi(x', x_n) = (x', x_n - \rho(x')) = (y', y_n)$, which has inverse $\Psi(y', y) = (y', y_n + \rho(x')) = (x', x_n)$. Setting $\Gamma = \partial U \cap B(x_0, r)$ for a small enough r, we obtain

$$\int_\Gamma |u|^p dS \leq C|u|^p_{W^{k,p}(U)} \tag{3.17}$$

by applying the estimate of step **2** to $v(y) := u(\Psi(y))$ and using the change of variable formula for integrals.

[7] Draw this setup.
[8] For the last inequality use $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ when $a, b \geq 0$ and $\frac{1}{p} + \frac{1}{q} = 1$. In this case take $q = \frac{p}{p-1}$.

4. Using step **3** and the compactness of ∂U, we can find points $x_i \in \partial U$ and balls $B(x_i, r_i)$ such that for $\Gamma_i := \partial U \cap B(x_i, r_i)$ we have $\partial U \subset \cup_{i=1}^N \Gamma_i$ and

$$|u|_{L^p(\Gamma_i)} \leq C_i |u|_{W^{1,p}(U)} \text{ for } u \in C^1(\overline{U}). \tag{3.18}$$

Defining $Tu = u|_{\partial U}$ for $u \in C^1(\overline{U})$, we sum over i to obtain the estimate (ii) for such u.

5. For $u \in W^{1,p}(U)$ we define Tu as follows. Choose $u_j \in \mathcal{D}(\mathbb{R}^n)$ such that $u_j \to u$ in $W^{1,p}(U)$.[9] The estimate (ii) implies

$$|Tu_j - Tu_k|_{L^p(U)} \leq C|u_j - u_k|_{W^{1,p}(U)},$$

so (Tu_j) is Cauchy in $L^p(\partial U)$. We define

$$Tu = \lim_{j\to\infty} Tu_j \text{ in } L^p(\partial U),$$

a limit that is independent of the choice of sequence (u_j) (why?). We have $|Tu_j|_{L^p(\partial U)} \leq C|u_j|_{W^{1,p}(U)}$ for all j, so letting $j \to \infty$ we obtain (ii). □

Remark 3.11. Problem 4 of section 3.5 shows that if $u \in W^{1,p}(U) \cap C(\overline{U})$, then $Tu = u|_{\partial U}$.

Proposition 3.12. *Let $U \subset \mathbb{R}^n$ be bounded and open with C^1 boundary and suppose $u \in W^{1,p}(U)$, where $1 \leq p < \infty$. Then $u \in W_0^{1,p}(U) \Leftrightarrow Tu = 0$.*

Partial proof. ($\Rightarrow$) We can choose $u_j \in \mathcal{D}(U)$ such that $u_j \to u$ in $W^{1,p}(U)$. By Theorem 3.10 this implies $0 = Tu_j \to Tu$ in $L^p(U)$, so $Tu = 0$.

The converse, while not surprising, is more difficult to prove; see [Evans], section 5.5. □

3.1.4 Dual of $H_0^1(U)$.

In the next section we will look for weak solutions of the problem 3.1 in the space $H_0^1(U)$. This will be accomplished using the Lax-Milgram theorem 8.45 and a knowledge of the dual of $H_0^1(U)$. For convenience we consider only real Sobolev spaces in this section.[10]

If $U \subset \mathbb{R}^n$ is open, the space $H_0^1(U) \subset H^1(U)$ is a Hilbert space with the H^1 inner product

$$(u, v)_{H^1(U)} := (u, v)_{L^2(U)} + \sum_{i=1}^n (u_{x_i}, v_{x_i})_{L^2(U)}. \tag{3.19}$$

Definition 3.13. *Suppose $U \subset \mathbb{R}^n$ is open. Let $H^{-1}(U) := (H_0^1(U))'$, the dual of $H_0^1(U)$, with the usual norm*

$$|f|_{H^{-1}(U)} := \sup_{u \in H_0^1(U), |u|_{H_0^1(U)} \leq 1} |f(u)|. \tag{3.20}$$

[9] By Exercise 3(a) of section 3.5 U has the segment property since ∂U is C^1.

[10] Then there is no need to distinguish linear functionals from antilinear functionals, for example.

Observe that if $f_0, f_1, \dots, f_n$ lie in $L^2(U)$, then the map $\phi : H_0^1 \to \mathbb{R}$ given by

$$\phi(v) := \int_U (f_0 v + \sum_{i=1}^n f_i v_{x_i}) dx \text{ for } v \in H_0^1 \tag{3.21}$$

is an element of $H^{-1}(U)$. We say that $\tilde{f} := (f_0, f_1, \dots, f_n)$ *represents* ϕ.

The next proposition shows that every element of $H^{-1}(U)$ has this form. To make this precise define an equivalence relation on $L^2(U, \mathbb{R}^{n+1})$ by $\tilde{f} \sim \tilde{g}$ if

$$\int_U (f_0 v + \sum_{i=1}^n f_i v_{x_i}) dx = \int_U (g_0 v + \sum_{i=1}^n g_i v_{x_i}) dx \text{ for all } v \in H_0^1(U), \tag{3.22}$$

that is, if $\tilde{f}$ and $\tilde{g}$ determine the same element of $H^{-1}(U)$. We define the norm of the equivalence class $[\tilde{f}]$ to be the quotient norm

$$|[\tilde{f}]| := \inf_{\tilde{g} \in [\tilde{f}]} \left(\int_U \sum_{i=0}^n |g_i|^2 dx \right)^{1/2} (= \inf_{\tilde{g} \in [\tilde{f}]} |\tilde{g}|_{L^2(U)} \text{ for short}). \tag{3.23}$$

Proposition 3.14 (Dual of $H_0^1(U)$)**.** *Let $U \subset \mathbb{R}^n$ be open and let $\phi \in H^{-1}(U)$. Then*

(a) there exist $f_0, f_1, \dots, f_n \in L^2(U)$ such that $\tilde{f}$ represents ϕ as in (3.21);

(b) $|\phi|_{H^{-1}(U)} = \inf_{\tilde{f} \text{ representing } \phi} |\tilde{f}|_{L^2(U)}$;

(c) the map $\phi \to [\tilde{f}]$, where $\tilde{f}$ represents ϕ, is an isometric isomorphism of H^{-1} with $\{[\tilde{f}] : \tilde{f} \in L^2(U, \mathbb{R}^{n+1})\}$, where the latter space has the norm $|[\tilde{f}]|$.

Proof. **(a)** Let $\phi \in H^{-1}(U)$. By the Riesz theorem there exists $u \in H_0^1(U)$ such that

$$\phi(v) = (u, v)_{H_0^1(U)} = \int_U (uv + \nabla u \cdot \nabla v) dx \text{ for } v \in H_0^1(U).$$

Thus, $\tilde{f} := (u, \nabla u)$ represents ϕ.

(b) The Riesz theorem also implies

$$|\phi|_{H^{-1}(U)} = |u|_{H_0^1(U)} = |\tilde{f}|_{L^2(U)} \text{ for } u, \tilde{f} \text{ as in (a)}. \tag{3.24}$$

If $\tilde{g} \in [\tilde{f}]$, then for $v \in H_0^1(U)$ we have by Cauchy-Schwarz

$$\phi(v) = \int_U (g_0 v + \sum_{i=1}^n g_i v_{x_i}) dx \le |\tilde{g}|_{L^2(U)} |v|_{H_0^1(U)} \Rightarrow |\phi|_{H^{-1}(U)} \le |\tilde{g}|_{L^2(U)}. \tag{3.25}$$

With (3.24) this implies (b).

(c) It just remains to check the surjectivity of the map $\phi \to [\tilde{f}]$. We observed below Definition 3.13 that each class $[\tilde{f}]$ determines an element of $H^{-1}(U)$.

□

Exercise 3.1.4a. Give an analogous characterization of the dual of $H^1(U)$.

3.2 Weak solutions to elliptic boundary problems

In this section we take $U \subset \mathbb{R}^n$ to be open and bounded and continue to work with real Sobolev spaces. Consider a second order operator in "divergence form" defined formally by

$$L(x,\partial)u = -\sum_{i,j=1}^{n}(a_{ij}(x)u_{x_i})x_j + \sum_{i=1}^{n} b_i(x)u_{x_i} + c(x)u \tag{3.26}$$

where for now we assume only that a_{ij}, b_i, c lie in $L^\infty(U)$. We define

$$B[u,v] = \int_U \left[\sum_{i,j=1}^{n} a_{ij}(x)u_{x_i}v_{x_j} + \sum_{i=1}^{n} b_i(x)u_{x_i}v + c(x)uv\right] dx \text{ on } H_0^1(U)\times H_0^1(U), \tag{3.27}$$

and call B the *bilinear form associated to* L.[11]

Consider the problem[12]

$$\begin{aligned} &L(x,\partial)u = f \text{ in } U \\ &u|_{\partial U} = 0. \end{aligned} \tag{3.28}$$

Definition 3.15. *Let $f \in H^{-1}(U)$. We say $u \in H_0^1(U)$ is a* weak solution *of (3.28) if*

$$B[u,v] = f(v) \text{ for all } v \in H_0^1(U). \tag{3.29}$$

Remark 3.16. (1) Definition 3.15 makes sense even though $(a_{ij}u_{x_i})_{x_j}$ does not, and even though the trace operator may not be defined.

(2) The reader might object that we have not actually defined the object $L(x,\partial)u$ that appears in (3.28). But since for a given $u \in H_0^1(U)$ the map $v \to B[u,v]$ defines an element, call it f, of $H^{-1}(U)$, we can define $L(x,\partial)u$ to be f.

Exercise 3.2a. Suppose $a_{ij} \in C^1(\overline{U})$, $b_i, c \in C(\overline{U})$, and $f \in C(\overline{U})$. Show that if $u \in H_0^1(U) \cap C^2(\overline{U})$ and $u|_{\partial U} = 0$, then u is a classical solution to (3.28) if and only if it is a weak solution.

Definition 3.17. *The operator $L(x,\partial)$ in (3.26) is* uniformly elliptic *in U if there exists $\theta > 0$ such that*

$$\theta|\xi|^2 \le \sum_{i=1}^{n} a_{ij}(x)\xi_i\xi_j \text{ for all } \xi \in \mathbb{R}^n \text{ and a.e. } x \in U. \tag{3.30}$$

[11] When the coefficients $a_{ij} \in C^1(\overline{U})$, any operator $L(x,\partial)$ in divergence form can be written in nondivergence form,

$$-\sum_{i,j=1}^{n} a_{ij}(x)u_{x_ix_j} + \sum_{i=1}^{n} b_i(x)u_{x_i} + c(x)u,$$

and vice versa (check). Divergence form is more convenient for energy methods that involve integrations by parts.

[12] Boundary problems with nonzero boundary data can often be reduced to problems of the form (3.28); see problem 5 in section 3.5.

Proposition 3.18. *Let $U \subset \mathbb{R}^n$ be bounded and open and suppose $L(x,\partial)$ is uniformly elliptic in U. Then there exist constants $\alpha, \beta > 0$ and $\gamma > 0$ such that for all $u, v \in H_0^1(U)$:*
(a) $|B[u,v]| \leq \alpha |u|_{H_0^1(U)} |v|_{H_0^1(U)}$ for B as in (3.27);
(b) $\beta |u|^2_{H_0^1(U)} \leq B[u,u] + \gamma |u|^2_{L^2(U)}$.

The estimate (3.18) is an example of an *a priori estimate*.
Observe that if we set

$$B_\gamma[u,v] := B(u,v) + \int_U uv dx, \tag{3.31}$$

then B_γ is the bilinear form associated to $L(x,\partial) + \gamma$.

Proof. **(a)** We have

$$\left| \int_U a_{ij} u_{x_i} v_{x_j} dx \right| \leq |a_{ij}|_{L^\infty} |\nabla u|_{L^2} |\nabla v|_{L^2} \leq C |u|_{H_0^1} |v|_{H_0^1}. \tag{3.32}$$

The other terms of $|B[u,v]|$ are estimated similarly.
(b) With θ as in (3.30) and setting $\xi = \nabla u(x)$, we estimate[13]

$$\begin{aligned} &\int_U \theta |\nabla u|^2 dx \leq \int_U \sum_{i,j=1}^n a_{ij}(x) u_{x_i}(x) v_{x_j}(x) dx = \\ &B[u,u] - \int_U \left[\sum_{i=1}^n b_i u_{x_i} u + c u^2 \right] dx \leq B[u,u] + C \int_U |\nabla u||u| dx + C \int_U |u|^2 dx \leq \\ &B[u,u] + C\epsilon \int_U |\nabla u|^2 dx + \frac{C}{4\epsilon} \int_U |u|^2 dx + C \int_U |u|^2 dx \leq = \\ &B[u,u] + C\epsilon \int_U |\nabla u|^2 dx + C_\epsilon \int_U |u|^2 dx. \end{aligned} \tag{3.33}$$

Now choose ϵ such that $C\epsilon < \frac{\theta}{2}$ and cancel a term to obtain

$$\frac{\theta}{2} \int_U |\nabla u|^2 dx \leq B[u,u] + C_\epsilon \int_U |u|^2 dx. \tag{3.34}$$

With $\beta = \min\{\frac{\theta}{2}, 1\}$ this implies

$$\beta |u|^2_{H_0^1} \leq \frac{\theta}{2} \int_U |\nabla u|^2 dx + \int_U |u|^2 dx \leq B[u,u] + (C_\epsilon + 1) \int_U |u|^2 dx := B_\gamma[u,u], \tag{3.35}$$

where $\gamma = C_\epsilon + 1$.

□

[13] The constant C in line two depends on $|b_i|_{L^\infty}$, $|c|_{L^\infty}$. For the third inequality we used $ab \leq \epsilon a^2 + \frac{1}{4\epsilon b^2}$.

With the above preparation the next theorem is a direct consequence of the Lax-Milgram theorem.

Theorem 3.19 (Existence of weak solutions)**.** *Suppose $U \subset \mathbb{R}^n$ be bounded and open and that $L(x,\partial)$ is uniformly elliptic in U. Let $\gamma > 0$ be as in Proposition 3.18. Then for $f \in H^{-1}(U)$ and any $\mu \geq \gamma$, there exists a unique weak solution $u \in H_0^1(U)$ of*

$$\begin{aligned} (L(x,\partial) + \mu)u &= f \\ u|_{\partial U} &= 0. \end{aligned} \tag{3.36}$$

Proof. Proposition 3.18 implies that $B_\mu = L + \mu$ satisfies the hypotheses of the Lax-Milgram theorem, Theorem 8.45. There there exists a unique $u \in H_0^1(U)$ subh that

$$B_\mu[u,v] = f(v) \text{ for all } v \in H_0^1(U).$$

□

Corollary 3.20. *For $L(x,\partial)$ and μ as in Theorem 3.19, the map $L + \mu : H_0^1(U) \to H^{-1}(U)$ is a homeomorphism.*[14]

Proof. By Theorem 3.19 the map $L + \mu$ is bijective. It remains to prove continuity of $L + \mu$ and its inverse. For $u \in H_0^1(U)$ set $f := (L + \mu)u \in H^{-1}(U)$. Continuity of $L + \mu$ follows from

$$|f(u)| = |B_\mu[u,u]| \leq C|u|_{H_0^1}^2 \Rightarrow |f|_{H^{-1}} \leq C|u|_{H_0^1}. \tag{3.37}$$

Continuity of $(L + \mu)^{-1}$ follows from

$$\begin{aligned} \beta|u|_{H_0^1}^2 \leq B_\mu[u,u] = f(u) \leq |f|_{H^{-1}}|u|_{H_0^1} \Rightarrow \beta|u|_{H_0^1} \leq |f|_{H^{-1}} \Leftrightarrow \\ |(L+\mu)^{-1}f|_{H_0^1} \leq \frac{1}{\beta}|f|_{H^{-1}}. \end{aligned} \tag{3.38}$$

□

3.2.1 Fredholm alternative

We will formulate and prove a Fredholm alternative for weak solutions to elliptic boundary problems. The analogue for $n \times n$ real matrices is the following basic fact from linear algebra.

Proposition 3.21. *Let $A : \mathbb{R}^n \to \mathbb{R}^n$ be an $n \times n$ real matrix. Then:*
(a) either $R(A) = \mathbb{R}^n$ or $N(A) \neq \{0\}$;
(b) $\dim N(A) = \dim N(A^t)$;
(c) The equation $Ax = b$ has a solution for $b \in \mathbb{R}^n \Leftrightarrow b \in (N(A^t))^\perp$.

[14] Recall Remark 3.16(2).

Proof. We have $R(A)^\perp = N(A^t)$. Indeed, since $(Au, v) = (u, A^t v)$ for all $u, v \in \mathbb{R}^n$ we have

$$v \in R(A)^\perp \Leftrightarrow (Au, v) = 0 \text{ for all } u \in \mathbb{R}^n \Leftrightarrow (u, A^t v) = 0 \text{ for all } u \in \mathbb{R}^n \Leftrightarrow v \in N(A^t).$$

Thus, we can write

$$\mathbb{R}^n = R(A) \oplus R(A)^\perp = R(A) \oplus N(A^t). \tag{3.39}$$

This implies (c). By the nullity plus rank theorem, $n = \dim R(A) + \dim N(A)$, so (3.39) also implies (b). Finally, to get (a) we observe that $R(A)$ is either $\mathbb{R}^n$ or a proper subspace of $\mathbb{R}^n$. In the second case (3.39) implies $N(A^t)$ is nontrivial, so $N(A)$ is nontrivial by (b). □

Assumption 3.22. *In the rest of this section we assume that U is bounded and open with C^1 boundary, and that L is uniformly elliptic in U with real coefficients in $C^1(\overline{U})$.*

As in section 2.2.1 we define the *formal adjoint* of $L(x, \partial)$ to be the operator $L(x, \partial)^*$ (or L^* for short) uniquely determined by the condition

$$\langle Lu, v\rangle = \langle u, L^* v\rangle \text{ for all } u, v \in \mathcal{D}(U) \text{ where } \langle \cdot, \cdot \rangle \text{ is the } L^2 \text{ pairing.} \tag{3.40}$$

Integration by parts as in Remark 2.16 shows that

$$(L(x, \partial))^* = -\sum_{i,j=1}^n (a_{ij} v_{x_j})_{x_i} - \sum_{i=1}^n b_i v_{x_i} + (c - \sum_{i=1}^n \partial_i b_i) v. \tag{3.41}$$

We note that L^* is uniformly elliptic in U if L is since $\sum_{i,j=1}^n a_{ji}\xi_i\xi_j = \sum_{i,j=1}^n a_{ij}\xi_i\xi_j$.

Proposition 3.23. *Let $B^* : H_0^1(U) \times H_0^1(U) \to \mathbb{R}$ be the bilinear form associated to L^*. Then*

$$B[u, v] = B^*[v, u] \text{ for all } u, v \in H_0^1(U). \tag{3.42}$$

Proof. For all $u, v \in \mathcal{D}(U)$ we have

$$B[u, v] = \langle Lu, v\rangle = \langle u, L^* v\rangle = B^*[v, u]. \tag{3.43}$$

By density and continuity (3.42) follows. □

The next definition is not new, but we state it anyway.

Definition 3.24. *Let $f \in H^{-1}(U)$. We say $v \in H_0^1(U)$ is a weak solution of*

$$L^* v = f, v|_{\partial U} = 0 \tag{3.44}$$

if $B^*[v, u] = f(u)$ for all $u \in H_0^1(U)$.

It is remarkable that a close analogue of Proposition 3.21 holds for elliptic boundary problems. Let $N = \{u \in H_0^1(U) : u \text{ is a weak solution of } Lu = 0,\ u|_{\partial U} = 0\}$ and $N^* = \{v \in H_0^1(U) : v \text{ is a weak solution of } L^*v = 0, v|_{\partial U} = 0\}$.

Theorem 3.25 (Fredholm alternative)**.** *Under Assumption 3.22:*
(a) either there exists a unique weak solution $u \in H_0^1(U)$ of $Lu = f$, $u|_{\partial U} = 0$ for all $f \in L^2$, or there exist nonzero weak solutions of $Lu = 0$, $u|_{\partial U} = 0$.
(b) $\dim N = \dim N^*$.
(c) the problem $Lu = f$, $u|_{\partial U} = 0$ has a weak solution $\Leftrightarrow \langle f, v\rangle_{L^2} = 0$ for all $v \in N^*$.

First we give a couple of results needed for the proof of Theorem 3.25.

Lemma 3.26. *The inclusion map $i : L^2(U) \to H^{-1}(U)$ is bounded.*

Proof. If $g \in L^2(U)$, then for any $u \in H_0^1(U)$ we have

$$|i(g)(u)| = |\int gu\, dx| \le |g|_{L^2}|u|_{L^2} \le |g|_{L^2}|u|_{H_0^1(U)} \Rightarrow |i(g)|_{H^{-1}} \le |g|_{L^2}.$$

□

Theorem 3.27 (Rellich's theorem)**.** *Suppose $U \subset \mathbb{R}^n$ is bounded and open with C^1 boundary. Then the inclusion map*

$$i : H_0^1(U) \to L^2(U)$$

is compact.

This is an immediate corollary of Theorem 8.71.

Proof of Theorem 3.22. **1. Part (a).** We begin the proof of (a) by choosing $\gamma > 0$ as in Theorem 3.19. Then the problem

$$L_\gamma u = (L + \gamma)u = f, u|_{\partial U} = 0 \tag{3.45}$$

has a unique weak solution $u \in H_0^1(U)$ for all $g \in L^2(U)$. This means

$$B_\gamma[u, v] = \langle g, v\rangle \text{ for all } v \in H_0^1(U). \tag{3.46}$$

That is $u = L_\gamma^{-1}g$, where $L_\gamma^{-1} : H^{-1}(U) \to H_0^1(U)$ is as in Corollary 3.20. In particular, for $g \in L^2(U)$ we have using Lemma 3.26:

$$|L_\gamma^{-1}g|_{H_0^1} \le C|g|_{H^{-1}} \le C|g|_{L^2}. \tag{3.47}$$

2. For $f \in L^2(U)$ observe that $u \in H_0^1(U)$ is a weak solution of

$$\begin{aligned} &Lu = f, u|_{\partial U} = 0 \Leftrightarrow B[u, v] = \langle f, v\rangle\ \forall v \in H_0^1(U) \Leftrightarrow \\ &\quad B_\gamma[u, v] = \langle \gamma u + f, v\rangle \Leftrightarrow u = L_\gamma^{-1}(\gamma u + f) := Ku + L_\gamma^{-1}f, \end{aligned} \tag{3.48}$$

where we set $K := \gamma L_\gamma^{-1}$. Thus we've shown that $u \in H_0^1(U)$ is a weak solution of

$$Lu = f, u|_{\partial U} = 0 \Leftrightarrow (I - K)u = L_\gamma^{-1} f. \tag{3.49}$$

3. By Corollary 3.20 $K : H^{-1}(U) \to H_0^1(U)$, but we can view K as a map $K : L^2(U) \to L^2(U)$ since the inclusions $i : L^2(U) \to H^{-1}(U)$ and $i : H_0^1(U) \to L^2(U)$ are continuous. The second inclusion is compact by Theorem 3.27, so $K : L^2(U) \to L^2(U)$ is compact (why?).

Corollary 8.55(i) tells us that either $(I - K)u = L_\gamma^{-1} f$ has a unique solution $u \in L^2(U)$ or $(I - K)u = 0$ has nonzero solutions $u \in L^2(U)$. Notice that if $u \in L^2(U)$, then $(I - K)u = L_\gamma^{-1} f$ implies $u \in H_0^1(U)$ and similarly $(I - K)u = 0$ implies $u \in H_0^1(U)$. So part (a) follows from (3.49) and the Fredholm alternative for $I - K : L^2(U) \to L^2(U)$.

4. Part (b). Recall the definitions of N and N^*:

$$\begin{aligned} N &= \{u \in H_0^1(U) : u \text{ is a weak solution of } Lu = 0, u|_{\partial U} = 0\} \\ N^* &= \{v \in H_0^1(U) : v \text{ is a weak solution of } L^* v = 0, v|_{\partial U} = 0.\} \end{aligned} \tag{3.50}$$

We can suppose that γ was chosen large enough so that

$$(L^* + \gamma)v := L_\gamma^* v = f,\ v|_{\partial U} = 0$$

has a unique weak solution for all $f \in L^2$. Set $\tilde{K} := \gamma (L_\gamma^*)^{-1}$, and observe that step **2** implies

$$L^* v = 0, v|_{\partial U} = 0 \Leftrightarrow (I - \tilde{K})v = 0. \tag{3.51}$$

Thus,

$$N^* = \{v \in H_0^1(U) : (I - \tilde{K})v = 0\} = \{v \in L^2(U) : (I - \tilde{K})v = 0\} \text{(why?)}.$$

Theorem 8.53 tells us that for $K : L^2(U) \to L^2(U)$ as in step **2**,

$$\dim N(I - K) = \dim N(I - K^*) < \infty,$$

where K^* is the Hilbert space adjoint of K. Thus, to prove (b) it is enough to show $\tilde{K} = K^*$ as an operator on $L^2(U)$. Indeed, letting $U, V \in L^2(U)$ and setting $u := KU, v = \tilde{K}V \in H_0^1(U)$, we have

$$\begin{aligned} B_\gamma[u, v] &= B_\gamma[KU, \tilde{K}V] = B_\gamma[\gamma L_\gamma^{-1} U, \tilde{K}V] = \langle \gamma U, \tilde{K}V \rangle \\ B_\gamma^*[v, u] &= B_\gamma^*[\tilde{K}V, KU] = B_\gamma^*[\gamma (L_\gamma^*)^{-1} V, KU] = \langle \gamma V, KU \rangle. \end{aligned} \tag{3.52}$$

Thus, (3.42) implies

$$\gamma \langle V, KU \rangle = \gamma \langle U, \tilde{K}V \rangle \Rightarrow \tilde{K} = K^*.$$

5. Part (c). In steps **2** and **3** we showed that $Lu = f, u|_{\partial U} = 0$ has a weak solution $u \in H_0^1(U)$ if and only if $(I - K)u = L_\gamma^{-1} f$ has a solution in $L^2(U)$. By Corollary 8.55(ii) the latter holds if and only if

$$\langle L_\gamma^{-1} f, v \rangle = 0 \text{ for all } v \in N(I - K^*) = N^*. \tag{3.53}$$

The property (3.53) is equivalent to (c), since

$$\langle L_\gamma^{-1} f, v\rangle = \frac{1}{\gamma}\langle Kf, v\rangle = \frac{1}{\gamma}\langle f, K^* v\rangle = \frac{1}{\gamma}\langle f, v\rangle \text{ for } v \in N(I - K^*) = N^*. \tag{3.54}$$

□

Consider now the "eigenvalue problem" of determining for which $\lambda \in \mathbb{R}$ the problem

$$(L - \lambda)u = 0,\ u|_{\partial U} = 0 \tag{3.55}$$

has nontrivial weak solutions $u \in H_0^1(U)$. Let us denote this set of "bad" λ by $\Sigma \subset \mathbb{R}$. Theorem 3.19 tells us that there exists $\gamma > 0$ such that $\Sigma \subset \{\lambda : \lambda > -\gamma\}$. The next Theorem gives much more information.

Theorem 3.28 (Eigenvalue problem)**.** *Under Assumption 3.22 the set*

$$\Sigma = \{\lambda \in \mathbb{R} : (3.55) \textit{ has nontrivial weak solutions } u \in H_0^1(U)\} \tag{3.56}$$

is at most countable. If it is infinite, then Σ *can be written* $\Sigma = \{\lambda_k, k \in \mathbb{N}\}$*, where* $\lambda \to +\infty$ *as* $k \to +\infty$*.*

Proof. As above fix $\gamma > 0$ such that $\Sigma \subset \{\lambda : \lambda > -\gamma\}$. We now derive a condition for λ not to lie in Σ. We know $\lambda = -\gamma \notin \Sigma$, so assume $\lambda \neq -\gamma$ henceforth. Now

$$\begin{aligned}
&\lambda \notin \Sigma \Leftrightarrow (3.55) \text{ has the unique weak solution } u = 0 \Leftrightarrow \\
&\quad (L + \gamma)u = (\gamma + \lambda)u, u|_{\partial U} = 0 \text{ has the unique weak solution } u = 0 \Leftrightarrow \\
&u = 0 \text{ is the unique solution in } L^2(U) \text{ of } u = (L + \gamma)^{-1}(\gamma + \lambda)u = \frac{\gamma + \lambda}{\gamma} Ku,\ (K := \gamma L_\gamma^{-1}) \\
&\quad \Leftrightarrow u = 0 \text{ is the unique solution in } L^2(U) \text{ of } Ku = \frac{\gamma}{\gamma + \lambda} u \Leftrightarrow \\
&\frac{\gamma}{\gamma + \lambda} \text{ is not an eigenvalue of the compact operator } K : L^2 \to L^2 \Leftrightarrow \frac{\gamma}{\gamma + \lambda} \notin \Sigma',
\end{aligned} \tag{3.57}$$

where $\Sigma' := \{\text{real eigenvalues of } K\}$. This shows

$$\Sigma = \{\lambda > -\gamma : \frac{\gamma}{\gamma + \lambda} \in \Sigma'\}. \tag{3.58}$$

By Theorem 8.50 and Remark 8.51 the set Σ' is either finite or can be written as a sequence converging to zero. Thus, (3.58) implies the result (check).

□

Remark 3.29. By Theorem 3.25(a) and Theorem 3.28 the problem $(L - \lambda)u = f$, $u|_{\partial U} = 0$ has a unique weak solution in $H_0^1(U)$ for each $f \in L^2(U)$ if and only if $\lambda \notin \Sigma$.

3.3 Elliptic regularity.

Our goal in this section is to show that weak solutions of elliptic boundary value problems have higher regularity than just $H_0^1(U)$ regularity.

Assumption 3.30. *Let U be a bounded open set and suppose $L(x,\partial)$ has the divergence form (3.26) with $a_{ij} \in C^1(U)$, $b_j, c \in L^\infty(U)$ and is uniformly elliptic in U (Definition 3.17).*

We will prove the following interior elliptic regularity theorem, which illustrates a gain of two Sobolev derivatives of the solution u over the regularity of the forcing term f.

Theorem 3.31. *Under Assumption 3.30, let $f \in L^2(U)$, and suppose $u \in H^1(U)$ is a weak solution of $Lu = f$ in U; that is, $B[u,v] = f(v)$ for all $v \in H_0^1(U)$. Then $u \in H^2_{loc}(U)$ and for $V \subset\subset U$ we have*

$$|u|_{H^2(V)} \leq C_V \left(|f|_{L^2(U)} + |u|_{H^1(U)} \right). \tag{3.59}$$

The next proposition states an a priori estimate whose proof suggests a strategy for the proof of Theorem 3.31.

Proposition 3.32. *Let U be a bounded open set and let $Lu = -\sum_{i,j=1}^n (a_{ij}u_{x_i})_{x_j}$ with $a_{ij} \in C^1(U)$ be uniformly elliptic in U. Suppose $V \subset\subset U$, let $u \in C_c^3(V)$ and define $f := Lu$. Then there exists $C_V > 0$ such that*

$$|u|_{H^2(V)} \leq C_V \left(|f|_{L^2(U)} + |u|_{H^1(U)} \right). \tag{3.60}$$

Proof. With $\langle \cdot, \cdot \rangle$ the $L^2(U)$ pairing, we integrate by parts twice to obtain

$$\begin{aligned} \langle Lu, -\partial_k^2 u \rangle &= \left\langle -\sum_{i,j=1} (a_{ij}u_{x_i})_{x_j}, -\partial_k^2 u \right\rangle = \left\langle \sum_{i,j=1} a_{ij}u_{x_i}, -\partial_j \partial_k^2 u \right\rangle \\ &\left\langle \sum_{i,j} a_{ij}\partial_i \partial_k u, \partial_j \partial_k u \right\rangle + \sum_{i,j} \langle (\partial_k a_{ij})\partial_i u, \partial_j \partial_k u \rangle := A_1 + A_2. \end{aligned} \tag{3.61}$$

Using uniform ellipticity (take $\xi = \partial_k \nabla u$) we obtain for some $\theta > 0$

$$\begin{aligned} \theta |\partial_k \nabla u|_{L^2}^2 \leq A_1 \leq |A_2| + |\langle f, -\partial_k^2 u \rangle| \leq \\ \left(C_\epsilon |f|_{L^2}^2 + \epsilon |\nabla^2 u|_{L^2} \right) + \left(C_\epsilon |\nabla u|_{L^2}^2 + \epsilon |\nabla^2 u|_{L^2}^2 \right). \end{aligned} \tag{3.62}$$

Summing over k, fixing $\epsilon > 0$ small enough, and absorbing the $\epsilon |\nabla^2 u|_{L^2}^2$ terms into the left side, we obtain for a new C_ϵ:

$$|\nabla^2 u|_{L^2}^2 \leq C_\epsilon \left(|f|_{L^2}^2 + |\nabla u|_{L^2}^2 \right). \tag{3.63}$$

Finally, add $|u|_{H^1}^2$ to both sides to obtain (3.60).

□

The manipulations in the proof of Proposition 3.32 used the C^3 regularity of u (where?), so we can't apply those manipulations directly to $u \in H^1(U)$ as in Theorem 3.31. One way around this difficulty is to "take derivatives of u without really taking them" by using finite difference operators D_k^h for $h > 0$, as we proceed to explain.

Proof of Theorem 3.31. **1.** Fix $V \subset\subset W \subset\subset U$. We have $B[u,v] = \int_U fv dx$ for all $v \in H_0^1(U)$, so

$$\int_U \sum_{i,j} a_{ij} u_{x_i} v_{x_j} dx = \int_U \tilde{f} v dx, \text{ where } \tilde{f} := f - \sum_i b_i u_{x_i} - cu \quad \forall v \in H_0^1(U). \tag{3.64}$$

Choose

$$\zeta \in \mathcal{D}(U) \text{ such that } \zeta = 1 \text{ on } V,\ \operatorname{supp}(\zeta) \subset \overline{W},\ 0 \le \zeta \le 1. \tag{3.65}$$

Set $v \in -D_k^{-h}(\zeta^2 D_k^h u) \in H_0^1(U)$. Substituting v in (3.64) we obtain[15]

$$A := -\sum_{i,j} \int_U a_{ij} u_{x_i} \left[D_k^{-h}(\zeta^2 D_k^h u) \right]_{x_j} dx = B := \int_U \tilde{f} v dx. \tag{3.66}$$

2. By Proposition 8.77 we have

$$\begin{aligned} A = \sum_{ij} \int_U D_k^h(a_{ij} u_{x_i})(\zeta^2 D_k^h u)_{x_j} dx = \\ \sum_{i,j} \int_U \left[a_{ij}^h (D_k^h u_{x_i})(\zeta^2 D_k^h u)_{x_j} + (D_k^h a_{ij}) u_{x_i} (\zeta^2 D_k^h u)_{x_j} \right] dx \end{aligned} \tag{3.67}$$

Taking x_j derivatives we obtain

$$\begin{aligned} A = \sum_{i,j} \int_U a_{ij}^h (D_k^h u_{x_i})(D_k^h u_{x_j}) \zeta^2 dx + A_2 := A_1 + A_2, \text{ where } A_2 := \\ \sum_{ij} \int_U \left[a_{ij}^h D_k^h u_{x_i} D_k^h u 2\zeta \zeta_{x_j} + (D_k^h a_{ij}) u_{x_i} (D_k^h u_{x_j}) \zeta^2 + (D_k^h a_{ij}) u_{x_i} D_k^h u) 2\zeta \zeta_{x_j} \right] dx. \end{aligned} \tag{3.68}$$

In each term of A_2 only *one* factor has two (regular or finite) "derivatives" acting on u. Using uniform ellipticity to estimate A_1 from below, we obtain with (3.66)[16]

$$\theta \int_U \zeta^2 |D_h^k Du|^2 dx \le A_1 \le |A_2| + |B|. \tag{3.69}$$

[15] Compare A to the last term in the first line of (3.61).
[16] Take $\xi = D_k^h \nabla u$ in (3.30). Compare (3.69) with the first line of (3.62).

3. Using the formulas for A_2 and B we find for some $C > 0$ depending on ζ and a_{ij}:

$$\begin{aligned}
|A_2| &\leq C \int_U \left[\zeta |D_k^h Du| D_k^h u| + \zeta |D_k^h Du||Du| + \zeta |D_k^h u||Du|\right] dx \leq \\
&\epsilon \int_W \zeta^2 |D_k^h Du|^2 dx + C_\epsilon \int_W (|D_k^h u|^2 + |Du|^2) dx \leq \\
&\epsilon \int_W \zeta^2 |D_k^h Du|^2 dx + C_\epsilon \int_U |Du|^2 dx; \\
|B| &\leq C \int_U (|f| + |Du| + |u|) \ |D_k^{-h}(\zeta^2 D_k^h u)| dx \leq \\
&C_\epsilon (|f|^2_{L^2(U)} + |u|^2_{H^1(U)}) + \epsilon \int_U |D(\zeta^2 D_k^h u)|^2 dx \leq \\
&C_\epsilon (|f|^2_{L^2(U)} + |u|^2_{H^1(U)}) + \epsilon \int_U \zeta^2 |D D_k^h u|^2 dx + C \int_U |Du|^2 dx.
\end{aligned} \tag{3.70}$$

Here we used Proposition 8.78 to replace a few D_k^h operators by D operators. The terms on the right in (3.70) in which two "derivatives" of u appear can be absorbed into the left side of (3.69) by taking ϵ small enough. This leaves

$$\int_U \zeta^2 |D_h^k Du|^2 dx \leq C_\epsilon (|f|^2_{L^2(U)} + |u|^2_{H^1(U)}). \tag{3.71}$$

Now apply Proposition 8.79 to conclude from (3.71):

$$\int_V |D^2 u|^2 dx \leq C_\epsilon (|f|^2_{L^2(U)} + |u|^2_{H^1(U)}), \tag{3.72}$$

and add $|u|^2_{H^1(V)}$ to both sides to obtain (3.59).

□

3.3.1 Higher regularity

In Theorem 3.31 we might expect that if we assume f has more regularity, then u should have more regularity. This is almost correct; we need to assume that the coefficients of L have more regularity as well.

Theorem 3.33. *(a) Let L be as in Assumption 3.30, but assume now that $a_{ij} \in C^2(U)$, $b_i, c \in C^1(U)$. Let $f \in H^1(U)$. Suppose $u \in H^1(U)$ is a weak solution of $Lu = f$. Then $u \in H^3_{loc}(U)$. Moreover, if $V \subset\subset W \subset\subset U$ we have*

$$|u|_{H^3(V)} \leq C \left(|f|_{H^1(U)} + |u|_{H^2(W)}\right). \tag{3.73}$$

(b) More generally, for $m \in \mathbb{N}$ if $a_{ij} \in C^{m+1}(U)$, $b_i, c \in C^m(U)$, and $f \in H^m(U)$, and if $u \in H^1(U)$ is a weak solution of $Lu = f$, then $u \in H^{m+2}_{loc}(U)$ and

$$|u|_{H^{m+2}(V)} \leq C \left(|f|_{H^m(U)} + |u|_{H^{m+1}(W)}\right). \tag{3.74}$$

The following corollary is immediate.

Corollary 3.34. *Let L be as in Theorem 3.33, but assume $a_{ij}, b_i, c \in C^\infty(U)$. Assume $f \in C^\infty(U)$. If $u \in H^1(U)$ is a weak solution of $Lu = f$, then $u \in C^\infty(U)$.*

As we did with Theorem 3.31, we first prove a simpler proposition whose proof suggests a strategy for proving Theorem 3.33.

Proposition 3.35. *Let L be as in Theorem 3.33, but assume $b_i = 0$, $c = 0$. Suppose $V \subset\subset U$, $u \in C_c^3(U)$, and define $f := Lu$. Then there exists $C_V > 0$ such that*

$$|u|_{H^3(V)} \leq C_V(|f|_{H^1(U)} + |u|_{H^1(U)}). \tag{3.75}$$

Proof. We have $Lu = -\sum_{i,j}(a_{ij}\partial_i u)_{x_j} = f$, so

$$\begin{aligned} \partial_k Lu = -\sum_{ij}(\partial_k a_{ij}\ \partial_i u)_{x_j} - \sum_{i,j}(a_{ij}\partial_i\partial_k u)_{x_j} = \partial_k f \Rightarrow \\ L(\partial_k u) = \partial_k f + \sum_{ij}(\partial_k a_{ij}\ \partial_i u)_{x_j} := \tilde{f}. \end{aligned} \tag{3.76}$$

Thus, $\partial_k u$ is a classical solution (hence a weak solution) of $L(\partial_k u) = \tilde{f}$. Applying the estimate of Theorem 3.31 we obtain

$$|\partial_k u|_{H^2(V)} \leq C_V\left(|\tilde{f}|_{L^2(U)} + |\partial_k u|_{H^1(U)}\right), \tag{3.77}$$

which obviously implies (3.75).

□

Proof of Theorem 3.33. We just prove (a) here, assuming Theorem 3.31, which is the case $m = 0$. Then (b) is proved by induction using the same argument for the induction step.

1. From Theorem 3.31 we know $u \in H^2_{loc}(U)$ and satisfies the estimate (3.59). For the moment let[17]

$$\tilde{v} \in \mathcal{D}(W), v = -D_k\tilde{v}\ \tilde{u} := D_k u \in H^1_{loc}(U), \tag{3.78}$$

Next we compute[18]

$$\begin{aligned} B[u,v] = \int_U \left[\sum_{i,j} a_{ij}u_{x_i}(-D_k\tilde{v}_{x_j}) + \sum_i b_i u_{x_i}(-D_k\tilde{v}) + cu(-D_k\tilde{v})\right] dx = \\ \langle f, -D_k v\rangle = (D_k f, \tilde{v}). \end{aligned} \tag{3.79}$$

[17] Here and in the rest of this proof $D_k := \partial_k$.

[18] Here $\langle\cdot,\cdot\rangle$ is the L^2 pairing. The integration by parts in the second line of (3.79) is justified by the usual approximation argument.

We also have

$$\begin{aligned} B[\tilde{u},\tilde{v}] &= \int_U \left[\sum_{i,j} a_{ij}\ D_k u_{x_i}\ \tilde{v}_{x_j} + \sum_i b_i\ D_k u_{x_i}\ \tilde{v} + cD_k u\ \tilde{v}\right] dx = \\ B[u,v] - \int_U &\left[\sum_{i,j} -(D_k a_{ij}\ u_{x_i})_{x_j}\tilde{v} + \sum_i D_k b_i\ u_{x_i}\tilde{v} + D_k c\ u\tilde{v}\right] dx = \\ &\langle D_k f,\tilde{v}\rangle - \langle h,\tilde{v}\rangle = \langle \tilde{f},\tilde{v}\rangle, \end{aligned} \tag{3.80}$$

where $h, \tilde{f} \in L^2(W)$ are defined by the last two equalities. To obtain the second equality in (3.80), we used

$$\begin{aligned} \int_U a_{ij}\ D_k u_{x_i}\ \tilde{v}_{x_j} dx = \int_U a_{ij} u_{x_i}(-D_k\tilde{v}_{x_j})dx - \int_U D_k a_{ij}\ u_{x_i}\tilde{v}_{x_j}dx = \\ \int_U a_{ij}u_{x_i}(-D_k\tilde{v}_{x_j})dx + \int_U (D_k a_{ij}\ u_{x_i})_{x_j}\tilde{v}dx. \end{aligned} \tag{3.81}$$

2. In (3.80) we have shown $B[\tilde{u},\tilde{v}] = \langle\tilde{f},\tilde{v}\rangle$ for all $\tilde{v} \in \mathcal{D}(W)$, so this holds for all $\tilde{v} \in H_0^1(W)$ as well (why?). Thus, $\tilde{u} \in H^1(W)$ is a weak solution of $L\tilde{u} = \tilde{f}$.[19] Applying Theorem 3.31 we conclude that $\tilde{u} \in H^2_{loc}(W)$ and

$$|\tilde{u}|_{H^2(V)} \le C(|\tilde{f}|_{L^2(W)} + |\tilde{u}|_{H^1(W)}) \Rightarrow |u|_{H^3(V)} \le C(|f|_{H^1(U)} + |u|_{H^2(W)}). \tag{3.82}$$

□

Exercise 3.3.1a. Show that the term $|u|_{H^{m+1}(W)}$ on the right in the estimate (3.74) can be replaced, possibly after changing the constant C, by $|u|_{L^2(U)}$. Hint: First show that this change can be made in the estimate (3.59) of Theorem 3.31 by setting $v = \zeta^2 u$ in the identity (3.64).

Exercise 3.3.1b. You may have noticed that while the proof of Theorem 3.31 required the use of finite difference operators, the proof of Theorem 3.33 did not. In part (a) of Theorem 3.33, although u did not have enough regularity for us to carry out the simple argument of Proposition 3.32, we were able to use the bilinear operator B to justify related calculations. What is the obstacle to using a similar strategy, suggested by the proof of Proposition 3.35, to prove Theorem 3.31 without the use of finite difference operators?

With more work one can obtain higher regularity of weak solutions up to the boundary.

Theorem 3.36 (Higher regularity up to the boundary)**.** *Let $m \in \mathbb{N}_0$, assume a_{ij}, b_i, $c \in C^{m+1}(\overline{U})$, and that L is uniformly elliptic on U. Suppose $f \in H^m(U)$ and that $u \in H_0^1(U)$ is a weak solution of $Lu = f$, $u|_{\partial U} = 0$. If in addition ∂U is C^{m+2}, then $u \in H^{m+2}(U)$ and for some C:*

$$|u|_{H^{m+2}(U)} \le C(|f|_{H^m(U)} + |u|_{L^2(U)}). \tag{3.83}$$

[19] Compare this with the second line of (3.76).

For a proof see [Eva10], section 6.3.2.

3.4 Elliptic theory with complex-valued functions.

In our treatment of elliptic theory we made the simplification of working just with real Hilbert spaces. The proof of Theorem 3.28 used some spectral theory, and when it comes to spectral theory, restricting attention to real vector spaces is unnatural. As you know, $N \times N$ real matrices can have eigenvalues that are nonreal, in fact for some real matrices all eigenvalues are nonreal. But if you insist on viewing a real matrix A with only nonreal eigenvalues as a map from $\mathbb{R}^N$ to $\mathbb{R}^N$ (rather than $\mathbb{C}^N$ to $\mathbb{C}^N$), then you must conclude that A has no eigenvalues at all.

We have studied elliptic differential operators $L(x, D)$ with real coefficients, and have insisted that these operators act only on real-valued functions. When considering the "eigenvalues" of such operators, that is, the values of λ such that the problem $(L - \lambda)u = 0$, $u|_{\partial U} = 0$ has nontrivial solutions, we are forced then to restrict our attention to real eigenvalues. That is why we demanded that the "bad" set Σ in Theorem 3.28 be a subset of $\mathbb{R}$:

$$\Sigma := \{\lambda \in \mathbb{R} : (L - \lambda)u = 0, u|_{\partial U} = 0 \text{ has nonzero weak solutions}\}. \tag{3.84}$$

It is more natural to view L as an operator on complex-valued functions. The point spectrum $\Sigma \subset \mathbb{R}$ is then in general *a proper subset* of the full point spectrum $\Sigma_{\mathbb{C}} \subset \mathbb{C}$:

$$\Sigma_{\mathbb{C}} := \{\lambda \in \mathbb{C} : (L - \lambda)u = 0, u|_{\partial U} = 0 \text{ has nonzero weak solutions}\}. \tag{3.85}$$

Let's recall a few definitions.

Definition 3.37. *Let H be a complex Hilbert space. The* spectrum *of a bounded operator $T : H \to H$ is $\sigma(T) = \{\lambda \in \mathbb{C} : T - \lambda I$ is not bijective$\}$. The* resolvent set *of T is $\mathbb{C} \setminus \sigma(T)$. When H is infinite dimensional the spectrum can contain elements that are not eigenvalues. The spectrum may be decomposed as follows:*

1) The point spectrum *consists of $\lambda \in \sigma(T)$ such that $T - \lambda I$ is not one-to-one. In this case λ is called an* eigenvalue *of T.*

2) The continuous spectrum *consists of $\lambda \in \sigma(T)$ such that $T - \lambda I$ is one-to-one, and such that the range of $T - \lambda I$ is dense in H, but not equal to H.*

3) The residual spectrum *consists of $\lambda \in \sigma(T)$ such that $T - \lambda I$ is one-to-one, and such that the range of $T - \lambda I$ is not dense in H.*

These definitions are the same when H is merely a Banach space.

Exercise: Let $H = L^2([0, 1], \mathbb{C})$ and define the multiplication operator $M : H \to H$ by $Mf(x) = xf(x)$. Show that $\sigma(M) = [0, 1]$ and that every $\lambda \in [0, 1]$ belongs to the continuous spectrum.

Here is the fact from spectral theory used at the end of the proof of Theorem 3.28; see [Wil26], Corollary 9.14.

Theorem 3.38. *Let H be a complex Hilbert space (or Banach space) and let $K : H \to H$ be a compact operator. Then, the nonzero elements of the spectrum $\sigma(K) \subset \mathbb{C}$ of K are eigenvalues. There are at most countably many eigenvalues, and, in the case of infinitely many, they form a sequence tending to 0.*

There are complex analogues of the Lax-Milgram lemma and the abstract Fredholm alternative for operators $I - K$ with K compact. Their proofs are quite close to the ones we gave (or at times did not give) in the real case. Using those results and theorem 3.38, one can formulate the complex analogues of the results we proved for elliptic operators in the real case.

Remark 3.39. 1) The second order operator $L(x, D)$ obviously fails to map $L^2(U)$ to itself. It does however map various subspaces of $L^2(U)$ into $L^2(U)$, such as $\mathcal{D}(U)$ for example, or $H_0^2(U)$. The operator L with the domain $H_0^2(U)$ (or $\mathcal{D}(U)$) is an example of an *unbounded operator on* $L^2(U)$. For γ large the operator $(L + \gamma)^{-1}$ is bounded, in fact compact, on $L^2(U)$, and we used this fact to derive information about the spectrum of the unbounded operator L.

2) The definition of the spectrum of an unbounded operator is similar to definition 3.37, but it depends on the choice of domain; see [Wil26], section 11.1, for example.

3). An important special case is when the uniformly elliptic operator L has the form

$$Lu = -\sum_{i,j=1}^{n} (a_{ij}(x)u_{x_i})_{x_j}, \tag{3.86}$$

where $a_{ij} \in C^1(\overline{U}, \mathbb{R})$ and $a_{ij} = a_{ji}$. It is easy to see that the associated bilinear form now satisfies $B[u, v] = B[v, u]$. The operator L is thus "formally symmetric". It is also easy to check for large γ that the associated compact operator $K = \frac{1}{\gamma} L_\gamma^{-1} : L^2(U) \to L^2(U)$ is *self-adjoint*, that is, $K = K^*$, where K^* is the Hilbert space adjoint of K. In this case the spectrum of K is entirely real (see [Wil26], section 10.1) so the set Σ we defined for L is equal to the full point spectrum $\Sigma_{\mathbb{C}}$.

Exercise. Show that when $L(x, D)$ is as in part (3) of the above remark, we have $\Sigma \subset (0, \infty)$. In particular, $0 \notin \Sigma$.

3.5 Exercises

1. Let $V = \{(x, y) \in \mathbb{R}^2 : |(x, y)| < 1, 0 < y < 1\}$ and define $u = 1$ on V. Check that $u \in W^{1,p}(U)$ but the zero extension $u_c \notin W^{1,p}(\mathbb{R}^n)$. Hint: compute $\partial_y u_c$ near $(0, 0)$.

2. Let $U \subset \mathbb{R}^n$ be open but not bounded. Suppose $1 \leq p < \infty$ and $u \in W^{k,p}(U)$. Show that there exist functions $u_n \in W^{k,p}(U)$ with bounded supports such that $u_n \to u$ in $W^{k,p}(U)$.

3. (a) Let $U \subset \mathbb{R}^n$ be open with C^1 boundary. Show that U has the segment property.
(b) Let $1 \leq p < \infty$, $k \in \mathbb{N}$. Show that if $U \subset \mathbb{R}^n$ is open and bounded with C^1 boundary, then any $u \in W^{k,p}(U)$ can be approximated in $W^{k,p}(U) \cap C(\overline{U})$ by functions in $\mathcal{D}(\mathbb{R}^n)$.[20]

4. Under the assumptions of problem 3, use that problem to show that if $u \in W^{1,p}(U) \cap C(\overline{U})$, then $Tu = u|_{\partial U}$.

5. Suppose $U \subset \mathbb{R}^n$ is bounded and open with C^1 boundary, $f \in H^{-1}(U)$, and that $g \in L^2(\partial U)$ is the trace of some $w \in H^1(U)$. For L as in (3.26) define what it means to find a weak solution $u \in H^1(U)$ of the problem

$$\begin{aligned} Lu &= f \\ u|_{\partial U} &= g, \end{aligned} \tag{3.87}$$

and then solve that problem by reducing it to a problem of the form (3.28).

6. Let $U \subset \mathbb{R}^n$ be open. Suppose $u \in W_0^{1,p}(U)$, where $1 \leq p \leq \infty$. Define the zero extension of u by $u_e = \begin{cases} u, & x \in U \\ 0, & x \in \mathbb{R}^n \setminus U \end{cases}$. Is it true that $u_e \in W^{1,p}(\mathbb{R}^n)$? Prove your answer.[21]

7. [Poincaré's inequality] Let $U \subset \mathbb{R}^n$ be bounded and open. Suppose $1 \leq p \leq \infty$. Show there exists $C > 0$ such that

$$|u|_{L^p(U)} \leq C|\nabla u|_{L^p(U)} \text{ for all } u \in W_0^{1,p}(U). \tag{3.88}$$

Hint: First do the case where $U = (0,1) \subset \mathbb{R}$ using the fundamental theorem of calculus.

[20] This means there exists $\phi_j \in \mathcal{D}(\mathbb{R}^n)$ such that $\phi_j \to u$ in the norm $|\cdot|_{W^{k,p}(U)} + |\cdot|_{C(\overline{U})}$.
[21] Recall problem 1.

4 Minimizers in the calculus of variations

This chapter provides a brief introduction to some methods in the calculus of variations. We take $U \subset \mathbb{R}^n$ to be bounded and open with a C^1 boundary.

We saw in Theorem 1.18 that if a minimizer $u \in \mathcal{A}$ of the functional

$$I_1(w) = \int_U \left(\frac{1}{2} |\nabla w|^2 \right) dx \tag{4.1}$$

exists on the "admissible set" $A = \{w \in C^2(\overline{U}) : w = g \text{ on } \partial U\}$, then u is a solution of the corresponding "Euler-Lagrange problem"

$$-\Delta u = 0 \text{ in } U, \; u|_{\partial U} = g. \tag{4.2}$$

But we did not discuss whether such a minimizer exists. Here we will show that a minimizer for I_1 does exist on the admissible set

$$A_1 = \{w \in H^1(U) : w = g \text{ on } \partial U\}. \tag{4.3}$$

The same argument will also prove existence of a minimizer for the functional

$$I_2(w) = \int_U \left(\frac{1}{4} |\nabla w|^4 \right) dx \tag{4.4}$$

on the admissible set

$$A_2 = \{w \in W^{1,4}(U) : w = g \text{ on } \partial U\}. \tag{4.5}$$

We'll show in both cases that a minimizer is a weak solution of the the corresponding Euler-Lagrange problem.[1] For I_2 this problem is nonlinear:

$$-\sum_{i=1}^{n} \left(|Du|^2 u_{x_i} \right)_{x_i} u = 0 \text{ and } u|_{\partial U} = g. \tag{4.6}$$

[1] Under appropriate assumptions on the regularity of g and ∂U, the regularity theory of section 3.3.1 implies that the minimizer $u \in A_1$ of I_1 actually lies in $C^2(\overline{U})$, for example. For I_2 and its associated nonlinear Euler-Lagrange problem, there is also a theory that implies higher regularity for weak solutions, but it is more difficult; see [Gia83].

4.1 Minimizers exist

Observe that I_1 and I_2 are functionals of the form

$$I(w) := \int_U L(Dw(x))dx \tag{4.7}$$

where $L(p)$ is a C^∞ function of $p \in \mathbb{R}^n$ such that for some $1 < q < \infty$:

$$\begin{aligned} &(a) \text{ for some } \alpha, \beta > 0, \quad \alpha|p|^q \le L(p) \le \beta(|p|^q + 1) \text{ for all } p \\ &(b) \text{ for some } C > 0, \quad |D_p L(p)| \le C(|p|^{q-1} + 1) \text{ for all } p \\ &(c) \sum_{i,j=1}^{n} L_{p_i p_j}(p)\xi_i\xi_j \ge 0 \text{ for all } p \in \mathbb{R}^n, \xi \in \mathbb{R}^n. \end{aligned} \tag{4.8}$$

The first inequality in (4.8)(a) is called *coercivity*. The condition (4.8)(c) is *convexity* of L. For a fixed g define the admissible set

$$A = \{w \in W^{1,q}(U) : w = g \text{ on } \partial U \text{ in the trace sense}\}. \tag{4.9}$$

Theorem 4.1. *Suppose A is nonempty and that L is C^∞ and satisfies (4.8), where $1 < q < \infty$. Then there exists $u \in A$ such that*

$$I(u) = \inf_{w \in A} I(w). \tag{4.10}$$

Proof. **1.** The second inequality of (4.8)(a) implies $I(w) < \infty$ for all $w \in A$. Letting

$$m = \inf_{w \in A} I(w), \tag{4.11}$$

we clearly have $0 \le m < \infty$.

2. Candidate for the minimizer. Use the definition of inf to choose a minimizing sequence, that is, a sequence $u_k \in A$ such that $I(u_k) \to m$. By coercivity we have

$$I(u_k) \ge \alpha \int_U |Du_k|^q dx = \alpha |Du_k|^q_{L^q(U)} \Rightarrow (Du_k) \text{ is bounded in } L^q(U), \tag{4.12}$$

since $(I(u_k))$ is bounded. To see that (u_k) is also bounded in $L^q(U)$ we'll use Poincarés inequality; recall problem 7 of section 3.5. Choose a particular $w_0 \in A$. Then $u_k - w_0 \in W_0^{1,q}(U)$ for all k by Proposition 3.12, so Poincaré's inequality implies

$$|u_k - w_0|_{L^q(U)} \le C|Du_k - Dw_0|_{L^q(U)}. \tag{4.13}$$

With (4.12) this implies (u_k) is bounded in $L^q(U)$, so (u_k) is bounded in $W^{1,q}(U)$. The Banach space $W^{1,q}(U)$ is separable and reflexive, so we apply weak sequential compactness

(Proposition 8.63) to conclude that there is a subsequence, also denoted (u_k), and an element $u \in W^{1,q}(U)$ such that

$$u_k \to u \text{ weakly in } W^{1,q}(U). \tag{4.14}$$

3. To finish we need to show that the trace $Tu = g$ and $I(u) = m$. The first statement would follow from the trace theorem, Theorem 3.10, if we knew that $u_k \to u$ in the $W^{1,q}(U)$ norm (why?). But we only have weak convergence.

The second statement would follow if we knew that $I(u_k) \to I(u)$, but we can't assert that since $I(w)$ is nonlinear and $u_k \to u$ weakly.[2]

We'll use Mazur's theorem to get around the first difficulty and weak lower semicontinuity of $I(w)$ to get around the second.

4. Show $u \in A$**.** Fix $w_0 \in A$. Then the sequence $(u_k - w_0)$ lies in $W_0^{1,q}(U)$ and we have

$$u_k - w_0 \to u - w_0 \text{ weakly in } W^{1,q}(U). \tag{4.15}$$

The subspace $W_0^{1,q}(U)$ is a *closed* subspace of $W^{1,q}(U)$ and obviously convex, so Mazur's theorem, Theorem 8.61, implies $W_0^{1,q}(U)$ is weakly closed. Thus

$$u - w_0 \in W_0^{1,q}(U) \text{ and hence } Tu = g. \tag{4.16}$$

5. Show $I(u) = m$**.** We have $I(u) \geq m$ since $u \in A$, so to finish we just need to show $I(u) \leq m$. This follows from the sequential *weak lower semicontinuity* of $I(w)$ on $W^{1,q}(U)$, that is, the property

$$u_k \to u \text{ weakly in } W^{1,q}(U) \Rightarrow I(u) \leq \liminf_{k\to\infty} I(u_k). \tag{4.17}$$

To show (4.17) we note first that Taylor's formula and the convexity of L imply for p_1, p_2 in $\mathbb{R}^n$ that

$$L(p_2) = L(p_1) + D_pL(p_1)(p_2 - p_1) + R(p_1, p_2) \text{ where } R(p_1, p_2) \geq 0. \tag{4.18}$$

Using (4.18) we obtain

$$I(u_k) = \int_U L(Du_k)dx \geq \int_U L(Du)dx + \int_U D_pL(Du)(Du_k - Du)dx. \tag{4.19}$$

Since $|Du| \in L^q(U)$, the condition (4.8)(b) implies $D_pL(Du) \in L^{q'}(U)$, where $q' = \frac{q}{q-1}$ (check). We know $Du_k \to Du$ weakly in $L^q(U)$, so letting $k \to \infty$ in (4.19) gives

$$m \geq \int_U L(Du)dx = I(u). \tag{4.20}$$

□

[2] If u_k converged to u in the $W^{1,q}(U)$ norm, we could conclude $I(u_k) \to I(u)$ (check).

Remark 4.2. (a)[Uniqueness] If we strengthen the convexity assumption on L to *uniform convexity*, that is:

$$\exists \theta > 0 \text{ such that } \sum_{i,j=1}^{n} L_{p_i p_j}(p)\xi_i\xi_j \geq \theta|\xi|^2 \text{ for all } p \in \mathbb{R}^n, \xi \in \mathbb{R}^n, \tag{4.21}$$

then a minimizer $u \in A$ of $I(w)$ is unique; see [Evans], section 8.2. To appreciate the need for a stronger convexity condition to guarantee the uniqueness of a minimizer, it is helpful to consider the difference between the graphs of the functions $z = x^2 + y^2$ and $z = x^2$ in $(x, y, z)-$space.

(b) With more work Theorem 4.1 and part (a) of this Remark can be extended to functionals of the form

$$I(w) = \int_U L(Du, u, x)dx$$

under appropriate generalizations for $L(p, z, x)$ of the hypotheses we have made on $L(p)$; see [Evans], section 8.2.

4.2 Minimizers satisfy the Euler-Lagrange problem

We let U, $I(w)$, $L(p)$, and A be as in Theorem 4.1.

Definition 4.3. *The Euler-Lagrange problem associated to* $I(w) = \int_U L(Dw)dx$ *is*

$$\begin{aligned} -\sum_{i=1}^{n}(L_{p_i}(Du))_{x_i} &= 0 \text{ in } U \\ u &= g \text{ on } \partial U. \end{aligned} \tag{4.22}$$

We say that $u \in A$ *is a weak solution of the Euler-Lagrange problem if*

$$\int_U \sum_{i=1}^{n} L_{p_i}(Du)v_{x_i}dx = 0 \text{ for all } v \in W_0^{1,q}(U). \tag{4.23}$$

Theorem 4.4. *If* $u \in A$ *satisfies*

$$I(u) = \inf_{w \in A} I(w), \tag{4.24}$$

then u *is a weak solution of the Euler-Lagrange problem* (4.22).

Proof. **1.** Parallel to what we did in the proof of Theorem 1.18, we define

$$i(\tau) = I(u + \tau v) \text{ for } v \in W_0^{1,q}(U). \tag{4.25}$$

By (4.8)(a) $i(\tau) < \infty$ for all τ. For $\tau \neq 0$ we consider

$$\frac{i(\tau) - i(0)}{\tau} = \int_U \frac{L(Du + \tau Dv) - L(Du)}{\tau} dx := \int_U L^\tau(x) dx. \tag{4.26}$$

The differentiability of $L(p)$ implies

$$L^\tau(x) \to \sum_{i=1}^n L_{p_i}(Du) v_{x_i} \text{ as } \tau \to 0 \text{ for a.e. } x. \tag{4.27}$$

2. To get a uniform dominator for $L^\tau(x)$ as $\tau \to 0$ we write

$$L^\tau(x) = \frac{1}{\tau} \int_0^\tau \frac{d}{ds} L(Du + sDv) ds = \frac{1}{\tau} \int_0^\tau \sum_{i=1}^n L_{p_i}(Du + sDv) v_{x_i} ds. \tag{4.28}$$

The estimate (4.8)(b) implies:[3]

$$\begin{aligned} &|L_{p_i}(Du + sDv) v_{x_i}| \leq C(|Du + sDv| + 1)^{q-1} |Dv| \leq \\ &\quad C\left[(|Du + sDv| + 1)^q + |Dv|^q\right] \leq C(|Du|^q + |Dv|^q + 1) \text{ for } s \in [0, 1]. \end{aligned} \tag{4.29}$$

The right side of (4.29) is independent of s, so

$$|L^\tau(x)| \leq C(|Du|^q + |Dv|^q + 1) \in L^1(U). \tag{4.30}$$

We can thus apply the dominated convergence theorem to conclude from (4.27) and (4.30) that $i'(0)$ exists and equals

$$i'(0) = \int_U \sum_{i=1}^n L_{p_i}(Du) v_{x_i} dx.$$

But $i(\tau)$ has a min at $\tau = 0$, so $i'(0) = 0$. Thus, u satisfies (4.23). □

Remark 4.5 (First and second variations)**.** The derivative $i'(0)$ is called the first variation of $I(\cdot)$ at u, while $i''(0)$ is the second variation of $I(\cdot)$ at u. When $u \in A$ is a minimizer of $I(w)$, one expects $i''(0) \geq 0$, and one can use this to see that the condition

$$\sum_{i,j=1}^n L_{p_i p_j}(Du) \xi_i \xi_j \geq 0 \text{ for } x \in U, \xi \in \mathbb{R}^n \tag{4.31}$$

is a necessary condition for u to minimize $I(w)$ over A; see [Eva10], section 8.1.3.

[3] For the second inequality we used $ab \leq \frac{a^{q'}}{q'} + \frac{b^q}{q}$, where $a, b \geq 0$ and $q' = \frac{q}{q-1}$.

5 Nonlinear first-order PDEs and the method of characteristics

In classical mechanics one sometimes needs to find solutions to first-order nonlinear initial value problems of the form

$$\phi_t = \lambda(t, x, \phi_x), \ \phi(0, y) = g(y), \tag{5.1}$$

for an unknown function $\phi(t, x)$. Here think of $t \in \mathbb{R}$ as a time variable and $x \in \mathbb{R}^d$ as a space variable, and take $g \in C^\infty$.[1] The real-valued function $\lambda(t, x, \xi)$ is a given C^∞ function of $(t, x, \xi) \in \mathbb{R}_t \times \mathbb{R}^d_x \times \mathbb{R}^d_\xi$. The problem (5.1) is usually referred to as a *Hamilton-Jacobi problem.*

Let $p = (\tau, \xi) \in \mathbb{R}^{1+d}$ be a placeholder for (ϕ_t, ϕ_x), set $q = (t, x)$, and define the *Hamiltonian*

$$F(q, p) = \tau - \lambda(t, x, \xi). \tag{5.2}$$

The corresponding *Hamiltonian vector field* $H(q, p)$ on $\mathbb{R}^{1+d}_{t,x} \times \mathbb{R}^{1+d}_{\tau,\xi}$ is

$$H(q, p) := \partial_p F \cdot \partial_q - \partial_q F \cdot \partial_p = (\partial_t - \partial_\xi \lambda \cdot \partial_x) - (-\lambda_t \partial_\tau - \lambda_x \cdot \partial_\xi). \tag{5.3}$$

We'll normally write this as $H = (\partial_p F, -\partial_q F) = (F_p, -F_q)$.

Integral curves of H are solutions $(q(s), p(s))$ of the following nonlinear, first-order system of ODEs:

$$(\dot{q}, \dot{p}) = H(p, q) = (F_p(q, p), -F_q(q, p)), \quad (q, p)(0) = (q_0, p_0), \tag{5.4}$$

where $\dot{q} = \frac{d}{ds} q$. Basic ODE theory implies that solutions exist on *some* time interval $[0, T]$, where $T > 0$. The curves $(q(s), p(s))$ lie in $\mathbb{R}^{1+d}_{t,x} \times \mathbb{R}^{1+d}_{\tau,\xi}$ and are often referred to as *bicharacteristics of* F. The curve $q(s)$ obtained by projecting a bicharacteristic onto the $\mathbb{R}^{1+d}_q$ factor is called a *characteristic*. We will explain how to use bicharacteristics (and characteristics) to solve the eikonal problem (5.1).[2]

[1] It will become clear later why it is helpful to use $y \in \mathbb{R}^d$ in (5.1) to denote the space variable on the initial surface $t = 0$.

[2] This method is sometimes called the Hamilton-Jacobi method. It is an elementary, but beautiful example of using ODEs to solve PDEs.

Remark 5.1. Let $(q(s), p(s))$ be a solution of the bicharacteristic equations (5.4). Then (5.4) implies

$$\frac{d}{ds}F(q(s),p(s)) = F_q(q(s),p(s))\cdot\dot{q}(s) + F_p(q(s),p(s))\cdot\dot{p}(s) = 0 \text{ for all } s\in[0,T]. \tag{5.5}$$

Thus, F is *constant* and equal to $F(q_0,p_0)$ along the bicharacteristic starting at (q_0,p_0). In classical mechanics, where the Hamiltonian F represents the energy of a physical system whose state at any time is described by position variables q and momentum variables p, a solution of (5.4) describes the time evolution of the system.[3] [4] In fact, equations (5.4) arise by rewriting Newton's second law, force $= ma$, appropriately. The fact that F is constant along bicharacteristics expresses *conservation of energy*.[5]

5.1 Procedure for solving the eikonal equation

In this section we give without justification a procedure for solving the eikonal problem (5.1). In the next section we motivate the steps and provide the justification.

Step 1. For g and λ as in (5.1) solve the bicharacteristic equations (5.4) with initial data

$$q_0(y) = (0,y),\ p_0(y) = (\lambda(0,y,g_y(y)), g_y(y)), \tag{5.6}$$

where y lies in an open set $\Omega \subset\subset \mathbb{R}^d$ containing some designated "basepoint" y_0. This produces functions [6]

$$\begin{aligned} &q(s,y) = (t(s,y),x(s,y)),\ p(s,y) = (\tau(s,y),\xi(s,y)) \text{ such that}\\ &t(0,y) = 0,\ x(0,y) = y,\ \tau(0,y) = \lambda(0,y,g_y(y)),\ \xi(0,y) = g_y(y). \end{aligned} \tag{5.7}$$

Step 2. Now that $q(s,y)$, $p(s,y)$ are determined, define the real-valued function $z(s,y)$ by solving

$$\dot{z}(s,y) = F_p(q(s,y),p(s,y))\cdot p(s,y),\ z(0,y) = g(y). \tag{5.8}$$

Step 3. Use the inverse function theorem to invert the map $G(s,y) := (t(s,y),x(s,y))$ in a neighborhood of $(0,y_0)$. We compute

$$G'(0,y_0) = \begin{pmatrix} 1 & 0 \\ * & I_{d\times d}\end{pmatrix},$$

[3] Note from (5.3) and (5.4) that $\dot{t} = 1$, so $t(s,y) = s$ if $t(0,y) = 0$.

[4] For more discussion of how problems (5.1) arise in mechanics, see Chapter 10 of Classical Mechanics by Goldstein, Poole, and Safko, 3rd edition. Such problems also arise in geometric optics, where (5.1) is called an *eikonal* problem. In that case the function $\lambda(t,x,\xi)$ is an eigenvalue of the matrix symbol $A(t,x,\xi)$ of some matrix differential operator $A(t,x,\partial_x)$.

[5] In physical problems one can often take the time T of existence of solutions to (5.4) to be ∞.

[6] We now write $q(s,y)$, $p(s,y)$ instead of $q(s)$, $p(s)$ to indicate that the solution depends on the parameter y as well as s.

so the inverse function theorem applies and yields functions $s = s(t,x)$, $y = y(t,x)$, the components of G^{-1}, which are defined for (t,x) near $(0,y_0)$.

Step 4. Define

$$\begin{aligned}&(a)\ \phi(t,x) = z(s(t,x),y(t,x))\\&(b)\ P(t,x) := p(s(t,x),y(t,x)) = (\tau(s(t,x),y(t,x)),\xi(s(t,x),y(t,x))).\end{aligned} \tag{5.9}$$

We claim that

$$P(t,x) = \phi_{t,x}(t,x), \tag{5.10}$$

and that $\phi(t,x)$ is a solution of (5.1) for (t,x) near $(0,y_0)$.[7]

5.2 Motivation and proof of the method

The procedure given in the last section has a clear geometric meaning that we will now explain.

The interior equation in (5.1) is the statement that $F(t,x,\phi_t(t,x),\phi_x(t,x)) = 0$. To motivate section 5.1, suppose for a moment that we are *given* some C^2 solution $\phi(t,x)$ of (5.1). Then

$$\phi_t(0,y) = \lambda(0,y,g_y(y)), \phi_x(0,y) = g_y(y),$$

so the functions $q(s,y)$, $p(s,y)$ from step **1** satisfy $(q(0,y),p(0,y)) = (0,y,\phi_t(0,y),\phi_x(0,y))$. Hence $F(q(0,y),p(0,y)) = 0$.

Thus, the initial point of each bicharacteristic curve $(q(s,y),p(s,y))$ lies in the $(2d+1)-$dimensional level set [8]

$$S = \{(q,p) \in \mathbb{R}^{1+d} \times \mathbb{R}^{1+d} : F(q,p) = 0\}, \tag{5.11}$$

and lies on the graph, $\mathcal{G} \subset \mathbb{R}^{1+d} \times \mathbb{R}^{1+d}$ of dimension $1+d$, of the map $(t,x) \to \phi_{t,x}(t,x)$. Remark 5.1 implies

$$F(q(s,y),p(s,y)) = 0 \text{ for all } s \in [0,T], \tag{5.12}$$

so the entire bicharacteristic curve lies in S. Does the entire bicharacteristic curve also lie in $\mathcal{G}$, or in other words is it true that

$$p(s,y) = \phi_{t,x}(q(s,y)) \text{ for all } s \in [0,T]? \tag{5.13}$$

This is possible, since the assumption that ϕ is a solution of (5.1) implies[9]

$$F(q(s,y),\phi_{t,x}(q(s,y))) = 0 \text{ for all } s \in [0,T]. \tag{5.14}$$

[7] Here $\phi_{t,x} = \nabla_{t,x}\phi$.

[8] The level set S is a C^2 hypersurface in $\mathbb{R}^{1+d} \times \mathbb{R}^{1+d}$.

[9] With step **3** of section 5.1, (5.14) implies $\mathcal{G} \subset S$ near $(0,y_0,\phi_t(0,y_0),\phi_x(0,y_0))$.

To see that (5.13) is true, by uniqueness for ODEs it will suffice to show for some function $\tilde{q}(s,y)$ such that $\tilde{q}(0,y) = (0,y)$, and for $\tilde{p}$ defined by

$$\tilde{p}(s,y) = \phi_{t,x}(\tilde{q}(s,y)) = \phi_q(\tilde{q}(s,y)), \tag{5.15}$$

that $(\tilde{q},\tilde{p})$ satisfy the bicharacteristic equations (5.4).[10] *Define* $\tilde{q}$ to be the solution of [11]

$$\dot{\tilde{q}}(s,y) = F_p(\tilde{q}(s,y),\tilde{p}(s,y)),\ \tilde{q}(0,y) = (0,y),\ \text{where } \tilde{p} \text{ has the form (5.15).} \tag{5.16}$$

Now differentiate (5.15) with respect to s and differentiate $F(\tilde{q},\phi_q(\tilde{q})) = 0$ with respect to $\tilde{q}$ to get

$$\begin{aligned} &\dot{\tilde{p}} = \phi_{qq}\dot{\tilde{q}} \\ &F_q(\tilde{q},\phi_q) + F_p(\tilde{q},\phi_q)\phi_{qq}(\tilde{q}) = 0, \end{aligned} \tag{5.17}$$

and observe that with (5.16) these imply

$$F_q(\tilde{q},\phi_q) + \dot{\tilde{p}} = 0,\ \text{that is } \dot{\tilde{p}} = -F_q(\tilde{q},\tilde{p}). \tag{5.18}$$

By uniqueness for ODEs we conclude that $(q(s,y),p(s,y)) = (\tilde{q}(s,y),\tilde{p}(s,y))$, so (5.13) follows from (5.15); henceforth, we drop the tildes.

Remark 5.2. *This shows that the* $(1+d)-$*dimensional graph of* $\phi_{t,x}$*, namely* $\mathcal{G}$*, is the* flowout *under the bicharacteristic flow of the* $d-$*dimensional initial submanifold*

$$\mathcal{G}_0 := \{(0,y,\phi_t(0,y),\phi_x(0,y)) : y \in \Omega\} \subset \mathbb{R}^{1+d} \times \mathbb{R}^{1+d}.$$

Thus, the graph of $\phi_{t,x}$ *is* foliated *by the family of disjoint bicharacteristic curves corresponding to distinct points in* $\mathcal{G}_0$*.*

Now *define* $z(s,y) = \phi(q(s,y))$, and compute

$$\dot{z} = \phi_q\dot{q} = p \cdot F_p(q,p), \tag{5.19}$$

where the second equality follows from (5.4) and (5.13). Since $z(0,y) = g(y)$, we have recovered (5.8).

Still working with the assumed solution ϕ of (5.1), we recover equations (5.9)(a) and (5.10) using G^{-1} from step **3**. For example, $\phi(t,x) = z(s(t,x),y(t,x))$ follows from

$$z(s,y) = \phi(q(s,y)) \text{ and } (q \circ G^{-1})(t,x) = (t,x),$$

and (5.10) follows from (5.13).[12] The next proposition summarizes what we have shown so far.

[10] Observe that $\tilde{q}(s,y),\tilde{p}(s,y)$ have the same initial data at $(0,y)$ as q,p. The first component of $\tilde{q}$ is the t component, the second is the $x \in \mathbb{R}^d$ component.

[11] The problem (5.16) is an ODE where the only unknown is $\tilde{q}$.

[12] This concludes our motivation of section 5.1.

Proposition 5.3. *Suppose ϕ is a given C^2 solution of the eikonal problem (5.1) for (t,x) near $(0,y_0)$. Then ϕ* must be *the function constructed by the procedure of section 5.1. In particular, a C^2 solution of (5.1) is uniquely determined for (t,x) near $(0,y_0)$.*

Now we drop the assumption that we are *given* a solution of the eikonal problem. We need to show that the function $\phi(t,x)$ constructed in section 5.1 *is* a solution of the eikonal problem. The uniqueness part of the following theorem has already been proved.

Theorem 5.4. *The function ϕ defined by the procedure of section 5.1 is the unique C^2 solution of the eikonal problem (5.1) for (t,x) near $(0,y_0)$.*

Proof. **1.** We use the notation of section 5.1.[13] From (5.12), which follows just from Remark (5.1) and the fact that $(q(0,y),p(0,y)) \in S$, we obtain using G^{-1} as above that

$$F(t,x,P(t,x)) = 0 \text{ for } P(t,x) \text{ as defined in (5.9).} \tag{5.20}$$

To finish the proof, we therefore only need to show (5.10), that is, $P(t,x) = \phi_{t,x}(t,x)$, where $\phi(t,x)$ is defined in (5.9). We can write this equality as $P(q) = \phi_q(q)$.

2. For $z(s,y)$ as defined in step **2** of section 5.1, recall that $\phi(t,x) := z(s(t,x),y(t,x))$. In order to compute ϕ_q we therefore need to compute z_s and z_y. We claim [14]

$$\begin{aligned} &(a)\dot{z}(s,y) = p(s,y)\cdot\dot{q}(s,y) \\ &(b)z_y(s,y) = p(s,y)\cdot q_y(s,y). \end{aligned} \tag{5.21}$$

Part (a) is immediate from (5.8) and (5.4). To prove (b) for a given y set

$$r(s) = z_y(s,y) - p(s,y)\cdot q_y(s,y). \tag{5.22}$$

Using (5.8) and (5.6) we see that

$$r(0) = g_y(y) - p(0,y)\cdot\begin{pmatrix}0\\ I_{d\times d}\end{pmatrix} = g_y(y) - g_y(y) = 0.$$

We'll show $\dot{r} = 0$ to conclude $r(s) = 0$ for all s, proving (b).

Differentiate (5.22) with respect to s and (5.21)(a) with respect to y to obtain

$$\begin{aligned} &(a)\ \dot{r} = z_{ys} - (\dot{p}\cdot q_y + p\cdot q_{ys}) \\ &(b)\ z_{sy} = p_y\cdot\dot{q} + \dot{p}\cdot q_{sy}. \end{aligned} \tag{5.23}$$

These imply

$$\dot{r} = p_y\cdot\dot{q} - \dot{p}\cdot q_y = p_y\cdot F_p + F_q\cdot q_y, \tag{5.24}$$

[13] This argument mostly follows the proof of Theorem 2 in [Eva10], section 3.2.4.
[14] Equation (8.23)(b) means that $z_{y_i} = p\cdot q_{y_i}$ for $i = 1,\dots,d$.

where for the second equality we used (5.4). But differentiating (5.12) with respect to y gives

$$F_q \cdot q_y + F_p \cdot p_y = 0, \text{ so } \dot{r} = 0. \tag{5.25}$$

3. Setting $v = (s, y)$, we can rewrite (5.21) as $z_v(v) = p(v) \cdot q_v(v)$ and use step **3** of section 5.1 to write $q = q(v(q))$.[15] Using (5.9)(a), which says $\phi(q) = z(v(q))$, and (5.21) we obtain[16]

$$\phi_q(q) = z_v(v(q))v_q(q) = p(v(q)) \cdot q_v(v(q))v_q(q) = p(v(q))q_q(q) = p(v(q)) = P(q). \tag{5.26}$$

□

[15] Observe $v(q) = G^{-1}(q)$ for G as in step **3** of section 5.1.

[16] Note that $q_q(q) = \frac{\partial(t,x)}{\partial(t,x)}(t, x) = I_{(1+d)\times(1+d)}$.

6 Introduction to microlocal analysis

To describe what microlocal analysis is about, we let $\Omega \subset \mathbb{R}^n$ be open and assume $u \in \mathcal{D}'(\Omega)$, $x_0 \in \Omega$, and $\xi_0 \in \mathbb{R}^n \setminus \{0\}$. We do a *local analysis* of u near x_0 by studying properties of ϕu for $\phi \in \mathcal{D}(\omega)$, where $\omega \subset\subset \Omega$ and $\phi(x_0) \neq 0$. We do a *microlocal analysis* of u near $(x_0, \xi_0) \in \Omega \times (\mathbb{R}^n \setminus \{0\})$ by *also* studying properties of $\widehat{\phi u}(\xi)$ for ξ near ξ_0. For reasons that will become clear below, we often focus on a *conic* neighborhood $\Gamma \ni \xi_0$, that is, an open set $\Gamma \subset \mathbb{R}^n \setminus \{0\}$ such that $t\Gamma = \Gamma$ for $t > 0$. We sometimes refer to $\omega \times \Gamma$ as a *microlocal neighborhood* of (x_0, ξ_0) in $\Omega \times (\mathbb{R}^n \setminus \{0\})$.

6.1 Singular support and wavefront set of $u \in \mathcal{D}'(U)$.

Definition 6.1 (Singular support)**.** *(a) Let $u \in \mathcal{D}'(\Omega)$ and suppose $\omega \subset \Omega$. We say u* is C^∞ in ω *if there exists $f \in C^\infty(\omega)$ such that $\langle u, \phi\rangle = \int f\phi dx$ for all $\phi \in \mathcal{D}(\omega)$.*

(b) The singular support of u, sing supp $u \subset \Omega$, *is the complement (in Ω) of the largest open set in which u is C^∞.*[1] *We'll sometimes write SSu for short.*

Example 6.2. 1. We have sing supp $\delta_0 =$ sing supp $(\delta_0 + e^x) = \{0\} \in \mathbb{R}$.

2. Define $u \in \mathcal{D}'(\mathbb{R}^2)$ by $u(x,y) = \begin{cases} 1, & y \geq x \\ 0, & y < x \end{cases}$. Then $SSu = \{(x,y) : x = y\}$.

Question. One of the earliest applications of microlocal analysis was to answer the following question. Let

$$P(x,D) = \sum_{|\alpha| \leq m} c_\alpha(x) D^\alpha, \text{ where } c_\alpha \in C^\infty(\Omega) \ \ \forall \alpha. \tag{6.1}$$

Suppose $u \in \mathcal{D}'(\Omega)$ satisfies $Pu \in C^\infty(\Omega)$ and $x_0 \in SSu$. Where else must u have singular support? In other words must the singularity at x_0 propagate, and if so, where does it go?

Examples 6.3. (a) For an example of propagation of singular support consider the equation

$$\Box u := u_{tt} - u_{xx} = (\partial_t + \partial_x)(\partial_t - \partial_x)u = 0 \text{ on } \mathbb{R}_x \times \mathbb{R}_t. \tag{6.2}$$

We saw earlier that if $f : \mathbb{R} \to \mathbb{R}$ lies in $C(\mathbb{R})$, then $u(x,t) = f(x-t)$ is a distribution solution of (6.2) which lies in $C(\mathbb{R}^2)$. Moreover, if $f(x)$ fails to be C^∞ near $x = 0$, then

[1] That open set is the union of all open sets in which u is C^∞.

$u(x,t)$ fails to be C^∞ near points (t,t). Thus, u has a singularity at $(0,0)$ that propagates along the line traced by the point (t,t) as $t \in \mathbb{R}$ varies.

(b) A very different situation occurs if we consider instead $\Delta = \partial_t^2 + \partial_x^2$, which is elliptic. Corollary 3.34 implies that if $u \in H^1(\Omega)$ is a distribution solution of $\Delta u = 0$, then $u \in C^\infty(\Omega)$, that is,

$$\text{sing supp } u = \emptyset.$$

There are no singularities to propagate.

In the remainder of the chapter we'll answer the above Question, and explain more generally what is going on in the above examples.

The next proposition explains how singularities of $u \in \mathcal{D}'(\Omega)$ can be detected using the Fourier transform. We showed in Proposition 8.83 that if $u \in \mathcal{E}'(\mathbb{R}^n)$ (distributions with compact support), then $\hat{u}(\xi)$ is C^∞ with polynomial growth in ξ.

Proposition 6.4. *Let $u \in \mathcal{D}'(\Omega)$ and suppose $x_0 \in \Omega$. Then $x_0 \notin$ sing supp $u \Leftrightarrow$ there exists $\phi \in \mathcal{D}(\Omega)$ with $\phi(x_0) \neq 0$ such that* [2]

$$\text{for all } N \text{ there exists } C_N \text{ such that } |\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \mathbb{R}^n. \tag{6.3}$$

Proof. **1. ($\Rightarrow$).** If $x_0 \notin$ sing supp u, then there exists ω with $x_0 \in \omega \subset \Omega$ such that for any $\phi \in \mathcal{D}(\omega)$ we have $\phi u \in \mathcal{D}(\omega)$. Then $\widehat{\phi u} \in \mathcal{S}(\mathbb{R}^n)$, so (6.3) holds.[3]

2. ($\Leftarrow$). Suppose there exists $\phi \in \mathcal{D}(\Omega)$ with $\phi(x_0) \neq 0$ such that (6.3) holds. By the Fourier inversion formula we have

$$\phi u(x) = \int e^{ix\xi}\, \widehat{\phi u}(\xi)\, d'\xi.$$

By differentiation under $\int$ we see that $\phi u \in C^\infty$. Hence u is C^∞ near x_0.

□

To motivate the definition of wavefront set, WFu, we rephrase Proposition 6.4:

$x_0 \in$ sing supp $u \Leftrightarrow$ for all $\phi \in \mathcal{D}(\Omega)$ with $\phi(x_0) \neq 0$, (6.3) fails.

Roughly speaking, failure of (6.3) means there are "bad directions" ξ_0 near which $\widehat{\phi u}(\xi)$ is *not* rapidly decreasing.

Definition 6.5 (Wavefront set of u). *Suppose $u \in \mathcal{D}'(U)$ and $(x_0, \xi_0) \in \Omega \times \mathbb{R}^n \setminus \{0\}$. Then*

$$\begin{aligned} (x_0,\xi_0) \notin WFu \Leftrightarrow\ & \text{there exists } \phi \in \mathcal{D}(\Omega) \text{ with } \phi(x_0) \neq 0, \\ & \text{and there exists a conic neighborhood } \Gamma \ni \xi_0 \text{ such that} \\ & \text{for all } N \text{ there exists } C_N \text{ such that } |\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \Gamma. \end{aligned} \tag{6.4}$$

[2] When $\widehat{\phi u}$ satisfies (6.3) for $\xi \in \Gamma$, we say it is *rapidly decaying* in Γ.

[3] As usual we can regard distributions with compact support in Ω as distributions on $\mathbb{R}^n$.

Thus, the wavefront set of u is a microlocal refinement of the singular support of u. The next proposition makes that precise.

Proposition 6.6. *Define the map* $\pi : \Omega \times (\mathbb{R}^n \setminus \{0\}) \to \Omega$ *by* $\pi(x,\xi) = x$. *If* $u \in \mathcal{D}'(\Omega)$, *then*

$$\pi(WFu) = \text{sing supp } u. \tag{6.5}$$

We will use the result of the following exercise in the proof.

Exercise 6.1a: Suppose $\phi \in \mathcal{D}(\Omega)$ with $\phi(x_0) \neq 0$ and there exists a conic neighborhood $\Gamma \ni \xi_0$ such that

$$\text{for all } N \text{ there exists } C_N \text{ such that } |\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \Gamma. \tag{6.6}$$

Suppose also that $\psi \in \mathcal{D}(\Omega)$ with $\text{supp } \psi \subset\subset \{x \in \Omega : \phi(x) \neq 0\}$. Then (6.6) holds with $\widehat{\psi u}$ in place of $\widehat{\phi u}$. Hint: Choose $\tilde{\phi} \in C^\infty(\Omega)$ such that $\tilde{\phi}\phi = 1$ on a neighborhood of supp ψ. Then $\psi u = \psi\tilde{\phi}\phi u = \beta\phi u$, where $\beta \in \mathcal{D}(\Omega)$. Next write $\widehat{\psi u}(\xi) = (\widehat{\phi u} * \hat{\beta})(\xi)$ and estimate the convolution for $\xi \in \Gamma$.

Proof of Proposition 6.6. **1. ($\subset$).** Suppose $x_0 \notin SSu$. Then there exists an open $\omega \ni x_0$ such that $u \in C^\infty(\omega)$. For any $\phi \in \mathcal{D}(\omega)$ we have $\phi u \in \mathcal{D}(\omega)$, so

$$\text{for all } N, \exists C_N \text{ such that } |\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \mathbb{R}^n \setminus \{0\}. \tag{6.7}$$

Thus $(x_0, \xi) \notin WFu$ for all $\xi \neq 0$. Hence $x_0 \notin \pi(WFu)$.

2. ($\supset$). Suppose $x_0 \notin \pi(WFu)$. Then for any $\xi_0 \in \mathbb{R}^n \setminus \{0\}$ there exists $\phi \in \mathcal{D}(\Omega)$ with $\phi(x_0) \neq 0$ and a conic neighborhood $\Gamma \ni \xi_0$, such that

$$\text{for all } N \text{ there exists } C_N \text{ such that } |\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \Gamma. \tag{6.8}$$

Using the compactness of S^{n-1}, cover $\mathbb{R}^n \setminus \{0\}$ with conic neighborhoods $\Gamma_i \ni \xi_{0,i}, i = 1, \dots M$ for which there exist $\phi_i \in \mathcal{D}(\Omega)$ with $\phi_i(x_0) \neq 0$ such that the obvious analogue of (6.8), call it $(6.8)_i$, holds for each i. Choose $\psi \in \mathcal{D}(\Omega)$ with $\psi(x_0) \neq 0$ such that

$$\text{supp } \psi \subset\subset \cap_{i=1}^M \{x \in \Omega : \phi_i(x) \neq 0\}.$$

Then the collection of estimates $(6.8)_i$ for $i = 1, \dots, M$ together with Exercise 6a imply

$$\text{for all } N, \exists C_N \text{ such that } |\widehat{\psi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \text{ for all } \xi \in \mathbb{R}^n \setminus \{0\}.$$

Hence $\psi u \in C^\infty(\Omega)$, which implies $x_0 \notin SSu$. □

Exercise 6.1b: (a) Let $\delta \in \mathcal{D}'(\mathbb{R}^n)$ be given by $\delta(\phi) = \phi(0)$ for $\phi \in \mathcal{D}(\mathbb{R}^n)$. Show that $WF\delta = \{(0,\xi) : \xi \in \mathbb{R}^n \setminus \{0\}\}$.

(b) Let $u \in \mathcal{D}'(\mathbb{R}^2)$ be given by $u(x,y) = \begin{cases} 1, & y \geq x \\ 0, & y < x \end{cases}$. Show that

$$WFu = \{(x, x, \xi, -\xi) : x \in \mathbb{R}, \xi \in \mathbb{R} \setminus \{0\}\}.$$

6.2 Microlocal elliptic regularity

Definition 6.7. *(a) For $P(x,D)$ as in (6.1), we define the* principal symbol of P *to be*

$$p_m(x,\xi) = \sum_{|\alpha|=m} c_\alpha(x)\xi^\alpha. \tag{6.9}$$

(b) Let $(x_0,\xi_0) \in \Omega\times(\mathbb{R}^n\setminus\{0\})$. We say that $P(x,D)$ is elliptic at (x_0,ξ_0) *if $p_m(x_0,\xi_0)\neq 0$. We say $P(x,D)$ is* elliptic *if it is elliptic at all $(x,\xi)\in\Omega\times\mathbb{R}^n\setminus\{0\}$.*

The next Theorem is a microlocal refinement of the earlier elliptic regularity result, Corollary 3.34.

Theorem 6.8 (Microlocal elliptic regularity)**.** *Let*

$$P(x,D) = \sum_{|\alpha|\le m} c_\alpha(x)D^\alpha, \quad c_\alpha \in C^\infty(\Omega).$$

Suppose $u\in\mathcal{D}'(\Omega)$ satisfies $P(x,D)u\in C^\infty(\Omega)$, and let $(x_0,\xi_0)\in\Omega\times(\mathbb{R}^n\setminus\{0\})$. Then

$$p_m(x_0,\xi_0)\neq 0 \Rightarrow (x_0,\xi_0)\notin WFu. \tag{6.10}$$

With Proposition 6.6 the following Corollary of Theorem 6.8 is immediate.

Corollary 6.9. *Let $P(x,D)$ be as in Definition 6.7, and suppose $u\in\mathcal{D}'(\Omega)$ satisfies $Pu\in C^\infty(\Omega)$. If $P(x,D)$ is elliptic, then $u\in C^\infty(\Omega)$.*

Remark 6.10. Here is a *strategy* for proving Theorem 6.8. Try to use the ellipticity of P at (x_0,ξ_0) to "invert P near (x_0,ξ_0) modulo a smoothing operator". This means finding an operator Q such that "$QP=I$ near (x_0,ξ_0)" in the sense that

$$v\in\mathcal{D}'(\Omega)\Rightarrow (x,\xi)\notin WF(QP-I)v$$

for x near x_0 and ξ in some conic neighborhood of ξ_0. In particular then, $(x,\xi)\notin WF(QP-I)u$ for such (x,ξ). Since $Pu\in C^\infty$, we expect this to imply $(x_0,\xi_0)\notin WFu$.

Carrying out the details of this strategy is nontrivial. The operator Q turns out to be a "pseudodifferential operator" of order $-m$, and the strategy is most readily implemented after a calculus of such operators has been developed; see [Taylor]. Instead, we will use this strategy to give a proof, using tools we already have, of a simpler version of Theorem 6.8 where $P(x,D)=p_m(D)$. That is, we'll assume $P(x,D)$ is homogeneous of degree m with constant coefficients.

Proof of Theorem 6.8 when $P(x,D)=p_m(D)$. **1.** For $x_0\in\Omega$ we'll show that if $p_m(\xi_0)\neq 0$, then $(x_0,\xi_0)\notin WFu$. We can reduce to the case $u\in\mathcal{E}'(\mathbb{R}^n)$ as follows. Let $\omega\subset\subset\Omega$ and choose $\chi\in\mathcal{D}(\Omega)$ such that $\chi=1$ on ω. Then

$$p_m(D)(\chi u) = \chi p_m(D)u + [p_m(D),\chi]u. \tag{6.11}$$

We have $\chi p_m(D)u \in C^\infty(\Omega)$ and $[p_m(D), \chi]u = 0$ in ω, since χ is constant on ω. Thus, we have $\chi u \in \mathcal{E}'(\mathbb{R}^n)$ and $p_m(D)(\chi u) \in C^\infty(\omega)$.

2. Since $p_m(\xi_0) \neq 0$, we can choose $q \in C^\infty(\mathbb{R}^n)$ and positively homogeneous of degree $-m$ in $\{|\xi| \geq 1\}$ such that for some conic neighborhood $\Gamma \ni \xi_0$ we have[4]

$$q(\xi)p_m(\xi) = 1 \text{ in } \Gamma \cap \{|\xi| \geq 1\}. \tag{6.12}$$

Equivalently, we have

$$r(\xi) := 1 - q(\xi)p_m(\xi) = 0 \text{ in } \Gamma \cap \{|\xi| \geq 1\}, \tag{6.13}$$

where $r \in C^\infty(\mathbb{R}^n)$ and is positively homogeneous of degree 0 in $|\xi| \geq 1$.

Recall that $p_m(D)u = \mathcal{F}^{-1}(p_m(\xi)\hat{u}(\xi))$. Similarly, for $v \in \mathcal{E}'(\mathbb{R}^n)$ *define*[5]

$$q(D)v = \mathcal{F}^{-1}(q(\xi)\hat{v}(\xi)). \tag{6.14}$$

Thus, we have

$$(q(D) \circ p_m(D))u = \mathcal{F}^{-1}(qp_m\hat{u}). \tag{6.15}$$

3. We now implement the strategy in Remark 6.10. From (6.13) setting $f := p_m(D)u$ we obtain

$$u - q(D)p_m(D)u = r(D)u \Leftrightarrow u = r(D)u + q(D)f. \tag{6.16}$$

Since f is C^∞ near x_0, Proposition 8.84 shows that $q(D)f$ is C^∞ near x_0. Thus, there is a neighborhood $\omega_1 \ni x_0$ such that for any $\phi \in \mathcal{D}(\omega_1)$ we have $\phi q(D)f \in \mathcal{D}(\omega_1)$. Thus, $\widehat{\phi q(D)f}$ is rapidly decaying in all directions.

From (6.16) we have

$$\phi u = \phi r(D)u + \phi q(D)f. \tag{6.17}$$

To finish we will show that for Γ as in (6.13), if $\Gamma' \ni \xi_0$ is a slightly smaller conic neighborhood such that

$$\Gamma' \cap S^{n-1} \subset\subset \Gamma \cap S^{n-1}, \tag{6.18}$$

then

$$\text{for all } N \text{ there exists } C_N \text{ such that } |\widehat{\phi r(D)u}(\xi)| \leq C_N\langle\xi\rangle^{-N} \text{ for all } \xi \in \Gamma'. \tag{6.19}$$

[4]The function q satisfies $q(t\xi) = t^{-m}q(\xi)$ for $t \geq 1$ and $|\xi| \geq 1$. One constructs the desired q by multiplying $1/p_m$ by an appropriate cutoff function that is nonvanishing in a conic neighborhood of ξ_0.

[5]We could write (6.14) formally in pseudodifferential operator form as $q(D)u = \int e^{ix\xi}q(\xi)\hat{u}(\xi)d'\xi$. The integral need not be absolutely convergent, since we know only that $\hat{u}$ has polynomial growth in ξ. On the other hand (6.14) is well-defined as a tempered distribution.

Now $\langle\xi\rangle^N \widehat{\phi r(D)u}(\xi) = \int \langle\xi\rangle^N \hat{\phi}(\xi-\eta) r(\eta)\hat{u}(\eta) d'\eta$, and we know that

$$\text{for all } N \text{ there exists } K_N \text{ such that } |\widehat{\phi}(\xi-\eta)| \le K_N \langle \xi-\eta\rangle^{-N} \text{ for all } \xi \in \mathbb{R}^n. \tag{6.20}$$

Thus, (6.19) will follow if we show that for all N there exists C_N such that[6]

$$\int \langle\xi\rangle^N \hat{\phi}(\xi-\eta) r(\eta)\hat{u}(\eta) d'\eta \le C_N \text{ for } \xi \in \Gamma'. \tag{6.21}$$

The inequalities (6.21) follow readily from (6.20) and the fact that for some $c > 0$ we have

$$\begin{aligned} &(a) |\xi-\eta| \ge c|\eta| \\ &(b) |\xi-\eta| \ge c|\xi| \end{aligned} \tag{6.22}$$

for $\xi \in \Gamma'$ and $\eta \in \operatorname{supp} r \cap \{|\eta| \ge 1\}$; here we used (6.13). For example, to see that (6.22)(a) holds, note that it is equivalent to

$$\left| \frac{\xi}{|\eta|} - \frac{\eta}{|\eta|} \right| \ge c.$$

This is clear from (6.18), since $\frac{\xi}{|\eta|} \in \Gamma'$, while $\frac{\eta}{|\eta|}$ lies on the unit sphere outside Γ if $\eta \in \operatorname{supp} r \cap \{|\eta| \ge 1\}$.[7] □

Exercise 6.2a: We have proved Theorem 6.8 in the case where $P(x,D) = \sum_{|\alpha|=m} c_\alpha D^\alpha$, with c_α constant. Extend the proof to the case where

$$P(x,D) = \sum_{|\alpha|\le m} c_\alpha D^\alpha, \ c_\alpha \text{ constant.}$$

Hint: Construct $q(D)$ to have total symbol of the form $q_{-m}(\xi) + q_{-m-1}(\xi) + \cdots + q_{-m-K}(\xi)$ for $K \in \mathbb{N}$ large, where the consecutively chosen q_{-m-j} for $j = 0, \ldots, K$ are positively homogeneous of degree $-m-j$ in $\{|\xi| \ge 1\}$.

6.3 Propagation of wavefront set

When $P(x,D$ is elliptic, Corollary 6.9 tells us that if $u \in \mathcal{D}'(\Omega)$ satisfies $P(x,D)u \in C^\infty(\Omega)$, then $u \in C^\infty(\Omega)$. That is, u can fail to be in $C^\infty(\Omega)$ only if $p_m(x,\xi)$ vanishes for some $(x,\xi) \in \Omega \times (\mathbb{R}^n \setminus \{0\})$.

Definition 6.11 (characteristic set of $P(x,D)$)**.** *The* characteristic set of $P(x,D)$ *is the set*

$$\mathcal{C}_P := \{(x,\xi) \in \Omega \times (\mathbb{R}^n \setminus \{0\}) : p_m(x,\xi) = 0\}. \tag{6.23}$$

[6] Note that $\langle\xi\rangle^N \widehat{\phi r(D)u}(\xi)$ is bounded on bounded sets (why?).
[7] Draw a picture. Also note that if $|\eta| \le 1$, the inequalities (6.22) hold for all $|\xi| \ge 2$.

Thus, Theorem 6.8 is the statement:

$$P(x,D)u \in C^\infty(\Omega) \Rightarrow WFu \subset \mathcal{C}_P. \tag{6.24}$$

We now refine the Question asked earlier.

Refined Question. For $u \in \mathcal{D}'(\Omega)$ such that $Pu \in C^\infty$, if $(x_0,\xi_0) \in WFu$, where else must u have wavefront set; that is, how does wavefront set propagate?

It turns out that WFu propagates along "null bicharacteristic curves of $P(x,D)$", which we now define.

Definition 6.12 (Bicharacteristics)**.** *Let $p_m : \Omega \times (\mathbb{R}^n \setminus \{0\}) \to \mathbb{R}$ be the principal symbol of $P(x,D)$:*

$$p_m(x,\xi) := \sum_{|\alpha|=m} c_\alpha(x)\xi^\alpha.$$

(a) The Hamiltonian vector field of P, *H_{p_m}, is the vector field on $\Omega \times (\mathbb{R}^n \setminus \{0\})$ given by*

$$H_{p_m}(x,\xi) := \sum_{j=1}^n \left(\frac{\partial p_m(x,\xi)}{\partial \xi_j}\partial_{x_j} - \frac{\partial p_m(x,\xi)}{\partial x_j}\partial_{\xi_j} \right). \tag{6.25}$$

(b) A bicharacteristic curve of P *through $(x_0,\xi_0) \in \Omega \times (\mathbb{R}^n \setminus \{0\})$ is a curve $\gamma : I \to (\Omega \times \mathbb{R}^n \setminus \{0\})$, where $I \ni 0$ is an open interval of $\mathbb{R}$, and $\gamma(t) = (x(t),\xi(t))$ satisfies the ODE initial value problem:*

$$\begin{aligned} &\dot\gamma(t) = (\dot x(t), \dot\xi(t)) = (\nabla_\xi p_m(x(t),\xi(t)), -\nabla_x p_m(x(t),\xi(t))) \text{ for } t \in I, \\ &\gamma(0) = (x_0,\xi_0). \end{aligned} \tag{6.26}$$

In other words $\dot\gamma = H_{p_m}$, $\dot\gamma(0) = (x_0,\xi_0)$. We call γ a bicharacteristic, for short.

(c) We say that a bicharacteristic γ is a null bicharacteristic *if $\{\gamma(t) : t \in I\} \subset \mathcal{C}_P$.*

Remark 6.13. Observe that $H_{p_m}p_m = 0$. Thus, p_m is constant along bicharacteristics. In particular, if γ is a bicharacteristic and $\gamma(0) = (x_0,\xi_0) \in \mathcal{C}_P$, then γ is a null bicharacteristic.

Exercise 6.3a: Compute the null bicharacteristic of $\Box = \partial_t^2 - \Delta$ through a point (x_0,t_0,ξ_0,τ_0), where $|\xi_0| = |\tau_0| \neq 0$.

In the mid-1960s Lars Hörmander proved the following beautiful result in answer to the refined question above.

Theorem 6.14 (Propagation of WFu)**.** *Suppose $P(x,D)$ as in (6.1) has* real *principal symbol p_m such that H_{p_m} is never 0 on $\Omega \times (\mathbb{R}^n \setminus \{0\})$. Let $u \in \mathcal{D}'(\Omega)$ and set $f := P(x,D)u$.*

(a) If $f \in C^\infty(\Omega)$, then $(x_0,\xi_0) \in WFu$ implies $\{\gamma(t) : t \in I\} \subset WFu$, where γ is the null bicharacteristic of P through (x_0,ξ_0).

(b) More generally, $WFu \setminus WFf$ is contained in $\mathcal{C}_P$ and is invariant under flow along null bicharacteristics.

For a proof see, for example, [Nir73].

7 Homogeneous distributions, pullbacks, and fundamental solutions

After introducing homogeneous distributions and pullbacks, we apply them to construct fundamental solutions of the wave operator, $\Box$, that are more explicit than the ones constructed using the Fourier transform in section 1.6.

7.1 Homogeneous distributions

A function $f \in L^1_{loc}(\mathbb{R}^n)$ is (positively) homogeneous of degree $a \in \mathbb{C}$, if

$$f(tx) = t^a f(x) \text{ for } t > 0 \text{ and a.e. } x. \tag{7.1}$$

In this case for $\phi \in \mathcal{D}(\mathbb{R}^n)$ we have:

$$\begin{aligned} &t^a \int f(x)\phi(x)dx = \int f(xt)\phi(x)dx = \int f(y)\phi(y/t)t^{-n}dy \text{ for } t > 0 \Leftrightarrow \\ &\langle f, \phi\rangle = t^{-a}\langle f(y), t^{-n}\phi(y/t)\rangle \text{ for } t > 0 \Leftrightarrow \langle f, \phi\rangle = t^a\langle f, t^n\phi(xt)\rangle \text{ for } t > 0. \end{aligned} \tag{7.2}$$

This suggests the following definition.

Definition 7.1. *A distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is said to be* homogeneous of degree $a \in \mathbb{C}$ *if*

$$\langle u, \phi\rangle = t^a\langle u, t^n\phi(xt)\rangle \text{ for } t > 0,\ \phi \in \mathcal{D}(\mathbb{R}^n). \tag{7.3}$$

Examples 7.2. (1) The Dirac delta distribution $\delta = \delta_0 \in \mathcal{D}'(\mathbb{R}^n)$ is homogeneous of degree $-n$.[1]

(2) When $\operatorname{Re} a > -1$, the distribution $x_+^a \in \mathcal{D}'(\mathbb{R})$ given by the locally integrable function

$$x_+^a = \begin{cases} x^a, & x > 0 \\ 0, & x \leq 0 \end{cases} = 1_{x>0}\ x^a \tag{7.4}$$

is homogeneous of degree a. By Exercise 7.1a we have

$$\frac{d}{dx}x_+^a = ax_+^{a-1} \text{ for } \operatorname{Re} a > 0, \tag{7.5}$$

[1] If $b \neq 0$, the distribution δ_b given by $\langle \delta_b, \phi\rangle = \phi(b)$ is not homogeneous.

and we can use this distributional equation to analytically extend x_+^a to $S = \mathbb{C}\backslash\{-1,-2,\dots\}$ as a distribution homogeneous of degree a.[2]

Indeed, for $\phi \in \mathcal{D}(\mathbb{R})$ the function $a \to f_a(\phi) := \langle x_+^a, \phi\rangle$ is analytic in $\operatorname{Re} a > -1$, since $x^a \log x$ is locally integrable on $(0,\infty)$. Moreover, (7.5) means that for $\phi \in \mathcal{D}(\mathbb{R})$:

$$f_a(\phi') = -af_{a-1}(\phi) \text{ for } \operatorname{Re} a > 0 \Leftrightarrow f_a(\phi) = -\frac{f_{a+1}(\phi')}{a+1} \text{ for } \operatorname{Re} a > -1. \tag{7.6}$$

By induction we obtain for $\operatorname{Re} a > -1$ and any $k \in \mathbb{N}$

$$f_a(\phi) = (-1)^k \frac{f_{a+k}(\phi^{(k)})}{(a+1)\cdots(a+k)}. \tag{7.7}$$

Observe that the right side of (7.7) is analytic for $\operatorname{Re} a > -k-1$, except for simple poles at $a = -1,\dots,-k$.[3] Thus, if we *define* $f_a(\phi)$ by (7.7) for such a, we obtain an analytic continuation of $a \to f_a(\phi)$ from $\{a : \operatorname{Re} a > -1\}$ to $\mathbb{C} \setminus -\mathbb{N}$.

We know that for $\operatorname{Re} a > -1$:

$$\langle x_+^a, \phi\rangle = t^a \langle x_+^a, t\phi(xt)\rangle. \tag{7.8}$$

By analytic continuation (7.8) continues to hold for $a \in \mathbb{C} \setminus -\mathbb{N}$.

(3) The factor of a in (7.5) prevented us from defining x_+^{-1}, or in fact x_+^{-k} for $k \in \mathbb{N}$, using the above method. We can use the gamma function to cancel that factor. The gamma function, defined by

$$\Gamma(a) = \int_0^\infty x^{a-1}e^{-x}dx \text{ for } \operatorname{Re} a > 0, \tag{7.9}$$

satisfies

$$\Gamma(a+1) = a\Gamma(a) \text{ for } \operatorname{Re} a > 0, \tag{7.10}$$

so can be analytically extended to $\mathbb{C} \setminus \{0,-1,-2,\dots\}$ with simple poles at $-k \in -\mathbb{N}_0$ by an argument like the one given above.[4] For $\operatorname{Re} a > -1$ define

$$\chi_+^a = \frac{x_+^a}{\Gamma(a+1)}. \tag{7.11}$$

The gamma function has no zeros, so χ_+^a is analytic for $\operatorname{Re} a > -1$. Now (7.6) implies that for $\operatorname{Re} a > 0$ we have

$$\chi_+^a(\phi') = \frac{f_a(\phi')}{\Gamma(a+1)} = \frac{-af_{a-1}(\phi)}{a\Gamma(a)} = -\frac{f_{a-1}(\phi)}{\Gamma(a)} = -\chi_+^{a-1}(\phi). \tag{7.12}$$

[2] This means that for any $\phi \in \mathcal{D}(\mathbb{R})$ we analytically extend the function $a \to f_a(\phi) := \langle x_+^a, \phi\rangle$, and that for any given a in the extended domain the map $\phi \to f_a(\phi)$ is a distribution.

[3] The residue at $a = -k$ is easily computed to be $\lim_{a\to -k}(a+k)(RHS) = \phi^{(k-1)}(0)/(k-1)!$.

[4] The pole at $-k \in -\mathbb{N}_0$ has residue $(-1)^k/k!$.

Thus, by an inductive argument like the one in (2), χ_+^a can be analytically continued to $\mathbb{C}$ so that

$$\frac{d}{dx}\chi_+^a = \chi_+^{a-1} \text{ for all } a \in \mathbb{C}. \tag{7.13}$$

Indeed, parallel to (7.7) we define

$$\chi_+^a(\phi) = (-1)^k \chi_+^{a+k}(\phi^{(k)}) \text{ for } k \in \mathbb{N},\ \operatorname{Re} a > -k-1. \tag{7.14}$$

Moreover, χ_+^a is homogeneous of degree a with support in $t \geq 0$ for all $a \in \mathbb{C}$.

For later use we note that for $k \in \mathbb{N}$:

$$\begin{aligned} &(a)\chi_+^0 = H, \text{ so } \chi_+^{-k} = \delta_0^{(k-1)}; \\ &(b)\chi_+^{-\frac{1}{2}-k}(x) = \frac{d^k}{dx^k}\chi_+^{-\frac{1}{2}} = \frac{1}{\sqrt{\pi}}\frac{d^k}{dx^k}x_+^{-\frac{1}{2}}, \end{aligned} \tag{7.15}$$

since $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Exercise 7.1a. Prove that (7.5) holds in the sense of distributions.

7.2 Pullback of a distribution by a function

Let $u \in \mathcal{D}'(\mathbb{R})$ and $U \subset \mathbb{R}^n$ an open set. We want to define the "composition" of a distribution $u \in \mathcal{D}'(\mathbb{R})$ with a smooth function $f \in C^\infty(U, \mathbb{R})$ as an element "f^*u" $\in \mathcal{D}'(U)$. When $u \in C(\mathbb{R})$, we just take $f^*u = u \circ f \in C(U) \subset \mathcal{D}'(U)$.

Assume now that $f'(x) \neq 0$ for all $x \in U$. Given $x_0 \in U$, suppose $\partial_n f(x_0) \neq 0$, and write $x = (x', x_n)$. By the implicit (or inverse) function theorem, the map $y = h(x)$ given by $y = (y', y_n) := (x', f(x)) = h(x)$ is a C^∞ diffeomorphism $h : \omega \to \Omega$ for some neighborhoods $\omega \ni x_0$ and $\Omega \ni h(x_0)$. Let $g : \Omega \to \omega$ be the inverse of h. If $u \in C(\mathbb{R})$, then for $\phi \in \mathcal{D}(\omega)$ we have

$$\int u(f(x))\phi(x)dx = \int u(y_n)\phi(g(y))|\det g'(y)|dy = \langle u(t), \int \phi(g(y', t))|\det g'(y', t)|dy'\rangle. \tag{7.16}$$

This suggests that for $u \in \mathcal{D}'(\mathbb{R})$, we should define $f^*u|_\omega$ by the right side of (7.16).[5] If we do that, the change of variable formula can be used to show that f^*u is defined consistently on overlapping coordinate neighborhoods. This defines $f^*u \in \mathcal{D}'(U)$ so that $f^*u = u \circ f$ when $u \in C(U)$.

[5] Observe that $\int \phi(g(y', t))|\det g'(y', t)|dy' \in \mathcal{D}(\mathbb{R})$.

A coordinate-independent and more computationally useful way to define $f^*u \in \mathcal{D}'(U)$ can be obtained by noticing that for $\phi \in \mathcal{D}(\omega)$:[6]

$$\int \phi(g(y',t))|\det g'(y',t)|dy' = \partial_t \int_{f(x)<t} \phi(x)dx \tag{7.17}$$

This suggests the following Proposition/Definition:

Proposition 7.3. *Let $u \in \mathcal{D}'(\mathbb{R})$, $f \in C^\infty(U,\mathbb{R})$, and suppose $f'(x) \neq 0$ for all $x \in U$. For $\phi \in \mathcal{D}(U)$ the equation*

$$\langle f^*u, \phi\rangle := \langle u, \partial_t \int_{f(x)<t} \phi(x)dx\rangle \tag{7.18}$$

defines an element $f^*u \in \mathcal{D}'(U)$ called the pullback of u by f. *Moreover, the map $f^* : \mathcal{D}'(\mathbb{R}) \to \mathcal{D}'(U)$ is the unique sequentially continuous map such that $f^*u = u \circ f$ when $u \in C(\mathbb{R})$.*

Proof. **1.** From (7.17) and a partition of unity argument we see that

$$\partial_t \int_{f(x)<t} \phi(x)dx \in \mathcal{D}(\mathbb{R}) \text{ for } \phi \in \mathcal{D}(U).$$

2. For $\phi \in \mathcal{D}(\omega)$ (7.17) implies that

$$\langle f^*u|_\omega, \phi\rangle = \langle u, \partial_t \int_{f(x)<t} \phi(x)dx\rangle = \langle 1(y') \otimes u(y_n), \phi(g(y',y_n))|\det g'(y',y_n)|\rangle. \tag{7.19}$$

The first equality in (7.19) implies that the locally defined distributions $f^*u|_\omega$ piece together to define an element $f^*u \in \mathcal{D}'(U)$.

3. The second equality in (7.19) implies that if $u_j \to u$ in $\mathcal{D}'(\mathbb{R})$, then

$$f^*u_j|_\omega \to f^*u|_\omega \text{ in } \mathcal{D}'(\omega),$$

which in turn implies that $f^*u_j \to f^*u$ in $\mathcal{D}'(U)$.[7]

4. The uniqueness follows from sequential continuity and the fact that if $u \in \mathcal{D}'(\mathbb{R})$, there exists $u_j \in \mathcal{D}(\mathbb{R})$ such that $u_j \to u$ in $\mathcal{D}'(\mathbb{R})$.[8] □

Corollary 7.4. *The derivatives of f^*u can be computed by the chain rule. For example,*

$$\partial_j(f^*u) = (\partial_j f)\ f^*(u'). \tag{7.20}$$

[6] This device is taken from [FJ98].

[7] The last assertion is checked by considering $\langle f^*u_j, \phi\rangle$ for $\phi \in \mathcal{D}(U)$, covering the support of ϕ by a finite number of neighborhoods like ω, and using a partition of unity.

[8] The u_j can be constructed by truncation and regularization: take $u_j = (\chi_j u) * \rho_j$, where χ_j is a suitable smooth cutoff function and (ρ_j) is an approximate identity.

Proof. This is clear for $u \in C^\infty(\mathbb{R})$ and extends to $u \in \mathcal{D}'(\mathbb{R})$ by sequential continuity of f^*. (check) □

Exercise 7.2a. Pullback of a distribution by a diffeomorphism. Let U and V be open subsets of $\mathbb{R}^n$ and suppose $f : U \to V$ is a C^∞ diffeomorphism with inverse $g : V \to U$. Show there is a unique sequentially continuous map $f^* : \mathcal{D}'(V) \to \mathcal{D}'(U)$ such that $f^*u = u \circ f$ when $u \in C(V)$, and that f^*u is given by

$$\langle f^*u, \phi\rangle := \langle u, \phi(g(y))|\det g'(y)|\rangle \text{ for } u \in \mathcal{D}'(V),\ \phi \in \mathcal{D}(U). \tag{7.21}$$

Show that the derivatives of f^*u can be computed by the chain rule.

Examples 7.5. 1. For $a \in \mathbb{C}$ let $u = \chi_+^a \in \mathcal{D}'(\mathbb{R})$ (recall (7.11), (7.14)) and define $f : \left(\mathbb{R}^{n-1}_{x'} \times \mathbb{R}_{x_n}\right) \setminus \{0\} \to \mathbb{R}$ by

$$f(x) = x_n^2 - |x'|^2. \tag{7.22}$$

Then $f'(x) \neq 0$ on $\mathbb{R}^n \setminus \{0\}$, so we can apply Proposition 7.3 to define $f^*\chi_+^a = \text{“}\chi_+^a(x_n^2 - |x'|^2)\text{”} \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\}$ by

$$\langle f^*\chi_+^a, \phi\rangle = \langle \chi_+^a, \partial_t \int_{f(x)<t} \phi(x)dx\rangle \tag{7.23}$$

for $\phi \in \mathcal{D}(\mathbb{R}^n \setminus \{0\})$. We have

$$\begin{aligned}\partial_t \int_{f(x)<t} \phi(x)dx = \partial_t \int \int_{-(|x'|^2+t)_+^{1/2}}^{(|x'|^2+t)_+^{1/2}} \phi(x', x_n)dx_n dx' = \\ \frac{1}{2}\int \phi\left(x', (|x'|^2+t)_+^{1/2}\right)(|x'|^2+t)_+^{-1/2}dx' + \frac{1}{2}\int \phi\left(x', -(|x'|^2+t)_+^{1/2}\right)(|x'|^2+t)_+^{-1/2}dx'\end{aligned} \tag{7.24}$$

Thus, we can write[9]

$$\begin{aligned}&\langle f^*\chi_+^a, \phi\rangle = \langle (f^*\chi_+^a)_+, \phi\rangle + \langle (f^*\chi_+^a)_-, \phi\rangle, \text{ where} \\ &\langle (f^*\chi_+^a)_\pm, \phi\rangle := \langle \chi_+^a, \frac{1}{2}\int \phi\left(x', \pm(|x'|^2+t)_+^{1/2}\right)(|x'|^2+t)_+^{-1/2}dx'\rangle.\end{aligned} \tag{7.25}$$

By Exercise 7.2b the distributions $f^*\chi_+^a$, $(f^*\chi_+^a)_\pm$ are homogeneous of degree $2a$ as elements of $\mathcal{D}'(\mathbb{R}^n \setminus \{0\})$. By Exercise 7.2c we have

$$\begin{aligned}&\text{supp } f^*\chi_+^a \subset \{x : |x_n| \geq |x'|\} \\ &\text{supp } (f^*\chi_+^a)_\pm \subset \{x : |x_n| \geq |x'|, \pm x_n > 0\}.\end{aligned} \tag{7.26}$$

[9] See Remark 7.6.

Some authors write

$$(f^*\chi_+^a)_\pm = H(\pm x_n) f^*\chi_+^a. \tag{7.27}$$

2. Let $n = 4$ and $a = -1$. We have $\chi_+^{-1} = \delta$, so (7.25) can be written

$$\begin{aligned} &\langle \delta(x_4^2 - |x'|^2), \phi\rangle = \frac{1}{2}\int \phi(x', |x'|)|x'|^{-1}dx' + \frac{1}{2}\int \phi(x', -|x'|)|x'|^{-1}dx', \text{ where} \\ &\langle \delta(f)_\pm, \phi\rangle = \frac{1}{2}\int \phi(x', \pm|x'|)|x'|^{-1}dx'. \end{aligned} \tag{7.28}$$

By inspection of (7.28) we see that the homogeneous distributions $\delta(f)_\pm \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ extend to homogeneous distributions of the same degree, namely -2, on $\mathbb{R}^n$.

Exercise 7.2b. Let $a \in \mathbb{C}$.
(a) Show that $f^*\chi_+^a \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ is a homogeneous distribution of degree $2a$.
(b) Show that the distributions $(f^*\chi_+^a)_\pm \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ defined by (7.25) are also homogeneous of degree $2a$.

Exercise 7.2c. For $a \in \mathbb{C}$ it is "formally obvious" that the distribution $f^*\chi_+^a = \chi_+^a(x_n^2 - |x'|^2) \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ has support contained in the closed cone $\{x : |x_n| \geq |x'|\}$, since χ_+^a has support in $t \geq 0$. (a) Prove this carefully using (7.23).
(b) Show that the distributions $(f^*\chi_+^a)_\pm$ have support in the half cones $\{x : |x_n| \geq |x'|, \pm x_n > 0\}$.

Remark 7.6. It is not obvious (to most of us I'd guess) that $(f^*\chi_+^a)_\pm$ as in (7.25) are distributions, or even make sense. For a given $\phi \in \mathcal{D}(\mathbb{R}^n \setminus \{0\})$ although $\partial_t \int_{f(x)<t} \phi(x)dx \in \mathcal{D}(\mathbb{R})$, the functions of t in (7.25) on which χ_+^a acts are not, in fact, smooth functions of $t \in \mathbb{R}$. Using Exercise 7.2b one can show that given $\phi \in \mathcal{D}_K$, where $K \subset\subset \mathbb{R}^n \setminus \{0\}$, the pairing

$$\langle \chi_+^a, \partial_t \int_{f(x)<t} \phi(x)dx\rangle$$

is unchanged if ϕ is replace by $\rho\phi$, where $\rho \in \mathcal{D}(\mathbb{R}^n)$ satisfies $\rho(x) = \begin{cases} 1, & \text{if } |x_n| \geq \frac{3}{4}|x'| \\ 0, & \text{if } |x_n| < \frac{1}{2}|x'| \end{cases}$ for $|x| > c$ if $c = c(K) > 0$ is small enough. If ϕ is replaced by $\rho\phi$ in (7.25), $|x'|^2 + t$ is positive and bounded away from zero on the support of $(\rho\phi)\left(x', (|x'|^2 + t)_+^{1/2}\right)$, so (7.25) makes sense independently of ρ chosen as above.

We state without proof the following result from [Hör00], which will allow us to construct fundamental solutions for $\Box$ on $\mathbb{R}^n$ as extensions of explicit homogenous distributions defined on $\mathbb{R}^n \setminus \{0\}$.

Theorem 7.7. *If $u \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ is homogeneous of degree a, where a is not an integer $\leq -n$, then u has a unique extension $\dot{u} \in \mathcal{D}'(\mathbb{R}^n)$, which is homogeneous of degree a. The map*

$$\mathcal{D}'(\mathbb{R}^n \setminus \{0\}) \ni u \to \dot{u} \in \mathcal{D}'(\mathbb{R}^n)$$

is continuous.

Hörmander gives an explicit formula for the extension $\dot{u}$:

$$\langle \dot{u}, \phi \rangle := \langle u, \psi R_a \phi \rangle, \ \phi \in \mathcal{D}(\mathbb{R}^n), \tag{7.29}$$

where $\psi \in \mathcal{D}(\mathbb{R}^n \setminus \{0\})$ is chosen so that

$$\int_0^\infty \psi(tx) dt/t = 1 \text{ for } x \neq 0$$

and

$$(R_a \phi)(x) := \langle t_+^{a+n-1}, \phi(tx) \rangle, \ \phi \in \mathcal{D}(\mathbb{R}^n), \ x \neq 0.$$

7.3 Fundamental solutions of $\Box$.

In this section we prove (most of) the following theorem after trying to motivate it. Recall that a distribution $E \in \mathcal{D}'(\mathbb{R}^n)$ is a fundamental solution of a constant coefficient operator $P(D)$ if $P(D)u = \delta$.

Theorem 7.8. *For $d \in \mathbb{N}$ let $n = d + 1$, $f(x) = x_n^2 - |x'|^2$, and set* [10]

$$E := c_d \, f^* \chi_+^{-\frac{d-1}{2}}, \ \text{where } c_d = \frac{1}{2} \pi^{\frac{1-d}{2}}. \tag{7.30}$$

The unique fundamental solution of $\Box$ on $\mathbb{R}^n$ with support in the forward cone $x_n \geq |x'|$ is $\dot{E}_+ \in \mathcal{D}'(\mathbb{R}^n)$, the unique homogeneous extension of degree $-\frac{d-1}{2}$ to $\mathbb{R}^n$ of E_+, where

$$E_\pm = c_d \left(\chi_+^{-\frac{d-1}{2}} (x_n^2 - |x'|^2) \right)_\pm \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\}) \tag{7.31}$$

are defined in (7.25). Here recall that

$$\chi_+^{-\frac{d-1}{2}} = \begin{cases} \delta^{\left(\frac{d-3}{2}\right)} & \text{for } d \text{ odd} \\ \frac{t_+^{-\frac{d-1}{2}}}{\Gamma(\frac{3-d}{2})} & \text{for } d \text{ even} \end{cases}. \tag{7.32}$$

We refer to $\dot{E}_+$ as the forward fundamental solution *of $\Box$.*

[10] The constant c_d is computed in [Hör00], Theorem 6.2.1 and [Oh14], Appendix B. We won't compute c_d here, but see step **2** of the proof of Theorem 7.8, where we show that c_0 in (7.47) is nonzero.

Remark 7.9. (1) From Theorem 7.8 we read off the following support properties of E_+:

$$\begin{aligned} \operatorname{supp}(\dot{E}_+) &= \{x \in \mathbb{R}^n : x_n = |x'|\} \text{ for } d = n-1 \text{ odd} \\ \operatorname{supp}(\dot{E}_+) &= \{x \in \mathbb{R}^n : x_n \geq |x'|\} \text{ for } d \text{ even.} \end{aligned} \tag{7.33}$$

(2) The unique backward fundamental solution of $\Box$ is $\dot{E}_-$, the unique homogeneous extension to $\mathbb{R}^n$ of

$$E_- := c_d \left(\chi_+^{-\frac{d-1}{2}}(x_n^2 - |x'|^2)\right)_- . \tag{7.34}$$

The support properties of $\dot{E}_-$ are the obvious backward analogues of those in (7.33). See Exercise 7.3b.

(3) Exercise 7.3b shows that $\dot{E}/2$, where $E = \chi_+^{-\frac{d-1}{2}}(x_n^2 - |x'|^2)$, is a fundamental solution of $\Box$ with support in $\{|x_n| \geq |x'|\}$.

In section 1.2.1 when we constructed a fundamental solution Φ of Δ, we were motivated by the invariance of Δ under the orthogonal group $O(n)$, that is, by the property

$$\Delta(\phi \circ R) = (\Delta\phi) \circ R \text{ for } R \in O(n), \tag{7.35}$$

to look for a radially symmetric distribution of the form $\Phi(x) = v(|x|)$, for some function $v : (0,\infty) \to \mathbb{R}$. Such a Φ is invariant under $R \in O(n)$ in the sense that the pullback satisfies $R^*\Phi = \Phi \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$. The condition $\Delta\Phi = \delta$ yielded an ODE for $v = v(r)$ in $r > 0$, and we constructed v as a homogeneous solution of that ODE of degree $-n+2$: $v(r) = c_n|r|^{-n+2}$ for $n > 2$. This homogeneity is to be expected from the equation $\Delta\Phi = \delta$, since $\delta \in \mathcal{D}'(\mathbb{R}^n)$ is homogeneous of degree $-n$. Note that we first constructed $\Phi \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ and then observed that it extended to a homogeneous distribution on $\mathbb{R}^n$ of degree $-n+2$.

Let's try to carry out an analogous construction for $\Box$. The group $O(n)$ is the group of linear maps that preserve the quadratic form $x \cdot x$ for $x \in \mathbb{R}^n$. The analogue in the study of the wave operator is the Lorentz group, $O(1,d)$, of linear maps of $\mathbb{R}^{d+1}$ that preserve the quadratic form[11]

$$f(x', x_n) = x_n^2 - |x'|^2.$$

By Exercise 7.3a the fact that $L^*f = f \circ L = f$ for $L \in O(1,d)$, implies the invariance property

$$\Box(\phi \circ L) = (\Box\phi) \circ L. \tag{7.36}$$

This suggests looking for a fundamental solution E such that $L^*E = E$ in $\mathbb{R}^n \setminus \{0\}$, $n = 1+d$.[12] If we take $E = f^*\chi$ for some $\chi \in \mathcal{D}'(R)$, then

$$L^*E = L^*(f^*\chi) = (f \circ L)^*\chi = f^*\chi = E.$$

[11] The Lorentz group is generated by reflections, Lorentz boosts, and (spatial) rotations.

[12] Observe that if E satisfies $\Box E = \delta$, then since $\det L = \pm 1$, we have:

$$\begin{aligned} \langle \Box L^*E, \phi\rangle = \langle L^*E, \Box\phi\rangle = \langle E, (\Box\phi)\circ L^{-1}\rangle = \langle E, \Box(\phi \circ L^{-1})\rangle = \langle \Box E, \phi \circ L^{-1}\rangle = \\ \langle L^*(\Box E), \phi\rangle = \langle L^*\delta, \phi\rangle = \langle \delta, \phi\rangle. \end{aligned} \tag{7.37}$$

Hence L^*E must also be a fundamental solution of $\Box$.

Observe that $\chi \in \mathcal{D}'(\mathbb{R})$ now plays a role like v above. Next we try to choose χ with the right homogeneity and satisfying an ODE that implies $\Box f^*\chi = 0$ in $\mathbb{R}^n \setminus \{0\}$.

Since $\Box$ is second order and $\delta \in \mathcal{D}'(\mathbb{R}^n)$ has homogeneity $-n$, the right homogeneity for E is $-n+2 = -(d-1)$, so the right homogeneity for χ is $-(d-1)/2$.

To find χ we use Corollary 7.4 to compute

$$\begin{aligned} &\Box(\chi \circ f) = \partial_{x_n}(2x_n\, \chi' \circ f) + \\ &\quad 4x_n^2\, \chi'' \circ f + 2\chi' \circ f - 4|x'|^2\, \chi'' \circ f + 2d\, \chi' \circ f = 4f\, \chi'' \circ f + 2(d+1)\, \chi' \circ f. \end{aligned} \tag{7.38}$$

Thus, $\chi \in \mathcal{D}'(\mathbb{R})$ should satisfy the ODE

$$2t\chi'' + (d+1)\chi' = 0 \text{ on } \mathbb{R}. \tag{7.39}$$

Let's solve this first in $t > \epsilon > 0$. Set $u = \chi'$ to obtain

$$2tu' + (d+1)u = 0, \tag{7.40}$$

which we solve by separation of variables to obtain $u = Ct^{-(\frac{d+1}{2})}$.Thus, $\chi = Ct^{-(\frac{d-1}{2})} + C'$, and we take $C' = 0$ to have the right homogeneity. This computation tells us we should be looking for a distribution $\chi \in \mathcal{D}'(\mathbb{R})$ that is homogeneous of degree $-\frac{d-1}{2}$ and which agrees with $Ct^{-(\frac{d-1}{2})}$ in $t > \epsilon > 0$. Based on Theorem 1.75, which was proved by energy methods, we expect the support of solutions to the wave equation to propagate at speed at most one. This suggests looking for a fundamental solution $f^*\chi$ with support in the wave cone $\{|x_n| \geq |x'|\}$. In that case $\chi \in \mathcal{D}'(\mathbb{R})$ should have support in $t \geq 0$.

Summarizing, we have derived the following constraints on $\chi \in \mathcal{D}'(\mathbb{R})$:

(a) χ satisfies (7.39);
(b) $\chi(t) = Ct^{-(\frac{d-1}{2})}$ in $t > \epsilon > 0$ for some constant C;
(c) $\operatorname{supp}(\chi) \subset \{t \geq 0\}$.

When d is even, we might expect the distribution $t_+^{-(\frac{d-1}{2})}$ to satisfy (a)-(c), but when d is odd it clearly does not.[13] Instead, when d is odd, we can try to use the distribution $\chi_+^{-(\frac{d-1}{2})}$. Since

$$\chi_+^{-(\frac{d-1}{2})} = \frac{t_+^{-(\frac{d-1}{2})}}{\Gamma\left(-\frac{d-3}{2}\right)} \text{ when } d \text{ is even},$$

we can try to use $\chi_+^{-(\frac{d-1}{2})}$ for all $d \in \mathbb{N}$. In the next lemma we prove that the distribution $\chi_+^{-\frac{d-1}{2}}$ satisfies the ODE (7.39), and thus satisfies (a)-(c).

Lemma 7.10. *Let $f(x) = x_n^2 - |x'|^2$ be as above and let $\chi := \chi_+^{-\frac{d-1}{2}} \in \mathcal{D}'(\mathbb{R})$ be as defined in (7.11), (7.14) Then χ satisfies the ODE (7.39), and hence the pullback distribution $f^*\chi =$ "$\chi \circ f$" $\in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ satisfies*

$$\Box f^*\chi = 0 \text{ in } \mathbb{R}^n \setminus \{0\}. \tag{7.41}$$

[13] We have not even defined t_+^{-k} as an element of $\mathcal{D}'(\mathbb{R})$ for $k \in \mathbb{N}$.

Proof. Observe that

$$t\chi_+^a(t) = (a+1)\chi_+^{a+1}(t) \text{ for all } a \in \mathbb{C}. \tag{7.42}$$

Indeed, for $\operatorname{Re} a > 0$ we have

$$t\chi_+^a(t) = t\frac{t_+^a}{\Gamma(a+1)} = \frac{t_+^{a+1}}{\Gamma(a+1)} = (a+1)\frac{t_+^{a+1}}{\Gamma(a+2)} = (a+1)\chi_+^{a+1}(t), \tag{7.43}$$

so (7.42) holds for all a by analytic continuation. Since

$$\chi' = \chi_+^{-\frac{d+1}{2}} \text{ and } \chi'' = \chi_+^{-\frac{d+3}{2}},$$

we can apply (7.42) with $a = -\frac{d+3}{2}$ to obtain

$$2t\,\chi'' = -2\,\frac{d+1}{2}\,\chi' = -(d+1)\,\chi', \tag{7.44}$$

so (7.39) holds.

□

We have constructed $E := f^*\chi_+^{-\frac{d-1}{2}} = \text{“}\chi_+^{-\frac{d-1}{2}}(x_n^2 - |x'|^2)\text{”} \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ such that (7.41) holds. We now finish the proof of Theorem 7.8 by showing that $\dot{E}_+$ has the stated properties.

End of the proof of Theorem 7.8. **1.** Write $E = E_+ + E_-$ for $E_\pm$ as in (7.31). Since $\Box E = 0 \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ and $E_\pm$ have disjoint supports, we deduce that $\Box E_\pm = 0$ on $\mathbb{R}^n \setminus \{0\}$. Thus, if $\dot{E}_\pm$ denotes the homogeneous extension of $E_\pm$ to $\mathcal{D}'(\mathbb{R}^n)$, we must have

$$\operatorname{supp} \Box \dot{E}_\pm \subset \{0\}. \tag{7.45}$$

By the result in Examples 1.33(1) we conclude in particular that for some $m \in \mathbb{N}_0$ and constants c_α:

$$\Box \dot{E}_+ = \sum_{|\alpha| \le m} c_\alpha \partial^\alpha \delta. \tag{7.46}$$

Since $\Box \dot{E}+$ is homogeneous of degree $(-n+2) - 2 = -n$, and the only term on the right with that homogeneity is $c_0\delta$, we must have

$$\Box \dot{E}_+ = c_0\delta. \tag{7.47}$$

2. It remains to check that $c_0 \neq 0$ in (7.47) and that $\dot{E}_+$ is the unique fundamental solution of $\Box$ with support in the forward light cone.[14] If $c_0 = 0$, Proposition 7.11 implies $\dot{E}_+ = 0$, contradiction. The uniqueness of $\dot{E}_+$ follows from Proposition 7.12 by taking $v = \delta$.

□

[14] The choice of c_d in (7.31) makes $c_0 = 1$.

Proposition 7.11. *Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and suppose $\Box u = 0$ in $\mathbb{R}^n$, $\operatorname{supp} u \subset \{x_n \geq |x'|\}$. Then $u = 0$.*

Proof. We give a proof "by duality". We'll show that $\langle u, \psi\rangle = 0$ for all $\psi \in \mathcal{D}(\mathbb{R}^n)$.

1. Fix $\psi \in \mathcal{D}(\mathbb{R}^n)$ and choose $T \in \mathbb{R}$ such that $\operatorname{supp} \psi \subset \{x_n < T\}$. Use the Fourier transform to solve the backward problem [15]

$$\Box\phi = \psi \text{ in } \mathbb{R}^n, \phi = 0 \text{ in } x_n > T. \tag{7.48}$$

Then $\phi \in C^\infty(\mathbb{R}^n)$ but is not necessarily compactly supported.

2. Observe that $K := \{x : x_n \leq T\} \cap \{x : x_n \geq |x'|\}$ is compact. Choose $\rho \in \mathcal{D}(\mathbb{R}^n)$ such that $\rho = 1$ in a neighborhood of the compact set $K \cup \operatorname{supp} \psi$. Now observe that

$$0 = \langle \Box u, \rho\phi\rangle = \langle u, \Box(\rho\phi)\rangle = \langle u, \rho\Box\phi\rangle + \langle u, \beta\rangle = \langle u, \psi\rangle + \langle u, \beta\rangle, \tag{7.49}$$

where $\beta \in \mathcal{D}(\mathbb{R}^n)$ satisfies $\operatorname{supp} \beta \cap \operatorname{supp} u = \emptyset$.(why?) Thus $\langle u, \beta\rangle = 0$, so we conclude $\langle u, \psi\rangle = 0$ for all $\psi \in \mathcal{D}(\mathbb{R}^n)$.

□

Proposition 7.12. *Let $v \in \mathcal{D}'(\mathbb{R}^n)$ have support in $\{x \in \mathbb{R}^n : x_4 \geq 0\}$. Then $\Box u = v$ has a unique solution $u \in \mathcal{D}'(\mathbb{R}^n)$ with support in $x_n \geq 0$.*[16]

Proof of Proposition 7.12. **1.** Clearly, $\dot{E}_+ * v \in \mathcal{D}'(\mathbb{R}^n)$ is a solution with support in $x_n \geq 0$; here we used Proposition 8.85 and Exercise 8.19a.

2. If $u \in \mathcal{D}'(\mathbb{R}^n)$, then $\Box u = \delta * \Box u = (\Box\delta) * u$.[17] Set $k := \Box\delta$. Then, if $\Box u_1 = \Box u_2 = v$, we have $k * u_1 = k * u_2 = v$. So (using Proposition 8.85) we obtain

$$u_1 = \delta * u_1 = (k * \dot{E}_+) * u_1 = \dot{E}_+ * (k * u_1) = \dot{E}_+ * (k * u_2) = \delta * u_2 = u_2. \tag{7.50}$$

□

Remark 7.13. In the proof of Proposition 7.11 we used solvability of the backward adjoint problem (here $\Box = \Box^*$) to prove uniqueness for the original forward problem. In the proof of Theorem 2.10 we used uniqueness for the backward adjoint problem, in the form of the estimate (2.46), to prove existence for the original forward problem.

Exercise 7.3a. Let $f(x) = x_n^2 - |x'|^2$. Show that for $L \in O(1, d)$, the fact that $L^* f = f \circ L = f$ implies the invariance property

$$\Box(\phi \circ L) = (\Box\phi) \circ L. \tag{7.51}$$

[15] The equation $\Box\phi = \psi$ transforms to the ODE $\partial_n^2\hat{\phi}(\xi', x_n) + |\xi'|^2\hat{\phi}(x', x_n) = \hat{\psi}(\xi', x_n)$. The solution is $\hat{\phi}(\xi', x_n) = \int_T^{x_n} \frac{\sin(x_n - s)|\xi'|}{|\xi'|}\hat{\psi}(\xi', s)ds$. (check)

[16] This is Theorem 7.3.2 in [FJ98].

[17] We used Exercise 8.19a here.

Hint: Show first that $L^t \eta L = \eta$, where $\eta = \mathrm{diag}(-1, \dots, -1, 1)$. Then use the chain rule.

Exercise 7.3b. Show that $E = f^* \chi_+^{-\frac{d-1}{2}} \in \mathcal{D}'(\mathbb{R}^n \setminus \{0\})$ extends to a distribution $\dot{E} \in \mathcal{D}'(\mathbb{R}^n)$ that satisfies $\Box \frac{\dot{E}}{2} = \delta$. In other words, $\dot{E}/2$ is also a fundmental solution of $\Box$. Hint: Let $R \in O(1, d)$ be the reflection map $R(x', x_n) = (x', -x_n)$. First show that $R^* E_+ = E_-$ and deduce $R^* \dot{E}_+ = \dot{E}_-$.

8 Supplementary material

In this chapter we gather, sometimes without proof, several results used in the main text. These include mainly standard results (from functional analysis, for example) that are needed in the proofs, and other more technical results used in the proofs. In sections 8.8 and 8.9, we explore a topic from the main text in more detail.

8.1 Existence and uniqueness for ODEs

Theorem 8.1. *Consider the $n \times n$ system of ODEs*

$$\begin{aligned} \dot{x} &= f(t,x) \\ x(0) &= x_0 \in \mathbb{R}^n, \end{aligned} \tag{8.1}$$

where for some $T > 0$, $f \in C([-T,T] \times \Omega, \mathbb{R}^n)$ and there exists $L > 0$ such

$$|f(t,x) - f(t,y)| \leq L|x-y| \text{ for all } (t,x),(t,y) \in [-T,T] \times \Omega. \tag{8.2}$$

Let $K \subset \Omega$ be compact. Then there exists $\delta = \delta(K)$ such that for all $x_0 \in K$ there exists a unique solution $x \in C^1([-\delta,\delta],\mathbb{R}^n)$ of (8.1).

8.2 Differentiation under the integral sign

Proposition 8.2. *Let $\phi(x) := \int_V f(x,y)dy$, where we assume*
(i) $f(x,y) \in C^1(I_x \times V_y)$, where $V \subset \mathbb{R}^m$ is Lebesgue measurable,
(ii) $\int_V |f(x,y)|dy < \infty$ for every $x \in I$,
(iii) there exists $g(y) \in L^1(V)$ such that $|f_x(x,y)| \leq g(y)$ for every $x \in I$.
Then we have $\phi'(x) = \int_V f_x(x,y)dy$.

Proof. We have

$$\begin{aligned} \frac{\phi(x+h)-\phi(x)}{h} = \int_V \frac{f(x+h,y)-f(x,y)}{h}dy = \\ \int_V f_x(c_h(x,y),y)dy \to \int_V f_x(x,y)dy \text{ as } h \to 0. \end{aligned} \tag{8.3}$$

We used the mean value theorem for the second equality and the dominated convergence theorem to take the limit as $h \to 0$.

□

8.3 Gauss-Green theorem

This theorem allows us to integrate by parts on certain bounded subsets of $\mathbb{R}^n$.

Definition 8.3. *Let $U \subset \mathbb{R}^n$ be a bounded open set and $k \in \mathbb{N}_0$. We say ∂U, the boundary of U, is C^k if for all $x_0 \in \partial U$ there exists $r > 0$ and a C^k function $\rho : \mathbb{R}^{n-1} \to \mathbb{R}$ such that, after choosing new axes if necessary,*

$$U \cap B(x_0, r) = \{x \in B(x_0, r) : x_n > \rho(x_1, \ldots, x_{n-1}).$$

Since $x_n = \rho(x_1, \ldots, x_{n-1})$ defines ∂U near x_0, we call ρ a defining function *for ∂U.*

Let's write $x = (x', x_n)$. Observe that if ∂U is C^k, where $k \geq 1$, there is a well-defined outward unit normal vector to ∂U at each $x \in \partial U$, namely

$$\nu(x) := \frac{(\nabla_{x'}\rho(x), -1)}{|(\nabla_{x'}\rho(x), -1)|}.$$

Theorem 8.4. *Assume $U \subset \mathbb{R}^n$ is bounded and open with a C^1 boundary. Let*

$$u \in C^1(\overline{U}) := \{u \in C^1(U) : \partial^\alpha u \text{ extends continuously to } \overline{U} \text{ for each } |\alpha| \leq 1\}.$$

Then

$$\int_U u_{x_i} dx = \int_{\partial U} u\nu_i dS. \tag{8.4}$$

Here $\nu = (\nu_1, \ldots, \nu_n)$ is the outward unit normal to ∂U, and the surface measure dS is locally given by

$$dS = \sqrt{1 + |\nabla_{x'}\rho|^2}\, dx', \tag{8.5}$$

when ρ is a defining function for ∂U.[1]

Using the above notation, we obtain the following corollaries.

Corollary 8.5. *Let $u, v \in C^1(\overline{U})$. Then*

$$\int_U u_{x_i} v\, dx = -\int_U u v_{x_i} dx + \int_{\partial U} uv\nu_i dS.$$

Proof. Write $(uv)_{x_i} = u_{x_i}v + uv_{x_i}$ and apply Theorem 8.4. □

Corollary 8.6. *Let $u, v \in C^2(\overline{U})$. Then*
(i) $\int_U \Delta u\, dx = \int_{\partial U} \frac{\partial u}{\partial \nu} dS$, *where* $\frac{\partial u}{\partial \nu} = \nabla u \cdot \nu$.

(ii) $\int_U \nabla u \cdot \nabla v dx = -\int_U u\Delta v dx + \int_{\partial U} u\frac{\partial v}{\partial \nu} dS$.

[1] For a proof see Theorem 4.4.2 in [Tay20].

Proof. To prove (ii) apply Corollary 8.5 to get

$$\int_U u_{x_i} v_{x_i} dx = -\int_U u v_{x_i x_i} dx + \int_{\partial U} u v_{x_i} \nu_i dS,$$

and then sum over i.

To prove (i) take $u = 1$ in (ii).

□

Corollary 8.7. *Let $u, v \in C^2(\overline{U})$. Then*

$$\int_U (u\Delta v - v\Delta u) dx = \int_{\partial U} \left(u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu} \right) dS.$$

Proof. Rewrite the equation in Corollary 8.6(ii) with u, v interchanged, and subtract the new equation from the first.

□

8.4 Polar coordinates on $\mathbb{R}^n$

Let $f : \mathbb{R}^n \to \mathbb{R}$ be Lebesgue integrable. Writing $x = r\omega$, where $r \in [0, \infty)$ and $\omega \in S^{n-1}$, we have[2]

$$\int_{\mathbb{R}^n} f(x) dx = \int_0^\infty \int_{S^{n-1}} f(\rho\omega) r^{n-1} dS_1 dr, \tag{8.6}$$

where dS_1 is the standard surface measure on S^{n-1}; recall 8.5. In particular,

$$\int_{S^{n-1}} 1 dS_1 = A(S^{n-1}) = n\alpha(n). \tag{8.7}$$

Example 8.8. *Consider $\int_{|x|<1} \frac{1}{|x|^\alpha} dx$, where $\alpha > 0$. Using (8.6), we see this equals*

$$\int_0^1 \int_{S^{n-1}} \frac{1}{r^\alpha} r^{n-1} dS_1 dr = n\alpha(n) \int_0^1 r^{n-1-\alpha} dr < \infty \Leftrightarrow n - 1 - \alpha > -1 \Leftrightarrow n > \alpha.$$

A similar computation shows $\int_{|x|>1} \frac{1}{|x|^\alpha} dx < \infty \Leftrightarrow n < \alpha$.

8.5 Approximate identities

Definition 8.9 (Friedrichs mollifier)**.** *Let $\eta \in C_c^\infty(\mathbb{R}^n)$ satisfy*

$$\eta \geq 0, \int_{\mathbb{R}^n} \eta dx = 1, \text{ and } \operatorname{supp} \eta \subset \{|x| \leq 1\}.$$

[2] For a proof see [Fol99], Theorem 2.49.

For $\epsilon > 0$ the function $\eta_\epsilon(x) := \frac{1}{\epsilon^n}\eta(\frac{x}{\epsilon})$ is called a Friedrichs mollifier *or an* approximate identity.[3]

Observe that $\int_{\mathbb{R}^n} \eta_\epsilon(x)dx = 1$ and supp $\eta_\epsilon \subset \{|x| \leq \epsilon\}$.

Definition 8.10. *Suppose $f \in L^1_{loc}(U)$ and for $\epsilon > 0$ let $U^\epsilon := \{x \in U : \text{dist}(x, \partial U) > \epsilon\}$. For $x \in U^\epsilon$ define*

$$f^\epsilon(x) := \int_U \eta_\epsilon(x-y)f(y)dy = \int_{B(x,\epsilon)} \eta_\epsilon(x-y)f(y)dy.$$

For $x \in U^\epsilon$ we have $B(x,\epsilon) \subset U$, or equivalently, $x - y \in U$ for $y \in B(0,\epsilon)$. Thus, a change of variable gives

$$f^\epsilon(x) = \int_{B(x,\epsilon)} \eta_\epsilon(x-y)f(y)dy = \int_{B(0,\epsilon)} \eta_\epsilon(y)f(x-y)dy. \tag{8.8}$$

Proposition 8.11. *(i) If $f \in L^1_{loc}(U)$, then $f^\epsilon \in C^\infty(U^\epsilon)$.*
(ii)If $f \in C(U)$ and $V \subset\subset U$, then $f^\epsilon \to f$ uniformly on V as $\epsilon \to 0$.
(iii) If $f \in L^1_{loc}(U)$, then $f^\epsilon \to f$ for almost every $x \in U$ as $\epsilon \to 0$.
(iv) If $1 \leq p < \infty$ and $f \in L^p_{loc}(U)$, then $f^\epsilon \to f$ in $L^p_{loc}(U)$; that is, for all $K \subset\subset U$ we have $|f^\epsilon - f|_{L^p(K)} \to 0$ as $\epsilon \to 0$.

Proof. **(i)** Apply Proposition 8.2 to differentiate under the integral sign in the definition of f^ϵ.

(ii) Fix $V \subset\subset U$ and set $V^\epsilon = V + B(0,\epsilon)$, so $V^\epsilon \subset U$ for small $\epsilon > 0$. If $x \in V$ then for small ϵ we have

$$f^\epsilon(x) - f(x) = \int_{B(0,\epsilon)} \eta_\epsilon(y)[f(x-y) - f(x)]dy. \tag{8.9}$$

Since f is uniformly continuous on $\overline{V}^\epsilon$, given $\delta > 0$ we can choose ϵ small enough so that

$$x, z \in V^\epsilon, |x - z| \leq \epsilon \Rightarrow |f(x) - f(z)| < \delta. \tag{8.10}$$

Then for $x \in V$ (8.9) implies

$$|f^\epsilon(x) - f(x)| \leq \delta \int_{B(0,\epsilon)} \eta_\epsilon(y)dy = \delta. \tag{8.11}$$

(iii) and (iv). See, for example, [Eva10], Appendix. C. □

[3] Peter Lax tells us that the name was suggested to K. Friedrichs by their colleague Donald Flanders, nicknamed "Moll".

The next exercise gives a variant of part (ii) of Proposition 8.11 with a similar proof that we use in the proof of Theorem 1.113. Let $\langle x\rangle := (1+|x|^2)^{1/2}$.

Exercise 8.5a. Suppose $\eta \in C^\infty(\mathbb{R}^n)$ and satisfies for some C_N:

$$\eta \geq 0, \int_{\mathbb{R}^n} \eta dx = 1, \text{ and } |\eta(x)| \leq C_N \langle x\rangle^{-N} \text{ for all } N.$$

For $\epsilon > 0$ let $\eta_\epsilon(x) := \frac{1}{\epsilon^n}\eta(\frac{x}{\epsilon})$. Then if $f \in C(\mathbb{R}^n)$ and is tempered in the sense that for some $p \geq 0$, there exists c_p such that $|f(x)| \leq c_p\langle x\rangle^p$, it follows that:

$$f^\epsilon(x) := \int_{\mathbb{R}^n} \eta_\epsilon(y) f(x-y)dy \text{ is } C^\infty \tag{8.12}$$

and converges to $f(x)$ uniformly on compact sets $K \subset \mathbb{R}^n$ as $\epsilon \to 0$.

Remark 8.12. Observe that if we take $\eta(x) = e^{-|x|^2/2}(2\pi)^{-n/2}$ and set $\epsilon = (2t)^{1/2}$ for $t > 0$ small, then

$$\eta_\epsilon(x) = (4\pi t)^{-n/2} e^{-|x|^2/4t},$$

the heat kernel, is an approximate identity as in Exercise 8.5a.

8.6 Computation of $\int_{\partial\mathbb{R}^n_+} K(x,y)dy$ for the Poisson kernel of $\mathbb{R}^n_+$.

Proposition 8.13. *The Poisson kernel for the upper half space* $\mathbb{R}^n_+$

$$K(x,y) = \frac{2x_n}{n\alpha(n)} \frac{1}{|x-y|^n}, \text{ where } y = (y',0) \in \partial\mathbb{R}^n_+, \tag{8.13}$$

satisfies $\int_{\mathbb{R}^{n-1}} K(x,y)dy' = 1$ *for* $x \in \mathbb{R}^n_+$.

Proof. **1.** Setting $y = (y',0)$, $\tilde{y} = y' - x'$, and $z = \tilde{y}/x_n$, we have

$$\int_{\mathbb{R}^{n-1}} K(x,y)dy' = \frac{2x_n}{n\alpha(n)} \int_{\mathbb{R}^{n-1}} \frac{1}{|(y'-x', x_n)|^n} dy' = \frac{2x_n}{n\alpha(n)} \int_{\mathbb{R}^{n-1}} \frac{1}{|(\tilde{y}, x_n)|^n} d\tilde{y} = \frac{2}{n\alpha(n)} \int_{\mathbb{R}^{n-1}} \frac{dz}{(|z|^2+1)^{n/2}}. \tag{8.14}$$

Now

$$2\int_{\mathbb{R}^{n-1}} \frac{dz}{(|z|^2+1)^{n/2}} = 2\int_0^\infty \int_{S^{n-2}} \frac{r^{n-2}}{(r^2+1)^{n/2}} dS_1 dr = 2(n-1)\alpha(n-1)\int_0^\infty \frac{r^{n-2}}{(r^2+1)^{n/2}} dr. \tag{8.15}$$

Using the substitution $r = \tan\theta$ the last term can be rewritten

$$2(n-1)\alpha(n-1)\int_0^{\pi/2}(\sin\theta)^{n-2}d\theta. \tag{8.16}$$

Thus, to see that $\int_{\partial\mathbb{R}^n_+} K(x,y)dy = 1$, it is enough in view of (8.14) to show that

$$2(n-1)\alpha(n-1)\int_0^{\pi/2}(\sin\theta)^{n-2}d\theta = n\alpha(n), \tag{8.17}$$

or, in other words, that

$$2A(S^{n-2})\int_0^{\pi/2}(\sin\theta)^{n-2}d\theta = A(S^{n-1}), \tag{8.18}$$

where $A(S^{m-1})$ denotes the "area" of the unit sphere in $\mathbb{R}^m$. But this follows easily from

$$A(S^{n-2}_{\sin\theta}) = (\sin\theta)^{n-2}A(S^{n-2}), \tag{8.19}$$

where $S^{n-2}_{\sin\theta}$ denotes the $(n-2)-$sphere of radius $\sin\theta$.[4]

□

Remark 8.14. The equality (8.18) is interesting in its own right, and allows us to compute areas of higher dimensional spheres explicitly.

8.7 Hölder spaces

Let $U \subset \mathbb{R}^n$ be open. Denote by $C^m(\overline{U})$ the space of functions u for which $\partial^\alpha u$ is bounded and uniformly continuous on U for $0 \le |\alpha| \le m$ with the norm

$$|u|_{C^m(\overline{U})} := \sup_{x\in\Omega, |\alpha|\le m} |\partial^\alpha u(x)|. \tag{8.20}$$

Definition 8.15. *Let $0 < \gamma \le 1$. Define $C^{0,\gamma}(\overline{U})$ to be the subspace of $C(\overline{U})$ consisting of functions such that the seminorm*

$$[u]_{C^{0,\gamma}(\overline{U})} := \sup_{x,y\in U, x\neq y} \frac{|u(x)-u(y)|}{|x-y|^\gamma} \tag{8.21}$$

is finite. The Hölder space $C^{m,\gamma}(\overline{U})$ consists of functions $u \in C^m(\overline{U})$ for which the norm

$$|u|_{C^{m,\gamma}(\overline{U})} : |u|_{C^m(\overline{U})} + \sum_{|\alpha|=m} |\partial^\alpha u|_{C^{0,\gamma}(\overline{U})} \tag{8.22}$$

is finite.

[4] Draw a picture in $n = 3$ to see (8.18). Observe that the upper hemisphere of S^2 is "filled out" by circles of radius $\sin\theta$, $\theta \in [0, \pi/2]$, where θ is the polar angle (with the z-axis) in spherical coordinates. A similar filling out of S^{n-1} by S^{n-2} spheres is evident in dimension n.

Exercise 8.7a.
(a) Show that if $u \in C^m(\overline{U})$, then $\partial^\alpha u$ has a unique bounded continuous extension to $\overline{U}$.
(b) Show that it makes no difference if $\sup_{x,y\in U, x\neq y}$ is replaced by $\sup_{x,y\in\overline{U},x\neq y}$ in (8.20) and (8.21).
(c) Suppose U is convex. Show that $C^{m,\gamma}(\overline{U}) \subset C^{m-1,\gamma}(\overline{U})$ if $m \geq 1$.

8.8 Removable singularity theorem for harmonic functions

Let $U \subset \mathbb{R}^n$ be an open set and suppose $x_0 \in U$. Suppose u is harmonic in $U \setminus x_0$. This is the situation in where U is $\mathbb{R}^n$ and $x_0 = 0$. You might have considered the possibility of trying to remove the singularity at 0 (and then applying Liouville's theorem to finish). The following question will then seem natural:

Question. If u *fails* to be extendible to U as a harmonic function, what can we say about the growth of $|u|$ near x_0?

The next theorem provides a sharp answer to this question and is of interest in its own right.[5]

Theorem 8.16. *Suppose $U \subset \mathbb{R}^n$ for $n \geq 3$, $x_0 \in U$, and $u \in C^2(U \setminus x_0)$ satisfies $\Delta u = 0$ in $U \setminus x_0$. Suppose also that* [6]

$$\lim_{x\to x_0} |u||x - x_0|^{n-2} = 0. \tag{8.23}$$

Then u extends to a harmonic function on U.

Proof. **1.** Without loss of generality take $x_0 = 0$. Choose $R > 0$ such that $B(0, R) \subset U$ and use Poisson's formula to solve

$$\Delta v = 0 \text{ in } B(0, R), \quad v|_{\partial B(0,R)} = u. \tag{8.24}$$

Observe that v is bounded on $B(0, R)$. To finish it suffices to show $v = u$ in $B(0, R) \setminus 0$.

2. The assumption (8.23) and the boundedness of v imply that given any $\epsilon > 0$ there exists a $\delta = \delta(\epsilon) > 0$ such that for all $0 < |x| \leq \delta$ we have

$$|u - v| \leq \epsilon|x|^{2-n}. \tag{8.25}$$

Now $u - v - \epsilon|x|^{2-n}$ and $v - u - \epsilon|x|^{2-n}$ are harmonic functions on $B(0, R) \setminus 0$ and from (8.25) we see that each of them is ≤ 0 on the boundary of the set $A(\delta, R) = \{x : \delta \leq |x| \leq R\}$. By the maximum principle both functions are ≤ 0 on $A(\delta, R)$ itself. Equivalently,

$$|u - v| \leq \epsilon|x|^{2-n} \text{ on } A(\delta(\epsilon), R). \tag{8.26}$$

[5] This is taken from Chapter 2 of [Fol99].
[6] This condition can be expressed in little-oh notation as $u = o(\Phi(x - x_0))$ as $x \to x_0$.

3. Finally, we show that since ϵ and δ in (8.26) can both be taken arbitrarily small, we may conclude that $u - v = 0$ in $B(0, R)$. Let $x_0 \in B(0, R) \setminus 0$ and suppose $|(u - v)(x_0)| > c > 0$. Choose ϵ such that $\epsilon |x_0|^{2-n} < c$ and choose $\delta(\epsilon) < |x_0|$ such that (8.26) holds. But (8.26) implies $|(u - v)(x_0)| \leq \epsilon |x_0|^{2-n} < c$, a contradiction.

□

Remark 8.17. *1. This result is sharp, since the harmonic function $|x - x_0|^{2-n}$ can* not *be extended to a harmonic function on U. Note that the theorem allows u to be much worse than bounded near x_0.*

2. In dimension $n = 2$ an analogous result is proved by replacing $|x - x_0|^{2-n}$ by $\ln |x - x_0|$.

The following Corollary is an immediate consequence of Theorem 8.16 and Liouville's theorem.

Corollary 8.18. *Suppose $u \in C^2(\mathbb{R}^n \setminus 0) \cap C(\mathbb{R}^n)$ is harmonic in $\mathbb{R}^n \setminus 0$ and bounded. Then u is constant on $\mathbb{R}^n$.*

Thus, the Dirichlet problem with a "boundary condition at ∞",

$$\Delta u = 0 \text{ in } \mathbb{R}^n \setminus 0, \; u(0) = 0 \text{ at } x = 0, \; u \text{ bounded},$$

has a unique solution in $C^2(\mathbb{R}^n \setminus 0) \cap C(\mathbb{R}^n)$, namely $u = 0$. [7]

8.9 Topology of $\mathcal{D}(U)$.

Let U be an open subset of $\mathbb{R}^n$. Although we have given criteria for determining when a linear functional on $\mathcal{D}(U)$ is continuous, we have not yet defined the topology on $\mathcal{D}(U)$. We explain now how the topology of $\mathcal{D}(U)$ is defined in terms of the Frechét topologies on the spaces $\mathcal{D}_K$, $K \subset U$ compact, that we defined in Remark 1.25.

Definition 8.19 (Convex, balanced sets)**.** *A set in a vector space $(V, \mathbb{C})$ is* convex *if it contains all points on the segment $tP + (1 - t)Q$, $t \in [0, 1]$, between any two of its points P, Q. A set B in V is* balanced *if $\alpha B \subset B$ for all $\alpha \in \mathbb{C}$ such that $|\alpha| \leq 1$.* [8]

Definition 8.20 (Topology on $\mathcal{D}(U)$)**.** *(a) Let β be the collection of all convex, balanced subsets W of $\mathcal{D}(U)$ such that for every compact set $K \subset U$ the set $W \cap \mathcal{D}_K$ is open in $\mathcal{D}_K$.*

(b) Let τ be the collection of all unions of subsets of $\mathcal{D}(U)$ of the form $\phi + W$, where $W \in \beta$ and $\phi \in \mathcal{D}(U)$.

(c) The collection τ can be shown to satisfy the conditions of a topology on $\mathcal{D}(U)$. The collection β is a "local base at 0" for that topology. [9] *The topology τ on $\mathcal{D}(U)$ is an example of an* inductive limit topology*, the inductive limit of the subspaces $\mathcal{D}_K$, $K \subset U$ compact.*

[7] The need for the extra boundary condition is illustrated by the example of the functions xy, $x^2 - y^2$ on $\mathbb{R}^2$.

[8] Note that a (nonempty) balanced set contains 0.

[9] This means that every open set containing 0 contains an element of β.

Remark 8.21. 1. The topology τ makes $\mathcal{D}(U)$ into a locally convex topological vector space (LCTVS). A topological vector space (TVS) is a vector space with a Hausdorff topology that makes the vector space operations continuous. A TVS is locally convex if there is a local base at 0 whose members are convex. Most of the spaces used in analysis are LCTVS's. This class includes Banach spaces and Frechét spaces. A Banach space is a complete normed vector space. A Frechét space is an LCTVS whose topology is induced by a complete, translation-invariant metric.

2. It is clear that when $\mathcal{D}(U)$ carries the topology τ, all of the inclusion maps $i_K : \mathcal{D}_K \to \mathcal{D}(U)$ are continuous, where $K \subset U$ is any compact set (why?). One can show that every convex neighborhood of 0 in a TVS contains a balanced convex neighborhood of 0. From this and Definition 8.20 it follows that τ is the strongest locally convex topology on $\mathcal{D}(U)$ that makes all the inclusion maps $i_K : \mathcal{D}_K \to \mathcal{D}(U)$ continuous. This is a characteristic feature of the inductive limit topology.

3. The convexity of the elements of β is used in proving that β is a local base for τ and that addition in $\mathcal{D}(U)$ is continuous with respect to τ. The proof of continuity of scalar multiplication uses the fact that the elements of β are both convex and balanced. [10]

Now that we have defined the topology on $\mathcal{D}(U)$, let us derive the criterion for continuity of linear functionals that we gave in class.

Proposition 8.22. *Suppose $u : \mathcal{D}(U) \to \mathbb{C}$ is linear, where $\mathcal{D}(U)$ carries the topology τ. Then u is continuous if and only if for all compact $K \subset U$ the restriction $u : \mathcal{D}_K \to \mathbb{C}$ is continuous.*[11]

Proof. $\Rightarrow$ This follows immediately from continuity of the inclusion maps $i_K : \mathcal{D}(K) \to \mathcal{D}(U)$.

$\Leftarrow$ Fix $\epsilon > 0$ and let $B_\epsilon = \{z \in \mathbb{C} : |z| < \epsilon\}$. Then $W = u^{-1}(B_\epsilon)$ is convex and balanced, and the assumption implies $W \cap \mathcal{D}_K$ is open in $\mathcal{D}_K$ for all compact $K \subset U$. Thus, $W \in \beta \subset \tau$. □

Since $\mathcal{D}(U) \subset C^\infty(U)$, you might well wonder why we don't just give $\mathcal{D}(U)$ the topology induced by the standard Frechét topology on $C^\infty(U)$. The reason is that $\mathcal{D}(U)$ is not complete with this topology.[12] On the other hand $\mathcal{D}(U)$ is complete with the τ topology.

Let us recall the standard Frechét topology on $C^\infty(U)$. Write $U = \cup_{j=1}^\infty K_j$, where

$$K_j = \{x \in U : \operatorname{dist}(x, \partial U) \geq \frac{1}{j} \text{ and } |x| \leq j\}. \tag{8.27}$$

[10] See Theorem 6.4 in [Rud91] for details.

[11] This condition for continuity is equivalent to that given in Definition 1.23.

[12] Consider, for example, the sequence $(\phi_n) \in \mathcal{D}(\mathbb{R}^m)$ given by $\phi_n(x) = \psi(x/n)$, where ψ is a test function that is identically 1 on $|x| \leq 1$. This sequence is Cauchy in $\mathcal{D}(\mathbb{R}^m)$ with the topology induced from $C^\infty(\mathbb{R}^m)$, but clearly this sequence has no limit in $\mathcal{D}(\mathbb{R}^m)$.

Clearly the K_j are compact and $K_j \subset K_{j+1}$ for all j. For $\phi \in C^\infty(U)$, define seminorms[13]

$$q_j(\phi) := \sup_{x \in K_j, |\alpha| \leq j} |\partial^\alpha \phi|, \tag{8.28}$$

and define the metric

$$d(\phi, \psi) := \sum_{j=1}^{\infty} \frac{q_j(\phi - \psi)}{1 + q_j(\phi - \psi)} 2^{-j}. \tag{8.29}$$

We observe $C^\infty(U)$ with this metric is complete. See Exercise 8.9a.

Remark 8.23. *Observe that for each compact subset $K \subset U$ the Fréchet topology on $\mathcal{D}_K$ defined in Remark 1.25 is the topology induced from $C^\infty(U)$.*

There is an analogue of Proposition 8.22 (with essentially the same proof) in the case when the target space is any Frechét space.

Proposition 8.24. *Let F be a Frechét space. Suppose $T : \mathcal{D}(U) \to F$ is linear, where $\mathcal{D}(U)$ carries the topology τ. Then T is continuous if and only if for all compact $K \subset U$ the restriction $T : \mathcal{D}_K \to F$ is continuous.*

From Proposition 8.24 and the definition (8.29) we obtain the following corollary:

Corollary 8.25. *A linear map $T : \mathcal{D}(U) \to C^\infty(U)$ is continuous if and only if given any q_j as in (8.28) and any compact set $K \subset U$, there exist constants $C(K)$, $N(K)$ such that for all $\phi \in \mathcal{D}_K$:*

$$q_j(T\phi) \leq C(K) \sup_{x \in K, |\alpha| \leq N(K)} |\partial^\alpha \phi|. \tag{8.30}$$

Proof. **1.** In view of Proposition 8.24 we must show: a linear map $T : \mathcal{D}_K \to C^\infty(U)$ is continuous if and only if for any q_j, there exist constants $C(K)$, $N(K)$ such that (8.30) holds. Recall that the metric on $\mathcal{D}_K$ is

$$d_K(\phi, \psi) = \sum_{j=1}^{\infty} \frac{p_j(\phi - \psi)}{1 + p_j(\phi - \psi)} 2^{-j}, \tag{8.31}$$

where $p_j(\phi) = \sup_{x \in K, |\alpha| \leq j} |\partial^\alpha \phi|$ for $\phi \in \mathcal{D}_K$.

2. First note that the maps $q_j : C^\infty(U) \to \mathbb{R}$ are continuous (why?). Thus, for $\epsilon > 0$ any set of the form $\{\phi \in C^\infty(U) : q_j(\phi) < \epsilon\}$ is open in $C^\infty(U)$. Similarly, any set of the form $\{\phi \in \mathcal{D}_K : p_j(\phi) < \epsilon\}$ is open in $\mathcal{D}_K$.

3. Observe next that any ball $B(0, \epsilon) := \{\phi \in C^\infty(U) : d(\phi, 0) < \epsilon\}$ (for d as in (8.29)) contains a set of the form $\{\phi \in C^\infty(U) : q_j(\phi) < \epsilon'\}$ for some j and ϵ'.(why?) Conversely,

[13] Seminorms are defined just like norms, *except* that the seminorm of a nonzero element can be zero.

given any j and ϵ, a set of the form $\{\phi \in C^\infty(U) : q_j(\phi) < \epsilon\}$ contains a ball $B(0, \epsilon')$ for some ϵ'.(why?) The analogous statements hold for balls $B_K(0, \epsilon) := \{\phi \in \mathcal{D}_K : d_K(\phi, 0) < \epsilon\}$ and open sets defined by p_j.

4. Steps **2** and **3** imply that if the estimate (8.30) holds for all $\phi \in \mathcal{D}_K$, then $T : \mathcal{D}_K \to C^\infty(U)$ is continuous. The proof that continuity implies (8.30) for some $C(K)$ and $N(K)$ is similar to an argument in the proof Proposition 1.21.

□

Exercise 8.9a (a) Show that if $u : C^\infty(U) \to \mathbb{C}$ is a linear functional, then u is continuous if and only if there exist N, C_N such that

$$|u(\phi)| \leq C_N q_N(\phi) \text{ for all } \phi \in C^\infty(U). \tag{8.32}$$

Hint: Recall part (a) of Exercise 1.3.3a.

(b) Show that a sequence $\phi_n \to \phi$ in the standard metric topology on $C^\infty(U)$ if and only if for any $K \subset\subset U$ and multiindex α we have $\partial^\alpha \phi_n \to \partial^\alpha \phi$ uniformly on K as $n \to \infty$. Deduce that $C^\infty(U)$ with this metric is complete.

8.10 A Fourier transform computation

The following proposition is needed in our discussion of the wave equation in section 1.6.2.

Proposition 8.26. *Let $\mathcal{F}$ denote the Fourier transform in x. For $s > 0$ we have*

$$\mathcal{F}^{-1}(e^{-s|\xi|}) := \int_{\mathbb{R}^n} e^{ix\xi} e^{-s|\xi|} d'\xi = \frac{2^{n/2}}{\sqrt{\pi}} \Gamma\left(\frac{n+1}{2}\right) \frac{s}{(s^2 + |x|^2)^{\frac{n+1}{2}}}, \tag{8.33}$$

where $\Gamma(u) = \int_0^\infty r^{u-1} e^{-r} dr$. Both sides of (8.33) extend analytically to the right half-plane, Re $s > 0$, *provided we use the branch of $\sqrt{z}$ cut along the negative real axis to compute $(s^2 + |x|^2)^{\frac{n+1}{2}}$.*[14] *Hence, the second equality in (8.33) holds for* Re $s > 0$.

We will use the following Lemma.

Lemma 8.27. *Let $\beta \geq 0$. Then*

$$e^{-\beta} = \int_0^\infty \frac{e^{-\tau}}{\sqrt{\pi\tau}} e^{-\beta^2/4\tau} d\tau. \tag{8.34}$$

Proof. We have

$$e^{-\beta} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{i\beta s}}{1+s^2} ds \text{ and } \frac{1}{1+s^2} = \int_0^\infty e^{-(1+s^2)\tau} d\tau, \tag{8.35}$$

[14] That is, we use $\sqrt{z} = r^{1/2} e^{i\theta/2}$, where $z = re^{i\theta}$ and $-\pi < \theta < \pi$.

where the first integral was computed by residues. Thus,[15]

$$e^{-\beta} = \frac{1}{\pi}\int_0^\infty \int_{-\infty}^\infty e^{i\beta s} e^{-\tau s^2} e^{-\tau} ds d\tau = \int_0^\infty \frac{e^{-\tau}}{\sqrt{\pi\tau}} e^{-\beta^2/4\tau} d\tau. \tag{8.36}$$

□

Proof of Proposition 8.26. Let $s > 0$. Applying Lemma 8.27 with $\beta = s|\xi|$ we obtain

$$\begin{aligned}\int_{\mathbb{R}^n} e^{ix\xi} e^{-s|\xi|} d'\xi &= \int_0^\infty \int_{\mathbb{R}^n} \frac{e^{-\tau}}{\sqrt{\pi\tau}} e^{-\frac{|\xi|^2 s^2}{4\tau}} e^{ix\xi} d'\xi d\tau = \int_0^\infty \frac{e^{-\tau}}{\sqrt{\pi\tau}} \frac{(2\tau)^{n/2}}{s^n} e^{-\frac{|x|^2\tau}{s^2}} d\tau = \\ &\frac{2^{n/2}}{\sqrt{\pi}} \frac{1}{s^n} \int_0^\infty e^{-\tau\left(1+\frac{|x|^2}{s^2}\right)} \tau^{\frac{n-1}{2}} d\tau = \frac{2^{n/2}}{\sqrt{\pi}} \frac{1}{s^n \left(1+\frac{|x|^2}{s^2}\right)^{\frac{n+1}{2}}} \int_0^\infty \sigma^{\frac{n-1}{2}} e^{-\sigma} d\sigma = \\ &\frac{2^{n/2}}{\sqrt{\pi}} \Gamma\left(\frac{n+1}{2}\right) \frac{s}{(s^2+|x|^2)^{\frac{n+1}{2}}}.\end{aligned} \tag{8.37}$$

Here we used the change of variable $\sigma = \tau\left(1+\frac{|x|^2}{s^2}\right)$ for the second to last equality.

The analytic extension of (8.33) to $\operatorname{Re} s > 0$ is clear. □

Remark 8.28. The function on the right in (8.37) is the Poisson kernel for the upper half space $\mathbb{R}^{n+1}_+$. See Problem 5 in section 1.7, which asks why we should *expect* the Poisson kernel to appear here.

8.11 Some functional analysis

Here we state, often without proof, some facts from functional analysis that we need in the main text. Proofs may be found in [Wil26], for example.

8.11.1 Normed spaces and inner product spaces

Suppose V and W are normed vector spaces. Let

$$\mathcal{L}(V,W) = \{T : V \to W : T \text{ is continuous and linear}\}.$$

A linear map $T : V \to W$ is said to be *bounded* if it maps bounded sets to bounded sets. Proposition 1.21 shows that T is continuous $\Leftrightarrow$ T is bounded. When $V = W$ we usually write $\mathcal{L}(V) := \mathcal{L}(V,V)$.

Theorem 8.29 (Hahn-Banach theorem)**.** *Let V be a normed vector space and $M \subset V$ a subspace. Suppose $f : M \to \mathbb{C}$ is linear and for some C satisfies $|f(m)| \le C|m|$ for all $m \in M$. Then f can be extended to a linear map $f : V \to \mathbb{C}$ such that $|f(v)| \le C|v|$ for all $v \in V$.*

[15] The ds integral is the integral of a Gaussian.

Definition 8.30. *A Banach space is a complete normed vector space.*

Definition 8.31 (Spectrum)**.** *The spectrum of $T \in \mathcal{L}(V)$ is $\sigma(T) := \{\lambda \in \mathbb{C} : T - \lambda I$ is not invertible$\}$.*

Remark 8.32. We have $\lambda \in \sigma(T)$ if and only if at least one of the following is true:
(a) $R(T - \lambda I) \neq V$,
(b) $T - \lambda I$ is not injective; that is, $N(T - \lambda I) \neq \{0\}$.
When (b) holds, we call λ and *eigenvalue* of T and call $N(T - \lambda I)$ the corresponding *eigenspace*. The set of eigenvalues of T is referred to as the *point spectrum* of T and denote $\sigma_p(T)$.

For linear operators $T : \mathbb{C}^n \to \mathbb{C}^n$, we have $N(T - \lambda I) = \{0\} \Leftrightarrow R(T - \lambda I) = \mathbb{C}^n$. This does not hold generally for operators on infinite dimensional spaces.

An inner product space is a special type of normed space.

Definition 8.33 (Inner product)**.** *Let $(H, \mathbb{F})$ be a vector space over the field $\mathbb{F}$, where $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$. An* inner product *on H is a map $(\cdot, \cdot) : H \times H \to \mathbb{F}$ such that for $u, v, u_1, u_2 \in H$ and $a \in \mathbb{F}$ we have:*
(a) $(u, u) \geq 0$ and $(u, u) = 0 \Leftrightarrow u = 0$;
(b) $(u, v) = \overline{(v, u)}$;
(c) $(au, v) = a(u, v)$;
(d) $(u_1 + u_2, v) = (u_1, v) + (u_2, v)$.

Remark 8.34. Observe that (b) and (c) imply $(u, av) = \overline{a}(u, v)$, while (b) and (d) imply $(u, v_1 + v_2) = (u, v_1) + (u, v_2)$. Hence, $(\cdot, \cdot)$ is *sesquilinear*, that is, linear in the first argument and antilinear in the second.

Proposition 8.35. *Let H be a vector space with an inner product $(\cdot, \cdot)$. Then the map $u \to |u| := \sqrt{(u, u)}$ is a norm.*

The main step in the proof of Proposition 8.35 is to check the triangle inequality. This follows easily from the Cauchy-Schwarz inequality:

Proposition 8.36 (Cauchy-Schwarz inequality)**.** *For $u, v \in H$ we have $|(u, v)| \leq |u||v|$.*

Definition 8.37 (Hilbert space)**.** *A Hilbert space $(H, \mathbb{F})$ is an inner product space with is complete with respect to the norm $|u| := \sqrt{(u, u)}$.*

Examples 8.38. The following spaces are Hilbert spaces with the given inner products:
(a) $\mathbb{C}^n$ with $(w, z) = \sum_{j=1}^n w_j \overline{z_j}$;
(b) $L^2(X, \mathcal{M}, \mu)$ with $(f, g) = \int_X f(x)\overline{g(x)}d\mu$. Here $(X, \mathcal{M}, \mu)$ is any measure space.

Remark 8.39. If H is a Hilbert space with inner product $(\cdot, \cdot)$, then $H \times H$ is a Hilbert space with inner product

$$(((u_1, u_2), (v_1, v_2))) := (u_1, v_1) + (u_2, v_2).$$

Proposition 8.40. *The composition of a bounded operator with a compact operator is compact, whenever it is well-defined (check).*

In the rest of section 8.11 let H be a Hilbert space.

If $g \in H$ Proposition 8.36 implies that the map given by $u \to (u, g)$ is a continuous linear functional. The next result, which turns out to be one of the most useful results in analysis, tells us that *every* continuous linear functional on H is of this form for some $g \in H$.

Theorem 8.41 (Riesz representation theorem)**.** *If $\phi : H \to \mathbb{C}$ linear and continuous, then there exists a unique $f \in H$ such that $\phi(u) = (u, f)$ for all $u \in H$.*

Proposition 8.42. *The map $T : H' \to H$ given by the Riesz representation theorem is an antilinear isometry of H' onto H.*

Proof. The bijectivity of T is immediate from Theorem 8.41. If $\phi \in H'$ and $T\phi = f$, we have using Cauchy-Schwarz:

$$|\phi| = \sup_{|u|\leq 1} |\phi(u)| = \sup_{|u|\leq 1} |(u, f)| \leq |f|. \tag{8.38}$$

Let $u = f/|f|$ to see that the sup in (8.38) equals $|f|$.

□

Remark 8.43 (anti-Riesz)**.** If $\phi : H \to \mathbb{C}$ is antilinear and continuous, we can apply Theorem 8.41 to $\overline{\phi}$ to see that there exists a unique $f \in H$ such that $\phi(u) = (f, u)$ for all $u \in H$.

The next theorem is a useful equivalent variant of the Riesz representation theorem used, for example, in the theory of elliptic problems. First we state a basic fact about Hilbert spaces that we'll need in the proof.

Proposition 8.44. *Let H be a Hilbert space and $M \subset H$ a closed subspace. Define $M^\perp = \{h \in H : (h, m) = 0$ for all $m \in M\}$. Then H can be written as the orthogonal direct sum*

$$H = M \oplus M^\perp$$

.

Theorem 8.45 (Lax-Milgram)**.** *Let $(H, \mathbb{R})$ be a real Hilbert space.*[16] *Assume $B : H \times H \to \mathbb{R}$ is a bilinear map for which there exist positive constants α, β such that*
(a)$|B[u, v]| \leq \alpha|u||v|$ (continuity)
(b) $|B[u, v]| \geq \beta|u||v|$ (coercivity)
for all $u, v \in H$. Then for any $f \in H'$ there exists a unique $u \in H$ such that

$$f(v) = B[u, v] \text{ for all } v \in H. \tag{8.39}$$

[16]The analogous theorem holds by the same argument for complex Hilbert spaces.

Proof. **1.** For any $u \in H$ (a) implies that that map $v \to B[u,v]$ is an element of H', so the Riesz theorem implies there exists a unique $w \in H$ such that

$$B[u,v] = (w,v) \text{ for all } v \in H. \tag{8.40}$$

If we define the map $A : H \to H$ by $w = Au$, then A is linear (why?) and (8.40) can be written $B[u,v] = (Au,v)$ for all v. The theorem is equivalent to the statement that A is bijective, since every $f \in H'$ has the form $f(v) = (w,v)$ for some $w \in H$.

2. To see that A is continuous we consider

$$|Au|^2 = (Au,Au) = B[u,Au] \leq \alpha|u||Au| \Rightarrow |Au| \leq \alpha|u|. \tag{8.41}$$

3. A **is injective with** $R(A)$ **closed.** Using coercivity we obtain for $u \in H$

$$\beta|u|^2 \leq B[u,u] = (Au,u) \leq |Au||u| \Rightarrow \beta|u| \leq \beta|u| \leq |Au|, \tag{8.42}$$

so A is injective.

Suppose $Au_n \to v$. Then (8.42) implies (u_n) is Cauchy, so there exists $u \in H$ such that $u_n \to u$ and hence $Au_n \to Au$. This implies $v = Au$, so $R(A)$ is closed.

4. Show $R(A) = H$**.** Use Lemma 8.44 to write $H = R(A) \oplus R(A)^\perp$. Suppose there exists $w \in R(A)^\perp \setminus \{0\}$. Then by coercivity

$$\beta|w|^2 \leq B[w,w] = (Aw,w) = 0,$$

a contradiction. □

Remark 8.46. (a) To see that Lax-Milgram implies Riesz take $B[u,v] := (u,v)$.

(b) If B is *symmetric*, that is $B[u,v] = B[v,u]$, then Lax-Milgram follows immediately from Riesz. Indeed, define a new inner product on H by $((u,v)) := B[u,v]$ and note that the associated norm $\|\cdot\|$ is equivalent to the original norm on H.(why?) This implies that a functional is continuous with respect to $|\cdot|$ if and only if it is continuous with respect to $\|\cdot\|$.

Suppose $T : H \to H$. We now define the *Hilbert space adjoint* T^* of T. Fix $v \in H$ and observe that the map $\ell : H \to \mathbb{C}$ given by $\ell(u) := (Tu,v)$ is continuous. By Theorem 8.41 there exists a unique $w \in H$ such that $\ell(u) = (u,w)$ for all $u \in H$. Define $T^*v := w$. Then T is linear (why?) and clearly satisfies

$$(Tu,v) = (u,T^*v) \text{ for all } u,v \in H. \tag{8.43}$$

Moreover,

$$|T^*v|^2 = (T^*v,T^*v) = (TT^*v,v) \leq |T||T^*v||v| \Rightarrow |T^*v| \leq |T||v|,$$

so T^* is bounded.

8.11.2 Compact operators

In this section V and W denote Banach spaces.

Definition 8.47 (Compact operators)**.** *Suppose $T \in \mathcal{L}(V, W)$. Let B_1 be the open unit ball in V. We say T is* compact *if*
(a) T maps bounded sets to relatively compact sets $\Leftrightarrow$
(b) $T(B_1)$ is relatively compact in W $\Leftrightarrow$
(c) $\overline{TB_1}$ is compact $\Leftrightarrow$
(d) whenever (v_n) is a bounded sequence in V, then (Tv_n) has a convergent subsequence.

Remark 8.48. A linear operator $T : V \to W$ is continuous $\Leftrightarrow$ T maps bounded sets to bounded sets. A compact operator has the stronger property that it maps bounded sets to relatively compact sets.

Examples 8.49. 1. If V is an infinite dimensional Banach space, the identity map $i : V \to V$ is not compact since the closed unit ball in V is not compact.[17]
2. If $U \subset \mathbb{R}^n$ is a bounded open set with a C^1 boundary, the inclusion map $i : H^1(U) \to L^2(U)$ is compact. This Rellich's theorem, see Theorem 8.71.
3. If $T \in \mathcal{L}(V, W)$ has finite rank, that is, $\dim R(T) < \infty$, then T is compact (check).

Exercise 8.11.2a. Suppose U is bounded and open with C^1 boundary.
(a) Show that the inclusion map $i : C^1(\overline{U}) \to C(\overline{U})$ is compact. (Hint: You may use Proposition 8.75 to deal with nonconvex U.)
(b) Suppose we drop the assumption that U is bounded. Is i still compact?

Theorem 8.50 (Spectrum $\sigma(K)$ of a compact operator)**.** *Let $K : V \to V$ be compact. Then*
(i) if $\lambda \in \sigma(K) \setminus \{0\}$, λ is an eigenvalue (i.e., $\lambda \in \sigma_p(K)$) and $N(K - \lambda I)$ is finite dimensional;
(ii) 0 is the only possible limit point of $\sigma(K)$, and if $\dim V = \infty$, then $0 \in \sigma(K)$;
(iii) $\sigma(K)$ is at most countable, and for any $r > 0$ the set $\{\lambda \in \sigma(K) : |\lambda| > r\}$ is finite.
(iv) $\sigma(K)$ is compact.

Remark 8.51. Theorem 8.50 implies that when $\sigma(K)$ is infinite, the set $\sigma_p(K) = \sigma(K) \setminus \{0\}$ can be written as a sequence converging to 0.

Now take H to be a Hilbert space.

Proposition 8.52. *Suppose $K : H \to H$ is compact. Then K^*, the Hilbert space adjoint of K, is compact.*

The next theorem is used in the proof of Theorem 3.25.

[17] For example, consider the sequence $(2\pi)^{-n/2} e^{in\theta}$ in the closed unit ball of $L^2([0, 2\pi])$. It fails to be Cauchy, so has no convergent subsequence.(check)

Theorem 8.53. *Suppose $K : H \to H$ is compact. We have: (i)* $\dim N(I-K) = \dim N(I-K^*) < \infty$*;*
(ii) the range of $I-K$ is closed and $R(I-K) = (N(I-K^*))^\perp$.

Observe that (ii) implies the analogue of (3.39). The space H can be decomposed as the orthogonal direct sum

$$H = R(I-K) \oplus N((I-K^*)). \tag{8.44}$$

Remark 8.54. The operator $I-K$ is an example of a Fredholm operator with index zero. An operator $T : H \to H$ is *Fredholm* if $\dim N(T) < \infty$ and $\operatorname{codim} R(T) := \dim R(T)^\perp < \infty$. The index of a Fredholm operator is defined to be

$$\operatorname{index}(T) := \dim N(T) - \operatorname{codim} R(T).$$

Fredholm operators of the form $I-K$ appeared in a 1903 paper of I. Fredholm on the Dirichlet problem for the Laplacian.

Parallel to Proposition 3.2.1 we have the following direct corollary of Theorem 8.53.

Corollary 8.55 (Fredholm alternative)**.** *Let $K : H \to H$ be compact. Then*
(i) either $(I-K)u = f$ has a unique solution for all $f \in H$ or $(I-K)u = 0$ has nonzero solutions;
(ii) the equation $(I-K)u = f$ has a solution if and only if $f \in (N(I-K^*))^\perp$.

8.11.3 Reflexivity and weak sequential compactness

Let $(V, \mathbb{F})$ be a Banach space, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The dual space of V is

$$V' := \{\omega : V \to \mathbb{F} \text{ is continuous}\} \tag{8.45}$$

with the norm $|\omega| = \sup_{|v|\le 1} |\omega(v)|$. The space V' is also a Banach space.
Setting $V'' := (V')'$, we see there is a natural map

$$J : V \to V'' \text{ given by } Jv(\omega) = \omega(v) \text{ for } v \in V,\ \omega \in V'. \tag{8.46}$$

The map J is clearly linear and

$$|Jv|_{V''} = \sup_{|\omega|\le 1} |Jv(\omega)| = \sup_{|\omega|\le 1} |\omega(v)| \le \sup_{|\omega|\le 1} |\omega||v| \le |v|. \tag{8.47}$$

Thus, $|Jv|_{V''} \le |v|_V$. The Hahn-Banach theorem implies more:

Proposition 8.56. *The map $J : V \to J(V) \subset V''$ is an isometry onto $J(V)$.*

Definition 8.57 (Reflexive Banach space)**.** *A Banach space is* reflexive *when the map J is surjective, that is, when J is an isometric isomorphism of V onto V''.*

Examples 8.58. Here are some examples of reflexive Banach spaces.

(i) finite dimensional normed spaces: Check!

(ii) Hilbert spaces V: The Riesz representation theorem gives a bijective isometry $T : V \to V'$, where $Tv(w) = (w, v)$ for $v, w \in V$. Let $f \in V''$, so $f : V' \to \mathbb{F}$. We have a continuous map $f \circ T : V \to \mathbb{F}$, so we can apply the Riesz theorem to find $u \in V$ such that $(f \circ T)(v) = (u, v)$ for all $v \in V$. I claim $Ju = f$. Indeed, for $v \in V$ we have

$$Ju(Tv) = Tv(u) = (u, v) = f(Tv).$$

Since T is surjective, we conclude $Ju = f$.

(iii) L^p spaces for $1 < p < \infty$: Let $U \subset \mathbb{R}^n$. If $q = \frac{p}{p-1}$ the map $T : L^p(U) \to (L^q(U))'$ given by

$$Tf(g) = \int_U fg dx \text{ for } f \in L^p, g \in L^q$$

is an isometric isomorphism. An argument like that in (ii) shows $J : L^p \to (L^p)''$ is surjective.

(iv) The L^p-based Sobolev spaces $W^{1,p}(U)$ for $1 < p < \infty$ are reflexive. This is proved using the reflexivity of L^p.

Definition 8.59 (Weak topology on V)**.** *A local base at* 0*, call it* $\mathcal{B}$*, for the* weak topology on V *is given by the set of all finite intersections of sets of the form* $\{v \in V : |\phi(V)| < \epsilon\}$*, where* $\phi \in V'$ *and* $\epsilon > 0$*. A local base at* $v \in V$ *is given by* $v + \mathcal{B}$*.*

The next proposition follows immediately from Definition 8.59.

Proposition 8.60. *A sequence* (v_k) *in* V *converges weakly (that is, in the weak topology) to* $v \in V$ *if* $\phi(v_k) \to \phi(v)$ *for all* $\phi \in V'$*.*

Notice that closed sets in V are not necessarily weakly closed (why?). We use the next result at one point in Chapter 4.

Theorem 8.61 (Mazur's theorem)**.** *Let* V *be a Banach space. Then any closed convex set is weakly closed.*

The next result follows is a consequence of the Hahn-Banach theorem.

Proposition 8.62. *Suppose* v_n *converges weakly to* v *in* V*. Then*

$$|v| \leq \liminf_{n\to\infty} |v_n|. \tag{8.48}$$

A bounded sequence in a Banach space V need not have a convergent subsequence.[18] But in special Banach spaces, bounded sequences always have *weakly convergent* subsequences. The next result used, for example, in the proof of Proposition 8.79, is a consequence of the Banach-Alaoglu theorem; see [Wil26], section 5.4.

Proposition 8.63 (Weak sequential compactness)**.** *Let* V *be a separable reflexive Banach space, and suppose the sequence* (v_n) *is bounded in* V*.*[19] *Then there exists a weakly convergent subsequence.*

[18] Consider for example the sequence $(e^{in\theta})$ in $L^2([0, 2\pi])$.

[19] Recall that V is *separable* if it has a countable dense subspace. The $L^p(U)$ spaces are separable for $1 \leq p < \infty$.

8.12 Tools used in the proof of the a priori estimate for hyperbolic systems

In this section we give proofs of a couple of technical tools used in the proof of the main a priori estimate for first-order symmetric hyperbolic systems.

8.12.1 A technical lemma

Lemma 8.64. *Let $s \in \mathbb{R}$ and $u \in H^s(\mathbb{R}^n)$. Then*

$$|e^{ix\eta}u(x)|_s \leq C\langle\eta\rangle^{|s|}|u|_s. \tag{8.49}$$

Proof. **1. The case $s \geq 0$.** We have

$$|e^{ix\eta}u(x)|_s = |\langle\xi\rangle^s\hat{u}(\xi-\eta)|_{L^2(\xi)} \tag{8.50}$$

and, since $\langle\xi\rangle^s \leq C_s(\langle\xi-\eta\rangle^s + \langle\eta\rangle^s)$ for $s \geq 0$, it follows that[20]

$$|e^{ix\eta}u(x)|_s \lesssim |\langle\xi-\eta\rangle^s\hat{u}(\xi-\eta)|_{L^2(\xi)} + |\langle\eta\rangle^s\hat{u}(\xi-\eta)|_{L^2(\xi)} = |u|_s + \langle\eta\rangle^s|u|_0 \leq \langle\eta\rangle^s|u|_s. \tag{8.51}$$

2. The case $s < 0$. Let $s = -t$. Suppose $u \in H^t$ and $v \in \mathcal{S}(\mathbb{R}^n)$. Using $\langle\cdot,\cdot\rangle$ to denote the extension of the L^2 inner product to $H^t \times H^{-t}$, we have using step **1**:

$$|\langle e^{-ix\eta}u, v\rangle| = |\langle u, e^{ix\eta}v\rangle| \leq |e^{-ix\eta}u|_t|v|_{-t} \lesssim \langle\eta\rangle^t|u|_t|v|_{-t}. \tag{8.52}$$

This implies that the norm of the functional on H^t defined by $l(u) = \langle u, e^{ix\eta}v\rangle$ satisfies

$$|l| = |e^{ix\eta}v|_{-t} \lesssim \langle\eta\rangle^t|v|_{-t}. \tag{8.53}$$

That is,

$$|e^{ix\eta}v|_s \lesssim \langle\eta\rangle^{|s|}|v|_s. \tag{8.54}$$

To finish let $v \in H^{-t}$ and choose $v_k \in \mathcal{S}$ such that $v_k \to v$ in H^{-t} to conclude that (8.54) holds for such v.[21] □

8.12.2 Commutator estimate

We will use the notation $\langle\xi\rangle := (1+|\xi|^2)^{1/2}$, $|u|_0 = |u|_{L^2}$, and

$$\Lambda^s u = \int e^{ix\xi}\langle\xi\rangle^s\hat{u}(\xi)d\xi. \tag{8.55}$$

As usual, we will suppress the t-dependence of $A(t,x,D_x)$.

[20] The notation $|A| \lesssim |B|$ means that there exists a constant C such that $|A| \leq C|B|$.

[21] It is a good exercise to think through this last step *carefully*. Also, why did we need to assume $v \in \mathcal{S}$ at first?

Proposition 8.65. *Let $A = A(x, D) = \sum_{j=1}^{d} A_j(x)D_j$, where the A_j are C^∞ and constant, equal to C_j, outside a compact set. Assume $u \in H^{s+1}(\mathbb{R}^d)$. Then*

$$|[A, \Lambda^s]u|_0 \leq C|u|_s. \tag{8.56}$$

Proof. **1. Reductions.** Set $v = \Lambda^s u \in H^1$. Then (8.56) takes the form

$$|[A, \Lambda^s]\Lambda^{-s}v|_0 \leq C|v|_0. \tag{8.57}$$

We have

$$\begin{aligned}[A_jD_j, \Lambda^s]\Lambda^{-s} = (A_jD_j\Lambda^s - \Lambda^s A_jD_j)\Lambda^{-s} = (A_j\Lambda^s D_j - \Lambda^s A_jD_j)\Lambda^{-s} = \\ - [\Lambda^s, A_j]D_j\Lambda^{-s},\end{aligned} \tag{8.58}$$

so it is enough to show

$$|[\Lambda^s, A_j]D_j\Lambda^{-s}v|_0 \leq C|v|_0. \tag{8.59}$$

Set $w = \Lambda^{s-1}D_j\Lambda^{-s}v$. Since $\Lambda^{s-1}D_j\Lambda^{-s}$ is a Fourier multiplier with bounded symbol, we have $|w|_0 \leq C|v|_0$, so (8.59) will follow from [22]

$$|[\Lambda^s, A_j]\Lambda^{-(s-1)}w|_0 \leq C|w|_0. \tag{8.60}$$

Since $A_j(x) = (A_j(x) - C_j) + C_j$, where $A_j(x) - C_j$ has compact support, it suffices henceforth to assume $A_j \in \mathcal{D}(\mathbb{R}^d)$. With

$$Tw := [\Lambda^s, A_j]\Lambda^{-(s-1)}w, \tag{8.61}$$

we must show

$$|Tw|_0 \leq C|w|_0, \text{ when } A_j \in \mathcal{D}(\mathbb{R}^d). \tag{8.62}$$

2. Compute $\widehat{Tw}$. Writing $\hat{g} = \mathcal{F}(g)$, we have

$$\begin{aligned}\mathcal{F}(\Lambda^s A_j\Lambda^{-(s-1)}w) = \langle\xi\rangle^s\mathcal{F}(A_j\Lambda^{-(s-1)}w) = \int \langle\xi\rangle^s\widehat{A_j}(\xi-\eta)\langle\eta\rangle^{-(s-1)}\hat{w}(\eta)\,d\eta \\ \mathcal{F}(A_j\Lambda^s\Lambda^{-(s-1)}w) = \mathcal{F}(A_j\Lambda w) = \int \widehat{A_j}(\xi-\eta)\langle\eta\rangle\hat{w}(\eta)\,d\eta,\end{aligned} \tag{8.63}$$

so

$$\widehat{Tw}(\xi) = \int (\langle\xi\rangle^s - \langle\eta\rangle^s)\widehat{A_j}(\xi-\eta)\langle\eta\rangle^{-(s-1)}\hat{w}(\eta)\,d\eta. \tag{8.64}$$

[22] An operator M is said to be a "Fourier multiplier" if its action on u is given by the inverse Fourier transform of $m(\xi)\hat{u}(\xi)$ for some function $m(\xi)$, which is referred to as the "symbol" of the operator.

With $f(\xi) := \langle\xi\rangle^s$ we can write[23]

$$\langle\xi\rangle^s - \langle\eta\rangle^s = (\xi-\eta)\int_0^1 \frac{\partial f}{\partial\xi}(\eta + t(\xi-\eta))dt := (\xi-\eta)g(\xi,\eta). \tag{8.65}$$

Thus,

$$|\widehat{Tw}(\xi)| \le \int |\xi-\eta||g(\xi,\eta)|\widehat{A_j}(\xi-\eta)\langle\eta\rangle^{-(s-1)}\hat{w}(\eta)\,d\eta = \int G(\xi,\eta)a(\xi-\eta)\hat{w}(\eta)\,d\eta, \tag{8.66}$$

where for some large N to be chosen,

$$\begin{aligned} G(\xi,\eta) &:= \langle\xi-\eta\rangle^{-N} g(\xi,\eta)\langle\eta\rangle^{-(s-1)} \\ a(\xi-\eta) &:= |\xi-\eta|\widehat{A_j}(\xi-\eta)\langle\xi-\eta\rangle^N. \end{aligned} \tag{8.67}$$

Observe that $a(\xi-\eta)$ is rapidly decreasing in $\langle\xi-\eta\rangle$ since $A_j \in \mathcal{D}$.

3. A Lemma. We will use the following lemma, whose proof is given in step **5**.[24]

Lemma 8.66. *Let f and g be functions in $\mathcal{S}(\mathbb{R}^d)$, and suppose*

$$h(\xi) = \int G(\xi,\eta)f(\xi-\eta)g(\eta)\,d\eta, \tag{8.68}$$

where $G(\xi,\eta)$ is any function that can be written as a finite sum of functions $G_i(\xi,\eta)$ such that either

$$\sup_\xi \int |G_i|^2 d\eta < \infty \text{ or } \sup_\eta \int |G_i|^2 d\xi < \infty. \tag{8.69}$$

Then

$$|h|_0 \le C|f|_0|g|_0. \tag{8.70}$$

Remark 8.67. *Since $\mathcal{S}$ is dense in L^2, the estimate (8.70) implies that the map $(f,g) \to h$ extends by continuity to a map $L^2 \times L^2 \to L^2$, and (8.70) holds for the extended map.*

4. Conclusion, assuming the Lemma. We now show that the function $G(\xi,\eta)$ in (8.67) satisfies (8.69). The lemma and (8.66) then imply

$$|\widehat{Tw}|_0 \le C|a|_0|\hat{w}|_0 \le C|\hat{w}|_0, \tag{8.71}$$

which is equivalent to (8.62).

[23] By the Fundamental Theorem of Calculus.

[24] This lemma goes back at least to the 1982 paper [RR82]. It is extremely useful and has been applied in many different contexts.

We write

$$G(\xi,\eta)=\frac{|g(\xi,\eta)|}{\langle\eta\rangle^{s-1}\langle\xi-\eta\rangle^{N}},\ \text{where from (8.65)},\ |g(\xi,\eta)|\le C\sup_{0\le t\le 1}\langle\eta+t(\xi-\eta)\rangle^{s-1}. \tag{8.72}$$

First consider the case $s-1\ge 0$. Then

$$\langle\eta+t(\xi-\eta)\rangle^{s-1}\le C(\langle\eta\rangle^{s-1}+\langle\xi-\eta\rangle^{s-1}), \tag{8.73}$$

so

$$|G|\le C\frac{\langle\eta\rangle^{s-1}}{\langle\eta\rangle^{s-1}\langle\xi-\eta\rangle^{N}}+C\frac{\langle\xi-\eta\rangle^{s-1}}{\langle\eta\rangle^{s-1}\langle\xi-\eta\rangle^{N}}. \tag{8.74}$$

The first term on the right satisfies (8.69) when $N>\frac{d}{2}$, and the second term when $N-s+1>\frac{d}{2}$.

Next consider the other case $s-1\le 0$. Setting $\zeta_t:=\eta+t(\xi-\eta)$ we define the sets

$$A=\{(\xi,\eta,t):|\zeta_t|\ge\frac{1}{2}|\eta|\},\ B=\{(\xi,\eta,t):|\zeta_t|<\frac{1}{2}|\eta|\}, \tag{8.75}$$

let χ_A and χ_B be the corresponding characteristic functions, define

$$g_A(\xi,\eta):=\int_0^1\chi_A(\xi,\eta,t)\frac{\partial f}{\partial\xi}(\eta+t(\xi-\eta))dt, \tag{8.76}$$

and similarly define g_B. Next define G_A and G_B as in (8.72) using g_A, g_B respectively.

Since $|\partial_\xi f(\zeta_t)|\le C\langle\zeta_t\rangle^{s-1}$ and $s-1\le 0$, we have $|g_A|\le C\langle\eta\rangle^{s-1}$, so G_A satisfies (8.69) when $N>d/2$.

When $|\zeta_t|<\frac{1}{2}|\eta|$ we have

$$|t(\xi-\eta)|>\frac{1}{2}|\eta|\Rightarrow\langle\xi-\eta\rangle>\frac{1}{2}\langle\eta\rangle, \tag{8.77}$$

and since $|g_B|\le 1$, we see that G_B satisfies (8.69) when $N+s-1>d/2$.

5. Proof of the Lemma. It is enough to consider the case where $G=G_i$, where say

$$\sup_\eta\int|G(\xi,\eta)|^2d\xi<\infty. \tag{8.78}$$

Then by Cauchy-Schwarz

$$\begin{aligned}|h(\xi)|^2=|\int G(\xi,\eta)f(\xi-\eta)g(\eta)d\eta|^2\le\int|G(\xi,\eta)g(\eta)|^2d\eta\cdot\int|f(\xi-\eta)|^2d\eta=\\ |f|_0^2\cdot\int|G(\xi,\eta)g(\eta)|^2d\eta.\end{aligned} \tag{8.79}$$

By Fubini we obtain

$$\int |h(\xi)|^2 d\xi \leq |f|_0^2 \cdot \int \left(\int |G(\xi,\eta) g(\eta)|^2 d\xi \right) d\eta = |f|_0^2 \cdot \int |g(\eta)|^2 \left(\int |G(\xi,\eta)|^2 d\xi \right) d\eta \leq$$
$$|f|_0^2 |g|_0^2 \cdot \sup_\eta \int |G(\xi,\eta)|^2 d\xi. \tag{8.80}$$

□

8.13 Gronwall's inequality

Proposition 8.68. *Let $T > 0$, $a \in \mathbb{R}$, $g \in C([0,T],\mathbb{R})$ and suppose*

$$y \in C^1([0,T],\mathbb{R}), \text{ and } y' - ay \leq g \text{ on } [0,T].$$

Then[25]

$$y(t) \leq e^{at} \left[y(0) + \int_0^t g(s) e^{-as} ds \right]. \tag{8.81}$$

Proof. We have

$$\frac{d}{dt}(ye^{-at}) = y'e^{-at} - aye^{-at} \leq g(t)e^{-at} \Rightarrow y(t)e^{-at} - y(0) \leq \int_0^t g(s)e^{-as} ds, \tag{8.82}$$

which implies (8.81).

□

8.14 Partitions of unity

Partitions of unity are useful for globalizing results that are initially local.

Definition 8.69. *Let $K \subset \mathbb{R}^n$ be compact and suppose $\{U_j, j = 1,\dots,N\}$ is an open cover of K. A C^∞* partition of unity on K subordinate to this covering *is a collection $\{\rho_j, j = 1,\dots N\}$ of C^∞ functions $\rho_j : \mathbb{R}^n \to \mathbb{R}$ with the properties:*

(a) $\operatorname{supp} \rho_j$ *is compact and contained in U_j for every j,*
(b) $\sum_{j=1}^N \rho_j = 1$ *on K.*[26]

Proposition 8.70. *Let $K \subset \mathbb{R}^n$ be compact and suppose $\{U_j, j = 1,\dots,N\}$ is an open cover of K. A C^∞ partition of unity on K subordinate to this covering exists.*

[25] Observe that the right side of (8.13) is the solution of $y' - ay = g$ with initial value $y(0)$.
[26] The support of ρ_j is the closure in $\mathbb{R}^n$ of the set where $\rho_j \neq 0$.

With K and the U_j as above, suppose that you have managed to construct C^∞ functions $b_j : U_j \to \mathbb{R}$ with some useful property. Then the function $\sum_j \rho_j b_j$ is a C^∞ function on $\mathbb{R}^n$, supported near K, which may have that useful property in a neighborhood of K.

The proof of Proposition 8.70 will use the following two facts, whose proofs are omitted (or left as exercises).

(i) If $U \subset \mathbb{R}^n$ is open and $K \subset U$ is compact, there exists a compact K' such that $K \subset \text{int } K'$ and $K' \subset U$.

(ii) (a) If $U \subset \mathbb{R}^n$ is open and $K \subset U$ is compact, there exists a nonnegative function $f \in C_c^\infty(U, \mathbb{R})$ such that $f > 0$ on K.[27]

(b) In fact, f can be chosen in $C_c^\infty(U, [0,1])$ to be one on K.

Proof of Proposition 8.70. **1.** First we find compact sets $K_i \subset U_i$ whose interiors cover K. Set

$$C_1 = K \setminus (U_2 \cup \cdots \cup U_N).$$

Then $C_1 \subset U_1$ (why?) is compact. By (i) we can find a compact set K_1 such that $C_1 \subset \text{int } K_1$ and $K_1 \subset U_1$. Suppose inductively that compact sets $K_1, \ldots, K_k$ have been chosen with $K_i \subset U_i$ so that

$$\{\text{int } K_1, \ldots, \text{int } K_k, U_{k+1}, \ldots, U_N\}$$

covers K. Let

$$C_{k+1} = K \setminus (\text{int } K_1 \cup \cdots \cup \text{int } K_k \cup U_{k+2} \cup \cdots \cup U_N)\,.$$

Then $C_{k+1} \subset U_{k+1}$ is compact, and by (i) we can find a compact K_{k+1} such that $C_{k+1} \subset \text{int } K_{k+1}$ and $K_{k+1} \subset U_{k+1}$.

2. Using (ii)(a), for each i choose a nonnegative function $\phi_i \in C_c^\infty(U_i, \mathbb{R})$ which is positive on K_i. Since $K \subset \cup_{k=1}^n \text{int } K_i := U$, the function $\phi_1 + \cdots + \phi_n$ is positive on the open set U. On U define

$$\psi_i = \frac{\phi_i}{\phi_1 + \cdots + \phi_N},\ i = 1, \ldots, N.$$

Use (ii)(b) to choose $f \in C_c^\infty(U, [0,1])$ which is one on K, and set $\rho_i = f\psi_i$ to get the desired partition of unity.

□

8.15 Rellich's theorem

In our existence theory for elliptic problems we used the following theorem:

Theorem 8.71 (Rellich A). *Let $U \subset \mathbb{R}^n$ be a bounded open set with C^1 boundary. In fact $i : H^1(U) \to L^2(U)$ is compact.*[28]

[27] Here $C_c^\infty(U, \mathbb{R})$ is the set of C^∞ functions $f : U \to \mathbb{R}$ with compact support contained in U.

[28] This implies immediately that $i : H_0^1(U) \to L^2(U)$ is compact.

We shall prove Theorem 8.71 by first proving a variant of it, Theorem 8.73, which follows easily from the Arzela-Ascoli theorem via the Fourier transform.

Definition 8.72. *Let $U \subset \mathbb{R}^n$ be an open set. For $s \in \mathbb{R}$ $H^s_{\overline{U}}(\mathbb{R}^n) := \{u \in H^s(\mathbb{R}^n) : \text{supp } u \subset \overline{U}\}$.*

Theorem 8.73 (Rellich B)**.** *Let $U \subset \mathbb{R}^n$ be bounded and open. Then $i : H^1_{\overline{U}}(\mathbb{R}^n) \to L^2_{\overline{U}}(\mathbb{R}^n)$ is compact.*

Proof. **1.** Let (f_k) be a bounded sequence in $H^1_{\overline{U}}(\mathbb{R}^n)$. That is, suppose there exists M such that

$$|f_k|_{H^1(\mathbb{R}^n)} \leq M \ \forall k. \tag{8.83}$$

It suffices to show there exists a subsequence of (f_k) that converges in $L^2(\mathbb{R}^n)$.

2. Choose $\phi \in \mathcal{D}(\mathbb{R}^n)$ such that $\phi = 1$ on a neighborhood of $\overline{U}$. Then $\phi f_k = f_k$ for all k, and hence

$$\hat{f}_k = \hat{\phi} * \hat{f}_k \text{ and } \partial_j \hat{f}_k = \partial_j \hat{\phi} * \hat{f}_k = -i(\widehat{x_j \phi}) * \hat{f}_k. \tag{8.84}$$

We claim that the sequences $(\hat{f}_k)$, $(\partial_j \hat{f}_k)$ are uniformly bounded on $\mathbb{R}^n$. Indeed, we have

$$|\hat{f}_k(\xi)| \leq \int |\hat{\phi}(\xi - \eta)| \hat{f}_k(\eta)| d\eta \leq |f_k|_{L^2} |\hat{\phi}|_{L^2} \leq M |\phi|_{L^2} \ \forall k, \tag{8.85}$$

and similarly for $\partial_j \hat{f}_k$.

3. Write $\mathbb{R}^n = \cup_{i=1}^\infty K_i$ where $K_i = \overline{B(0,i)}$. Step **2** implies that for each i, the sequence $(\hat{f}_k)$ is uniformly bounded and equicontinuous on K_i. Hence, Arzela-Ascoli implies there is a subsequence $(\widehat{f_{K_i,j}})_{j=1}^\infty$ that converges uniformly on K_i. The diagonal sequence $(\widehat{f_{K_i,i}})_{i=1}^\infty$, which we now relabel $(\hat{f}_i)$, has the property that for any $K \subset \mathbb{R}^n$ compact $(\hat{f}_i)$ converges uniformly on K. We claim that (f_i) converges in $L^2(\mathbb{R}^n)$. Indeed, for any $R > 0$ and i, j we have

$$\begin{aligned} |f_i - f_j|^2_{L^2(\mathbb{R}^n)} = \int_{|\xi| \leq R} |\hat{f}_i - \hat{f}_j|^2(\xi) d\xi + \int_{|\xi| \geq R} \langle \xi \rangle^{-2} \langle \xi \rangle^2 |\hat{f}_i - \hat{f}_j|^2(\xi) d\xi \leq \\ \sup_{|\xi| \leq R} |\hat{f}_i - \hat{f}_j|^2 \int_{|\xi| \leq R} d\xi + \langle R \rangle^{-2} |f_i - f_j|^2_{H^1(\mathbb{R}^n)} = A + B. \end{aligned} \tag{8.86}$$

For any fixed $\epsilon > 0$ by fixing R large enough, we make $|B| < \epsilon/2$. Then taking i, j large enough, we make $A < \epsilon/2$. □

Remark 8.74. (1) It is only when choosing R in (8.86) that we used the full strength of (8.83). In step **2** we used only the uniform boundedness of the $|f_k|_{L^2}$ for $k \in \mathbb{N}$.

(2) A similar proof shows that if $s, t \in \mathbb{R}$ with $s > t$, then the inclusion $i : H^s_{\overline{U}}(\mathbb{R}^n) \to H^t_{\overline{U}}(\mathbb{R}^n)$ is compact; see [Fol95].

To obtain Theorem 8.71 from Theorem 8.73 we use the following extension lemma, which has many additional applications.

Proposition 8.75. *Let $U \subset \mathbb{R}^n$ be bounded and open with C^1 boundary. Then there exists a continuous extension operator $E : H^1(U) \to H^1(\mathbb{R}^n)$.*

Proof. **1.** The main step is to define the extension for functions in $C^\infty(\overline{U})$. The result then follows by approximating elements of $H^1(U)$ by functions in $C^\infty(\overline{U})$.

2. Let $f \in C^\infty(\overline{U})$. Assume first that $x_0 \in \partial U$, that the boundary of U is flat near x_0, and that U is given by $x_n \geq 0$ near x_0. Let $\psi \in \mathcal{D}(\mathbb{R}^n)$ be a test function identically equal to one near x_0 and supported in a small ball B centered at x_0. Set $u = \psi f$. Define the extension u^e by

$$u^e = \begin{cases} u(x', x_n), \ x_n \geq 0 \\ -3u(x', -x_n) + 4u(x', -\frac{1}{2}x_n), \ x_n < 0 \end{cases} \tag{8.87}$$

It is easy to check that $u^e \in C^1(B)$ and that (with obvious notation)

$$|u^e|_{H^1(B)} \leq C|u|_{H^1(B^+)}. \tag{8.88}$$

3. If ∂U is not flat near x_0, use the fact that ∂U is C^1 to make a change of variables that reduces to the case considered in step 2. Use a partition of unity to define an extension f^e and to derive the estimate

$$|f^e|_{H^1(\mathbb{R}^n)} \leq C|u|_{H^1(U)} \text{ for } f \in C^\infty(\overline{U}) \tag{8.89}$$

by piecing together estimates similar to (8.88). To obtain the estimate (8.89) for an arbitrary $f \in H^1(U)$, choose a sequence $f_m \in C^\infty(\overline{U})$ converging to f in $H^1(U)$. The corresponding extensions f_m^e converge to an extension of f that satisfies (8.89). □

Proof of Theorem 8.71. Let (f_k) be a bounded sequence in $H^1(U)$. We must show that a subsequence converges in $L^2(U)$. Let (F_k) denote the corresponding bounded sequence (Ef_k) in $H^1(\mathbb{R}^n)$, where E is the extension operator of Proposition 8.75. Choose a cutoff function $\psi \in \mathcal{D}(\mathbb{R}^n)$ such that $\psi = 1$ on a neighborhood of $\overline{U}$ and supp $\psi \subset V \subset\subset \mathbb{R}^n$. By Theorem 8.73 a subsequence of (ψF_k) converges in $L^2_{\overline{V}}(\mathbb{R}^n)$. This implies the result.

□

8.16 Finite difference operators

Finite difference operators provide a way to "take derivatives without actually taking them". We use them in section 3.3.1 to carry out rigorously arguments that would otherwise fail because the solution being considered has insufficient regularity.

Definition 8.76. *Let $U \subset \mathbb{R}^n$ be open. Suppose $w \in L^2(U)$ has* supp $w \subset\subset U$. *For $1 \leq k \leq n$ and $|h|$ small define*

$$D_k^h w(x) = \frac{w(x + he_k) - w(x)}{h}, \tag{8.90}$$

where e_k is the $k-$th standard basis vector of $\mathbb{R}^n$.

Proposition 8.77. *Let $v, w \in L^2(U)$ and suppose* supp $w \subset\subset U$. *Then for $|h$ small:*

$$\begin{aligned} &(\alpha)\ \int_U v D_k^{-h} w \, dx = -\int_U w D_k^h v \, dx; \\ &(\beta)\ D_k^h(vw) = v^h D_k^h w + w D_k^h v, \ \textit{where } v^h(x) = v(x + he_k). \end{aligned} \tag{8.91}$$

Proof. The proofs are by direct computation. For example, to prove (α) we observe:

$$\begin{gathered} \int_U w D_k^h v dx = \int_U w(x) \left[\frac{v(x + he_k) - v(x)}{h} \right] dx \text{ and} \\ \int_U \frac{w(x) v(x + he_k)}{h} = \int_U \frac{v(x) w(x - he_k)}{h}. \end{gathered} \tag{8.92}$$

□

Proposition 8.78. *Let $1 \leq p < \infty$ and $V \subset\subset U \subset \mathbb{R}^n$. Set $D^h u = (D_1^h u, \ldots, D_n^h u)$ and write $D = \nabla u$. If $u \in W^{1,p}(U)$ and $0 < |h| < \frac{1}{2}\text{dist}(V, \partial U)$, then*

$$|D^h u|_{L^p(V)} \leq C |Du|_{L^p(U)}. \tag{8.93}$$

Proof. **1.** First suppose $u \in W^{1,p}(U) \cap C^1(U)$. Then for $x \in V$ and $q = \frac{p}{p-1}$:

$$\begin{aligned} &u(x + he_i) - u(x) = h \int_0^1 u_{x_i}(x + the_i) dt \Rightarrow \\ &\quad |D_i^h u(x)| \leq \int_0^1 |Du(x + the_i)| dt \leq |Du(x + the_i)|_{L^p([0,1])} 1_{L^q([0,1])} = |Du(x + the_i)|_{L^p([0,1])} \Rightarrow \\ &\qquad |D_i^h u(x)|^p \leq \int_0^1 |Du(x + the_i)|^p dt. \end{aligned} \tag{8.94}$$

Thus, for some C:

$$\int_V |D^h u|^p dx \leq C \sum_{i=1}^n \int_V \int_0^1 |Du(x + the_i)|^p dt dx \leq nC \int_U |Du|^p dx. \tag{8.95}$$

2. Now suppose $u \in W^{1,p}(U)$ and choose W such that $V \subset\subset W \subset\subset U$. By Proposition 3.3 we can choose $u_j \in C^1(U)$ such that $u_j \to u$ in $W^{1,p}(W)$. This implies

$$\begin{aligned} &D^h u_j \to D^h u \text{ in } L^p(W) \text{ for each } h, \\ &Du_j \to Du \text{ in } L^p. \end{aligned} \tag{8.96}$$

By step **1**

$$|D^h u_j|_{L^p(V)} \leq C|Du_j|_{L^p(W)} \text{ for all } j.$$

With (8.96) this implies

$$|D^h u|_{L^p(V)} \leq C|Du|_{L^p(W)} \leq C|Du|_{L^p(U)}.$$

□

Proposition 8.79. *Let $1 < p < \infty$, $V \subset\subset U \subset \mathbb{R}^n$, and $u \in L^p(U)$. Suppose there exists $0 < \delta \leq \frac{1}{2}\text{dist}(V, \partial U)$ and $C > 0$ such that for $0 < |h| < \delta$ we have $|D^h u|_{L^p(V)} \leq C$. Then $Du \in L^p(V)$ and $|Du|_{L^p(V)} \leq C$.*

In the proof we will use Propositions 8.62 and 8.63.

Proof. For some $C > 0$ we have

$$\sup_{|h| \leq \delta} |D_i^{-h} u|_{L^p(V)} \leq C.$$

Choose a sequence $h_k \to 0$. By Proposition 8.63 there is a subsequence (also labeled by h_k) such that

$$D_i^{-h_k} u \to v_i \text{ weakly for some } v_i \in L^p(V). \tag{8.97}$$

Proposition 8.62 implies

$$|v_i|_{L^p(V)} \leq C. \tag{8.98}$$

Thus, to finish it suffices to show $v_i = u_{x_i}$ as elements of $\mathcal{D}'(V)$. Let $\phi \in \mathcal{D}(V)$. Then

$$\begin{aligned} &\int_V u\phi_{x_i} dx = \lim_{h_k \to 0} \int_V u D_i^{h_k} \phi \, dx = -\lim_{h_k \to 0} \int_V D_i^{-h_k} u\phi \, dx = -\int_V v_i \phi \, dx \Rightarrow \\ &v_i = u_{x_i}. \end{aligned} \tag{8.99}$$

Here we used (8.97) for the last equality before $\Rightarrow$. □

8.17 Compactly supported distributions $\mathcal{E}'(U) \subset \mathcal{D}'(U)$.

We show that compactly supported distributions $u \in \mathcal{D}'(\mathbb{R}^n)$ are tempered distributions, and have Fourier transforms that are C^∞ functions with polynomial growth.

Definition 8.80. *Let $U \subset \mathbb{R}^n$ be open. Define*

$$\mathcal{E}'(U) = \{u \in \mathcal{D}'(U) : \operatorname{supp} u \subset K_u \text{ for some compact set } K_u \subset U\}.$$

We will show that $\mathcal{E}'(U)$ can be identified with the dual space of $C^\infty(U)$, when $C^\infty(U)$ is given its standard Fréchet topology; recall (8.29).

Proposition 8.81. *There exists a bijective linear map $T : (C^\infty(U))' \to \mathcal{E}'(U)$.*

Proof. **1.** Let $i : \mathcal{D}(U) \to C^\infty(U)$ be the inclusion map. Then i is injective and continuous (by Proposition 8.24) with dense range. To see the last point, let $f \in C^\infty(U)$, choose K_j as in (8.27), and for each j choose $\rho_j \in \mathcal{D}(U)$ such that $0 \le \rho_j \le 1$ and $\rho_j = 1$ on K_j. Then $f_j := \rho_j f$ converges to f in $C^\infty(U)$. (check)

2. Define T to be the transpose map

$$i^t : (C^\infty(U))' \to \mathcal{D}'(U) \tag{8.100}$$

given by

$$(i^t u)(\phi) := (u \circ i)(\phi) = u(i(\phi)) \text{ for } u \in (C^\infty(U))' \text{ and } \phi \in \mathcal{D}(U).$$

Then T is injective, since i has dense range; hence T is bijective onto its range.[29] Since T is injective, we can identify $(C^\infty(U))'$ with a subspace of $\mathcal{D}'(U)$. This justifies calling the elements of $(C^\infty(U))'$ distributions.

3. Now we show $R(T) \subset \mathcal{E}'(U)$. If $u \in \mathcal{D}'(U)$ is in the range of T, then u is the restriction to $\mathcal{D}(U)$ of a continuous linear functional on $C^\infty(U)$.[30] Elements of $(C^\infty(U))'$ were characterized in Exercise 8.9a. Thus, (8.32) implies

$$\exists C_N, N \text{ such that } |u(\phi)| \le C_N \sup_{|\alpha| \le N, x \in K_N} |\partial^\alpha \phi| \ \ \forall \phi \in \mathcal{D}(U) \subset C^\infty(U). \tag{8.101}$$

The inequality (8.101) shows that $u(\phi) = 0$ if $\operatorname{supp} \phi \subset K_N^c$. Thus, $\operatorname{supp} u \subset K_N$.

4. To finish showing $R(T) = \mathcal{E}'(U)$, we suppose $u \in \mathcal{E}'(U)$ has $\operatorname{supp} u \subset K \subset\subset U$, and show that u can be extended to an element of $(C^\infty(U))'$. Choose $\chi \in \mathcal{D}(U)$ with $\chi = 1$ on a neighborhood of K, and define an extension u_e of u by

$$u_e(\phi) := u(\chi\phi) \text{ for all } \phi \in C^\infty(U). \tag{8.102}$$

It is easy to check that $u_e(\phi)$ is independent of the choice of χ as above, and that u_e defines an element of $(C^\infty(U))'$ that extends u. □

[29] We have $i^t u = 0 \Leftrightarrow u|_{\mathcal{D}(U)} = 0 \Leftrightarrow u = 0$ on $C^\infty(U)$.

[30] Equivalently, u can be *extended* from $\mathcal{D}(U)$ to be a continuous linear functional on $C^\infty(U)$.

An argument similar to the one in steps **1** and **2** of the above proof, but which now uses the injective continuous inclusion with dense range $j : \mathcal{S}(\mathbb{R}^n) \to C^\infty(\mathbb{R}^n)$, shows that the transpose map $j^t : (C^\infty(\mathbb{R}^n))' \to \mathcal{S}'(\mathbb{R}^n)$ is injective. With Proposition 8.81 this immediately implies

Proposition 8.82. *There exists an injective linear map $K : \mathcal{E}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$. Thus, $\mathcal{E}'(\mathbb{R}^n)$ can be identified with a subspace of $\mathcal{S}'(\mathbb{R}^n)$.*

Indeed, just take $K = j^t \circ T^{-1}$. Note that Proposition 8.82 implies that the Fourier transform $\hat{u} \in \mathcal{S}'(\mathbb{R}^n)$ is defined for $u \in \mathcal{E}'(\mathbb{R}^n)$.

Proposition 8.83. *If $u \in \mathcal{E}'(\mathbb{R}^n)$, then*
(a) $\hat{u} \in C^\infty(\mathbb{R}^n)$;
(b) $\hat{u}(\xi) = \langle u, e^{-ix\xi}\rangle (2\pi)^{-n/2}$;
(c) there exist N, C_N such that $|\hat{u}(\xi)| \le C_N \langle \xi \rangle^N$ for all $\xi \in \mathbb{R}^n$.

Proof. **1.** Part (a) follows directly from (b), and (c) follows from (b) if we take $\phi(x) = e^{-ix\xi}$ in (8.32) .

2. Proof of (c). Suppose supp $u \subset B(0, R)$ for some $R > 0$. Choose $\psi \in \mathcal{D}(\mathbb{R}^n)$ such that $\psi = 1$ on $B(0, R+1)$. Then we have

$$u = \psi u \text{ and } \hat{u} = \widehat{\psi u} = \hat{u} * \hat{\psi}. \tag{8.103}$$

Choosing $\phi \in \mathcal{S}'(\mathbb{R}^n)$ such that $\hat{\phi} = \psi$, we compute

$$\begin{aligned} \hat{u}(\xi) = (\hat{u} * \hat{\psi})(\xi) = (\hat{u} * \hat{\hat{\phi}}))(\xi) = (\hat{u} * \tilde{\phi})(\xi) \ \ (\text{here } \tilde{\phi}(\xi) = \phi(-\xi)) = \\ \langle \hat{u}, \tilde{\phi}(\xi - \cdot)\rangle (2\pi)^{-n/2} = \langle \hat{u}, \phi(\cdot - \xi)\rangle (2\pi)^{-n/2} = \\ \langle u, \widehat{\phi(\cdot - \xi)}\rangle (2\pi)^{-n/2} = \langle u, e^{-ix\xi}\hat{\phi}(x)\rangle (2\pi)^{-n/2} = \langle u, e^{-ix\xi}\rangle (2\pi)^{-n/2}. \end{aligned} \tag{8.104}$$

□

8.18 Pseudolocality lemma used in Chapter 6

The proposition below was used in the proof of microlocal elliptic regularity, Theorem 6.8.

Proposition 8.84. *Let $q(\xi) \in C^\infty(\mathbb{R}^n)$ be homogeneous of degree $-m$ in $|\xi| \ge 1$, where $m \in \mathbb{N}$. If $f \in \mathcal{E}'(\Omega)$ and f is C^∞ near $x_0 \in \Omega$, then $q(D)f$ is C^∞ near x_0. Equivalently, we have* sing supp $q(D)f \subset$ sing supp f.[31]

[31] Here $q(D) : \mathcal{S}' \to \mathcal{S}'$ is defined by $\widehat{q(D)u} := q(\xi)\hat{u}$. Proposition 8.84 (pseudolocality) holds for more general pseudodifferential operators $q(x, D)$ with symbols $q(x, \xi)$ by a proof close to that given here. Note that in general it is *not* true that supp $q(x, D)f \subset$ supp f (locality), although this is true for differential operators.

Proof. **1. Preparation.** For any $u \in \mathcal{E}'(\mathbb{R}^n)$ we define $q(D)u \in \mathcal{D}'(\mathbb{R}^n)$ by

$$\langle q(D)u, \phi\rangle = \int\int \langle e^{i(x-y)\xi}, u(y)\rangle_y \, q(\xi)\phi(x)dxd\xi, \ \phi \in \mathcal{D}(\mathbb{R}^n) \tag{8.105}$$

where $\langle\cdot,\cdot\rangle_y$ is the distributional pairing in the y variable, and the integrals are absolutely convergent if $\int dx$ is done first. Let $\chi(\xi)$ be a test function such that $\chi = 1$ on $|\xi| \leq 1$. Then we have[32]

$$\langle q(D)u, \phi\rangle = \lim_{\epsilon\to 0}\int \left\langle \int e^{i(x-y)\xi} q(\xi)\chi(\epsilon\xi)d\xi, u(y)\right\rangle_y \phi(x)dx. \tag{8.106}$$

For use below observe also that for $x \neq y$:

$$Le^{i(x-y)\xi} = e^{i(x-y)\xi}, \text{ where } L := \frac{1}{|x-y|^2}\sum_{i=1}^n D_{\xi_i}^2. \tag{8.107}$$

2. Extending f by 0 outside Ω, we can view it as an element of $\mathcal{E}'(\mathbb{R}^n)$. Choose open sets ω_1, ω such that $x_0 \in \omega_1 \subset\subset \omega \subset\subset \Omega$ and $f \in C^\infty(\omega)$, and choose $\psi \in \mathcal{D}(\omega)$ such that $\psi = 1$ on a neighborhood of $\overline{\omega_1}$. We will show $q(D)f \in C^\infty(\omega_1)$.

We can write

$$q(D)f = q(D)(\psi f) + q(D)[(1-\psi)f] := A + B, \tag{8.108}$$

where $A \in C^\infty(\mathbb{R}^n)$ since ψf is a test function. We will finish the proof by showing that there exists a function $b \in C^\infty(\omega_1)$ such that $\langle B, \phi\rangle = \langle b, \phi\rangle$ for all $\phi \in \mathcal{D}(\omega_1)$.

3. Let $\phi \in \mathcal{D}(\omega_1)$. Using (8.106) and then (8.107) and the fact that $\overline{\omega_1} \cap \operatorname{supp}(1-\psi) = \emptyset$, we have for k large:[33]

$$\begin{aligned}
&\langle q(D)[(1-\psi)f], \phi\rangle = \lim_{\epsilon\to 0}\int \left\langle \int e^{i(x-y)\xi} q(\xi)\chi(\epsilon\xi)d\xi, (1-\psi(y))f(y)\right\rangle \phi(x)dx = \\
&\lim_{\epsilon\to 0}\int \left\langle \int e^{i(x-y)\xi} (L^t)^k[q(\xi)\chi(\epsilon\xi)]d\xi, (1-\psi(y))f(y)\right\rangle \phi(x)dx = \\
&\int \left\langle \int e^{i(x-y)\xi} (L^t)^k q(\xi)d\xi, (1-\psi(y))f(y)\right\rangle \phi(x)dx
\end{aligned} \tag{8.109}$$

Now (8.109) shows that the function

$$K(x,y) := \int e^{i(x-y)\xi}(L^t)^k q(\xi)d\xi$$

is independent of k large. Thus, K is C^∞ for (x,y) in a neighborhood of $\overline{\omega_1} \times \operatorname{supp}(1-\psi)f$, and we can take

$$b(x) = \langle K(x,y), (1-\psi(y))f(y)\rangle$$

as the function $b \in C^\infty(\omega_1)$ sought in step **2**.

□

[32] Insert $\chi(\epsilon\xi)$ in (8.105) and use LDC to justify the limit as $\epsilon \to 0$. For ϵ fixed use Fubini to switch dx, $d\xi$; then use Riemann sums to move u outside the $d\xi$ integral.

[33] Here $L^t = L$, where L^t is the formal transpose of L. We use the fact that $\partial_\xi^\alpha q$ is homogenous of degree $-m-|\alpha|$ in $|\xi| \geq 1$.

8.19 Extended definition of convolution

In our proof of Proposition 7.12 in section 7.3, we used convolutions $u * v$ where u and v are distributions neither of which necessarily has compact support or is tempered. The next proposition shows that such convolutions sometimes make sense.

Proposition 8.85. *Let $u, v \in \mathcal{D}'(\mathbb{R}^n)$ have supports S_u, S_v such that the map*

$$\mu : S_u \times S_v \to \mathbb{R}^n \text{ given by } \mu(x, y) = x + y \tag{8.110}$$

is proper.[34] *Then $u * v$ is well-defined, agrees with the earlier definition when $u \in \mathcal{D}'(\mathbb{R}^n)$, $v \in \mathcal{D}(\mathbb{R}^n)$, and satisfies*

$$\begin{aligned} &u * v = v * u \\ &\partial_j(u * v) = \partial_j u * v = u * \partial_j v \\ &\operatorname{supp}(u * v) \subset \overline{\operatorname{supp}(u) + \operatorname{supp}(v)} \\ &u * (v * w) = (u * v) * w. \end{aligned} \tag{8.111}$$

The associative property holds under the condition that the map

$$\mu : S_u \times S_v \otimes S_w \to \mathbb{R}^n \text{ given by } \mu(x, y, z) = x + y + z$$

is proper.

Sketch of proof. **1. Definition.** Let $\epsilon > 0$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. Let S_u^ϵ, S_v^ϵ be closed $\epsilon-$neighborhoods of S_u, S_v. It is easy to check that the map

$$\mu : S_u^\epsilon \times S_v^\epsilon \to \mathbb{R}^n \text{ given by } \mu(x, y) = x + y$$

is also proper, and thus

$$K^\epsilon(\phi) := (S_u^\epsilon \times S_v^\epsilon) \cap \operatorname{supp} \phi(x + y) \tag{8.112}$$

is compact. Choose functions $\rho_u(x)$, $\rho_v(y)$ such that $\rho(x, y) := \rho_u(x) \otimes \rho_v(y)$ is supported in $K^\epsilon(\phi)$ with $\rho = 1$ in a neighborhood of $K_0(\phi)$. Now define

$$\langle u * v, \phi \rangle := \langle u \otimes v, \rho(x, y)\phi(x + y) \rangle. \tag{8.113}$$

By slight variants of earlier arguments, one checks that this definition is independent of the choice of ρ as above, that $u * v \in \mathcal{D}'(\mathbb{R}^n)$, and that the properties 8.111 hold.

2. Consistency. Suppose $u \in \mathcal{D}'(\mathbb{R}^n)$, $v \in \mathcal{D}(\mathbb{R}^n)$. Applying the earlier definition we have for $\phi \in \mathcal{D}(\mathbb{R}^n)$:[35]

$$\begin{aligned} \langle u * v, \phi \rangle = \int \langle u(y), v(x - y) \rangle_y \phi(x) dx = \langle u(y), \int v(x - y)\phi(x) dx \rangle = \\ \langle u(y), \int v(z)\phi(y + z) dz \rangle = \langle \rho_u(y)u(y), \int \rho_u(z)v(z)\phi(y + z) dz \rangle = \\ \langle u \otimes v, \rho(y, z)\phi(y + z) \rangle, \end{aligned} \tag{8.114}$$

for ρ_u, ρ_v, ρ as chosen above. □

[34] This means $\mu^{-1}(K)$ is compact when K is compact.
[35] See (1.152) and Remark 1.68.

The result of the next exercise is used in the proof Proposition 7.12.

Exercise 8.19a. (a) Show that if $u \in \mathcal{D}'(\mathbb{R}^n)$ and $v \in \mathcal{E}'(\mathbb{R}^n)$, then the map μ as in (8.110) is proper, and thus $u * v$ is well-defined.
(b) Suppose $u, v \in \mathcal{D}'(\mathbb{R}^n)$ with

$$\operatorname{supp} u \subset \{x = (x', x_n) \in \mathbb{R}^n : x_n \geq |x'|\} \text{ and } \operatorname{supp} v \subset \{x_n \geq 0\}. \tag{8.115}$$

Show that the map μ as in (8.110) is proper, and thus $u * v$ is well-defined.

9 Additional exercises.

Problem 1: Recall the fundamental existence and uniqueness theorem for ODEs.

Theorem 9.1. *Let $D \subset \mathbb{R}^n$ be an open, connected set, let $T > 0$, and suppose $f : [0,T] \times D \to \mathbb{R}^n$ is continuous. Suppose also that there exists $K > 0$ such that*[1]

$$|f(t,x) - f(t,y)| \leq K|x-y| \text{ for all } (t,x),(t,y) \in [0,T] \times D. \tag{9.1}$$

Consider the initial value problem for the unknown function $x(t)$:

$$x' = f(t,x), \quad x(0) = x_0 \in D. \tag{9.2}$$

There exists a $\delta > 0$ such that (9.2) has a unique C^1 solution on $[0,\delta]$. Moreover, if D' is a compact subset of D, the time of existence δ can be chosen uniformly for $x_0 \in D'$.

Prove Theorem 9.1 using the contraction mapping theorem, after first writing (5.2) as an equivalent integral equation

$$x(t) = \int_0^t f(s, x(s))ds + x_0.$$

The proof can be found in many references, but see if you can do it without looking. It's not too hard.

Problem 2: Consider the vector field $X = \partial_t + a_1(x)\partial_{x_1} + a_2(x)\partial_{x_2} + \cdots + a_n(x)\partial_{x_n}$, where the $a_j \in C^1(\mathbb{R}^n, \mathbb{R})$.[2] Show that the initial value problem for the real-valued unknown function $u(t,x)$:

$$Xu = 0, \; u|_{t=0} = g \in C_c^1(\mathbb{R}^n)$$

has a unique solution $u \in C^1([0,\delta] \times \mathbb{R}^n)$ for some $\delta > 0$. This generalizes the result we proved in class for the special case where the a_j are constant functions.

Problem 3: Let $f : \mathbb{R} \to \mathbb{R}$ be continuous but not C^1. Show that $u(x,t) := f(x-t)$ is a weak solution of $(\partial_t^2 - \partial_x^2)u = 0$ on $\mathbb{R}^2$.

Problem 4: Suppose $u \in C^2(\mathbb{R}^n)$, $\Delta u = 0$ on $\mathbb{R}^n$, and $|u(x)| \leq C|x|^m$ for some $m \in \mathbb{N}$. Show u is a polynomial of degree $\leq m$.

[1] The constant K is referred to as a "Lipschitz constant".
[2] This is the space of C^1 functions on $\mathbb{R}^n$ taking values in $\mathbb{R}$.

Problem 5: Let $u \in C(U)$. We showed in class that if $u(x) = \frac{\int_{\partial B(x,r)} u(y)dS(y)}{A(\partial B(x,r))}$ for all $B(x,r) \subset U$, then $u(x) = \frac{\int_{B(x,r)} u(y)dy}{V(B(x,r))}$ for all $B(x,r) \subset U$. Prove the converse.

Problem 6: Suppose $u \in C^2(\mathbb{R}^n \setminus 0) \cap C(\mathbb{R}^n)$ is harmonic in $\mathbb{R}^n \setminus 0$ and bounded. Show that u is constant on $\mathbb{R}^n$. (This is a challenging and interesting problem. If you can solve it, please show me your solution.)

Problem 7: Metric on $C^k(\Omega, \mathbb{R})$ for Ω open. In this problem we consider the natural and standard way of defining a metric on $C^k(\Omega, \mathbb{R})$ when $\Omega \subset \mathbb{R}^n$ is an *open* set. This metric makes $C^k(\Omega, \mathbb{R})$ a Fréchet space.[3]

Let $\Omega \subset \mathbb{R}^n$ be open. Write $\Omega = \cup_{j=1}^{\infty} K_j$ as the union of an increasing sequence of compact subsets K_j with the property that for any compact $K \subset \Omega$, there exists j such that $K \subset K_j$. For example, one can take[4]

$$K_j = \left\{ x \in \Omega : \operatorname{dist}(x, b\Omega) \geq \frac{1}{j} \right\} \cap \overline{B(0,j)}.$$

For each j define a "seminorm" on $C^k(\Omega, \mathbb{R})$ by[5]

$$\rho_j(f) := \sup_{x \in K_j,\ |\alpha| \leq k} |\partial^\alpha f(x)|. \tag{9.3}$$

Finally, for $f, g \in C^k(\Omega, \mathbb{R})$ define

$$d(f,g) = \sum_{j=1}^{\infty} 2^{-j} \frac{\rho_j(f-g)}{1 + \rho_j(f-g)}. \tag{9.4}$$

(a) Show that d is a metric. Check all three conditions, but write out only the proof of the triangle inequality.

(b) Let (f_n) be a sequence in $C^k(\Omega, \mathbb{R})$ and $f \in C^k(\Omega, \mathbb{R})$. Show that $f_n \to f$ in the metric space $C^k(\Omega, \mathbb{R})$ if and only if given any compact set $K \subset \Omega$ and multi-index α with $|\alpha| \leq k$, the sequence $\partial^\alpha f_n \to \partial^\alpha f$ uniformly on K.[6]

(c) Show that $C^k(\Omega, \mathbb{R})$ with the metric d is a complete metric space.

Remark. (1) Every Banach space is a Fréchet space, but not vice versa. In particular, it turns out that the space $C^k(\Omega, \mathbb{R})$ is not normable. That is, there is no norm $|\cdot|$ on $C^k(\Omega, \mathbb{R})$ such that $d(f,g) = |f - g|$.

[3] A Fréchet space is a locally convex topological vector space with a topology given by a complete, translation-invariant metric.

[4] If $\Omega = \mathbb{R}^n$ ignore the first set in the intersection that defines K_j.

[5] A "seminorm" $\rho : V \to \{x \geq 0\}$ on a vector space V is defined just like a norm, except that we remove the condition: $|v| = 0 \Rightarrow v = 0$. Observe that ρ_j in (9.3) is not a norm on $C^k(\Omega, \mathbb{R})$. (why?)

[6] You might want to focus first on the case $k = 0$ or $k = 1$.

(2) One can similarly make $C^\infty(\Omega, \mathbb{R})$ a complete metric space. In that case we modify the definition of ρ_j as follows:

$$\rho_j(f) := \sup_{x \in K_j,\, |\alpha| \le j} |\partial^\alpha f(x)|,$$

and define the metric as in (9.4). The obvious analogue of the conclusion in **(b)** above holds here.

(3) The metric topologies just defined on $C^k(\Omega, \mathbb{R})$ and $C^\infty(\Omega, \mathbb{R})$ are independent of the *choice* of the increasing sequence of compacts (K_j) with union Ω such that any compact $K \subset \Omega$ is contained in some K_j. (why?) The latter property holds if we require, for example, that $K_j \subset \text{int } K_{j+1}$ for all j (why?). Paul Hamrick has shown that independence of choice and also part (b) can fail if the sequence of compacts (K_j) is merely increasing with union Ω.

Problem 8: Let Φ be the fundamental solution for the Laplacian on $\mathbb{R}^n$ defined in Definition 1.1. We showed that for each $x \in \mathbb{R}^n$, the function $y \to \Phi(x - y)$ is locally integrable, and thus defines a distribution. Show that $\Delta_y \Phi(x - y) = \delta_x$ in the sense of distributions. (Here $\delta_x(\phi) = \phi(x)$ for any test function ϕ.) So in particular $\Delta\Phi = \delta$, where $\delta(\phi) = \phi(0)$.

Problem 9: Let $U \subset \mathbb{R}^n$ be open, suppose $K \subset U$ is compact, and recall the subspace $\mathcal{D}_K \subset \mathcal{D}(U)$ defined in Definition 1.22. Show that a linear functional $u : \mathcal{D}_K \to \mathbb{C}$ is continuous if and only if there exist constants C_K, N_K such that

$$|u(\phi)| \le C_K \sup_{x \in K, |\alpha| \le N_K} |\partial^\alpha \phi| \text{ for all } \phi \in \mathcal{D}_K.$$

Problem 10: Fundamental solution of the heat operator. Let

$$\Phi(x, t) = \frac{1}{(4\pi t)^{n/2}} e^{-|x|^2/4t}$$

be the heat kernel on $\mathbb{R}^n \times (0, \infty)$. Define the extension

$$\Phi_e(x, t) = \begin{cases} \Phi, & t > 0 \\ 0, & t \le 0 \end{cases}.$$

(a) Show that $\Phi_e \in L^1_{loc}(\mathbb{R}^n \times \mathbb{R})$, and thus $\Phi_e \in \mathcal{D}'(\mathbb{R}^n \times \mathbb{R})$.
(b) Show that Φ_e is a fundamental solution for the heat operator; that is,

$$(\partial_t - \Delta)\Phi_e = \delta \in \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}).$$

Hint: For $\eta > 0$ define $\Phi_\eta(x, t) = \begin{cases} \Phi_e, & t > \eta \\ 0, & t \le \eta \end{cases}$, and
a) show $\Phi_\eta \to \Phi_e$ in $\mathcal{D}'(\mathbb{R}^n \times \mathbb{R})$ as $\eta \to 0$;

b) show $(\partial_t - \Delta)\Phi_\eta \to \delta$ in $\mathcal{D}'(\mathbb{R}^n \times \mathbb{R})$ as $\eta \to 0$.

Problem 11: Let $U \subset \mathbb{R}^n$ be open. In section 8.9 we defined the standard Fréchet topology on $C^\infty(U)$. Show that a linear function $u : C^\infty(U) \to \mathbb{C}$ is continuous if and only if there exist a compact set K, $N \in \mathbb{N}$, and $C > 0$ such that

$$|u(\phi)| \leq C \sup_{x \in K, |\alpha| \leq N} |\partial^\alpha \phi| \text{ for all } \phi \in C^\infty(U).$$

(Caution: For any compact $K \subset U$ and $N \in \mathbb{N}$, the function

$$\rho_{K,N}(\phi) := \sup_{x \in K, |\alpha| \leq N} |\partial^\alpha \phi|$$

is a *seminorm*, but not a norm, on $C^\infty(U)$.)

Problem 12. Jump formula. Recall that if

$$H(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

is the Heaviside function and $u \in C(\mathbb{R}_t, \mathcal{S}'(\mathbb{R}^n))$, we define $H(t)u(t,x) \in \mathcal{D}'(\mathbb{R}_t \times \mathbb{R}^n)$ by

$$\langle H(t)u(t,x), \phi(t,x) \rangle := \int_0^\infty \langle u(t,\cdot), \phi(t,\cdot) \rangle dt \text{ for } \phi \in \mathcal{D}(\mathbb{R}_t \times \mathbb{R}^n).$$

Suppose $u(t,x) \in C^1(\mathbb{R}_t, \mathcal{S}'(\mathbb{R}^n))$. Show that [7]

$$\partial_t(H(t)u) = \delta(t) \otimes u(0,x) + H(t)\partial_t u. \tag{9.5}$$

Problem 13. Fundamental solutions. (a) Let $\Phi_t(x) = \mathcal{F}^{-1}(\frac{\sin(t|\xi|)}{|\xi|})$ as in (1.176). Use the jump formula to show that

$$H(t)\Phi_t(x)(2\pi)^{-n/2} \in \mathcal{D}'(\mathbb{R}^n_x \times \mathbb{R}_t)$$

is a fundamental solution for the wave operator $\partial_t^2 - \Delta$. This is called the "forward fundamental solution" since its support is contained in the forward light cone $\{(t,x) : t \geq 0, |x| \leq |t|\}$.

(b) Let $K_t(x) = \mathcal{F}^{-1}(e^{-t|\xi|^2})$.[8] Use the jump formula to show that

$$H(t)K_t(x)(2\pi)^{-n/2} \in \mathcal{D}'(\mathbb{R}^n_x \times \mathbb{R}_t)$$

is a fundamental solution for the heat operator $\partial_t - \Delta$.

[7] The formula (1.162) works just as well if $u \in C^1([0,\infty)_t, \mathcal{S}'(\mathbb{R}^n))$.

[8] So $K_t(x)(2\pi)^{-n/2} = (4\pi t)^{-n/2} e^{-|x|^2/4t}$ is the classical heat kernel.

Problem 14. 1. Use Lemma 8.66 and Remark 8.67 of the supplementary notes to show that if $s > \frac{n}{2}$, then $H^s(\mathbb{R}^n)$ is an algebra. That is, show that there exists $C_s > 0$ such that for any $u, v \in H^s$ the product $uv \in H^s$ and satisfies

$$|uv|_s \leq C_s |u|_s |v|_s.$$

Problem 15. Consider the semilinear initial value problem on $\mathbb{R}_t \times \mathbb{R}^n_x$ for the unknown $u(t,x)$:

$$P(t,x,\partial_{t,x})u = Q(u),\ u(0,x) = g(x),$$

where P is an $N \times N$ first-order symmetric hyperbolic system (as defined in class) and $g \in H^s(\mathbb{R}^n)$, where $s > n/2$. Here $Q(u) = B(u,u)$, where $B : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}^N$ bilinear.[9]

Use Picard iteration to prove that for $T > 0$ small enough, this problem has a solution $u \in C([0,T], H^s(\mathbb{R}^n))$.

Here are some comments on Picard iteration. Consider the sequence of problems

$$Pu_{n+1} = Q(u_n), u_{n+1}(0,x) = g(x),\ \text{where } u_0 := g.$$

Each such problem is linear in u_{n+1}. Show that by solving these problems we obtain a sequence (u_n) that is Cauchy in $C([0,T], H^s(\mathbb{R}^n))$ provided $T > 0$ is small enough, and use that fact to obtain the desired solution. The main tools for doing this are the estimate you proved in Problem 14 and our main existence/uniqueness theorem (including the energy estimate) for first-order symmetric hyperbolic systems.

Problem 16. Use the contraction mapping theorem to obtain the result of problem 15. Give explicit definitions of the complete metric space M and the contraction $T : M \to M$ that you use.

Before doing this problem, it might be helpful for you to review the proof of existence of solutions to initial value problems for nonlinear first-order ODEs.

Problem 17. Prove that the Fourier transform of a nonzero function in $\mathcal{D}(\mathbb{R}^n)$ cannot have compact support. (Hints: (a) Do the case $n = 1$ first. (b) Use some complex analysis.)

[9] Thus, Q is a quadratic form.

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